COM5231 · Introduction to Wireless and Ad Hoc Networking

# Complete Exam Preparation Pack

Overview · Study Notes (Topics 1–11) · Exam Pattern Analysis · Answers to All Past Papers · Practice Questions · Revision Guide

**SUPPLEMENTARY** content not found in the uploaded lecture notes

**NOTES ERROR** a mistake in the lecture notes, with the verified correction

**EXAM** how the topic is or could be examined

★ high-priority, frequently examined content

Exam preparation pack, built from the uploaded lecture materials and past papers

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- TEST 2 2025 (T2-2025; 100 marks)
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# Overview — Document Inventory, Topic Outline and Study Plan

## A. What was analysed

All 23 files in *networking.zip* were opened and read. The scanned PDF and the image-only slide deck were read visually, page by page.

### Lecture notes (14 files)

| # | File | Main topics | Status |
|---|---|---|---|
| 1 | Chapter 2 Transmission Fundamentals.pptx (50 slides) | Signals, analog/digital, Nyquist, Shannon, media, FDM/TDM. Slide 1 = **COM5231 course outline** | Read (formula slides are images; the PDF has the same example) |
| 2 | CHAPTER 2 Transmission Fundamentals_1.pdf (28 pp) | Textbook version of #1 with derivations | Read |
| 3 | The Wireless Channel.pptx (74 slides) | Spectrum/ISM/ICASA, propagation, antennas, free-space loss, path-loss exponent, noise, fading, channel correction, ASK/FSK/PSK, BER, review questions | Read |
| 4 | Wireless Communication.pdf (19 pp) | Textbook Ch. 5: detailed version of #3 | Read |
| 5 | Wireless Communication Problems.pdf (8 pp) | Ten wireless problems; worked power/distance/SNR calculations | Read (3 errors found, see Part 1) |
| 6 | Multiple access techniquesV2.pdf (31 pp, scanned) | FDMA, TDMA, CDMA, SDMA, comparison table, 1G–5G | Read visually |
| 7 | Fundamentals of Communications Access Technologies.pdf | FDMA, TDMA (T1, GSM), CDMA, OFDMA, SDMA, CSMA | Read |
| 8 | Multiplexing notes.pdf (4 pp) | FDMA, SDMA, TDMA (fixed/dynamic), CDMA, cellular | Read |
| 9 | OFDM.pdf (11 pp) | Orthogonality, OFDMA subcarrier methods, scheduling, SC-FDMA | **Pages 241–244 missing** (IFFT, PAPR) |
| 10 | Spread spectrum.pdf (5 pp) | FHSS, DSSS | **Ends mid-sentence** |
| 11 | Signal encoding techniques.pdf (11 pp) | ASK, FSK, PSK, QPSK, QAM, exercises | Read |
| 12 | WLANS_v2.pptx (47 slides) | 802.11 architecture, CSMA/CA, RTS/CTS/NAV, PCF, hidden/exposed station, PHY | Read |
| 13 | Chapter 15 WLANS by Forouzan.pdf (10 pp) | Wired vs wireless LAN, CSMA/CA, NAV, exposed station | Physical-layer table missing |
| 14 | BLUETOOTH AND IEEE 802.15_V2.pdf + Bluetooth and IEEE 802.15.pptx (90 image-only slides) | Bluetooth stack, profiles, piconets, radio, baseband, links, packets, FEC/ARQ, BT 3.0/4.0, 802.15.3/.4, ZigBee | Read (slides read visually) |

### Worksheets (2 files)

| # | File | Content | Status |
|---|---|---|---|
| 15 | Error detection and correction_v2.docx | Links, plus worked CRC examples and unsolved Hamming/checksum/CRC/parity exercises | No theory text; worked CRC examples verified correct |
| 16 | Bluetooth_Exercises_Reordered_20_Questions.pdf | 20 Bluetooth review questions (no answers) | Read |

### Assessments (7 photos = 5 papers)

| Paper | Marks | Note |
|---|---|---|
| **Exam paper** (green; no title or year; pages 2–4) | 5 questions × 20 = 100 | Page 1 not photographed |
| **Test 1, version A** (T1A) | 14 questions | |
| **COM5231 Test 1, version B** (T1B) | 9 questions | |
| **Test 2 2025** (T2-2025) | 100 | |
| **COM5231 Test 2 2026** (T2-2026) | 50 | |

No memorandums were provided, so all answers are model answers, not the lecturer's marking guide.

## B. Coverage gap: routing

Routing is **30% of the exam paper's marks** and **25–32% of the Test 2 papers** (about 16% of all 450 past-paper marks, since the Test 1 papers don't cover it). The topics are Dijkstra, Bellman-Ford, link-state vs distance-vector, count-to-infinity, MANET routing categories, and routing vs forwarding vs switching. **No uploaded notes cover these.** The course outline lists Ch. 3, Ch. 4, Ch. 10 and Ch. 13 (Agrawal, Ad hoc Networks), none of which were provided.

| Paper | Routing questions | Marks |
|---|---|---|
| Exam | Q2(d), Q3(a)–(d) | 30 / 100 |
| T2-2025 | Q1–Q3 | 25 / 100 |
| T2-2026 | Q1, Q9 | 16 / 50 |

These answers will be clearly labelled **SUPPLEMENTARY** and based on standard sources (Kurose & Ross; Agrawal).

## C. Module topic outline

1. Transmission fundamentals
2. The wireless channel
3. Channel correction mechanisms and capacity
4. Signal encoding techniques
5. Multiple access (FDMA, TDMA, CDMA, SDMA, cellular generations)
6. Spread spectrum (FHSS, DSSS)
7. OFDM, OFDMA, SC-FDMA

8. Error detection and correction (parity, checksum, Hamming, CRC)
9. Wireless LANs (IEEE 802.11)
10. Bluetooth and IEEE 802.15 (incl. ZigBee)
11. Routing and ad hoc networks (supplementary)

## D. Most-repeated past-paper questions

| Question | Appears in |
|---|---|
| Bluetooth profiles as an advantage for applications | Exam, T2-2025, T2-2026, BT exercises (4) |
| Path loss vs multipath vs interference | Exam, T1A, T1B (3) |
| FHSS and DSSS principles | Exam, T1A, T1B (3) |
| CDMA and why it is difficult to jam | Exam, T1A, T1B (3) |
| Two ISPs in a café on channel 11, then channels 1 and 11 | Exam, T1A, T2-2025 (3) |
| CRC calculation | Exam, T1B, T2-2026 (3) |
| One device as controller of two piconets | Exam, T2-2025, BT exercises (3) |
| Scatternet collision probability (80 frequencies) | Exam, T2-2025, BT exercises (3) |
| Dijkstra on the same Figure 1 network (from x, t and w) | Exam, T2-2025 (2) |
| OFDMA subcarrier approaches; three propagation mechanisms; hidden node and CSMA/CA (Figure 2); 802.11 ACKs; mobile node moving away; interference troubleshooting; Hamming; checksum; MANET routing; count-to-infinity | 2 papers each |

## E. Study plan

| Part | Content |
|---|---|
| **Part 1** | Study notes: Topics 1–4 |
| **Part 2** | Study notes: Topics 5–8 |
| **Part 3** | Study notes: Topics 9–11 |
| **Part 4** | Examination pattern analysis |
| **Parts 5–6** | Full answers to every past-paper question |
| **Part 7** | Additional practice questions with model answers |
| **Part 8** | Final revision guide and checklist |

At the end, everything is also combined into one complete PDF.

# Part 1 — Detailed Study Notes: Topics 1–4

### Topics in this part

1. Transmission Fundamentals
2. The Wireless Channel (propagation, path loss, noise, fading) + wireless calculations
3. Channel Correction Mechanisms and Capacity (FEC, equalisation, diversity, MIMO, AMC, bandwidth expansion)
4. Signal Encoding Techniques (ASK, FSK, PSK, QPSK, QAM, BER)

## How to read these notes

### Source codes used in citations

| Code | File |
|---|---|
| **[TF-S n]** | *Chapter 2 Transmission Fundamentals.pptx*, slide n |
| **[TF-B p]** | *CHAPTER 2 Transmission Fundamentals_1.pdf* (Beard & Stallings textbook), book page p |
| **[WC-S n]** | *The Wireless Channel.pptx*, slide n |
| **[WC-B §]** | *Wireless Communication.pdf* (textbook Chapter 5), section § |
| **[WCP]** | *Wireless Communication Problems.pdf* (Basics of Wireless Networks, §2.3) |
| **[SE]** | *Signal encoding techniques.pdf* (Forouzan Ch. 5) |
| **[MA p]** | *Multiple access techniquesV2.pdf*, page p |

### Labels

- Everything **without a label** comes from your lecture material.
- SUPPLEMENTARY marks explanation or content that is *not* in your uploaded notes. Use it to understand, but in the exam prefer the lecturer's wording.
- NOTES ERROR marks a place where your notes contain a mistake. The correction is shown and verified.
- EXAM marks how the topic has been or could be examined. "Asked in:" lists real past-paper questions.

**Past-paper short names**: *Exam* (green paper, pp. 2–4), *T1A* (Test 1, 14 questions), *T1B* (COM5231 Test 1, 9 questions), *T2-2025*, *T2-2026*.

# TOPIC 1 — TRANSMISSION FUNDAMENTALS

## 1.1 Introduction

Every wireless system moves information by sending an **electromagnetic signal** from a transmitter to a receiver. Before you can understand fading, OFDM, CDMA or Wi-Fi, you need the basic vocabulary: what a signal is, how it can be described in **time** and in **frequency**, what **bandwidth** is, and what limits the **maximum data rate** of a channel (Nyquist and Shannon). This topic is the foundation of the whole module. It is the first chapter of the textbook (Beard & Stallings, *Wireless Communication Networks and Systems*) that the module follows [TF-S 1].

**Learning objectives** [TF-S 3]:

- Distinguish between digital and analog information sources.
- Explain how audio, data, image and video can be represented by electromagnetic signals.
- Discuss the characteristics of analog and digital waveforms.
- Explain the roles of frequencies and frequency components in a signal.
- Identify the factors that affect channel capacity.
- Compare and contrast various forms of wireless transmission.

## 1.2 Key definitions

| Term | Definition (lecturer's wording) | Source |
|---|---|---|
| **Electromagnetic signal** | A function of time that can also be expressed as a function of frequency; the signal consists of components of different frequencies. | [TF-S 4] |
| **Analog signal** | Signal intensity varies in a smooth fashion over time; no breaks or discontinuities. | [TF-S 5] |
| **Digital signal** | Signal intensity maintains a constant level for some period of time and then changes to another constant level. | [TF-S 5] |
| **Periodic signal** | An analog or digital signal pattern that repeats over time: $s(t + T) = s(t)$, $-\infty < t < +\infty$, where T is the period. | [TF-S 5] |
| **Aperiodic signal** | An analog or digital signal pattern that does not repeat over time. | [TF-S 7] |
| **Peak amplitude (A)** | Maximum value or strength of the signal over time; typically measured in volts. | [TF-S 7] |
| **Frequency (f)** | Rate, in cycles per second or hertz (Hz), at which the signal repeats. | [TF-S 7] |
| **Period (T)** | Amount of time it takes for one repetition of the signal; **$T = 1/f$**. | [TF-S 8] |
| **Phase (φ)** | Measure of the relative position in time within a single period of a signal. | [TF-S 8] |
| **Wavelength (λ)** | Distance occupied by a single cycle of the signal, or the distance between two points of corresponding phase of two consecutive cycles; **$\lambda = vT$**, and for radio **$\lambda = c/f$**. | [TF-S 8] |

| Term | Definition (lecturer's wording) | Source |
|---|---|---|
| **Fundamental frequency** | When all frequency components of a signal are integer multiples of one frequency, that frequency is the fundamental frequency. (The other components are called **harmonics** [TF-B 13].) | [TF-S 13] |
| **Spectrum** | Range of frequencies that a signal contains. | [TF-S 13] |
| **Absolute bandwidth** | Width of the spectrum of a signal. | [TF-S 13] |
| **Effective bandwidth (or just bandwidth)** | Narrow band of frequencies in which most of the signal's energy is contained. | [TF-S 13] |
| **Data** | Entities that convey meaning, or information. | [TF-S 17] |
| **Signals** | Electric or electromagnetic representations of data. | [TF-S 17] |
| **Transmission** | Communication of data by the propagation and processing of signals. | [TF-S 17] |
| **Channel capacity** | The maximum rate at which data can be transmitted over a given communication path, or channel, under given conditions. | [TF-S 26] |
| **Noise** | Unwanted signal, e.g. crosstalk, thermal noise, impulse noise from lightning; it degrades signal quality. | [TF-S 26] |
| **Signal-to-noise ratio (SNR, S/N)** | Ratio of the power in a signal to the power contained in the noise present at a particular point in the transmission; typically measured at a receiver. | [TF-S 30] |
| **Transmission medium** | Physical path between transmitter and receiver. | [TF-S 34] |
| **Multiplexing** | Carrying multiple signals on a single medium. | [TF-S 43] |

## 1.3 Time-domain concepts

### The sine wave: the basic analog signal

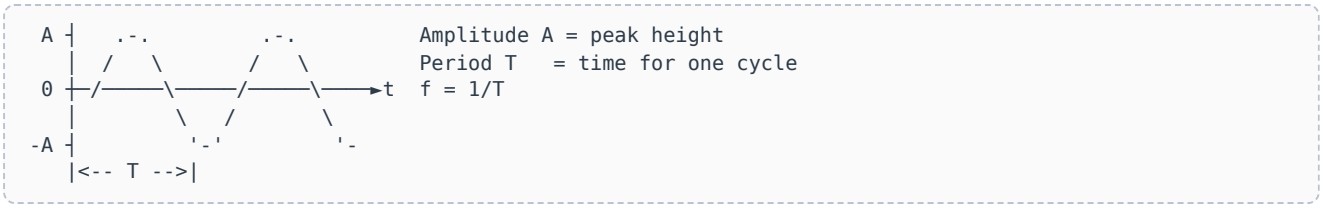
The general sine wave is [TF-S 10]:

$$s(t) = A \sin(2\pi ft + \varphi)$$

It has three parameters. Changing each one changes the wave differently (Figure 2.3 in the notes):

| Case | Parameters | Effect |
|---|---|---|
| (a) | A = 1, f = 1 Hz, φ = 0 | Reference wave; T = 1 s |
| (b) | A = 0.5 | Same shape, **half the height** (reduced peak amplitude) |
| (c) | f = 2 Hz | **Twice as many cycles** per second; T = ½ s |
| (d) | φ = π/4 rad (45°) | Wave is **shifted in time** by 1/8 of a period |

Remember: **2π radians = 360° = 1 period** [TF-S 10].




### Time vs distance [TF-S 12]

- With **time** on the horizontal axis, the graph shows the signal's value **at one point in space** as time passes.
- With **distance** on the horizontal axis, the graph shows the signal's value **at one instant in time** at different distances from the source.
- At a particular instant, the intensity of a radio wave varies sinusoidally with distance from the antenna. One cycle in space is the **wavelength λ**.

### Worked example: period, frequency, wavelength SUPPLEMENTARY practice

*A Wi-Fi signal has f = 2.4 GHz. Find T and λ.*

- $T = 1/f = 1/(2.4 \times 10^9)$ = **$4.17 \times 10^{-10}$ s (0.417 ns)**
- $\lambda = c/f = (3 \times 10^8)/(2.4 \times 10^9)$ = **0.125 m (12.5 cm)**

## 1.4 Frequency-domain concepts

**Key idea**: any electromagnetic signal can be shown to consist of a **collection of periodic analog signals (sine waves) at different amplitudes, frequencies and phases**. The **period of the total signal equals the period of the fundamental frequency** [TF-S 15]. This is Fourier analysis [TF-B 14].

**Example from the notes** [TF-B 13–14, TF-S 14]:

$$s(t) = (4/\pi) \times [\ \sin(2\pi ft) + (1/3)\ \sin(2\pi(3f)t)\ ]$$

- The components are at **f** and **3f**. Since 3f is an integer multiple of f, **f is the fundamental** and 3f is a harmonic.
- The spectrum extends from f to 3f, so the **absolute bandwidth = 3f – f = 2f**.
- Adding more odd harmonics (5f, 7f, …) makes the waveform look more and more like a **square wave** (Figure 2.5). A perfect square wave needs infinitely many harmonics, so it has **infinite bandwidth** [TF-B 15].

## 1.5 Relationship between data rate and bandwidth

**The greater the bandwidth, the higher the information-carrying capacity** [TF-S 16].

**Conclusions the lecturer lists** [TF-S 16]:

1. Any digital waveform (rectangular pulses) has **infinite bandwidth**.
2. **BUT** the transmission system will limit the bandwidth that can be transmitted.
3. **AND**, for any given medium, the greater the bandwidth transmitted, the **greater the cost**.
4. **HOWEVER**, limiting the bandwidth creates **distortions**.

Analogy from the slide: **bandwidth is the width of a pipe; data rate is the actual amount of digital data transmitted** [TF-S 16].

### Worked illustration: the three cases [TF-B 15–17]

A square wave of alternating 0s and 1s: each pulse lasts 1/(2f), so **data rate = 2f bps**.

| Case | Approximation used | f | Bandwidth | Data rate |
|---|---|---|---|---|
| I | f + 3f + 5f components | 1 MHz | 5f − f = 4 MHz | 2f = **2 Mbps** |
| II | f + 3f + 5f components | 2 MHz | 5f − f = 8 MHz | 2f = **4 Mbps** |
| III | f + 3f only (rougher wave) | 2 MHz | 3f − f = 4 MHz | 2f = **4 Mbps** |

**Lessons**:

- Case I → II: **doubling the bandwidth doubles the data rate**.
- Case I → III: the **same bandwidth (4 MHz) can carry different data rates**, depending on how well the receiver can tell a 0 from a 1 in the presence of noise. A rougher waveform uses less bandwidth per bit but is harder to detect.

## 1.6 Analog and digital data, signals and transmission

### Data [TF-S 18]

- **Analog data** take continuous values in some interval: video, audio (and sensor data such as temperature and pressure [TF-B 17]).
- **Digital data** take discrete values: text, integers.

### Signals [TF-S 19–20]

| | Analog signal | Digital signal |
|---|---|---|
| What it is | A continuously varying electromagnetic wave | A sequence of voltage pulses |
| Media | Copper (twisted pair, coax), fibre, atmosphere/space | Copper wire medium |
| Carries | Analog **and** digital data | Analog **and** digital data |
| Advantages | — | Generally **cheaper**; **less susceptible to noise** |
| Disadvantages | — | **Suffers more from attenuation** |

Attenuation of a digital signal: pulses become **rounded and smaller** as higher-frequency components are attenuated, which can quickly lead to loss of information [TF-S 22, TF-B 18].

### The four data/signal combinations and why each is used [TF-S 21]

| Combination | Device | Reason for choosing it |
|---|---|---|
| **Digital data → digital signal** | Encoder | Encoding equipment is **less expensive** than digital-to-analog equipment |
| **Analog data → digital signal** | **Codec** (coder–decoder) | Permits use of **modern digital transmission and switching equipment** |
| **Digital data → analog signal** | **Modem** (modulator–demodulator) | **Some media only propagate analog signals**, e.g. optical fibre and satellite |

| Combination | Device | Reason for choosing it |
|---|---|---|
| **Analog data → analog signal** | (e.g. telephone handset) | Analog data are **easily converted** to an analog signal |

Voice example: sound covers 20 Hz–20 kHz, speech energy is mostly 100 Hz–7 kHz, and the standard telephone channel is **300–3400 Hz** [TF-B 20].

### Analog vs digital transmission [TF-S 24–25]

| | Analog transmission | Digital transmission |
|---|---|---|
| Concern | Transmits analog signals **without regard to content** | **Concerned with the content** of the signal |
| Distance limit | Attenuation limits link length | Attenuation endangers data integrity |
| Long-distance device | **Amplifiers** (cascaded) | **Repeaters** |
| What the device does | Boosts energy, **but also boosts noise and causes distortion** | **Recovers** the 1s and 0s and **retransmits a new, clean signal**, so noise is not accumulated |
| Effect on data | Analog data tolerate distortion; **digital data get errors** | Works for digital signals, and for analog signals carrying digital data (the retransmission device recovers the data and generates a new clean analog signal) |

## 1.7 Channel capacity

### Four related concepts [TF-S 28]

1. **Data rate**: rate at which data can be communicated (bps).
2. **Bandwidth**: bandwidth of the transmitted signal, as constrained by the transmitter and the medium (Hz).
3. **Noise**: average level of noise over the path.
4. **Error rate**: rate at which errors occur (transmit 1, receive 0, or vice versa).

The goal: get **as high a data rate as possible at a given error rate for a given bandwidth**. The main constraint is **noise** [TF-B 22].

### 1.7.1 Nyquist bandwidth (noiseless channel) [TF-S 29, TF-B 22–23]

For a **noiseless** channel, the Nyquist formula gives the **theoretical maximum bit rate**.

- **Binary signals (2 levels): $C = 2B$**
- **Multilevel signalling: $C = 2B \log_2 M$**

where B = channel bandwidth (Hz), M = number of signal levels, C = bit rate (bps).

**Why it holds**: given bandwidth B, the highest **signal (symbol) rate** is 2B, a limit caused by intersymbol interference. Each symbol carries $\log_2 M$ bits [TF-B 22].

**Trade-off**: increasing M increases the data rate for the same bandwidth, **but** the receiver must now distinguish one of M signals instead of two. **Noise limits the practical value of M** [TF-B 23].

#### Worked example N1 (lecturer's Example 1) [TF-S 29]

*Noiseless channel, B = 4000 Hz, two signal levels. Maximum bit rate?*

1. Formula: $C = 2B \log_2 M$
2. Substitute: $C = 2 \times 4000 \times \log_2 2 = 2 \times 4000 \times 1$
3. **C = 8000 bps** ✔ (matches the slide's answer)

#### Worked example N2 (lecturer's Example 2) [TF-S 29]

*Send 250 kbps over a noiseless channel with B = 30 kHz. How many signal levels?*

1. $250\,000 = 2 \times 30\,000 \times \log_2 M$
2. $\log_2 M = 250\,000 / 60\,000 = 4.167$
3. M = 2^4.167 = 17.96
4. **M ≈ 18 levels** ✔ (the slide's answer)

**SUPPLEMENTARY**: practical systems use a **power of 2** for M (each level must map to a whole number of bits). With 16 levels you get only 240 kbps, which is too little, so a real design would use **M = 32**. If the question says "how many levels are needed", give 18 as the lecturer does, and add one line noting that 32 would be used in practice.

#### Worked example N3 (end-of-chapter exercise) [TF-S 50]

*Maximum data rate of a channel with B = 200 kHz and four levels of digital signalling?*

- $C = 2 \times 200\,000 \times \log_2 4 = 2 \times 200\,000 \times 2 =$ **800 kbps**

#### Worked example N4 (textbook) [TF-B 23]

*Voice channel, B = 3100 Hz.* Binary: C = 2B = **6200 bps**. With M = 8: C = 2 × 3100 × 3 = **18 600 bps**.

### 1.7.2 Signal-to-noise ratio (SNR) [TF-S 30, TF-B 23]

$$\text{SNR_dB} = 10 \log_{10} ( \text{signal power} / \text{noise power} )$$

- A **high SNR** means a **high-quality signal** and **fewer intermediate repeaters** are needed.
- **SNR sets the upper bound on the achievable data rate.**

Converting dB back to a ratio: **SNR = 10^(SNR_dB / 10)**.

### 1.7.3 Shannon capacity (noisy channel) [TF-S 31, TF-B 24]

Real channels are always noisy. Shannon's formula gives the **theoretical highest data rate for a noisy channel**:

$$C = B \log_2(1 + \text{SNR}) \text{ (SNR as a ratio, NOT in dB)}$$

**Limitations** [TF-S 31]:

- It is a **theoretical maximum**; in practice only much lower rates are achieved.
- It assumes **white (thermal) noise** only.
- **Impulse noise** is not accounted for.
- **Attenuation distortion and delay distortion** are not accounted for.

**Further observations** [TF-B 25]:

- C is the **error-free capacity**: if the actual information rate is below C, it is theoretically possible to find a code that gives error-free transmission. Shannon does not say how to find such codes, but the formula is a yardstick for practical schemes.
- One might think the rate can be increased just by raising **signal strength** or **bandwidth**. But stronger signals increase **nonlinearities and intermodulation noise**, and because noise is white, a **wider bandwidth admits more noise**, so **as B increases, SNR decreases**.

#### Worked example S1: Nyquist and Shannon together (lecturer's slides 32–33; textbook Example 2.1)

*The spectrum of a channel is between 3 MHz and 4 MHz; SNR_dB = 24 dB. (a) Find the Shannon capacity. (b) How many signalling levels are needed to reach it?*

**Step 1: bandwidth.** B = 4 MHz − 3 MHz = **1 MHz**

**Step 2: convert SNR from dB to a ratio.** 24 = 10 $\log_{10}$(SNR) → SNR = 10^2.4 = **251** (more precisely 251.19)

**Step 3: Shannon.** C = $10^6 \times \log_2(1 + 251) = 10^6 \times \log_2(252) \approx 10^6 \times 7.98 \approx$ **8 Mbps**

**Step 4: Nyquist, to find M.** $8 \times 10^6 = 2 \times 10^6 \times \log_2 M \rightarrow \log_2 M = 4 \rightarrow$ **M = 16 levels**

**Interpretation**: 8 Mbps is a theoretical limit that is unlikely to be reached. If it were reached, you would need 16-level signalling.

> **EXAM TIP**: The commonest mistake is putting the dB value straight into Shannon's formula. **Always convert dB → ratio first.**

## 1.8 Transmission media and the spectrum

### Guided vs unguided media [TF-S 34–35]

- **Guided media**: waves are guided along a solid medium (copper twisted pair, coaxial cable, optical fibre).
- **Unguided media**: provide a means of transmission but do not guide the signal. Usually called **wireless transmission** (atmosphere, outer space). Transmission and reception are **by means of an antenna**.
  - **Directional**: focused beam.
  - **Omnidirectional**: radiates in all directions.

### General frequency ranges [TF-S 38]

| Range | Frequencies | Characteristics / uses |
|---|---|---|
| **Microwave** | 1 GHz – 100 GHz | Directional beams possible; point-to-point; satellite communications |
| **Radio** | 30 MHz – 1 GHz | Suitable for **omnidirectional** applications |
| **Infrared** | ≈ $3 \times 10^{11}$ – $2 \times 10^{14}$ Hz | Local point-to-point and multipoint applications within confined areas |

### Common wireless technologies and bands [TF-S 39]

| Technology | Frequency range | Typical applications | Characteristics |
|---|---|---|---|
| AM radio | 530–1700 kHz (MF) | Broadcast radio | Very long range; lower audio quality |
| FM radio | 88–108 MHz (VHF) | Music, local broadcasting | High-quality audio; shorter range |
| Wi-Fi | 2.4, 5 and 6 GHz | Home/office Internet | High data rates; moderate range |
| Bluetooth | 2.4 GHz | Headsets, keyboards, IoT | Short range; low power |
| 4G/LTE | 700 MHz – 2.6 GHz | Mobile broadband | Good coverage and speed |
| 5G | Sub-6 GHz and 24–100 GHz | Ultra-fast mobile, IoT | Very high speed; low latency |
| Microwave | 1–30 GHz | Satellite links, radar, point-to-point, backhaul | Line-of-sight; high capacity |
| Infrared | 300 GHz – 400 THz | TV remotes, short-range sensors, thermal imaging | Very short range; cannot penetrate walls |

### Wireless transmission types [TF-S 40–42]

| | **Terrestrial microwave** | **Satellite microwave** | **Broadcast radio** |
|---|---|---|---|
| Antenna | Parabolic dish (~3 m), rigidly fixed, narrow beam, mounted high | Satellite acts as a **microwave relay station** | **Omnidirectional**; no dish; no precise alignment needed |
| How it works | **Line-of-sight** to the receiving antenna | Receives on one band (**uplink**), amplifies/repeats, transmits on another band (**downlink**) | 30 MHz – 1 GHz (VHF and part of UHF) |
| Applications | Long-haul telecoms; short point-to-point links between buildings | TV distribution; long-distance telephone; private business networks | FM radio; UHF and VHF television |

## 1.9 Multiplexing (introduction)

**Multiplexing** means carrying multiple signals on a single medium. It is used because the capacity of a medium usually **exceeds** what a single signal needs, so multiplexing makes **more efficient use of the medium** [TF-S 43]. A **MUX** combines n inputs onto one link with n channels, and a **DEMUX** separates them again [TF-B 31].

**Reasons for its widespread use** [TF-S 45]:

1. The **cost per kbps** of a transmission facility declines as the data rate increases.
2. The **cost of transmission and receiving equipment** (per kbps) declines with increased data rate.
3. Most individual devices need only **modest data rates**.

**Two main techniques** [TF-S 46, TF-B 32]:

| | **FDM** (frequency division) | **TDM** (time division) |
|---|---|---|
| Exploits | Useful **bandwidth** of the medium exceeds the bandwidth a single signal needs | Achievable **bit rate** of the medium exceeds the data rate a single digital signal needs |

| | **FDM** (frequency division) | **TDM** (time division) |
|---|---|---|
| How | Each signal is modulated onto a **different carrier frequency**; carriers are far enough apart that the bands don't overlap; **guard bands** (unused spectrum) separate channels | Portions of each signal are **interleaved in time** (bit level or blocks); one cycle of slots is a **frame** |
| Example | Voice: 300–3400 Hz fits in a 4 kHz channel; standard = **twelve 4-kHz voice channels from 60 to 108 kHz** | Six 9.6 kbps sources fit on one **57.6 kbps** line |

- **Synchronous TDM**: time slots are **preassigned and fixed** [TF-B 32].
- **Asynchronous (statistical) TDM**: time on the medium is allocated **dynamically** [TF-B 32].

### Worked example M1 (end-of-chapter exercise) [TF-S 50]

*A voice channel occupies 4 kHz. Multiplex 10 voice channels using FDM with guard bands of 500 Hz. Required bandwidth?*

1. Ten channels need 10 × 4 kHz = 40 kHz.
2. Guard bands are needed **between** adjacent channels: 10 − 1 = **9 guard bands** → 9 × 0.5 kHz = 4.5 kHz.
3. Total = 40 + 4.5 = **44.5 kHz**.

SUPPLEMENTARY: the common mistake is counting 10 guard bands (45 kHz). Only the gaps *between* channels need guarding. If the question explicitly says guard bands at both edges as well, add those too. Compare the FDMA formula N = (Bt − 2Bg)/Bc in [MA p3], which includes edge guard bands; it will be covered in Part 2.

## 1.10 Examination focus — Topic 1

**Lecturer's review questions** [TF-S 50]. These are very likely exam material:

1. Relationship between period and frequency → **T = 1/f**.
2. What do amplitude, frequency and phase measure? → strength (peak value); repetition rate (cycles/s); relative position in time within a period.
3. How can a composite signal be decomposed into its frequencies? → **Fourier analysis**: any signal is a sum of sine waves of different amplitudes, frequencies and phases [TF-S 15].
4. Name three transmission impairments → **attenuation, distortion, noise** (Topic 2 gives the wireless list).
5. Baseband vs broadband transmission SUPPLEMENTARY: *baseband* sends the digital signal directly, without a carrier, and needs a **low-pass** channel. *Broadband* modulates the signal onto a carrier and needs a **bandpass** channel.
6. Low-pass vs bandpass channel SUPPLEMENTARY: a low-pass channel passes frequencies from 0 to B. A bandpass channel passes frequencies between $f_1$ and $f_2$ ($f_1 > 0$).
7. What does Nyquist have to do with communications? → max bit rate of a **noiseless** channel: $C = 2B \log_2 M$.
8. What does Shannon capacity have to do with communications? → max (error-free) bit rate of a **noisy** channel: $C = B \log_2(1 + SNR)$.
9. Synchronous vs statistical TDM → fixed preassigned slots vs dynamically allocated slots (statistical TDM wastes no slots on idle sources).
10. 200 kHz, 4 levels → **800 kbps** (N3).
11. 10 voice channels, 500 Hz guard bands → **44.5 kHz** (M1).

### Possible question types

- *Definition (2 marks)*: define bandwidth / channel capacity / SNR.
- *Comparison (4–6 marks)*: analog vs digital signals; amplifiers vs repeaters; Nyquist vs Shannon; FDM vs TDM.
- *Calculation (4–6 marks)*: Nyquist levels/bit rate; dB conversion; Shannon + Nyquist combined (Example S1 pattern); FDM bandwidth.

## 1.11 Topic 1 revision summary

- A signal can be described in **time** (A, f, T, φ, λ) and in **frequency** (spectrum, bandwidth). **$T = 1/f$, $\lambda = c/f$.**
- Any signal is a **sum of sine waves** (Fourier). Square waves have **infinite bandwidth**; real systems limit bandwidth, which causes **distortion**.
- **More bandwidth → higher data rate**, but higher cost.
- Digital signals: cheaper and less noise-sensitive, but **attenuate more**. **Repeaters regenerate**; **amplifiers amplify noise too**.
- **Nyquist (noiseless): $C = 2B \log_2 M$. Shannon (noisy): $C = B \log_2(1 + SNR)$**, with SNR as a ratio. **$SNR_{dB} = 10 \log_{10}(S/N)$.**
- Media: guided vs unguided. Microwave 1–100 GHz (directional), radio 30 MHz–1 GHz (omnidirectional), infrared.
- Multiplexing: **FDM** (frequency + guard bands) and **TDM** (time slots, frames; synchronous vs statistical).

# TOPIC 2 — THE WIRELESS CHANNEL

## 2.1 Introduction

The **wireless channel** is the path through the air between transmitter and receiver. Unlike a cable, it is **shared, unpredictable and constantly changing**. Signals weaken with distance, bounce off buildings, arrive by many paths, interfere with each other and fade as users move. This topic explains *what goes wrong* with a wireless signal. Topic 3 explains *how systems fix it*.

**Why it matters for the exam**: this is the **most heavily examined theory topic** in your Test 1 papers and the exam. "Three propagation mechanisms", "path loss vs multipath vs interference", "problems of signal propagation" and the distance/power calculations all come from here.

**Learning objectives** [WC-S 2]: explain the importance of unlicensed frequencies; compute path loss for free space and real-world environments using the path loss exponent; describe the effects of multipath propagation and fading; describe the approaches used to correct channel impairments; describe three bandwidth approaches.

## 2.2 Key definitions

| Term | Definition | Source |
|---|---|---|
| **Spectrum** | A compact representation of the frequency content of a signal composed of sinusoids. | [WC-S 3] |
| **ISM bands** | Industrial, Scientific and Medical bands that can be used **without a licence**, as long as transmitters stay within power limits and use a spread-spectrum technique. | [WC-S 9] |
| **Antenna** | An electrical conductor or system of conductors that **radiates** electromagnetic energy into space (transmission) or **collects** it from space (reception). | [WC-S 25] |
| **Radiation pattern** | Graphical representation of the radiation properties of an antenna as a function of space coordinates, usually shown as a 2-D cross-section. | [WC-S 27] |
| **Isotropic antenna** | Idealised point antenna that radiates power **equally in all directions** (spherical pattern). | [WC-B 5.2] |
| **Attenuation** | Strength of a signal falls off with distance over the transmission medium. | [WC-S 29] |
| **Free space loss** | Attenuation caused because the signal spreads over a larger and larger area (sphere) as distance increases, even with no obstructions. | [WC-B 5.2] |
| **Path loss exponent (n)** | Parameter that describes **how fast a radio signal's power decreases with distance** in a particular environment. | [WC-S 33] |
| **Noise** | Unwanted signals inserted somewhere between transmission and reception. | [WC-S 36] |
| **Multipath** | Obstacles reflect signals, so **multiple copies with varying delays** are received. | [WC-S 39] |
| **Intersymbol interference (ISI)** | One or more **delayed copies of a pulse** arrive at the same time as the primary pulse for a **subsequent bit**. | [WC-S 40] |
| **Fading** | The fluctuation or variation in received signal strength due to **multipath propagation, shadowing, and movement** of the transmitter, receiver or surrounding objects. (Textbook: time variation of received power caused by changes in the medium or path.) | [WC-S 42], [WC-B 5.3] |

| Term | Definition | Source |
|---|---|---|
| **Coherence time (Tc)** | Time over which the channel stays relatively constant. | [WC-B 5.3] |
| **Coherence bandwidth (Bc)** | Range of frequencies over which the channel response is relatively constant. | [WC-B 5.3] |

## 2.3 Spectrum considerations

### What differentiates one signal from another? [WC-S 4–6]

1. **Carrier frequency**: each signal is shifted from its base frequency up to a carrier. *Example*: a 22 MHz IEEE 802.11 signal centred on **2.412 GHz** occupies **2.401–2.423 GHz**.
2. **Signal power**: signals are limited in range by the allowed transmit power, so users far enough apart can **reuse the same spectrum**.
3. **Multiple access scheme**: users in the same spectrum share it by time slots (**TDMA**), frequency bands (**FDMA**) or different signal encodings (**CDMA**).

### Regulation [WC-S 7–8]

- In the USA the **FCC** regulates spectrum (military, broadcasting, public safety, mobile, amateur, ...). Bands can be government-exclusive, non-government-exclusive, or shared.
- In **South Africa**, most of the work is done by **ICASA** (Independent Communications Authority of South Africa).
- **Problem**: licensing leaves spectrum **underutilised**. **Solutions**:
  - **Dynamic Spectrum Access (DSA)**: users share spectrum. **Secondary users** may use it while **primary users** are inactive, and must release it as soon as a primary user needs it [WC-B 5.1].
  - **Cognitive radio**: devices **sense** when the spectrum is busy.

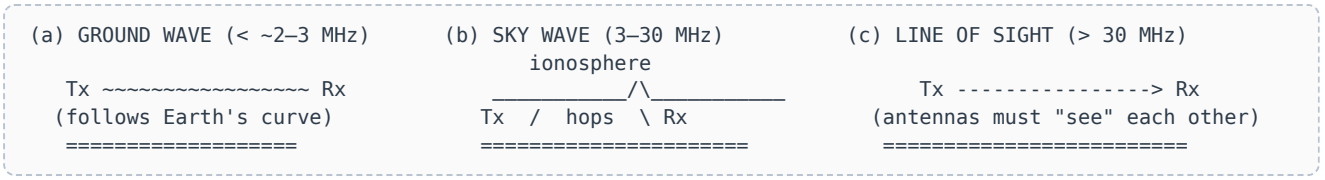
### ISM bands [WC-S 9–11]

- Usable **without a licence** if transmitters stay within **power limits** and use **spread spectrum**.
- Used for **WLANs, WPANs and the Internet of Things**.
- Examples: **915 ± 13 MHz, 2450 ± 50 MHz, 5.8 ± 0.75 GHz, 57–64 GHz**.
- **The ISM issue**: powerful emissions from ISM devices (e.g. microwave ovens) can cause electromagnetic interference. Equipment in these bands **must tolerate any interference** from ISM applications, and users have **no regulatory protection**.

> **EXAM** This is why 2.4 GHz Wi-Fi and Bluetooth suffer interference from microwave ovens. That idea comes back in T1A Q12, T1A Q13 and T2-2025 Q6.

## 2.4 Propagation modes [WC-S 12–17, WC-B 5.1]

A signal radiated from an antenna travels by one of three routes:




| Mode | Frequencies | How it works | Examples | Notes |
|---|---|---|---|---|
| **Ground wave** | Up to ~3 MHz | Follows the **contour of the Earth**; can travel well beyond the visual horizon | **AM radio** | Suffers from atmospheric noise; affected by reflection, refraction and scattering by ground objects |
| **Sky wave** | 3–30 MHz | Signal is **refracted by the ionosphere** back to Earth; can make several hops between ionosphere and ground | Amateur radio, CB radio, international broadcasting (BBC World Service, Voice of America) | Can be received **thousands of km** away |
| **Line of sight (LOS)** | Above 30 MHz | Needed when neither ground nor sky wave works; antennas must be in line of sight | **Satellite** (above 30 MHz the ionosphere does not reflect); most technologies in this module (100s of MHz to a few GHz) | LOS does not always mean completely free space; the ability to pass through objects **degrades as frequency increases** |

**Refraction** [WC-S 19]: the bending of microwaves by the atmosphere. The speed of an EM wave depends on the **density of the medium**; when the wave changes medium its speed changes, and it **bends at the boundary**.

## 2.5 Line-of-sight transmission impairments [WC-S 18, WC-B 5.2]

The received signal differs from the transmitted signal because of transmission impairments. For analog signals these cause **random modifications** that degrade quality; for digital data they cause **bit errors**. The most significant LOS impairments:

1. **Attenuation and attenuation distortion**
2. **Free space loss**
3. **Noise**
4. **Atmospheric absorption**: water vapour and oxygen contribute to attenuation [WC-S 39]
5. **Multipath**: obstacles reflect signals so that multiple copies with varying delays are received [WC-S 39]
6. **Refraction**: bending of radio waves as they propagate through the atmosphere [WC-S 39]

### Attenuation [WC-S 29]

- The received signal must have **sufficient strength** for the receiver circuitry to interpret it.
- It must stay **sufficiently higher than noise** to be received without error.
- **Attenuation is greater at higher frequencies**, which causes **distortion**.

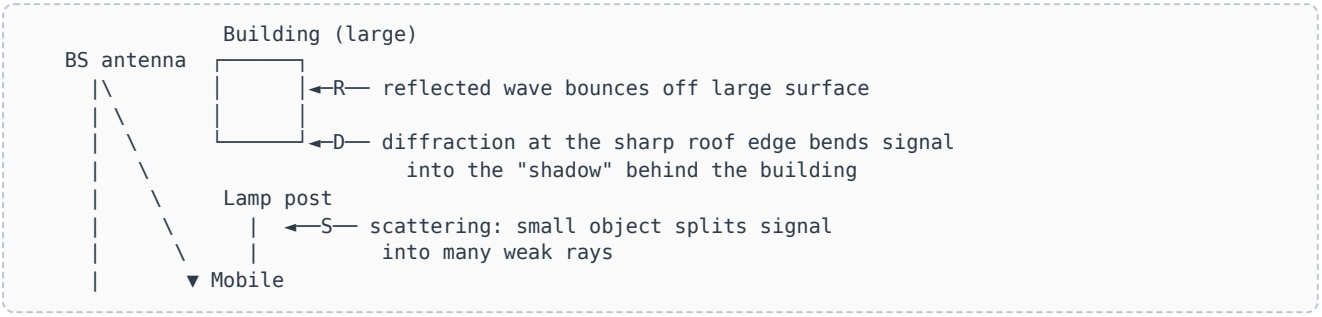
## 2.6 The five basic propagation mechanisms [WC-S 20–24, WC-B 5.2]

| # | Mechanism | What happens | Trigger |
|---|---|---|---|
| 1 | **Free-space propagation** | Wave travels with **no obstructions**; signal strength **decays as a function of distance** | Nothing in the way |
| 2 | **Transmission** | Signal **penetrates into and through a medium**; it is **refracted** at the surface to a different angle | Wall, window, any penetrable medium |
| 3 | **Reflection** | Wave bounces off a surface | Surface **large compared to the wavelength** (walls, buildings, ground) |
| 4 | **Diffraction** | **Secondary waves** form behind sharp edges and deliver the signal to a possibly **shadowed** receiver | Object with **sharp edges** (building corner, rooftop) |
| 5 | **Scattering** | Signal splits into many weaker outgoing signals | **Many objects small relative to the wavelength**: rough surfaces, foliage, street signs, lamp posts |

The last four involve interacting objects; the objects' **dielectric and conducting properties** affect the strength and angle of the signal [WC-B 5.2].

### The THREE basic propagation mechanisms in a mobile system ★ [WC-B 5.3, Figure 5.5]

**EXAM** **Asked in: T1A Q1 (12), T1B Q3 (12); also the lecturer's review question [WC-S 72].** The answer is **reflection, diffraction and scattering**, the three that cause multipath in a mobile (cellular) environment.




*(Text version of Figure 5.5: R = reflection, D = diffraction, S = scattering.)*

1. **Reflection**: occurs when an EM signal meets a surface that is **large relative to its wavelength**. E.g. a **ground-reflected wave** near the mobile may **cancel** the LOS wave and cause high signal loss. Because the mobile antenna is lower than most buildings, **multipath interference** occurs, and reflected waves may combine **constructively or destructively** at the receiver.
2. **Diffraction**: occurs at the **edge of an impenetrable body that is large compared to the wavelength**. The edge acts as a **new source**, so waves spread in different directions and the signal can be received **even with no unobstructed LOS**.
3. **Scattering**: occurs when the obstacle is **on the order of the wavelength or smaller**. The incoming signal is **scattered into several weaker outgoing signals**. At cellular microwave frequencies there are many such objects (lamp posts, traffic signs), so **scattering effects are difficult to predict**.

**How they affect performance** [WC-B 5.3]:

- With a **clear LOS**, diffraction and scattering are generally **minor**, although **reflection may have a significant impact**.

- With **no clear LOS** (e.g. urban street level), **diffraction and scattering are the primary means of signal reception**.

## 2.7 Antennas and radiation patterns [WC-S 25–28, WC-B 5.2]

- In two-way communication, **the same antenna can be used for transmission and reception** [WC-S 25].
- An antenna radiates in all directions but **does not perform equally well in all directions**. This is described by its **radiation pattern**.

| Term | Meaning [WC-S 27] |
|---|---|
| **Beam width (half-power beam width)** | Measure of the **directivity** of the antenna |
| **Reception pattern** | The receiving antenna's equivalent of the radiation pattern |
| **Main lobe** | Direction of strongest radiation |
| **Sidelobes** | Extra energy in directions **outside the main lobe** |
| **Nulls** | Very low energy, between the main lobe and sidelobes |

- **Omnidirectional (isotropic)** pattern: a circle in 2-D; vectors A and B in any directions are **equal length**, so equal power.
- **Directional** pattern (e.g. a 4-antenna linear array spaced λ/2, main lobe ~60° wide): the vector toward the main lobe is longer, so **more power** goes in that direction.
- The pattern shows **relative antenna gain** in each direction, **not** relative coverage distance (although they are related).

## 2.8 Free space loss ★

### The formula [WC-S 30–31, WC-B 5.2 Eq. 5.1–5.2]

For an ideal isotropic antenna:

$$Pt/Pr = (4\pi d)^2 / \lambda^2 = (4\pi fd)^2 / c^2$$

| Symbol | Meaning |
|---|---|
| Pt | Signal power at the transmitting antenna |
| Pr | Signal power at the receiving antenna |
| λ | Carrier wavelength (**λ = c/f**) |
| f | Carrier frequency |
| d | Propagation distance between antennas (**same units as λ**, e.g. metres) |
| c | Speed of light = $3 \times 10^8$ m/s |

**In decibels**:

$$L_dB = 10 \log(Pt/Pr) = 20 \log(4\pi d/\lambda) = -20 \log(\lambda) + 20 \log(d) + 21.98 \text{ dB}$$
$$= 20 \log(4\pi fd/c) = 20 \log(f) + 20 \log(d) - 147.56 \text{ dB}$$

**Why it happens** (physical explanation, [WC-B 5.2]): energy radiates outward as a **growing sphere**. A receiving antenna of fixed area captures energy from only part of the sphere's surface. As d increases the sphere's area grows with **$d^2$**, so there is less energy per unit area. **Doubling distance → ¼ of the power (−6 dB). Doubling frequency → also −6 dB.** For satellite communication this is the **primary** source of loss.

## The same formula in the "Problems" notes [WCP Eq. 2.1–2.3]

The *Wireless Communication Problems* notes write the same law in three forms. **These are the forms used in the calculation questions**:

| Equation | Use it to find |
|---|---|
| **(2.1) $Pr = Pt / (4\pi d/\lambda)^2$** | Received ("captured") power, or rearranged for Pt, d, λ or f |
| **(2.2) SNR = Pr / Pn** | Signal-to-noise ratio (Pn = average noise power at the receiver) |
| **(2.3) $Lp = (4\pi d/\lambda)^2 = Pt/Pr$** | Path loss (a ratio, no units) |

**Rearrangements to memorise**:

```
Pt = Pr × (4πd/λ)²
d  = (λ / 4π) × √(Pt/Pr)
λ  = 4πd / √(Pt/Pr)        then  f = c/λ
f  = c × √Lp / (4πd)
```

## Path loss exponent in practical systems [WC-S 33–34]

Real environments have reflection, scattering and so on, so power falls off faster or slower than $d^2$. Generalising, **$Pr \propto 1/d^n$**, where **n is the path loss exponent** (based on many measurement studies):

| Environment | Path loss exponent n |
|---|---|
| Free space | **2** |
| Urban area cellular radio | 2.7 – 3.5 |
| Shadowed cellular radio | 3 – 5 |
| In-building line-of-sight | 1.6 – 1.8 (less than 2: corridors act like waveguides) |
| Obstructed in building | 4 – 6 |
| Obstructed in factories | 2 – 3 |

SUPPLEMENTARY (formula not shown in your text): in dB, **$L_{dB}(d) = L_{dB}(d_o) + 10\,n \log_{10}(d/d_o)$**. Each **tenfold** increase in distance adds **10n dB** of loss: 20 dB in free space, ~35 dB in a city.

## Empirical models [WC-S 35]

Systems must be designed from **empirical data** for a specific environment (to set power levels, tower heights and mobile antenna heights). **Okumura** developed a model from detailed measurements in Tokyo, later **refined by Hata**. It has among the best accuracy across many situations and predicts path loss for **urban (small/medium city, large city), suburban and rural** environments.

## 2.9 ★ Wireless communication calculations: full worked set

All of these use **$Pr = Pt/(4\pi d/\lambda)^2$**, **$\lambda = c/f$**, **$c = 3 \times 10^8$ m/s**, **$SNR = Pr/Pn$** and **$Lp = (4\pi d/\lambda)^2$** [WCP]. Every answer below has been **recomputed and verified**.

**Method for every problem**:

1. List the givens and **convert units** (mW → $\times 10^{-3}$ W; µW → $\times 10^{-6}$ W; MHz → $\times 10^6$ Hz; km → $\times 10^3$ m).
2. Compute **$\lambda = c/f$**.
3. Pick the right rearrangement and substitute.
4. Compute, giving units, and sanity-check (received power should be tiny; distance should be plausible).

### ★ Problem 2.7 = T1A Question 14 (6 marks)

*A transmitter operates at 850 MHz with a power of 125 mW. The receiver has a received power of 1 µW. What is the distance between transmitter and receiver?*

**Given**: Pt = 125 mW = $125 \times 10^{-3}$ W; Pr = 1 µW = $1 \times 10^{-6}$ W; f = $850 \times 10^6$ Hz.

**Step 1: path loss (ratio).** Lp = Pt/Pr = $(125 \times 10^{-3})/(1 \times 10^{-6})$ = **$125 \times 10^3 = 125\,000$**

**Step 2: wavelength.** $\lambda = c/f = (3 \times 10^8)/(850 \times 10^6)$ = **0.3529 m**

**Step 3: set $Lp = (4\pi d/\lambda)^2$.** $125\,000 = (4\pi d / 0.3529)^2$

**Step 4: square root of both sides.** $\sqrt{125\,000} = 353.55 = 4\pi d / 0.3529$

**Step 5: solve for d.** $d = (353.55 \times 0.3529) / (4\pi) = 124.77 / 12.566$ = **9.93 m**

**Answer: d ≈ 9.93 m.**

> **NOTES ERROR**: the worked solution in *Wireless Communication Problems* p. 62 writes the step as "$125 \times 10^{-3} = (4 \times 3.14 \times d / 0.3529)^2$". The left side must be the **path loss $125 \times 10^3$**, not $125 \times 10^{-3}$ (which is Pt in watts). Their final answer, 9.93 m, is correct because they actually used $125 \times 10^3$. **In the exam, write $125 \times 10^3$**, otherwise you will get d ≈ 0.01 m and lose the marks.

### Problem 2.1: find Pt [WCP]

*$f = 90$ MHz, $d = 650$ m, $Pr = 1 \times 10^{-6}$ W. Find Pt.*

- $\lambda = 3 \times 10^8 / 90 \times 10^6$ = **3.33 m**
- $(4\pi d/\lambda) = 4\pi \times 650 / 3.33 = 2452.9$ → squared = $6.017 \times 10^6$
- $Pt = Pr \times (4\pi d/\lambda)^2 = 1 \times 10^{-6} \times 6.017 \times 10^6$ = **≈ 6 W** ✔

### Problem 2.2: find d [WCP]

*Pt = 10 mW, f = 85 MHz, Pr = 0.1 µW. Find d.*

- $\lambda = 3 \times 10^8 / 85 \times 10^6$ = **3.53 m**
- $Pt/Pr = 10 \times 10^{-3} / 0.1 \times 10^{-6} = 10^5$ → √ = 316.2
- $d = (\lambda/4\pi) \times 316.2 = (3.53/12.566) \times 316.2$ = **≈ 88.8 m** ✔ (notes: ≈ 88 m)

### Problem 2.3: find Pr [WCP]

*Pt = 20 mW, f = 75 MHz, d = 500 m. Find Pr.*

- $\lambda = 3 \times 10^8 / 75 \times 10^6$ = **4 m**

- $(4\pi d/\lambda) = 4\pi \times 500 / 4 = 1570.8 \rightarrow$ squared = **$2.467 \times 10^6$**
- $Pr = 20 \times 10^{-3} / 2.467 \times 10^6$ = **$8.1 \times 10^{-9}$ W ≈ 0.0081 µW**

> **NOTES ERROR**: the notes give **0.01 µW**. The correct value is **0.0081 µW (8.1 nW)**. Write $8.1 \times 10^{-9}$ W.

### Problem 2.4: SNR in dB [WCP]

*Pn = 25 µW, Pr = 100 mW. Find SNR in dB.*

- $SNR = Pr/Pn = 100 \times 10^{-3} / 25 \times 10^{-6}$ = **4000**
- $SNR_dB = 10 \log_{10}(4000)$ = **36.0 dB** ✔

### Problem 2.5: path loss [WCP]

*d = 5 km, f = 750 MHz. Find path loss.*

- $\lambda = 3 \times 10^8 / 750 \times 10^6$ = **0.4 m**
- $4\pi d/\lambda = 4\pi \times 5000 / 0.4$ = **157 080**
- $Lp = (157\,080)^2$ = **$2.47 \times 10^{10}$** (in dB: $10 \log(2.47 \times 10^{10}) \approx$ **103.9 dB**)

> **NOTES ERROR**: the notes give **Lp = $2.46 \times 10^8$**, which is wrong by a factor of 100. $157\,080^2 = 2.467 \times 10^{10}$. Check: $1.57 \times 10^5$ squared = $2.47 \times 10^{10}$.

### Problem 2.6: find frequency [WCP]

*Lp = $10^9$, d = 3 km. Find f.*

- $Lp = (4\pi df/c)^2 \rightarrow \sqrt{Lp} = 4\pi df/c$
- $\sqrt{10^9} = 31\,623 = 4\pi \times 3000 \times f / 3 \times 10^8$
- $f = 31\,623 \times 3 \times 10^8 / (4\pi \times 3000)$ = **251.6 MHz** ✔

### End-of-section unsolved problems (WCP p. 72), solved and verified

| # | Given | Find | Working | Answer |
|---|---|---|---|---|
| 1 | f = 600 MHz, Pt = 100 mW, Pr = 0.1 µW | d | $\lambda$ = 0.5 m; Pt/Pr = $10^6$, $\sqrt{}$ = 1000; d = 0.5 × 1000 / 4π | **39.8 m** |
| 2 | d = 2.5 km, f = 450 MHz, Pt = 75 mW | Pr | $\lambda$ = 0.667 m; 4πd/$\lambda$ = 47 124; $^2$ = $2.22 \times 10^9$; Pr = 0.075/$2.22 \times 10^9$ | **$3.38 \times 10^{-11}$ W (33.8 pW)** |
| 3 | Pt = 175 mW, d = 2 km, Pr = 0.25 µW | f | $\sqrt{}$(Pt/Pr) = $\sqrt{}$($7 \times 10^5$) = 836.7; f = 836.7 × 3 × $10^8$ / (4π × 2000) | **≈ 9.99 MHz** |
| 4 | f = 550 MHz, d = 1.5 km, Pr = 0.01 µW | Pt | $\lambda$ = 0.5455 m; 4πd/$\lambda$ = 34 558; $^2$ = $1.194 \times 10^9$; Pt = $10^{-8} \times 1.194 \times 10^9$ | **≈ 11.9 W** |
| 5 | f = 550 MHz, d = 3 km | Lp | $\lambda$ = 0.5455 m; 4πd/$\lambda$ = 69 115; $^2$ | **$4.78 \times 10^9$ (≈ 96.8 dB)** |
| 6 | Pt = 10 mW, f = 950 MHz, d = 5.5 km | Pr | $\lambda$ = 0.3158 m; 4πd/$\lambda$ = 218 868; $^2$ = $4.79 \times 10^{10}$; Pr = 0.01/$4.79 \times 10^{10}$ | **$2.09 \times 10^{-13}$ W** |
| 7 | Pn = 10 µW, Pr = 10 mW | SNR | $10 \times 10^{-3} / 10 \times 10^{-6}$ = 1000 | **1000 = 30 dB** |
| 8 | SNR = 25 dB, Pn = 20 µW | Pr | SNR = 10^2.5 = 316.2; Pr = 316.2 × 20 × $10^{-6}$ | **6.32 mW** |

| # | Given | Find | Working | Answer |
|---|---|---|---|---|
| 9 | Lp = $10^9$, d = 3 km | f | Same as Problem 2.6 | **251.6 MHz** |
| 10 | Pt = 95 mW, Pr = 10 µW | Lp | $95 \times 10^{-3} / 10 \times 10^{-6}$ | **9500 (≈ 39.8 dB)** |

### Free-space loss in dB: worked example SUPPLEMENTARY practice

*Free-space loss at 2.4 GHz over 100 m?* $L_{dB} = 20 \log(2.4 \times 10^9) + 20 \log(100) - 147.56 = 187.60 + 40 - 147.56 =$ **80.0 dB**

## 2.10 Noise [WC-S 36–38, WC-B 5.2]

The received signal = transmitted signal + distortions + **noise** (unwanted signals inserted between transmission and reception). **Noise is the major limiting factor in communications system performance.** Four categories:

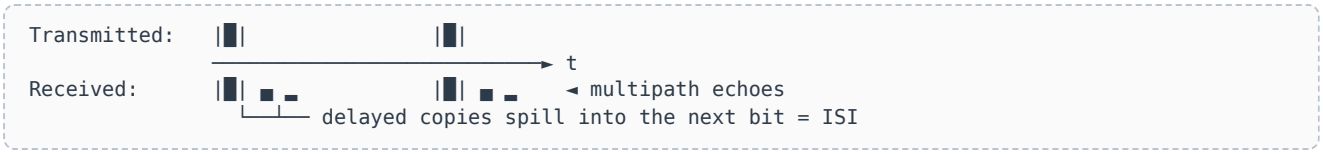
| Type | Cause | Key facts |
|---|---|---|
| **Thermal noise** | Thermal agitation of electrons | Present in **all** electronic devices and media; a **function of temperature**; uniformly distributed across frequency (**white noise**); **cannot be eliminated**, so it places an **upper bound** on performance; particularly significant for **satellite** communication |
| **Intermodulation noise** | Signals at different frequencies share the same medium | Produces signals at the **sum or difference** of the original frequencies (or multiples) |
| **Crosstalk** | **Unwanted coupling between signal paths** | E.g. hearing another conversation on the phone |
| **Impulse noise** | External EM disturbances (**lightning**), faults and flaws in the system | **Irregular pulses/spikes**, short duration, relatively high amplitude, unpredictable; the **primary source of error in digital data**. A 0.01 s spike barely affects voice but **wipes out ~10 000 bits at 1 Mbps** |

## 2.11 Multipath propagation and its effects ★ [WC-S 40–41, WC-B 5.3]

**What it is**: where antennas can't be placed with a clear path (e.g. mobile telephony), the signal is **reflected** (also diffracted and scattered) by obstacles, so **multiple copies with varying delays** reach the receiver. In extreme cases the receiver gets **only reflected signals**. Depending on path-length differences, the composite signal can be **larger or smaller** than the direct signal (**reinforcement or cancellation**).

**Effects of multipath** [WC-S 40, WC-B 5.3]:

1. **Signal level drops**: copies arriving in different phases may add **destructively**. The resulting power can be lower by a factor of 100 or 1000 (**20–30 dB**), the signal falls relative to noise, and detection becomes harder.
2. **Intersymbol interference (ISI)**: delayed copies of a pulse arrive **at the same time as the primary pulse of the next bit**, acting as noise to it. As the mobile moves, the **number, magnitude and timing** of secondary pulses change, which makes ISI hard to filter out.
3. **Rapid signal fluctuations** over a few **centimetres**.




*Multipath in a room* [WCP Fig 2.11]: signals reflect off ceiling, floor and obstructions, and the final received signal is the **sum of all received copies with different time delays**. Multipath can cause **high RF signal strength but poor signal quality**. WLAN vendors fight it with **equalisation and antenna diversity**, and may need to operate at **lower data rates** if multipath is severe [WCP].

## 2.12 Fading ★ [WC-S 42–43, WC-B 5.3, WCP]

**Fading** is the variation in received signal strength due to multipath, shadowing and movement. In a **fixed** environment it is affected by atmospheric changes such as **rainfall**. In a **mobile** environment the positions of obstacles change over time, creating complex and sometimes rapid effects. The textbook calls it *perhaps the most challenging technical problem facing communications systems engineers*.

**Types listed by the lecturer** [WC-S 43]: fast, slow, flat, selective, Rayleigh, Rician. The slide marks this as **"H/W: study the different types of fading"** [WC-S 42].

### Large-scale vs small-scale fading [WC-B 5.3]

| | Large-scale fading | Small-scale fading |
|---|---|---|
| Distance scale | Distances **well beyond a wavelength** (passing buildings, vacant lots, intersections) | About **half a wavelength** (at 900 MHz, λ = 0.33 m) |
| Cause | **Shadowing** and changes in distance from the transmitter | **Multipath** combination and **Doppler** (movement) |
| Appearance (Fig 5.7) | **Slowly changing** average power | **Rapid** variations, up to **20–30 dB** over a short distance |

### Two small-scale effects (independent of each other)

**1. Doppler spread → fast vs slow fading** (a **time** effect)

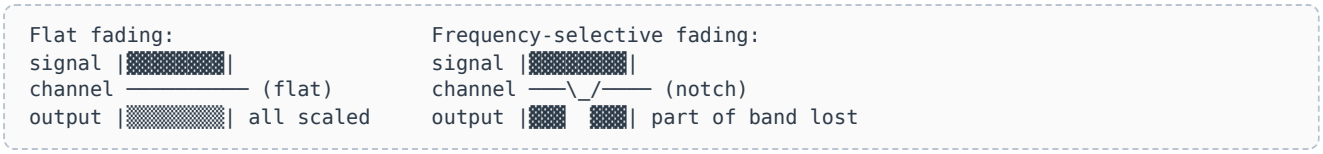
- Characterised by **coherence time Tc**: how long the channel stays roughly constant (e.g. **70 ms pedestrian, 5 ms vehicle at highway speed**).
- Bit time **Tb = 1/rb**.
- **Slow fading**: Tc ≫ Tb, so the channel hardly changes during one bit.
- **Fast fading**: Tc ≤, ≈ or only slightly > Tb, i.e. Tc is **not much greater** than Tb.
- Formula (footnote in textbook): **Tc = 0.423c / (v f)**.

*Worked example* SUPPLEMENTARY practice: a car at v = 30 m/s, f = 900 MHz → Tc = 0.423 × 3 × $10^8$ / (30 × 9 × $10^8$) = **4.7 ms**, which agrees with the "5 ms at highway speed" figure.

**2. Multipath fading → flat vs frequency-selective** (a **frequency** effect)

- Characterised by **coherence bandwidth Bc**: the range of frequencies over which the channel response is constant.
- **Flat fading**: **Bc ≫ Bs** (signal bandwidth, Bs ≈ rb). **All frequency components fluctuate in the same proportions simultaneously** ("rise and fall in unison" [WCP]).
- **Frequency-selective fading**: flat fading is **not** present. Attenuation affects **only part** of the signal bandwidth; different frequencies are attenuated and phase-shifted **differently**. This creates **notches**

in the frequency response and needs an **equaliser** [WCP].




**Four combinations**: fast-flat, slow-flat, fast-frequency-selective, slow-frequency-selective [WC-B 5.3].

SUPPLEMENTARY (Rayleigh and Rician are listed on [WC-S 43] but not explained in your notes):

- **Rayleigh fading**: a statistical model for **no dominant LOS path**; the signal is the sum of many scattered paths (dense urban, indoor non-LOS).
- **Rician fading**: a statistical model for when there **is a dominant LOS path** plus scattered paths. It is less severe than Rayleigh.

## 2.13 ★ Wireless communication problems: the full list [WCP §2.3]

EXAM **This list answers "What are the main problems of signal propagation?" (T1A Q2, 12 marks).**
The *Basics of Wireless Networks* notes list ten problems:

1. **Shared media**: the wireless medium offers **lower bandwidth**, so it must be used efficiently.
2. **Increased bit error rate (BER)**: obstacles between Tx and Rx and interference from neighbouring transmitters cause errors, **frequent disconnections** and data loss, especially annoying for voice and video.
3. **Lower radio transmission power**: battery-powered mobiles limit power, and power also limits interference. Signal strength decreases with the **inverse square of distance**. Higher frequencies (3–5 GHz) increase attenuation and reduce range.
4. **Scattering**: the medium contains objects **small compared to the wavelength**, many per unit volume (rough surfaces, small objects); needs correct deployment (placement, antenna selection).
5. **Reflection**: the wave strikes an object **very large compared to the wavelength** (walls, furniture, building structure).
6. **Diffraction**: the path is obstructed by a surface with **sharp irregularities (edges)**; secondary waves exist even **behind** the obstacle.
7. **Multipath propagation**: reflection and scattering mean waves arrive from many directions with different delays and **combine vectorially**. Movement causes **Doppler shift**. The result can be high RF strength but poor quality. **Indoor is especially bad.** Countered by **equalisation, antenna diversity, lower data rates**.
8. **Fading**: often caused by multipath; **degrades BER**. Flat vs frequency-selective (above).
9. **Path loss**: key for design. It determines **transmit power, receiver sensitivity and SNR**. Indoors it is roughly **20 dB per 100 ft**. Multipath fading effectively increases it. **Main countermeasure: add more access points** for coverage.
10. **RF signal interference**: vulnerable to **atmospheric noise** and **transmissions from other systems**.
    - **Inward interference** (**co-channel**): from harmonics of other transmitters or products using similar frequencies, e.g. **microwave ovens at 2.4 GHz**, which block transmissions or cause bit errors.
    - **Outward interference** (**adjacent channel**): a WLAN's signal disrupts **other** systems (adjacent WLANs, aircraft navigation).
    - Interference is uncommon in spread-spectrum WLAN bands because power is low (< 1 W); devices must be **very close and in the same band**.

## 2.14 ★ Model answer framework: Path loss vs multipath vs interference

**EXAM** **Asked in: Exam Q1(a) (6), T1A Q3 (9), T1B Q4 (12).** Full answers will be in Part 5, but here is the core comparison you must know:

| | Path loss | Multipath propagation | Interference from other sources |
|---|---|---|---|
| **What it is** | Reduction of signal power as it travels from Tx to Rx | Signal reaches the receiver by **several paths** (reflection, diffraction, scattering) with different delays and phases | **Unwanted signals from other transmitters** add to the wanted signal |
| **Cause** | Signal **spreading over distance** (free-space loss ∝ $(4\pi d/\lambda)^2$); absorption, obstruction | Obstacles: buildings, walls, ground, ceiling, small objects | Other devices using the **same or nearby frequencies**: co-channel (same channel), adjacent channel, microwave ovens, other WLANs/Bluetooth |
| **Depends on** | **Distance** and **frequency** (path loss exponent n of the environment) | **Environment geometry** and **movement** | **Other users/devices** nearby; not related to the wanted signal's path |
| **Behaviour** | **Predictable, deterministic**, smooth decrease with distance | **Random, time-varying**; constructive/destructive addition, rapid fluctuations (fading), ISI, Doppler | **Random/bursty**, depends on others' activity |
| **Effect** | Weak signal → low SNR → limited range | **Fading** (20–30 dB dips), **ISI**, distortion, high strength but poor quality | Lower **SINR**, bit errors, collisions, blocked transmissions |
| **Countermeasures** | Higher Tx power, higher antenna gain, more APs/cells, lower frequency | **Equalisation, diversity/MIMO, OFDM, spread spectrum, FEC** | **Channel planning** (use different channels), spread spectrum, directional antennas, power control, remove the source |

## 2.15 Examination focus — Topic 2

### Real past-paper questions from this topic

| Question | Paper(s) | Marks |
|---|---|---|
| Explain the three basic propagation mechanisms in a mobile communication system | T1A Q1, T1B Q3 | 12, 12 |
| Main problems of signal propagation? Why don't radio waves always follow a straight line? Why is reflection both useful and harmful? | T1A Q2 | 12 |
| Differences between path loss, multipath propagation and interference from other sources | Exam Q1(a), T1A Q3, T1B Q4 | 6, 9, 12 |
| Distance between Tx (850 MHz, 125 mW) and Rx (1 µW) | T1A Q14 | 6 |
| A receiving site has occasional loss of signal due to interference; steps to correct it | T1A Q12, T2-2025 Q6 | 10, 10 |
| Why many WiFi networks have a high error rate caused by co-channel interference | T1A Q13 | 4 |

**Lecturer's own review questions** [WC-S 72]: three propagation mechanisms; **attenuation, dispersion and noise**; FD-MIMO and beamforming; SU-MIMO vs MU-MIMO; channel correction mechanisms; three bandwidth expansion approaches.

**Quick prep for "Why don't radio waves always follow a straight line? Why is reflection both useful and harmful?"** (T1A Q2). Draft answer, which Part 5 will expand:

- **Not a straight line**: waves **refract** (bend when the density of the medium changes; ionospheric refraction for sky waves), **diffract** around sharp edges, **scatter** off small objects, **reflect** off large surfaces, and ground waves follow the Earth's curvature.
- **Reflection useful**: it lets the signal reach a receiver with **no LOS** (around buildings, indoors) and provides **multiple copies** that diversity/MIMO can combine.
- **Reflection harmful**: reflected copies arrive **delayed and phase-shifted**, causing **destructive interference (fading)**, **ISI** and **distortion**. A ground reflection can cancel the LOS wave.

SUPPLEMENTARY: **dispersion** (asked in the lecturer's review list but not defined in your notes) is the **spreading of a pulse in time**. Different components travel at different speeds or by different path lengths (e.g. multipath delay spread), so pulses widen and overlap, causing ISI.

## 2.16 Topic 2 revision summary

- Spectrum is regulated (FCC; **ICASA** in SA). Licensing leaves spectrum underused, hence **DSA and cognitive radio**. **ISM bands** are licence-free but have **no protection from interference**.
- Propagation modes: **ground wave (<3 MHz, AM)**, **sky wave (3–30 MHz, ionosphere)**, **LOS (>30 MHz)**.
- **Five mechanisms**: free space, transmission, reflection (large objects), diffraction (sharp edges), scattering (small objects). **The three mobile mechanisms: reflection, diffraction, scattering.**
- Antennas: radiation pattern, main lobe, sidelobes, nulls, beam width; isotropic vs directional.
- **Free space loss: $Pt/Pr = (4\pi d/\lambda)^2 = (4\pi fd/c)^2$.** In dB: 20 log f + 20 log d − 147.56. **Path loss exponent n**: free space 2, urban 2.7–3.5, obstructed in-building 4–6. Okumura–Hata model.
- Calculations: **$Pr = Pt/(4\pi d/\lambda)^2$; $\lambda = c/f$; $SNR = Pr/Pn$; $Lp = Pt/Pr$.** Watch out for the errors in the notes (Problems 2.3, 2.5, 2.7).
- **Noise**: thermal (cannot be eliminated), intermodulation, crosstalk, impulse (main cause of digital errors).
- **Multipath** → destructive addition (20–30 dB), **ISI**, rapid fluctuations.
- **Fading**: large-scale (shadowing) vs small-scale. **Fast/slow** depends on coherence time vs bit time; **flat/frequency-selective** depends on coherence bandwidth vs signal bandwidth.
- Ten wireless problems: shared media, high BER, low power, scattering, reflection, diffraction, multipath, fading, path loss, interference (inward/co-channel, outward/adjacent).

# TOPIC 3 — CHANNEL CORRECTION MECHANISMS AND CAPACITY

## 3.1 Introduction

Topic 2 showed what goes wrong in the wireless channel. This topic covers the **techniques systems use to compensate for the errors and distortions introduced by multipath fading**, and the ways to get **more capacity** when one channel is not enough [WC-S 44, WC-B 5.4]. In the mobile environment, **techniques from all categories are combined** [WC-B 5.4].

**The lecturer's list of channel correction mechanisms** [WC-S 44]:

1. Forward error correction (FEC)
2. Adaptive equalization
3. Adaptive modulation and coding (AMC)
4. Diversity techniques and MIMO
5. OFDM
6. Spread spectrum
7. Bandwidth expansion

(OFDM and spread spectrum get full treatment in Part 2.)

## 3.2 Key definitions

| Term | Definition | Source |
|---|---|---|
| **Forward error correction (FEC)** | The receiver, **using only information in the incoming transmission**, corrects bit errors. | [WC-B 5.4] |
| **Backward error correction (ARQ)** | The receiver **only detects** errors and **requests retransmission**. | [WC-B 5.4], [WC-S 69] |
| **Adaptive equalization** | Gathering **dispersed symbol energy back into its original time interval**; used to combat ISI. | [WC-S 47] |
| **Diversity** | Combats fading by providing **multiple independent copies** of the same signal to the receiver, because individual channels experience **independent fading** events. | [WC-S 48] |
| **MIMO** | A transmitter and receiver with **multiple antennas**. | [WC-B 5.4] |
| **Adaptive modulation and coding (AMC)** | **Dynamically adjusting** the modulation scheme and coding rate based on the current channel conditions. | [WC-S 51] |
| **Bandwidth expansion** | Approaches used when the needed throughput exceeds what the bps/Hz of a channel allows: carrier aggregation, frequency reuse, mmWave. | [WC-S 52] |

## 3.3 Forward error correction (FEC) [WC-S 45–46, WC-B 5.4]

**Applicable to**: digital transmission carrying digital data, digitized voice or video.

**How it works** (Figure 5.15, the FEC process):

1. The transmitter uses a coding algorithm to **add redundant bits** (an error-correcting code) to each data block. These bits are a **function of the data bits**.
2. For each incoming block, the receiver **calculates a new code** from the received data bits.
3. If the calculated code **matches** the received code, it assumes **no error**.
4. If they **don't match**, one or more bits are in error. If the number of errors is **below a threshold** (which depends on code length and algorithm), the receiver can **locate and correct** the bits.

```
Tx: data ──► [FEC encoder] ──► data + check bits ──► channel (errors) ──►
Rx: ──► [FEC decoder: recompute, compare] ──► corrected data (no request back to Tx)
```

**Overhead**: in mobile wireless the ratio **total bits / data bits is typically 2–3**, so capacity is cut to ½ or ⅓. This looks extravagant, but the mobile channel is so hostile that the redundancy is necessary.

### ★ Why backward error correction (ARQ) is not used in wireless [WC-S 45]

Backward error correction is not practical in many wireless applications:

- **Satellite communications**: the **delay** is so large that retransmission is undesirable.
- **Mobile communications**: **error rates are so high** that a retransmitted block is **likely to contain errors too**.

So **FEC is required** in these applications.

SUPPLEMENTARY nuance: ARQ is still used in wireless at higher layers or alongside FEC (802.11 uses ACK + retransmission; Bluetooth uses both FEC and ARQ, see Part 3). The point is that *ARQ alone* is not enough.

## 3.4 Adaptive equalization [WC-S 47, WC-B 5.4]

- Can be applied to **analog** (voice, video) **or digital** (data, digitized voice/video) information.
- Used to **combat intersymbol interference (ISI)**.
- **Gathers the dispersed symbol energy back into its original time interval.**
- **Techniques**: **lumped analog circuits**, or **sophisticated digital signal processing (DSP) algorithms**.
- "Adaptive": the equaliser **continually adjusts** as the channel changes. Equalisation tries to restore the flat-fading nature of the channel, which allows **higher data rates** before ISI appears [WCP].

SUPPLEMENTARY: equalisers become complex and expensive as ISI becomes more severe. This is one reason OFDM (Part 2) is attractive: with long OFDM symbols, an equaliser may not be needed [OFDM notes].

## 3.5 Diversity techniques ★ [WC-S 48, WC-B 5.4]

**Principle**: individual channels experience **independent fading**, e.g. antennas spaced far enough apart. So provide **multiple logical channels** and send part of (or a copy of) the signal on each. Diversity **does not eliminate errors, but reduces the error rate**, because it avoids being exposed to the worst error rate. FEC and equalisation then cope with the reduced error rate.

| Type | Idea | Examples |
|---|---|---|
| **Space diversity** | Uses the **physical transmission path**: multiple antennas spaced apart | Several nearby antennas whose signals are combined; **collocated directional antennas** pointing at different angles |

| Type | Idea | Examples |
|---|---|---|
| **Frequency diversity** | Spread the signal over a **larger bandwidth** or **multiple carriers** | **OFDM**, **spread spectrum (DSSS, FHSS)** |
| **Time diversity** | Spread the data **out over time** so a **noise burst affects fewer bits** | **Interleaving**, **Rake receiver** |

**Using the multiple received signals**:

1. **Selection diversity**: choose the **best** (or an acceptable) signal.
2. **Combining diversity**: **combine** the signals, adjusting gain and phase so they **add together** to improve the output.

## 3.6 MIMO (multiple-input multiple-output) ★ [WC-S 49–50, WC-B 5.4]

A system where the transmitter **and** receiver have **multiple antennas**. Antenna arrays are used for **four purposes** (Figure 5.9):

| Use | What it does | Benefit (Fig 5.9 caption) |
|---|---|---|
| **1. Diversity** | Multiple received copies via multiple Tx and/or Rx antennas | **Improved system performance** |
| **2. Beamforming** | Antennas configured to create **directional patterns**, focusing energy on intended recipients | **Improved coverage** (fewer cells for a given area) |
| **3. Multi-user MIMO (MU-MIMO)** | With enough antennas, **directional beams to multiple users simultaneously** (spatial division multiple access) | **Improved capacity** (more users per cell) |
| **4. Multilayer transmission (SU-MIMO)** | **Multiple parallel data streams** between **one pair** of transmitter and receiver | **Higher data rates** in a given bandwidth |

**Sizes**: modern systems use **4 × 4** (4 Tx, 4 Rx) and **8 × 8**; 2-D arrays of **64 antennas** are envisioned; the future is **massive MIMO**. MIMO and OFDM together are the **cornerstone of broadband wireless** (802.11n/ac/ax Wi-Fi, 4G LTE).

### ★ SU-MIMO vs MU-MIMO

**EXAM Asked in: T1B Q9 (8 marks); also lecturer's review list [WC-S 72].**

| | Single-user MIMO (SU-MIMO) | Multi-user MIMO (MU-MIMO) |
|---|---|---|
| Who | All spatial streams go between the base station/AP and **one user** at a time | Streams/beams go to **several different users simultaneously** |
| Technique | **Multilayer transmission**: parallel data streams over the same frequency/time | **Directional beams (beamforming)** to each user = **SDMA** |
| Goal | **Higher data rate** (peak throughput) for that user | **Higher capacity**: more users per cell |
| Requirement | The user device must have **multiple antennas** | Users can have **single antennas**; the BS/AP needs **enough antennas** to form a beam per user and must know each user's channel |
| Fig 5.9 label | "Multilayer transmission (SU-MIMO) for higher data rates in a given bandwidth" | "Spatial division multiple access (MU-MIMO) for improved capacity (more users per cell)" |

SUPPLEMENTARY: **FD-MIMO** (full-dimension MIMO, asked in the lecturer's review list) uses a **2-D antenna array** (e.g. the 64-antenna arrays in the notes) so beams can be steered in **both horizontal and vertical (elevation)** directions, e.g. to different floors of a tall building. **Beamforming** = adjusting the phase/amplitude of each antenna's signal so they add up constructively in the wanted direction and cancel elsewhere.

## 3.7 Adaptive modulation and coding (AMC) [WC-S 51, WC-B 5.4]

**Definition**: dynamically adjusting the **modulation scheme** and **coding rate** based on the **current channel conditions**.

- **Modulation** formats the signal to best transmit bits: to **overcome noise** and to **send as many bits as possible**.
- **Coding detects and corrects** errors.
- The channel can change **hundreds of times per second** (e.g. **200 times/s for a 5 ms coherence time**), so AMC must adapt equally fast.

**Two features needed in the protocol** [WC-B 5.4]:

1. **Mechanisms to measure channel quality**: packet loss rates, or **pilot signals** sent specifically for measurement.
2. **Messaging mechanisms** to send channel-quality indicators between Tx and Rx and to communicate the **new modulation and coding formats**.

*Practical meaning*: good channel → high-order modulation (e.g. 64-QAM) and light coding → high rate. Bad channel → robust modulation (e.g. BPSK/QPSK) and heavy coding → lower rate but reliable. This is the same idea as **rate adaptation** in 802.11 [WLAN slide 23].

## 3.8 Spread spectrum and OFDM: preview [WC-B 5.4]

Traditional systems just shift a baseband signal up to a carrier. Two methods **significantly modify the signal** to beat channel impairments:

- **OFDM** splits a signal into **many lower bit-rate streams** on **carefully spaced frequencies**. Each stream uses **much lower bandwidth with longer bit times**, which overcomes **frequency-selective fading**.
- **Spread spectrum** makes the signal use **100 times or more** wider bandwidth with **lower energy density** at each frequency. Even if some frequencies are poor, **average performance** is good.

*(Both are covered in full in Part 2.)*

## 3.9 Bandwidth expansion ★ [WC-S 52–53, WC-B 5.4]

All the correction mechanisms try to increase efficiency in **bps/Hz**, but **Shannon limits** the efficiency for a given SNR. If throughput must exceed what a bandwidth allows, use bandwidth expansion:

| Approach | How it works | Examples / issues |
|---|---|---|
| **1. Carrier aggregation** | **Combine multiple channels** into one wider channel | 802.11n/ac combine 20 MHz channels into **40, 80 or 160 MHz**; **4G LTE combines carriers** (Fig 14.14) |

| Approach | How it works | Examples / issues |
|---|---|---|
| **2. Frequency reuse** | **Limit propagation range** to an area and **reuse the same frequencies** where devices are far enough apart that interference is low | Traditional **macro cells** (km diameter); now **small cells**: indoor **femtocells**, outdoor **picocells/relays**, called **network densification** |
| **3. Millimeter wave (mmWave)** | Use **higher carrier frequencies (30–300 GHz)**, where wider channels are available; wavelengths 10 mm–1 mm (λ = c/f) | More bandwidth, **but** expensive, and **susceptible to obstructions and atmospheric absorption**; 802.11ad uses mmWave within a single room |

EXAM **Lecturer's review question [WC-S 72]: "Discuss three bandwidth expansion approaches"**. The answer is this table.

### Capacity: "Three methods to increase the capacity of a wireless communication system"

EXAM **Asked in: T1B Q5 (12 marks).** Your notes support these answers. Choose **three** and explain each well (Part 5 will give the full model answer):

1. **Bandwidth expansion**: carrier aggregation, mmWave (above).
2. **Frequency reuse / smaller cells (network densification)**: the same frequencies are reused many times [WC-B 5.4]. Note also the SDMA advantage: moving from a 7-cell to a 4-cell reuse pattern **nearly doubles capacity** [MA p9].
3. **MIMO**: SU-MIMO multilayer transmission (higher data rate in the same bandwidth) and MU-MIMO/SDMA (more users per cell) [WC-B 5.4].
4. **Efficient multiple access**: CDMA gives capacity **8–10× analog** and **4–5× other digital** systems [MA p6]; OFDMA [Part 2].
5. **Adaptive modulation and coding**: send as many bits as possible for the current channel [WC-S 51].

## 3.10 Examination focus — Topic 3

| Question | Source |
|---|---|
| What distinguishes SU-MIMO from MU-MIMO? (8) | T1B Q9; [WC-S 72] |
| Discuss three methods to increase the capacity of a wireless system (12) | T1B Q5 |
| As a mobile node moves farther from a BS, what two actions could the BS take so that the frame loss probability does not increase? (4 / 2) | T1A Q7, Exam Q5(a) |
| Explain FEC, adaptive equalization, AMC, diversity with MIMO | [WC-S 72] |
| Describe FD-MIMO and beamforming | [WC-S 72] |
| Three bandwidth expansion approaches | [WC-S 72] |
| Why is backward error correction not used in wireless? | [WC-S 45] |

**Preview for the "mobile node moves away" question** (T1A Q7 / Exam Q5(a)), which draws on this topic. Full answer in Part 5:

1. **Increase transmission power** to compensate for the larger path loss (free-space loss grows with $d^2$).
2. **Reduce the transmission rate**: switch to a **more robust modulation and stronger coding** (AMC / rate adaptation), which tolerates a lower SNR. *(Other valid options from your notes: use beamforming/directional antennas toward the node, or add FEC/retransmissions.)* SUPPLEMENTARY:

this question comes from Kurose & Ross (Ch. 7). Their model answer is exactly "increase transmit power" and "reduce the transmission rate".

## 3.11 Topic 3 revision summary

- **Seven correction mechanisms**: FEC, adaptive equalisation, AMC, diversity/MIMO, OFDM, spread spectrum, bandwidth expansion.
- **FEC**: add redundant bits; receiver corrects without retransmission; overhead ratio 2–3. **ARQ is impractical**: satellite delay; high mobile error rates mean retransmissions also fail.
- **Equalisation**: combats **ISI** by gathering dispersed energy back into the symbol's interval (analog circuits or DSP).
- **Diversity**: independent fading. **Space, frequency (OFDM, SS), time (interleaving, Rake).** Selection vs combining.
- **MIMO, four uses**: diversity (performance), beamforming (coverage), **MU-MIMO (capacity, more users)**, **SU-MIMO multilayer (data rate)**. 4×4, 8×8, massive MIMO.
- **AMC**: adapt modulation/coding hundreds of times per second; needs channel measurement and signalling.
- **Bandwidth expansion**: carrier aggregation, frequency reuse (small cells/densification), mmWave (30–300 GHz).

# TOPIC 4 — SIGNAL ENCODING TECHNIQUES (DIGITAL-TO-ANALOG MODULATION)

## 4.1 Introduction

Wireless channels carry **analog** electromagnetic waves, but our data are **digital** bits. **Digital-to-analog conversion (modulation)** changes a characteristic of a high-frequency **carrier** wave according to the data, so bits can travel through the air [SE]. Modulation also lets many transmitters **share a channel** (FDM) and makes **antennas of practical size** possible [WC-S 55–56].

## 4.2 Key definitions

| Term | Definition | Source |
|---|---|---|
| **Digital-to-analog conversion** | Changing one of the characteristics of an analog signal based on the information in digital data. | [SE] |
| **Modulation** | Shifts the spectrum of a baseband signal to some high frequency. **Bandpass modulation** varies some characteristic of a sinusoidal waveform according to the message signal. | [WC-S 55] |
| **Carrier signal** | A high-frequency signal produced by the sender that acts as a **base for the information signal**; the receiver is tuned to it. | [SE] |
| **Data element** | Smallest piece of information exchanged: **the bit**. | [SE] |
| **Signal element** | Smallest unit of a signal that is **constant**. | [SE] |
| **Bit rate (N)** | Number of bits per second. | [SE] |
| **Baud rate (S)** | Number of **signal elements per second**. In analog transmission of digital data, **baud rate ≤ bit rate**. | [SE] |
| **Bit error rate (BER)** | Better called **bit error probability**. It is **not a rate in bits/s but a probability**. | [WC-S 67] |

## 4.3 Why modulate? [WC-S 55–56]

1. **Antenna size**: antenna size is inversely proportional to frequency (an efficient antenna is about λ/4).
   - Speech f = 3 kHz → λ = c/f = $(3 \times 10^8)/(3 \times 10^3) = 10^5$ m → λ/4 = **25 km (≈ 15 miles)**, which is practically unrealisable.
   - Same speech modulated onto **fc = 900 MHz** → λ = 0.333 m → λ/4 ≈ **8 cm**.
2. **Channel sharing**: modulation onto different carriers lets several transmitters share a channel (**FDM**).
3. Most channels require the baseband signal to be **shifted to a higher frequency**.

## 4.4 Bit rate vs baud rate [SE]

$$S = N / r \text{ where } r = \log_2 L$$

- **S** = baud rate (signal elements/s); **N** = bit rate (bps); **r** = number of bits carried per signal element; **L** = number of different signal elements.
- Analogy from the notes: **a baud is a vehicle, a bit is a passenger**. Put more people in each car (more bits per signal element) to reduce traffic (baud rate, and therefore bandwidth).

#### Worked example B1 SUPPLEMENTARY practice

*An analog signal carries 4 bits per signal element and the bit rate is 8000 bps. Find the baud rate.* r = 4 → S = N/r = 8000/4 = **2000 baud**.

#### Worked example B2 SUPPLEMENTARY practice

*16-QAM at 1000 baud: bit rate?* L = 16 → r = $\log_2 16$ = 4 → N = S × r = **4000 bps**.

**Bandwidth** [SE]: the required bandwidth for analog transmission of digital data is **proportional to the signal (baud) rate**, except for **FSK**, where the **difference between the carriers** must be added.

## 4.5 The three basic methods

A sine wave has **three characteristics: amplitude, frequency and phase**. Changing any one creates a different version of the wave, which gives three basic modulation methods. A fourth, **QAM**, combines amplitude and phase changes and is the **most efficient** and most common today [SE, WC-S 57].




### 4.5.1 Amplitude-shift keying (ASK) [WC-S 59–60, SE]

**Definition**: the **amplitude** of the carrier is varied to create signal elements; frequency and phase stay constant.

**Binary ASK / on-off keying (OOK)**: one binary digit = **presence of the carrier at constant amplitude**, the other = **absence of the carrier**:

$$s(t) = A\cos(2\pi f_c t) \quad \text{for binary 1}$$
$$= 0 \quad \text{for binary 0}$$

- **Implementation**: multiply a **unipolar NRZ** signal (1 V / 0 V) by the carrier from an oscillator [SE].
- **Disadvantages**: **susceptible to sudden gain changes** (noise easily changes amplitude); **inefficient**.
- **Use**: transmitting digital data over **optical fibre** (the light source on/off).

### 4.5.2 Frequency-shift keying (FSK) [WC-S 61–63, SE]

**Definition**: the **frequency** of the carrier is varied to represent data; amplitude and phase stay constant. The frequency is constant for one signal element and changes when the data changes.

**Binary FSK (BFSK)**, the most common form: two binary values = **two different frequencies near the carrier**:

$$s(t) = A\cos(2\pi f_1 t) \quad \text{for binary 1}$$
$$= A\cos(2\pi f_2 t) \quad \text{for binary 0}$$

$f_1$ and $f_2$ are **offset from fc by equal but opposite amounts** (fd). The two bands are centred on $f_1$ and $f_2$, each Δf from the midpoint, so the **difference between the frequencies is 2Δf** [SE].

- **Advantages**: **less susceptible to error than ASK**.
- **Uses**: high-frequency radio (**3–30 MHz**); higher frequencies on LANs using **coaxial cable**.
- **Implementation** [SE]:
  - **Noncoherent BFSK**: phase may be **discontinuous** at element boundaries; implemented as **two ASK modulations with two carriers**.
  - **Coherent BFSK**: phase **continues** across boundaries; implemented with one **voltage-controlled oscillator (VCO)**, whose frequency rises when the NRZ input is positive.

**Multiple FSK (MFSK)**: more than two frequencies are used [WC-S 63]:

```
sᵢ(t) = A cos(2πfᵢt),   1 ≤ i ≤ M
fᵢ = fc + (2i − 1 − M) fd
```

fc = carrier frequency; fd = difference frequency; **$M = 2^L$** (number of signal elements); L = bits per element.

- **More bandwidth efficient but more susceptible to error** [WC-S 63].
- Forouzan's version [SE]: MFSK **uses more bandwidth** than other techniques and should be used **when noise is a serious issue**.

> SUPPLEMENTARY (reconciling the two statements): Stallings means MFSK sends more bits per *signal element* than BFSK. Forouzan compares MFSK to ASK/PSK, where MFSK needs more *total* bandwidth because each frequency needs its own band. If asked, say: "MFSK carries more bits per signal element, but needs a wider band than PSK/QAM; it is robust to noise."

#### Worked example F1 SUPPLEMENTARY practice, using the lecturer's formula

*MFSK with fc = 250 kHz, fd = 25 kHz, M = 8 (L = 3 bits). List the frequencies.* $f_i$ = 250 + (2i − 1 − 8) × 25 kHz = 250 + (2i − 9) × 25

- i = 1: 250 − 175 = **75 kHz**; i = 2: **125**; i = 3: **175**; i = 4: **225**; i = 5: **275**; i = 6: **325**; i = 7: **375**; i = 8: **425 kHz**.
- The frequencies are spaced 2fd = 50 kHz apart; total bandwidth ≈ 2M·fd = **400 kHz**.

### 4.5.3 Phase-shift keying (PSK) [WC-S 64–65, SE]

**Definition**: the **phase** of the carrier is shifted to represent data; amplitude and frequency stay constant. **Today PSK is more common than ASK or FSK** [SE].

**Binary PSK (BPSK)**: two phases, **0° and 180°**:

```
s(t) = A cos(2πfc t)        for binary 1
     = A cos(2πfc t + π)    for binary 0     (= −A cos(2πfc t))
```

- **Advantages** [SE]:
  - **Less susceptible to noise than ASK**: noise changes amplitude more easily than phase.
  - **Superior to FSK**: does not need two carriers.
  - Bandwidth is **the same as binary ASK and less than BFSK**, since no bandwidth is wasted separating two carriers.
- **Disadvantage**: needs **more sophisticated hardware** to distinguish phases.

- **Implementation**: multiply a **polar NRZ** signal (+V = 1, −V = 0) by the carrier [SE].

**Quadrature PSK (QPSK)**: **four phases, each element represents 2 bits** [WC-S 65]:

```
s(t) = A cos(2πfc t + π/4)     for 11
     = A cos(2πfc t + 3π/4)    for 01
     = A cos(2πfc t − 3π/4)    for 00
     = A cos(2πfc t − π/4)     for 10
```

- Built from **two BPSK modulators**, one **in-phase (I)** and one **quadrature (Q, 90° out of phase)**. A serial-to-parallel converter sends alternate bits to each [SE].
- Each BPSK branch gets bits of duration **2T** (half the rate). The sum of the two is a sine wave with one of **four phases: ±45°, ±135°** [SE].
- **L = 4, r = 2**, so **baud rate = half the bit rate**, which lowers the required bandwidth.

### 4.5.4 Quadrature amplitude modulation (QAM) [SE, WC-S 66]

- PSK is limited by how well equipment can **distinguish small phase differences**.
- **QAM = combination of ASK and PSK**: two carriers (in-phase and quadrature), each with **different amplitude levels**.
- Variations [SE Fig 5.14]: 4-QAM with unipolar NRZ (like OOK); **4-QAM with polar NRZ = QPSK**; 4-QAM with two positive levels; **16-QAM** (8 levels, 4 positive and 4 negative), shown in the **16-QAM constellation diagram** [WC-S 66].
- **Most efficient** digital-to-analog method; **dominant method** today; used by 802.11a/g OFDM subcarriers (Part 3).

```
16-QAM constellation (each dot = one 4-bit symbol)
      Q
  •  •  |  •  •
  •  •  |  •  •
 ———————┼——————— I
  •  •  |  •  •
  •  •  |  •  •
```

## 4.6 Comparison of modulation schemes SUPPLEMENTARY (assembled from [SE] and [WC-S 59–65])

| Scheme | What varies | Bits/symbol | Noise robustness | Bandwidth | Notes |
|---|---|---|---|---|---|
| ASK (OOK) | Amplitude | 1 | **Worst**: sensitive to gain changes | Same as BPSK | Simple; optical fibre |
| BFSK | Frequency | 1 | Better than ASK | **More** (two carriers + gap 2Δf) | HF radio 3–30 MHz, coax LANs |
| MFSK | Frequency (M values) | $\log_2 M$ | Robust to noise | Wide | Use when noise is serious |
| BPSK | Phase (0°/180°) | 1 | **Better than ASK** | Same as ASK, less than BFSK | Needs complex hardware |
| QPSK | Phase (4 values) | 2 | Good | Half of BPSK for the same bit rate | Two BPSK in quadrature |
| QAM | Amplitude **and** phase | $\log_2 L$ (e.g. 4 for 16-QAM) | Less robust as L grows (points closer) | Most efficient | Dominant today |

EXAM **"Which of ASK, FSK, PSK, QAM is most susceptible to noise? Defend your answer."** [SE Exercise 13] **Answer: ASK.** Its data is carried **only by amplitude**, and noise is additive, so it directly changes the amplitude. FSK and PSK carry data in frequency and phase, which noise disturbs much less. *(QAM also uses amplitude, so it is more noise-sensitive than PSK at high orders, but ASK is the classic answer because it relies on amplitude alone.)*

## 4.7 Bit error rate (BER) and Eb/N0 [WC-S 67–68]

- Performance must be assessed **in the presence of noise**.
- "**Bit error probability**" is a clearer term: **BER is not bits/s but a probability**.
- Plotted with **log-scale BER on the y-axis** and **Eb/N0 in dB on the x-axis** (Eb/N0 = energy per bit / noise power spectral density).
- **As Eb/N0 increases, BER drops.**
- **Curves to the lower left are better**: **lower BER at the same Eb/N0**, or **lower Eb/N0 for the same BER**.
- Figure 5.12 compares multilevel FSK, PSK and QAM. SUPPLEMENTARY reading of that figure: for **PSK and QAM**, increasing M (more bits per symbol) **needs more Eb/N0** for the same BER. For **MFSK**, increasing M **reduces** the Eb/N0 needed, at the cost of more bandwidth. This trade-off is why AMC picks the modulation order according to channel quality.

## 4.8 Coding and error control: preview [WC-S 69–71]

- **Error detection codes** detect the **presence** of an error.
  - With **ARQ (automatic repeat request)**, a block with an error is **discarded** and the transmitter **retransmits** it.
- **Error correction codes (FEC)** are designed to **detect and correct** errors.
- **Error detection process** [WC-S 70]:
  - **Transmitter**: for each frame, compute an error-detecting code (**check bits**) from the data bits and **append** them.
  - **Receiver**: separate data and check bits, **recompute** the check bits from the received data, **compare**. A **mismatch = detected error**.

*(Parity, checksum, Hamming and CRC calculations are fully covered in Part 2, Topic 8.)*

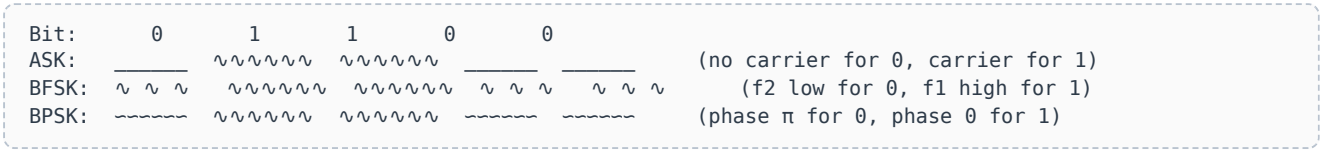
## 4.9 Examination focus — Topic 4

**Lecturer's exercise list** [SE] (likely short questions):

1. Define carrier signal and its role → see 4.2.
2. Function of **scrambling** in digital-to-digital encoding SUPPLEMENTARY: replaces long runs of 0s with special patterns so the receiver keeps **synchronisation**, without increasing bandwidth (e.g. B8ZS, HDB3).
3. What does a **modem** do? → converts digital data to an analog signal by modulating a carrier, and back again [TF-B 20].
4. How are binary values represented in ASK, and the limitation → presence/absence of carrier; susceptible to sudden gain changes, inefficient.
5. QPSK vs **offset QPSK** SUPPLEMENTARY: OQPSK delays the Q stream by **half a symbol (one bit time)**, so I and Q never change together. This avoids 180° phase jumps and gives smaller envelope fluctuations (better for power amplifiers).

6. What is QAM? → combination of ASK and PSK (4.5.4).
7. Sampling theorem SUPPLEMENTARY: sample at **at least twice the highest frequency** (Nyquist rate). Compare the T1 example: voice is sampled at 8 kHz [Access Tech article].
8. Angle modulation, PM and FM SUPPLEMENTARY: angle modulation varies the carrier's angle. **FM** varies frequency in proportion to the message amplitude; **PM** varies phase in proportion to the message. Both are forms of angle modulation.
9. **DPSK** SUPPLEMENTARY: differential PSK. The bit is represented by a **change or no change in phase** relative to the previous symbol, so no absolute phase reference is needed. (Bluetooth EDR uses π/4-DQPSK and 8DPSK, Part 3.)
10. Encode **01100** with ASK, BFSK and BPSK → see the sketch below. 11–14. PCM vs DM, modem vs codec SUPPLEMENTARY: **PCM** samples and quantises each sample into n bits. **DM (delta modulation)** sends 1 bit per sample (up or down a step). PCM gives better quality; DM is simpler but suffers slope overload and granular noise. A modem and a codec are **not functional inverses**: a modem maps *digital data → analog signal*; a codec maps *analog data → digital signal*.
11. Most susceptible to noise → **ASK** (4.6).

**Exercise 10 sketch: encode 01100**




In BPSK, draw a **phase reversal** (the wave flips upside down) at each 0↔1 boundary: between bits 1–2 and 3–4. There is no reversal between bits 2–3 or 4–5.

## 4.10 Topic 4 revision summary

- Modulation shifts a baseband signal to a carrier: **small antennas** (λ/4: 25 km at 3 kHz vs 8 cm at 900 MHz) and **channel sharing**.
- **S = N/r, r = $\log_2$L**: baud ≤ bit rate. Bandwidth ∝ baud rate (FSK adds the carrier separation).
- **ASK/OOK**: amplitude; simple, noise-sensitive, inefficient; optical fibre.
- **BFSK**: two frequencies ±fd; better than ASK; HF radio. **MFSK**: $f_i$ = fc + (2i−1−M)fd.
- **BPSK**: 0°/180°; better than ASK in noise; same bandwidth as ASK. **QPSK**: 4 phases (±45°, ±135°), 2 bits/symbol, I and Q branches.
- **QAM** = ASK + PSK; most efficient; 16-QAM constellation.
- **BER** is a probability. BER vs Eb/N0: lower-left curves are better.
- **Error detection**: append check bits, recompute, compare. **ARQ** retransmits; **FEC** corrects.

## What comes next

**Part 2** covers Topic 5 (multiple access: FDMA, TDMA, CDMA, SDMA, cellular generations), Topic 6 (spread spectrum: FHSS, DSSS, why CDMA is hard to jam), Topic 7 (OFDM, OFDMA, SC-FDMA) and Topic 8 (error detection and correction: parity, checksum, Hamming and CRC, all worked step by step, including the exact past-paper numbers).

## Errors found in your notes (Part 1 topics)

| Location | Error | Correct |
|---|---|---|
| *Wireless Communication Problems*, Problem 2.7 (p. 62) | "125 × $10^{-3}$ = (4 × 3.14 × d/0.3529)²" | Path loss = **125 × $10^{3}$**; final answer 9.93 m is correct |
| Same file, Problem 2.3 | Pr = 0.01 µW | **8.1 × $10^{-9}$ W (0.0081 µW)** |
| Same file, Problem 2.5 | Lp = 2.46 × $10^{8}$ | **Lp = 2.47 × $10^{10}$** |

# Part 2 — Detailed Study Notes: Topics 5–8

**Topics in this part** 5. Multiple Access Techniques (FDMA, TDMA, CDMA, SDMA) and Cellular Generations 6. Spread Spectrum (FHSS, DSSS, CDMA anti-jamming) 7. OFDM, OFDMA and SC-FDMA 8. Error Detection and Correction (parity, Internet checksum, Hamming code, CRC)

### Source codes used in this part

| Code | File |
|---|---|
| **[MA p]** | *Multiple access techniquesV2.pdf*, page p (scanned slides) |
| **[AT]** | *Fundamentals of Communications Access Technologies.pdf* (Frenzel article) |
| **[MX]** | *Multiplexing notes.pdf* |
| **[SS]** | *Spread spectrum.pdf* (Forouzan §6.2) |
| **[OF]** | *OFDM.pdf* (Beard & Stallings Ch. 8) |
| **[ED]** | *Error detection and correction_v2.docx* (worksheet) |
| **[WC-S n] / [WC-B §]** | *The Wireless Channel.pptx* / *Wireless Communication.pdf* |
| **[BT]** | Bluetooth PDF and slides |

# TOPIC 5 — MULTIPLE ACCESS TECHNIQUES

## 5.1 Introduction

*"Sharing the limited bandwidth efficiently among many users is one of the main objectives of multiple access schemes"* [MA p1].

Cellular systems divide a region into **cells**, and a mobile in each cell communicates with a **base station**. The main objective of cellular design is to **handle as many calls as possible (capacity) in a given bandwidth with reliability**. Success depends on the availability of **radio frequency spectrum**, a **finite natural resource** defined in terms of **bandwidth**. The technologies that let many users share a finite amount of spectrum simultaneously and efficiently are called **multiple access** [MA p2].

The choice of multiple access technology **can significantly enhance or lower the service quality** delivered to end users [MA p2].

**The four multiple access techniques** [MA p2]: **FDMA, TDMA, CDMA, SDMA**. The article [AT] adds **OFDMA** as a fifth, and mentions CSMA as "other methods".

**Multiplexing vs multiple access**: multiplexing creates channels in frequency, time or code (FDM, TDM, CDM) [MA p3]. **Access methods are multiplexing techniques that provide communication services to multiple users** in a single-bandwidth medium [AT].

**Highway analogy** [MA p3]: the medium is the **highway**, the users are **cars**, and interference is **accidents**. Separate lanes = SDMA or FDMA; cars taking turns on one lane = TDMA.

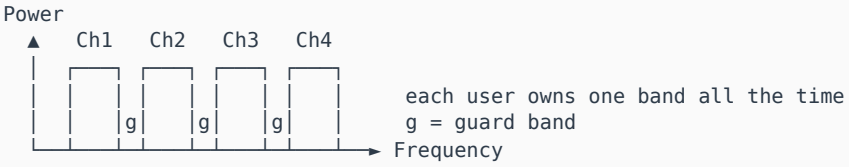
## 5.2 Key definitions

| Term | Definition | Source |
|---|---|---|
| **Multiple access** | Allowing many mobile users to share a finite amount of radio spectrum simultaneously and efficiently, to maximise call-handling capacity in a given bandwidth | [MA p2] |
| **FDMA** | Radio spectrum divided into several frequency bands separated by **guard bands**; each band is used by one user simultaneously with the others | [MA p3] |
| **Guard band** | Unused frequency slots that separate neighbouring channels to minimise adjacent channel interference (crosstalk) | [MA p4] |
| **TDMA** | Several users share the same frequency band by dividing time into **slots** periodically allocated to each user | [MA p5] |
| **CDMA** | Transmissions **occupy the entire bandwidth at the same time** without interference, separated by **unique codes** | [MA p7] |
| **Spread-spectrum signal** | A signal with an **extra modulation** that expands its bandwidth **beyond what the data modulation requires** | [MA p7] |
| **Chip rate** | The rate of a spreading code (instead of "bit rate") | [MA p8] |
| **Pseudorandom code sequence** | Unique code shared by the mobile and the base station, used to spread and despread a user's data | [MA p8] |
| **SDMA** | Uses the **spatial dimension**: the same bandwidth is shared simultaneously by users in **different geographical locations** or **different beam directions** | [MA p9] |

# 5.3 Frequency Division Multiple Access (FDMA) ★

## Principle [MA p3–4, MX, AT]

- The **simplest** scheme; used for multiple access in **analogue** transmission.
- The spectrum is **divided into several frequency bands separated by guard bands**. Each band can be used simultaneously.
- The bandwidth is divided into channels and distributed among users, each with a **finite portion of bandwidth for permanent use**.
- **Only one user per channel**, who can use the channel **100% of the time**. Only the **frequency dimension** defines channels.
- **Guard bands** between adjacent channels minimise **crosstalk**.
- Each FDMA channel has **two frequencies**: a **forward channel** (base → mobile) and a **reverse channel** (mobile → base).
- While a user is "in conversation", **no other user can access the same spectrum**.
- Additional points [MX]: compatible with **both digital and analog** signals; needs **highly efficient filters**; **no timing issues** (unlike TDMA); **not prone to the near-far problem** of CDMA (thanks to frequency filtering); its weakness is **crosstalk** between frequencies.




## Number of channels ★ [MA p4]

$$N = (Bt - 2Bg) / Bc$$

- **Bt** = allocated frequency spectrum (total bandwidth)
- **Bg** = guard band allocated **at the edge** of the spectrum
- **Bc** = channel bandwidth

### Worked example FD1 SUPPLEMENTARY (standard textbook numbers, using the lecturer's formula)

*A US AMPS-type system is allocated Bt = 12.5 MHz per direction, with Bg = 10 kHz at each edge and Bc = 30 kHz. How many channels?*

1. Formula: N = (Bt − 2Bg)/Bc
2. Substitute in kHz: N = (12 500 − 2 × 10)/30 = 12 480/30
3. **N = 416 channels**

### Worked example FD2 SUPPLEMENTARY

*Bt = 5 MHz, Bg = 25 kHz, Bc = 200 kHz.* N = (5000 − 50)/200 = 24.75 → **24 channels** (always round **down**: a partial channel can't be used).

### Advantages of FDMA [MA p4]

1. **Continuous transmission**, so lower complexity than TDMA; synchronisation requirements are not severe.
2. **Simple hardware**: users are isolated with simple **band-pass filters**.
3. Fairly efficient with a **small user population** and **constant traffic**.
4. **No channel equalization** required.
5. Capacity can be increased by **reducing the information bit rate** and using efficient digital speech coding.

### Disadvantages of FDMA [MA p4]

1. An idle channel **sits idle** and cannot be used by others → **inefficient use of spectrum**.
2. Needs **tight RF filtering** to minimise adjacent channel interference.
3. **Network and spectrum planning are intensive**.
4. **Frequency planning is time-consuming**.
5. **Guard bands waste bandwidth**; bandwidth is also wasted when transmission isn't continuous.

### Applications [MA p4, AT]

- **Walkie-talkies** and mobile networks for **closed user groups**.
- **AM/FM radio broadcasting** (each station has its own channel).
- **Early cellular (1G)**, analogue.
- **Cable TV** (coax 4 MHz–1 GHz divided into 6 MHz channels) [AT].
- Fibre optics: **WDM/WDMA** (wavelength division) [AT].
- The old analog telephone hierarchy: 12 voice channels on subcarriers from 60–108 kHz (SSB) [AT].

## 5.4 Time Division Multiple Access (TDMA) ★

### Principle [MA p5–6, MX, AT]

- Digital systems don't need continuous transmission because users don't use their bandwidth all the time.
- Several users share the **same frequency band** by **dividing time into slots** allocated periodically to each user for the duration of a call.
- Possible because TDMA systems **transfer digital data**.
- **Different users can use the same frequency in the same cell, at different times.**
- 2G TDMA systems: **GSM, IS-136, PDC, DECT**.
- The US cellular TDMA standard divides a channel into **six slots, each signal using two**, so **three users per channel**, a **3:1 capacity gain over AMPS**.
- **Registration**: when the user presses push-to-talk (PTT), a control channel registers the radio with the nearest base station, which assigns **a pair of channels** (transmit/receive) **and a time slot**. The user can **only send/receive in its slot**.
- **Information flow is not continuous**: it is sent in **bursts**, re-assembled at the receiver fast enough to sound continuous.
- Analogue calls are **sampled** into digital format and placed into the assigned slot.
- Users need **synchronisation via a global clock** [MX].

**Two types** [MX]:

- **Fixed TDMA**: slots are statically assigned to data streams, changed only for large traffic variations. Slot size = **T/N** (T = frame time, N = users).
- **Dynamic TDMA**: a **scheduling algorithm** reserves a **variable number of slots** per frame according to traffic demand.




## Number of users ★ [MA p6]

$$\text{Users supported} = \text{Number of channels in the spectrum} \times \text{Time slots per channel}$$

Example in the notes' figure: **4 channels × 3 slots = 12 users**.

### Worked example TD1 SUPPLEMENTARY (GSM numbers from [AT])

*GSM uses 200 kHz channels with 8 slots each. With 25 MHz (one direction) and no guard bands, how many users?*

1. Channels = 25 000/200 = **125**
2. Users = 125 × 8 = **1000 simultaneous users** *(With one channel reserved as a guard band, as in real GSM: 124 × 8 = 992.)*

### Worked example TD2: combining FDMA and TDMA SUPPLEMENTARY

*Bt = 12.5 MHz, Bg = 10 kHz, Bc = 30 kHz, 3 slots/channel (IS-136).* N = 416 channels (FD1) → users = 416 × 3 = **1248**.

## The T1 and GSM examples [AT]

- **T1**: 24 voice calls; each voice signal (300–3000 Hz) is sampled at **8 kHz** (just above the Nyquist rate) × 8 bits = **64 kbps**. 24 bytes + 1 framing bit = **193 bits per frame**, giving **1.544 Mbps**.
- **GSM**: 200 kHz channels, **8 users per channel**, frame sent at **270 kbps** using **GMSK** (a form of FSK); one frame = 1248 bits.

## Advantages of TDMA [MA p7]

1. Data are transmitted in **discrete bursts**, giving:
   - **extended battery life** and talk time compared to FDMA;
   - a **simpler handoff**: the phone can listen to other base stations during idle slots.
2. **More efficient use of spectrum** than FDMA:
   - **accommodates more users** in the same spectrum (capacity in high-traffic metropolitan areas);
   - efficient use of **hierarchical cell structures** (pico, micro, macro).
3. Different slots are used for transmit and receive, so **duplexers are not required**.

## Disadvantages of TDMA [MA p7]

1. **Requires synchronisation**: if slot sync is lost, channels may collide.

2. On the uplink, handsets need **high peak power** in transmit mode, which shortens battery life. With n channels the transmit power is **10 log n** dB higher than FDMA.
3. Network and spectrum planning are intensive.
4. **Dropped calls** are possible when users move between cells.
5. **Higher costs** from equipment sophistication.
6. **Equalization is required**, because transmission rates are much higher than in FDMA channels.

## 5.5 Code Division Multiple Access (CDMA) ★★

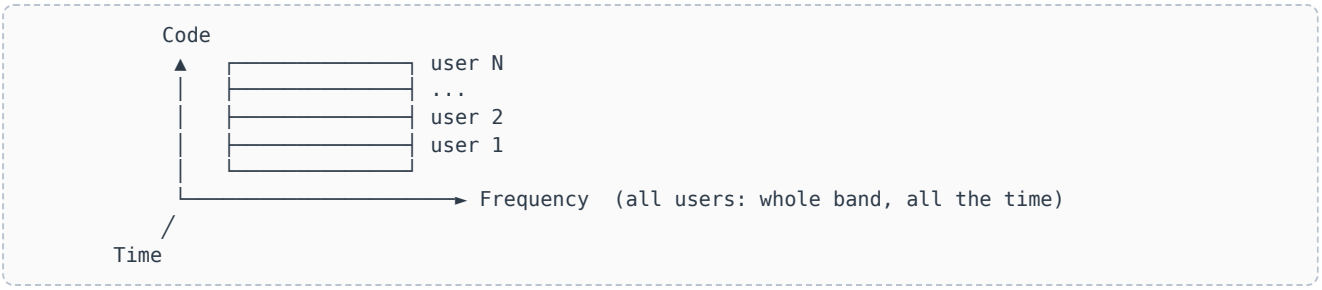
### Principle [MA p7–8, MX, AT]

- CDMA allows transmissions to **occupy the entire bandwidth at the same time without interference**.
- Based on **spread spectrum**: an extra modulation expands the bandwidth well beyond what the data need.
- The dominant spread-spectrum methods are **direct-sequence (DSSS)** and **frequency hopping (FHSS)**.
- **Spread spectrum is useful for** [MA p7]:
  1. **Suppressing interference**
  2. **Making interception difficult**
  3. **Accommodating fading and multipath channels**
  4. **Providing a multiple-access capability**
- History: originally **IS-95**, competing with GSM; operates in **800 MHz and 1.9 GHz PCS** bands; developed by **Qualcomm**. Compared to GSM, it needs **fewer towers** and gives **up to 5× the calling capacity**; **>10× the voice traffic of AMPS**; the basis for **3G** [MA p7].
- **Military origin**: first used in **WWII** by the English allies **to foil German attempts at jamming** [MX].

**How it works** [MA p8]:

1. Each user gets a **unique code sequence** (a **pseudorandom code**) used to code data before transmission; the code is shared by the mobile and the base station.
2. The baseband data are **spread** with this code, which runs at a **much higher rate** (the **chip rate**) than the data.
3. **All users access the entire spectrum all of the time.** The **unique code separates each call** from all others.
4. The transmitted signal sits **below the noise level**.
5. The receiver uses a **correlator** to **despread the wanted signal**, which then passes through a **narrowband filter**.
6. **Unwanted signals are not despread**, so they do not pass the filter; they stay as low-level noise.
7. **Power control**: phones and base stations use the **minimum power** needed, with precise power control to reduce users' power. This also gives **battery life, talk time and smaller batteries** [MA p8].

[AT] adds: data are **XORed with a chipping signal**. IS-95 uses a **1.2288 Mchip/s** chipping signal to spread **13 kbps** voice over **1.25 MHz**. **64 codes = 64 users** in one 1.25 MHz channel. **WCDMA** (3G) uses **3.84 Mchip/s** in **5 MHz**. In [MX], each subscriber is assigned an **orthogonal code called a "chip"**.




## Advantages of CDMA [MA p8]

1. **Greatest spectrum efficiency**: capacity about **8–10× analogue** and **4–5× other digital systems**; best in high-traffic areas with limited spectrum.
2. **Improves call quality** by filtering out background noise, crosstalk and interference.
3. **Soft handoffs**: because of multiple diversities, handoffs between cells are **undetected by the user**.
4. **Simplified frequency planning**: all users use the **same frequency spectrum**, so no detailed frequency plans, no re-tunes for expansion, and **fewer cells** for coverage.
5. **Random Walsh codes enhance user privacy**.
6. **Precise power control** increases talk time and battery life.

## Disadvantages of CDMA [MA p9]

1. **Backwards-compatibility** techniques are costly.
2. Equipment is **expensive**.
3. **Difficult to optimise** for maximum performance.
4. **Low-traffic areas** make inefficient use of spectrum and equipment.
5. (From the comparison table) **complex receivers**; needs **complicated power control** for senders [MA p10].

## Worked example: how CDMA codes separate users SUPPLEMENTARY (illustration of the principle in [MA p8])

Use bipolar values: bit 1 → +1, bit 0 → −1. Two users get **orthogonal (Walsh) codes**:

- User A code: **[+1 +1 +1 +1]**
- User B code: **[+1 −1 +1 −1]**
- Orthogonality check: A·B = (1)(1) + (1)(−1) + (1)(1) + (1)(−1) = **0** ✔

**Step 1: spread.** A sends bit **1** (+1) → +1 × A = [+1 +1 +1 +1]. B sends bit **0** (−1) → −1 × B = [−1 +1 −1 +1]. **Step 2: the channel adds them.** Received = [0 +2 0 +2]. **Step 3: despread for A.** Received · A = 0 + 2 + 0 + 2 = 4 → divide by 4 chips = **+1 → bit 1 ✔ Step 4: despread for B.** Received · B = 0 − 2 + 0 − 2 = −4 → /4 = **−1 → bit 0 ✔**

Each receiver recovers **only its own data**, even though both signals used the **same frequency at the same time**. A receiver without the right code sees only noise.

## Why is CDMA difficult to jam? ★ (Asked in: Exam Q1(d), T1A Q6, T1B Q7)

From the notes:

1. **The signal is spread over a very wide bandwidth** with low power density, **below the noise level** [MA p8]. A narrowband jammer only hits a small part of the band.

2. **Despreading spreads the jammer**: the receiver's correlator multiplies the received signal by the user's code. The wanted signal collapses back to narrowband, while the **jammer (not coded with the user's code) is spread out** and mostly removed by the **narrowband filter** [MA p8].
3. **Pseudorandom code is unknown to the jammer**: without knowing the code (or the hop sequence in FH), a jammer cannot concentrate energy where the signal is. In FHSS, *"a malicious sender may be able to send noise to jam the signal for one hopping period (randomly), but not for the whole period"* [SS].
4. **Designed for this purpose**: spread spectrum's stated uses include **suppressing interference** and **making interception difficult** [MA p7]. It was used in **WWII to foil German jamming** [MX]; Forouzan describes SS as wrapping the message in a "protective envelope" against **jamming from a malicious intruder** [SS].

SUPPLEMENTARY: this resistance is measured by the **processing gain** Gp = Bss/B (= chip rate / bit rate). E.g. IS-95: 1.2288 Mchip/s ÷ 9.6 kbps = 128, i.e. **≈ 21 dB**. A jammer is suppressed by roughly this factor.

## 5.6 Space Division Multiple Access (SDMA) ★

### Principle [MA p9–10, MX, AT]

- Besides frequency, time and code, the **spatial dimension** can separate users: data streams are sent over **different, non-overlapping spatial channels**.
- **SDMA lets users share the same bandwidth simultaneously in different geographical locations.**
- It solves the capacity problem by **exploiting the spatial dimension**: users can be identified **even when they share the same time/frequency/code**.
- Achieved by **beamforming** or **sectorization**.
- **Primitive form**: the **same frequency reused in different cells** (frequency reuse) [MX, AT]. **Sectorized antennas** (three 120° sectors per site) are a primitive form of SDMA [MA p10, AT].
- **Advanced form**: **smart antennas / adaptive arrays** with **dynamic beamforming**, focusing narrow beams on specific users and excluding others [AT]; **spot beams** [MX].
- Areas covered by different beams can use the **same frequency (TDMA or CDMA)** or **different frequencies (FDMA)** [MA p10, MX].
- In practice more than one transmitter may fall inside a beam, so **SDMA is combined with other multiple access techniques** [MX].

**Why SDMA was developed: omnidirectional vs adaptive antennas** [MA p9–10]

| Omnidirectional (traditional) | Adaptive antenna / SDMA |
|---|---|
| Base station has **no information on the mobile's position**; radiates in **all directions** | Uses **adaptive antenna arrays (smart antennas)** to send/receive **only within a limited angular range**, following the directional nature of multipath |
| **Wastes power** where there are no mobiles; causes **interference to co-channel cells** | Improves coverage/link quality when **noise-limited**; **enhances capacity** when **interference-limited** |
| Receives noise and interference from all directions | A BS with a **P-element antenna array** serves several single-antenna mobiles at once |

**Polarization division (PDMA)** [AT]: a variant of SDMA. Two signals on the same frequency use **vertical and horizontal polarization**; the antennas don't respond to the opposite polarization. Widely used in **satellites**.

### Advantages of SDMA [MA p10]

1. **Range extension**: fewer cells needed to cover an area.
2. **Interference suppression**: interference from other systems and cells is greatly reduced by exploiting each user's **unique channel impulse response**.
3. **Multipath effect elimination**: multipath copies can be mitigated, or even used to **reinforce** the signal.
4. **Capacity increase**, in two ways:
   - **tighter reuse patterns** (e.g. moving from a **7-cell to a 4-cell pattern nearly doubles capacity**);
   - **intracell reuse**: separate spatial channels on the **same conventional channel within one cell**.
5. **Compatibility** with most existing modulation schemes, carrier frequencies and specifications.

## 5.7 OFDMA (from the access article) [AT]

- Used in **LTE**. OFDM divides a channel into many narrow **orthogonal** subcarriers (**15 kHz** each) that don't interfere.
- Data are divided into many **lower-speed streams** modulated onto the subcarriers.
- **Each user is assigned a group of subcarriers and time slots.** The smallest group is **12 subcarriers = one resource block (RB)**.
- Very **spectrally efficient** and **less affected by multipath**. (Full detail in Topic 7.)

## 5.8 Other methods: CSMA [AT]

- **CSMA/CD**: the classical **Ethernet** method. Listen first, transmit if idle; if a collision occurs it is **detected**, and nodes wait a **random time** before retrying.
- **CSMA/CA**: similar, but a scheduling algorithm determines when to transmit. **CSMA/CD is used in wired networks; CSMA/CA is preferred in wireless.** (Full detail in Topic 9.)

## 5.9 ★ Comparison table [MA p10]

| | SDMA | TDMA | FDMA | CDMA |
|---|---|---|---|---|
| **Idea** | Segment **space** into cells/sectors | Segment **sending time** into disjoint slots, demand-driven or fixed | Segment the **frequency band** into disjoint sub-bands | **Spread the spectrum** using orthogonal codes |
| **Terminals** | Only one terminal active in one cell/sector | All terminals active for short periods on the same frequency | Every terminal has its **own frequency**, uninterrupted | **All terminals active at the same place at the same moment**, uninterrupted |
| **Signal separation** | Cell structure, directed antennas | **Synchronization in the time domain** | **Filtering in the frequency domain** | **Code plus special receivers** |
| **Advantages** | Very simple; increases capacity per km² | Established, fully digital, very flexible | Simple, established, robust | Flexible, less planning needed, **soft handover** |
| **Disadvantages** | Inflexible; antennas typically fixed | **Guard space** needed (multipath); synchronization difficult | Inflexible; frequencies are a scarce resource | **Complex receivers**; needs complicated **power control** |

| | SDMA | TDMA | FDMA | CDMA |
|---|---|---|---|---|
| **Comment** | Only useful combined with TDMA, FDMA or CDMA | Standard in fixed networks; with FDMA/SDMA in many mobile networks | Typically combined with TDMA (frequency hopping) and SDMA (frequency reuse) | Used in many 3G systems; higher complexity; integrated with TDMA/FDMA |

## 5.10 Principles of cellular networks [MX, MA p2]

- Cellular technology is the basis of mobile wireless communication. It supports users where wired networks are hard to provide.
- It was developed to **increase capacity** for mobile radio telephone service, **replacing high-power transmitter/receiver systems**.
- It uses **lower power, shorter range and more transmitters**.
- The area is divided into **cells**, each served by a **base station** (network tower).
- The aim: **increase channel capacity**, i.e. handle as many calls as possible in a given bandwidth with sufficient quality of service.

## 5.11 Cellular network generations [MA p11–16]

| Generation | Key description [MA p11] |
|---|---|
| **0G** | Briefcase-size mobile radio telephones |
| **1G** | **Analog** cellular telephony |
| **2G** | **Digital** cellular telephony |
| **3G** | High-speed digital cellular telephony (including **video telephony**) |
| **4G (LTE)** | **IP-based** "anytime, anywhere" voice, data and multimedia at faster rates than 3G |

**Technologies per generation** [MA p11]: 1G: **AMPS, NMT, TACS**. 2G: **D-AMPS, GSM/GPRS, cdmaOne**. 3G: **CDMA2000/EV-DO, WCDMA/HSPA+, TD-SCDMA**. 4G: **LTE, LTE-Advanced**. **Services**: 1G analog voice; 2G digital voice + simple data (<0.5 Mbps); 3G mobile broadband (63+ Mbps); 4G faster and better (300+ Mbps; richer content, more connections) [MA p11]. **Evolution summary** [MA p12]: 1G established **seamless mobile voice**; 2G **increased voice capacity** ("mobile to the masses"); 3G **optimised mobile for data** (mobile broadband); 4G LTE delivers **more capacity** for faster broadband.

### Details per generation [MA p13–15]

**1G**: circuit-switched; **FDMA**; analog; basic mobility; **poor voice quality**; **poor security**. Frequency **150 / 900 MHz**; analog bandwidth (**30 kHz**); data rate **2 kbps**; **1980–1990**; poor battery, big phones; "better than nothing".

**2G**: frequency **1.8 GHz (900 MHz)**; bandwidth 900 MHz (25 MHz); **digital**; **GSM**; **64 kbps**; **1991–2000**; **text messaging**; the signal must be strong or the digital signal is weak. **2.5G**: 2G with **GPRS**: e-mail, web browsing, camera phones.

**3G**: frequency **1.6–2.0 GHz**; bandwidth **100 MHz**; digital broadband with increased speed; **CDMA, UMTS, EDGE**; **144 kbps – 2 Mbps**; **2000–2010**; smartphones, **video calls**, mobile TV; phones expensive.

**4G**: frequency **2–8 GHz**; bandwidth 100 MHz; high speed, **all-IP**; **LTE, WiFi**; **100 Mbps – 1 Gbps**; **2010–(2020)**; **MAGIC** = **M**obile multimedia, **A**nytime anywhere, **G**lobal mobile support, **I**ntegrated wireless

solutions, **C**ustomized personal service; good QoS + high security; **bigger battery usage**.

**5G**: **2020–2030**; capacity **1 Gbps – "unlimited"**; high speed and capacity; faster than 4G; supports interactive multimedia, voice streaming; more efficient; frequency **3–300 GHz**; **1000× bandwidth per unit area**; technology **WWWW** (World Wide Wireless Web).

### Comparison table [MA p16]

| | 1G | 2G | 3G | 4G | 5G |
|---|---|---|---|---|---|
| Deployment | 1970–80 | 1990–2004 | 2004–10 | Now | Soon (≈2020) |
| Data bandwidth | 2 kbps | 64 kbps | 2 Mbps | 1 Gbps | > 1 Gbps |
| Technology | Analog | Digital | CDMA 2000, UMTS, EDGE | Wi-Max, Wi-Fi, LTE | WWWW |
| Core network | PSTN | PSTN | Packet network | Internet | Internet |
| **Multiplexing** | **FDMA** | **TDMA/CDMA** | **CDMA** | **CDMA** | **CDMA** |
| Switching | Circuit | Circuit, packet | Packet | All packet | All packet |
| Primary service | Analog phone calls | Digital calls + messaging | Calls, messaging, data | All-IP service (incl. voice) | High speed, high capacity, Gbps broadcasting |
| Key differentiator | Mobility | Secure, mass adoption | Better Internet experience | Faster broadband, lower latency | Better coverage, no dropped calls, much lower latency |
| Weakness | Poor spectral efficiency, major security issue | Limited data rates; hard to support Internet/e-mail | Real performance fails to match hype; failure of WAP | More battery use; complicated, expensive hardware | ? |

**NOTES ERROR**: the multiplexing row lists **CDMA for 4G and 5G**. That is inaccurate. **4G LTE uses OFDMA** (downlink) and **SC-FDMA** (uplink), as your own OFDM notes state [OF §8.2–8.3] and the article confirms [AT]. **5G NR also uses OFDM-based access.** If asked about 4G multiple access, answer **OFDMA**. Also note that the two "period" tables in your notes disagree (1G "1980–1990" vs "1970–80"). Quote the period with "approximately".

## 5.12 Examination focus — Topic 5

- Define multiple access; list the four techniques.
- FDMA/TDMA/CDMA/SDMA: principle, advantages, disadvantages, applications (4–10 marks each).
- **Describe CDMA and explain why it is difficult to jam** (4–10 marks; asked **3 times**).
- Compare FDMA, TDMA, CDMA and SDMA (the comparison table).
- Calculations: FDMA channels N = (Bt − 2Bg)/Bc; TDMA users = channels × slots.
- 1G–5G comparison; MAGIC.

## 5.13 Topic 5 revision summary

- **FDMA**: frequency bands + guard bands; one user per channel, 100% of the time; simple filters; wastes idle channels and guard bands. **N = (Bt − 2Bg)/Bc.**
- **TDMA**: one frequency, time slots, bursts; needs **sync** and **equalization**; better spectrum use, battery life, simple handoff; no duplexer. **Users = channels × slots.**
- **CDMA**: whole band, all the time, **unique pseudorandom codes**, chip rate ≫ bit rate, correlator despreads; **power control**; soft handoff; 4–5× capacity of other digital systems; **hard to jam/intercept**.
- **SDMA**: space; beamforming, sectorization, frequency reuse; range extension, interference suppression, capacity.
- Generations: 1G analog FDMA → 2G digital GSM TDMA → 3G CDMA/UMTS → 4G all-IP LTE (OFDMA) → 5G.

# TOPIC 6 — SPREAD SPECTRUM

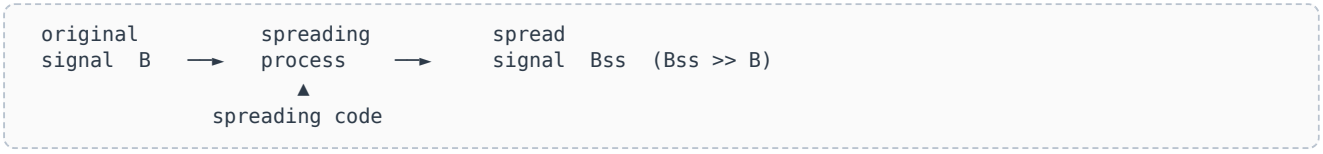
## 6.1 Introduction

**Multiplexing** combines signals to achieve **bandwidth efficiency**. **Spread spectrum** also combines signals into a larger bandwidth, but with **different goals**: in wireless, all stations share the air, so they must share the medium **without interception by an eavesdropper** and **without being jammed by a malicious intruder** (e.g. military) [SS].

To achieve this, spread spectrum **adds redundancy**: the bandwidth B each station needs is **expanded to Bss, where Bss ≫ B**. Forouzan's analogy: like putting a delicate, expensive gift in a **protective box** and using a superior delivery service [SS].

## 6.2 Two principles of spread spectrum ★ [SS]

1. The bandwidth allocated to each station must be **far larger than what is needed**. This allows **redundancy**.
2. The expansion from B to Bss must be done **by a process independent of the original signal**: the spreading happens **after** the source creates the signal, using a **spreading code** (a series of numbers that **look random but are actually a pattern**).



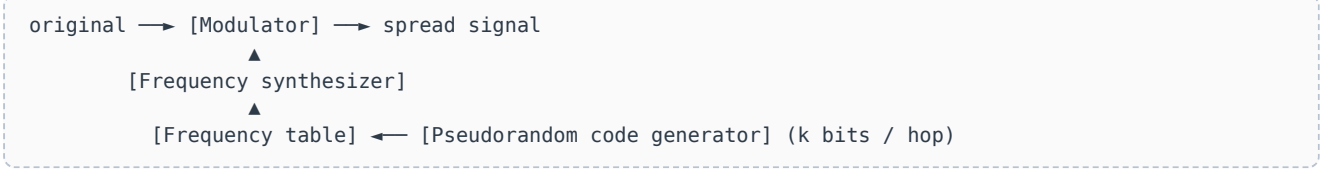

**Two techniques**: **FHSS** and **DSSS** [SS].

Wireless benefits (also from [WC-B 5.4]): spread spectrum uses **100× or more bandwidth** with **lower energy density** per frequency. Even if some frequencies are poor (frequency-selective fading), average performance is good. It is a form of **frequency diversity** [WC-S 48].

## 6.3 Frequency Hopping Spread Spectrum (FHSS) ★

### How it works [SS Fig 6.28]

- Uses **M different carrier frequencies** modulated by the source signal. At any moment the signal modulates **one** carrier; at the next moment, **another**. Over time all M are used, so **B_FHSS ≫ B**.
- Components:
  1. A **pseudorandom code generator (PN)** creates a **k-bit pattern every hopping period Th**.
  2. A **frequency table** maps the pattern to a frequency.
  3. A **frequency synthesizer** creates that carrier.
  4. The **modulator**: the source signal modulates the carrier.



### The notes' worked example [SS Fig 6.29–6.30]

- **M = 8** hopping frequencies, so **k = 3** bits ($2^3 = 8$). The original signal needs **100 kHz**.
- Frequency table: 000→200 kHz, 001→300, 010→400, 011→500, 100→600, **101→700**, 110→800, **111→900 kHz**.
- PN pattern: **101, 111, 001, 000, 010, 110, 011, 100**, then it **repeats after 8 hops**.
- Hop 1: 101 → **700 kHz**; hop 2: 111 → **900 kHz**; ...; hop 8: 100 → **600 kHz**; then back to 101.

| Hop | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|
| Pattern | 101 | 111 | 001 | 000 | 010 | 110 | 011 | 100 |
| Carrier (kHz) | 700 | 900 | 300 | 200 | 400 | 800 | 500 | 600 |

> **NOTES ERROR (minor)**: the text in [SS] lists the pattern as "101, 111, 001, 000, 010, 011, 100" (only 7 patterns, missing **110**). The figure shows all eight: 101 111 001 000 010 **110** 011 100. Use the eight-pattern version.

### Why FHSS achieves the goals [SS]

- **Privacy**: with many k-bit patterns and a short hopping period, an intruder can only catch a **small piece** of data, because they don't know the sequence and can't follow the hops.
- **Anti-jamming**: a jammer may jam **one hopping period** (randomly), **not the whole transmission**.

### Bandwidth sharing [SS Fig 6.31]

- With **M hopping frequencies**, **M channels can be multiplexed** into the same Bss: each station uses one frequency per hop, leaving M − 1 for M − 1 other stations (with a suitable modulation such as MFSK).
- **FHSS vs FDM**: in **FDM** each station uses **1/M of the band, fixed**. In **FHSS** each station also uses 1/M, but **the allocation changes from hop to hop**.

### FHSS in the module's technologies

- **IEEE 802.11 FHSS**: 2.4 GHz ISM; 2- or 4-level FSK; 1 or 2 Mbps; band divided into **79 sub-bands of 1 MHz**; a pseudorandom generator selects the hop sequence [WLAN slides 23–24].
- **Bluetooth**: **79 × 1 MHz channels**, **1600 hops/s** (slot 0.625 ms); piconet access **FH-TDD-TDMA**; scatternet access **FH-CDMA** [BT].

## 6.4 Direct Sequence Spread Spectrum (DSSS) ★

### How it works [SS Fig 6.32]

- Also expands bandwidth, but differently: **each data bit is replaced by n bits (chips)** using a spreading code. The **chip rate is n times the data bit rate**.
- A **chips generator** feeds a **modulator** together with the original signal.

### The notes' example: the Barker sequence [SS Fig 6.33]

- Used in **wireless LANs**: the **Barker sequence, n = 11**: **10110111000**.

- With polar NRZ, **spread signal = original × chips**: a data bit 1 sends the code as is; a data bit 0 sends the **inverted** code.
- If the original rate is N, the spread rate is **11N**, so the spread signal needs **11 times the bandwidth**.

**Worked example DS1: spreading data 1 0 1 with the Barker code**

| Data bit | Chips sent (11 chips per bit) |
|---|---|
| 1 | 1 0 1 1 0 1 1 1 0 0 0 (code as is) |
| 0 | 0 1 0 0 1 0 0 0 1 1 1 (code inverted) |
| 1 | 1 0 1 1 0 1 1 1 0 0 0 |

SUPPLEMENTARY: in binary (0/1) form, spreading = **data XOR chip** with the convention "data 1 ↔ chip unchanged". Using bipolar values (+1/−1) it is plain multiplication. At the receiver, multiplying by the same code again recovers the data (correlation).

- **Benefits**: **privacy** if the intruder doesn't know the code; **immunity against interference** if each station uses a different code [SS].
- **Bandwidth sharing in DSSS: "no and yes"** [SS]: if the code's spread signals **cannot be combined and separated**, the bandwidth **cannot** be shared (e.g. some WLANs using the Barker code). With **special codes that allow combining and separating** (orthogonal codes, i.e. **CDMA**), it **can** be shared.

### DSSS in the module's technologies

- **802.11 DSSS**: 2.4 GHz ISM; **BPSK/QPSK**; 1 or 2 Mbps. **802.11b HR-DSSS**: **5.5 and 11 Mbps** [WLAN slides 24–25].
- **CDMA (IS-95)**: 1.2288 Mchip/s, 1.25 MHz channel [AT].
- **802.15.4/ZigBee**: DSSS PHY at 868/915 MHz and 2.4 GHz [BT].

## 6.5 ★ FHSS vs DSSS comparison (assembled from [SS], [MA p7])

| | FHSS | DSSS |
|---|---|---|
| Spreading method | Carrier **hops** among M frequencies in a pseudorandom sequence | Each bit is replaced by **n chips** (code at n× the bit rate) |
| Bandwidth at any instant | **Narrow** (one channel) | **Wide** (whole spread band all the time) |
| Code determines | **Which frequency** to use in each hop period | **The chip pattern** that multiplies each bit |
| Key components | PN generator, frequency table, frequency synthesizer, modulator | Chip generator, modulator (XOR/multiply) |
| Effect of narrowband interference/jamming | Only the hops that land on the jammed frequency are hit | The receiver's despreading **spreads the jammer**, which is filtered out |
| Sharing | M stations can share Bss (like time-varying FDM) | Shareable only with orthogonal codes (CDMA) |
| Examples | 802.11 FHSS, **Bluetooth** | 802.11 DSSS/b, **IS-95 CDMA**, WCDMA, 802.15.4 |

**FH-CDMA vs DS-CDMA** (Bluetooth exercise 15): in **FH-CDMA**, users are separated because they follow **different hopping sequences**; they occasionally collide on the same frequency (e.g. Bluetooth piconets

in a scatternet [BT]). In **DS-CDMA**, users transmit **at the same time over the same wide band**, separated by **different (orthogonal/pseudorandom) chip codes** and correlation at the receiver (e.g. IS-95) [MA p7–8, AT].

## 6.6 Examination focus — Topic 6

- **Explain the spread spectrum principles of (i) FHSS and (ii) DSSS** (asked in **Exam Q1(b), T1A Q4, T1B Q6**; 4–10 marks).
- **Describe CDMA and why it is difficult to jam** (3 times).
- Spread spectrum vs multiplexing; two principles of SS; the Barker code; the FHSS example table; FH-CDMA vs DS-CDMA.

## 6.7 Topic 6 revision summary

- SS goals: **privacy and anti-jamming**, not bandwidth efficiency. **Bss ≫ B**, added **redundancy**; spreading is **independent of the signal**, using a **pseudorandom code**.
- **FHSS**: PN generator → frequency table → synthesizer → modulator; hops every Th; M frequencies, k = $\log_2 M$ bits. Jammer hits only some hops.
- **DSSS**: each bit → n chips; chip rate = n × bit rate; **Barker 11 chips = 10110111000**; bandwidth × n.
- **CDMA** = DS spread spectrum with orthogonal codes, so many users share the band.

# TOPIC 7 — OFDM, OFDMA AND SC-FDMA

## 7.1 Introduction

**OFDM** has *"created great expansion in the capacity of wireless networks"*. It is the main air-interface technology used to move from **3G to 4G**, allows the expansion of **IEEE 802.11** data rates, and is critical to **WiMAX** broadband [OF]. Together with **MIMO**, OFDM is the **cornerstone of broadband wireless** [WC-B 5.4].

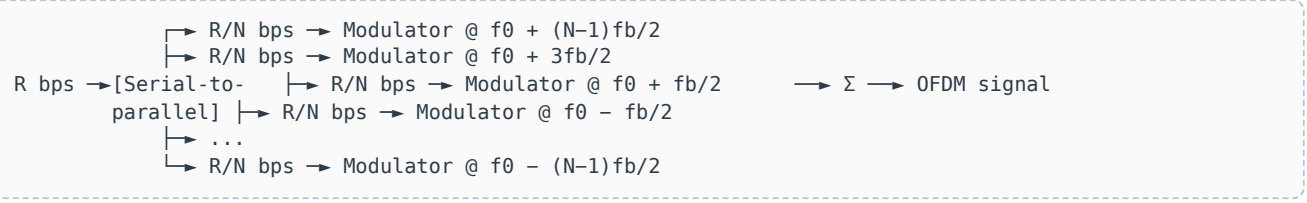
**Learning objectives** [OF]: overview of OFDM, OFDMA and SC-FDMA; how OFDM **combats frequency-selective fading**; the value of **orthogonal carriers**; the reduction in oscillators using the **IFFT**; problems of **PAPR**; how multiple access is achieved using OFDM and SC-FDMA.

## 7.2 Key definitions

| Term | Definition | Source |
|---|---|---|
| **OFDM** (multicarrier modulation) | Uses **multiple carrier signals at different frequencies**, sending some of the bits on each. Similar to FDM, **but many subcarriers are dedicated to a single data source** | [OF §8.1] |
| **Orthogonality** | Two signals $s_1(t)$, $s_2(t)$ are orthogonal if the **average over a bit time of $s_1(t)\cdot s_2(t) = 0$** | [OF §8.1] |
| **OFDMA** | Combination of **FDMA and TDMA**: different users use **subsets of the subcarriers** at **different times** | [OF §8.2] |
| **Subchannel** | A **group of subcarriers** (scheduling individual subcarriers is not feasible or sufficient); called a **Resource Block** in LTE | [OF §8.2] |
| **SC-FDMA** | Single-carrier FDMA: similar structure and performance to OFDMA, but with **lower PAPR**; performs an extra **DFT** before the IFFT | [OF §8.3] |

## 7.3 How OFDM works ★ [OF §8.1, Fig 8.1]

- A binary stream at **R bps**, with available bandwidth **N·fb** centred on $f_o$.
- **Option 1**: use the whole bandwidth for one stream, so each bit lasts **1/R**.
- **OFDM**: a **serial-to-parallel converter** splits the stream into **N substreams**, each at **R/N bps**, each on its own **subcarrier**, spaced **fb** apart. Now each bit lasts **N/R**, which is **much longer**. This gives special capabilities to **overcome multipath fading**.




### Orthogonality ★ [OF §8.1, Fig 8.2]

- The **base frequency fb** is the lowest subcarrier; all others are **integer multiples**: 2fb, 3fb, ...
- The spectra **overlap substantially**, but **the peak of each subcarrier's spectrum falls where every other subcarrier's power is zero**, so there is **no interference** between them.

- **Traditional FDM** spaces carriers apart to (1) avoid overlap and (2) add **guard bands**. In the notes' comparison (Fig 8.2b vs 8.2c, subcarriers at fb, 2fb, 3fb vs fb, 7fb, 13fb), **OFDM supports 6 times as many signals** in the same spectrum.

**Mathematical condition** [OF]:

$$\text{Average over bit time of } s_1(t)\cdot s_2(t) = 0$$

At the transmitter $s(t) = s_1(t) + s_2(t)$. To extract $s_1$, the receiver multiplies by $s_1(t)$ and averages:

$$\text{avg}[s_1(t)\cdot s(t)] = \text{avg}[s_1^2(t)] + \text{avg}[s_1(t)\cdot s_2(t)] = \text{avg}[s_1^2(t)] + 0$$

Only $s_1$ remains; all other orthogonal signals are removed.

**Requirement for OFDM subcarriers**: if the subcarrier bit time is **T**, then **fb must be a multiple of 1/T**, and every other subcarrier is M·fb = M/T for an integer M [OF]. **Examples**: **LTE subcarrier spacing = 15 kHz**. **802.11n**: **48 subcarriers in 20 MHz** (108 in 40 MHz), **fb = 0.3125 MHz**, then translated to 2.4 or 5 GHz [OF].

## Benefits of OFDM ★ [OF §8.1]

1. **Frequency-selective fading only affects some subcarriers**, not the whole signal. With **FEC**, this is easily handled.
2. **Overcomes intersymbol interference (ISI)** (the most important benefit). The data rate per carrier drops by **N**, so the **symbol time grows by N**: from **Ts to N·Ts**. Choose N so that **N·Ts ≫ the RMS delay spread** of the channel, and multipath delays hardly matter.
3. As a result **equalizers may not be needed**. Equalizers are complex and expensive, and their complexity grows with ISI severity.
4. **Spectral efficiency**: orthogonal subcarriers pack tightly, with no guard bands between them (6× in the example).
5. [WLAN slide notes] **Degradations are easier to handle per subcarrier**: badly degraded subcarriers can be **excluded** in favour of good ones.

## Difficulties of OFDM SUPPLEMENTARY (pages 241–244 of [OF] are missing from your upload; the learning objectives name these topics)

1. **Peak-to-average power ratio (PAPR)**: adding many subcarriers occasionally produces **very high peaks**, so the transmitter's power amplifier must be **linear over a wide range**. This is expensive and **power-inefficient**, which is bad for battery-powered phones. This is why LTE uses **SC-FDMA on the uplink**.
2. **Intercarrier interference (ICI)**: orthogonality depends on exact spacing; **frequency offsets and Doppler** break it, so subcarriers leak into each other. Tight synchronisation is needed.
3. **IFFT implementation**: building N separate oscillators would be impractical. The **inverse fast Fourier transform (IFFT)** generates all subcarriers digitally at once (and the FFT at the receiver), which is why OFDM is practical. A **cyclic prefix** (copy of the end of each symbol placed at its start) acts as a guard interval against ISI [OF Fig 8.8 shows "Add CP"].

# 7.4 OFDMA ★ [OF §8.2]

## Principle

- Multiple access options: scheduled times (**TDMA**), random access (**CSMA**), scheduled frequencies (**FDMA**), coded spreading (**DSSS**), coded hopping (**FHSS**).
- **OFDMA = FDMA + TDMA**: users get **subsets of subcarriers at different times**.
- **Not all OFDM systems use OFDMA**: e.g. some 802.11 versions use OFDM for transmission but **CSMA** for access, and the transmitter uses **all** subcarriers when sending. **LTE uses OFDMA only on the downlink** and a **single-carrier approach (SC-FDMA) on the uplink**.
- Subcarriers are grouped into **subchannels**. In the **downlink**, subchannels may go to **different receivers**. In the **uplink**, a transmitter may be assigned one or more subchannels.
- **Uplink subchannelization saves transmit power**: the device concentrates its power on its own subchannels, which is useful for **battery-powered 4G devices**.

### ★ OFDM vs OFDMA (Asked in: T1B Q8, 10 marks; OF review Q 8.5)

| | OFDM | OFDMA |
|---|---|---|
| What it is | A **modulation / transmission technique** (multicarrier) | A **multiple access technique** built on OFDM |
| Users per symbol | **All subcarriers carry one user's data** at a time | Subcarriers are **divided among several users** at the same time |
| Sharing between users | Needs another method, e.g. **TDMA or CSMA** (802.11a/g) | Built in: **FDMA + TDMA** (subchannels × time slots) |
| Allocation unit | Whole channel | **Subchannel / resource block** (LTE: 12 subcarriers) |
| Scheduling | None per subcarrier | **Opportunistic scheduling** by channel quality |
| Power (uplink) | Device spreads power over all subcarriers | Device concentrates power on **its subchannels** |
| Examples | 802.11a/g/n | **LTE downlink**, WiMAX |

# 7.5 ★ The three subchannel allocation methods (Asked in: Exam Q1(c), T1A Q5; OF review Q 8.10)

| Method | Description | **Benefits** | Drawbacks | Used by |
|---|---|---|---|---|
| **(i) Adjacent subcarriers** | All subcarriers of a subchannel in a **contiguous block** | All **SINRs ≈ equal**. Gives an **opportunity**: the system can choose among many blocks the **best block for each user**, optimising **channel efficiency, user requirements, priority and fairness** (exploits frequency-selective fading for multi-user gain) | If those frequencies are poor, the whole subchannel suffers. Requires accurate knowledge of each user's channel over the **full bandwidth → pilot symbols**, prediction (information may already be **out of date**) | LTE, WiMAX |

| Method | Description | **Benefits** | Drawbacks | Used by |
|---|---|---|---|---|
| **(ii) Regularly spaced subcarriers** ("distributed" / "comb") | Subcarriers spread at **regular intervals** across the band | **Frequency diversity**: there are enough **good subcarriers regardless of which are chosen**; **channel estimation burden is much lower** | Can't exploit the single best block for a user | LTE |
| **(iii) Randomly spaced subcarriers** | Subcarriers chosen **randomly** across the band | **Same frequency-diversity benefits** as regularly spaced, **plus reduced adjacent-cell interference** (neighbouring cells' random patterns rarely collide fully) | | WiMAX |

## 7.6 Opportunistic scheduling [OF §8.2]

Subchannels (**Resource Blocks** in LTE) are allocated to subscriber stations by **channel conditions and data requirements**, possibly with prescribed power levels. Time-varying, multipath channels create **opportunities** that can be exploited. Most standards **don't prescribe** the scheduler (vendors compete). Considerations:

1. **Efficiency**: schedule users with the **highest SINR**; they use AMC for high throughput, which maximises total capacity.
2. **Fairness**: pure efficiency starves users far from the base station. **Proportional fairness**: each user computes $\mathbf{r_i / \bar{r}_i}$ (rate achievable now ÷ average rate obtained). Users are compared **against themselves**; those with a good opportunity **for them** get scheduled.
3. **Requirements**: delay and jitter for audio and video.
4. **Priority**: police, fire, ambulance and public-safety users in emergencies.

### Worked example PF1 SUPPLEMENTARY (applying the notes' formula)

User 1: achievable now $r_1$ = 6 Mbps, average $\bar{r}_1$ = 4 Mbps → metric **1.5**. User 2 (cell edge): $r_2$ = 2 Mbps, average $\bar{r}_2$ = 1 Mbps → metric **2.0**. **User 2 is scheduled**, even though user 1's absolute rate is higher, because user 2 currently has an unusually good channel *for them*.

## 7.7 SC-FDMA [OF §8.3]

- Similar structure and performance to OFDMA. Its prominent advantage is a **lower PAPR**, which gives **battery life, power efficiency and lower cost** for mobiles.
- Not the same as basic OFDM: it adds an **extra DFT** at the transmitter and **frequency equalization** at the receiver.
- Used on the **uplink** (LTE); **OFDMA stays on the downlink** for greater multiple-access flexibility.
- **Chain** (Fig 8.8): bit stream → **N-point DFT** → subcarrier mapping → **Q-point IFFT** → add **CP** → channel → remove CP → FFT → de-mapping → **EQ** → **IDFT** → bits. (OFDMA omits the DFT/IDFT.)
- The DFT **spreads each data symbol over all subcarriers**, giving a **virtual single-carrier** structure. **Every data symbol is carried by every subcarrier.**
- **Multiple access**: in OFDMA, different users can share subcarriers **in the same interval**. In SC-FDMA, at any instant **all subcarriers carry the same stream, so they must belong to one user**; multiple access is only **over time**. A better name would be "SC-OFDM-TDMA".

## 7.8 Examination focus — Topic 7

- Benefits of the three OFDMA approaches (**Exam Q1(c) 6, T1A Q5 9**).
- Difference between OFDM and OFDMA (**T1B Q8 10**).
- Review questions [OF]: define OFDM, OFDMA, SC-FDMA; the **fundamental requirement of OFDM** (**orthogonality**: subcarrier spacing a multiple of 1/T); main strengths; main problems (**PAPR, ICI**); how orthogonality increases capacity (overlapping without interference, ~6×); the purpose of SC-FDMA (**lower PAPR on the uplink**).

## 7.9 Topic 7 revision summary

- OFDM: split R bps into **N streams of R/N** on **orthogonal subcarriers** (multiples of fb; spacing = 1/T). Symbol time N·Ts ≫ delay spread → **no ISI**, often no equalizer. Selective fading hits only a few subcarriers (FEC fixes them). Tight packing.
- LTE 15 kHz spacing; 802.11n 48 subcarriers/20 MHz, fb = 312.5 kHz.
- OFDMA = FDMA + TDMA over **subchannels/resource blocks**; LTE **downlink**. Allocation: **adjacent** (best-block selection, needs full CSI), **regular** (diversity, low estimation burden), **random** (diversity + less adjacent-cell interference).
- Scheduling: efficiency, **proportional fairness $r_i/\bar{r}_i$**, requirements, priority.
- SC-FDMA: extra DFT, **low PAPR**, LTE **uplink**; one user at a time.

# TOPIC 8 — ERROR DETECTION AND CORRECTION

## 8.1 Introduction

Wireless channels have high bit error rates (Topic 2). Receivers must at least **detect** errors (then use ARQ retransmission) or, better, **correct** them (FEC). The worksheet [ED] lists the four methods examined in this module: **parity checks, the Internet checksum, the Hamming code and the cyclic redundancy check (CRC)**. The calculation questions on these appear in **every** Test 1/Test 2/Exam paper.

> **Source note**: the worksheet [ED] contains web links and worked CRC examples, but **no written theory** for parity, checksum or Hamming. Theory below marked SUPPLEMENTARY follows the standard sources the worksheet links to (UNB ECE4253 pages; Kurose & Ross; Forouzan). The **error detection process** and FEC/ARQ come from your notes [WC-S 69–71; WC-B 5.4].

## 8.2 Key definitions

| Term | Definition | Source |
|---|---|---|
| **Error detection code** | Detects the **presence** of an error | [WC-S 69] |
| **ARQ (automatic repeat request)** | The block with an error is **discarded** and the transmitter **retransmits** it | [WC-S 69] |
| **Error correction code (FEC)** | Designed to **detect and correct** errors | [WC-S 69] |
| **Single-bit error** | **Only one bit** of the data unit (byte, frame, packet) is changed (1→0 or 0→1) | SUPPLEMENTARY (Forouzan) |
| **Burst error** | **Two or more bits** in the data unit have changed (not necessarily consecutive) | SUPPLEMENTARY (Forouzan) |
| **Burst length** | Measured **from the first corrupted bit to the last corrupted bit**, inclusive; bits in between may or may not be corrupted | SUPPLEMENTARY (Forouzan) |
| **Codeword** | Data bits plus the redundant (check) bits actually transmitted | SUPPLEMENTARY |
| **Modulo-2 arithmetic** | Binary addition/subtraction **without carries or borrows**, identical to **XOR** | SUPPLEMENTARY |

## 8.3 The error detection process [WC-S 70–71]

**Transmitter**: for a given frame, an error-detecting code (**check bits**) is calculated from the data bits and **appended** to them. **Receiver**: separates the frame into data and check bits, **recalculates** the check bits from the received data, **compares** them with the received check bits. A **mismatch = detected error**.

```
Tx:  data ──► f(data) = E ──► send [data | E]
Rx:  receive [data' | E'] ──► compute f(data') = E'' ──► compare E'' with E'
                                                          equal → accept
                                                          differ → error detected (ARQ / discard)
```

## 8.4 ★ Single-bit errors vs burst errors (Asked in: T1B Q1 [10], T2-2026 Q5

## [3])

SUPPLEMENTARY theory (standard Forouzan/Stallings definitions):

| | Single-bit error | Burst error |
|---|---|---|
| Definition | **Only 1 bit** in the data unit is changed | **2 or more bits** changed |
| Typical cause | Short, isolated noise; rare in serial transmission | **Impulse noise**, fading dips. A noise burst lasts longer than one bit, so it hits many bits. In wireless, **burst errors dominate** |
| Example | Sent 0000 0010 → received 0000 **1**010 | Sent 0100 0100 0100 0011 → received 0101 1101 0100 0011 (errors at positions 4, 5 and 8 → burst length 8 − 4 + 1 = 5) |
| Length | 1 bit | From the **first to the last corrupted bit** inclusive |

Your notes support the cause: **impulse noise is the primary source of error in digital data**. A 0.01 s spike wipes out **~10 000 bits at 1 Mbps**, which is a long burst [WC-B 5.2]. Fading in mobile channels causes 20–30 dB drops lasting many bits [WC-B 5.3].

**Measuring burst length (worked)**: sent 1011 0110, received 10**0**1 0**0**10. Errors at bit positions 3 and 6 (from the left). Burst length = 6 − 3 + 1 = **4 bits** (positions 4 and 5 are inside the burst but happen to be correct).

## 8.5 ★ Parity checking

SUPPLEMENTARY theory (standard; the worksheet's first link covers it).

**Single (even) parity**: append **one bit** so the total number of 1s (data + parity) is **even**. (Odd parity: make it odd.)

- Data 1011001 (four 1s) → even parity bit **0** → send 1011001**0**.
- Data 1011000 (three 1s) → even parity bit **1** → send 1011000**1**.
- **Receiver**: counts the 1s. Odd count → **error detected**.

### ★ What parity can and cannot do (Asked in: T1B Q1 [10], T2-2026 Q6 [5])

1. **Single-bit errors: always detected** ✔. One flip changes the count of 1s by one, so even becomes odd.
2. **Burst errors: NOT guaranteed.** Parity detects **any odd number** of flipped bits but **misses any even number** (2, 4, ...), because two flips leave the count's parity unchanged.
   - Example: send 1011001**0** (four 1s). A burst flips bits 2 and 3 → 1**10**10010 still has four 1s → **passes the check** → undetected.
   - A burst flips 3 bits → odd count → detected. **About half of random bursts escape detection.**
3. **Correction: impossible** with one parity bit. The receiver knows *that* an odd number of bits changed, but **not which bit**; any of the n bits could be the culprit. Only ARQ (retransmission) can recover.

**Conclusion on appropriateness**: parity is **appropriate for single-bit errors** (cheap, 1 extra bit, always detects them) but **inappropriate for burst errors**, which are the common case in wireless/noisy channels. For bursts, use **CRC** (which detects all bursts of length ≤ r, where r = number of check bits) or **two-dimensional parity**.

**Two-dimensional parity** SUPPLEMENTARY: arrange data in rows and add a parity bit to **every row and every column**. It can **detect and correct any single-bit error** (the failing row and column intersect at the

bad bit) and detects many burst errors, but some patterns of **4 errors in a rectangle** go undetected.

```
          data        row parity
        1 0 1 1   |   1
        0 1 1 0   |   0
        1 1 0 1   |   1
        ----------
col par 0 0 0 0   |   0
A single error at row 2, col 3 makes row 2 AND column 3 fail → bit located → flip it.
```

## 8.6 ★ The Internet checksum

SUPPLEMENTARY theory (standard RFC 1071 / Kurose method, at the 4-bit word size used in your questions).

**Sender**:

1. Divide the message into **k-bit words** (here k = 4).
2. Add all words using **one's-complement addition**: any **carry out of the top bit is wrapped around** and added to the result (**end-around carry**).
3. The **checksum = one's complement of the sum** (flip every bit).
4. Send the data words + the checksum.

**Receiver**: adds **all words including the checksum** with end-around carry. If there are no errors, the result is **all 1s (1111)**, and its complement is 0000. Anything else → error.

### ★ Worked example CK1 (Exam Q2(c), 2 marks; also the worksheet [ED])

*Message 1001 1100 1010 0011, Internet checksum with 4-bit words. Value of the checksum?*

| Step | Operation | Binary result | Carry? | After wrap |
|---|---|---|---|---|
| 1 | 1001 + 1100 | 1 0101 | carry 1 | 0101 + 1 = **0110** |
| 2 | 0110 + 1010 | 1 0000 | carry 1 | 0000 + 1 = **0001** |
| 3 | 0001 + 0011 | 0100 | none | **0100** |

- **Sum = 0100**
- **Checksum = one's complement of 0100 = 1011**

**Check (decimal)**: 9 + 12 + 10 + 3 = 34. One's-complement 4-bit sum: 34 = 2 × 15 + 4 → 4 = 0100 ✔ → complement **1011**. **Receiver check**: 0100 (sum of data) + 1011 = **1111** ✔ (no error).

### ★ Worked example CK2 (T2-2026 Q4: (a) 3 marks, (b) 1 mark)

*Message 1010 1100 0110 1001, 4-bit words, one's-complement addition with end-around carry.*

| Step | Operation | Binary result | Carry? | After wrap |
|---|---|---|---|---|
| 1 | 1010 + 1100 | 1 0110 | carry 1 | 0110 + 1 = **0111** |
| 2 | 0111 + 0110 | 1101 | none | **1101** |
| 3 | 1101 + 1001 | 1 0110 | carry 1 | 0110 + 1 = **0111** |

- **Sum = 0111 → Checksum = 1000**
- **Check (decimal)**: 10 + 12 + 6 + 9 = 37 = 2 × 15 + 7 → 0111 ✔.

- **(b)** The receiver adds all message words **and** the checksum: 0111 + 1000 = **1111**. So the expected result is **1111** (all ones; its complement is 0000).

## 8.7 ★ Hamming code

SUPPLEMENTARY theory (the worksheet links the UNB Hamming page; **this is the method T2-2026 Q2 specifies**).

**Purpose**: an **error-correcting code** (FEC) that can **detect and correct any single-bit error** by using several parity bits, each covering a different subset of positions. The pattern of failing checks **points to the bad bit**.

### Step 1: how many check bits? The Hamming inequality

For m data bits and r check bits:

$$2^r \geq m + r + 1$$

Why: the r check bits can give 2^r different "syndromes". These must name every one of the **m + r positions** where a single error could be, **plus 1** for "no error".

For **m = 8**:

- r = 3: $2^3 = 8 \geq 8 + 3 + 1 = 12$? **No ✘**
- r = 4: $2^4 = 16 \geq 8 + 4 + 1 = 13$? **Yes ✔**
- **Minimum r = 4 check bits**, so the codeword has **12 bits**.

### Step 2: positions

- Number positions **1 to 12 from the left**.
- **Check bits go at positions that are powers of 2: 1, 2, 4, 8.**
- **Data bits fill the remaining positions 3, 5, 6, 7, 9, 10, 11, 12, left to right.**

### Step 3: which positions each check bit covers

Check bit P at position 2^k covers every position whose **binary number has that bit set**:

| Check bit | Covers positions (rule) | Data positions it checks |
|---|---|---|
| **P1** (pos 1) | binary xxx**1**: 1, 3, 5, 7, 9, 11 | **3, 5, 7, 9, 11** |
| **P2** (pos 2) | binary xx**1**x: 2, 3, 6, 7, 10, 11 | **3, 6, 7, 10, 11** |
| **P4** (pos 4) | binary x**1**xx: 4, 5, 6, 7, 12 | **5, 6, 7, 12** |
| **P8** (pos 8) | binary **1**xxx: 8, 9, 10, 11, 12 | **9, 10, 11, 12** |

**Even parity**: each check bit = XOR (sum mod 2) of the data bits it covers, so the group's total number of 1s is even.

### ★ Worked example HM1 (Exam Q2(a), 4 marks; also the worksheet [ED]): data 10101111

**Check bits needed**: m = 8 → **r = 4** (inequality above).

**Placement:**

| Position | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Type | P1 | P2 | D | P4 | D | D | D | P8 | D | D | D | D |
| Data bit | | | **1** | | **0** | **1** | **0** | | **1** | **1** | **1** | **1** |

**Check bits (even parity):**

- **P1** = bits at 3, 5, 7, 9, 11 = 1, 0, 0, 1, 1 → three 1s → **P1 = 1**
- **P2** = bits at 3, 6, 7, 10, 11 = 1, 1, 0, 1, 1 → four 1s → **P2 = 0**
- **P4** = bits at 5, 6, 7, 12 = 0, 1, 0, 1 → two 1s → **P4 = 0**
- **P8** = bits at 9, 10, 11, 12 = 1, 1, 1, 1 → four 1s → **P8 = 0**

**Codeword:**

| Position | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Bit | **1** | **0** | 1 | **0** | 0 | 1 | 0 | **0** | 1 | 1 | 1 | 1 |

**Transmitted codeword = 1010 0100 1111** (12 bits).

> SUPPLEMENTARY convention warning: some textbooks (e.g. Stallings' *Data and Computer Communications*) number positions **from the right** and place data in a different order, which gives a different-looking codeword. Exam Q2(a) does not state a convention, so **write your convention in one line** ("positions numbered 1–12 from the left; check bits at 1, 2, 4, 8; data bits in order in 3, 5, 6, 7, 9–12") and the marker can follow your working. T2-2026 Q2 specifies exactly this convention.

## ★ Worked example HM2 (T2-2026 Q2(a)–(c), 8 marks): data 11001010

**(a) Minimum check bits.** Inequality **2^r ≥ m + r + 1** with m = 8. r = 3: 8 ≥ 12 ✘. r = 4: 16 ≥ 13 ✔. **r = 4.** Verify: $2^4 = 16 \geq 13$ ✔ while $2^3 = 8 < 12$.

**(b) Placement and check bits:**

| Position | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Type | P1 | P2 | D | P4 | D | D | D | P8 | D | D | D | D |
| Data bit | | | **1** | | **1** | **0** | **0** | | **1** | **0** | **1** | **0** |

- **P1** covers 1, 3, 5, 7, 9, 11 → data 1, 1, 0, 1, 1 = four 1s → **P1 = 0**
- **P2** covers 2, 3, 6, 7, 10, 11 → data 1, 0, 0, 0, 1 = two 1s → **P2 = 0**
- **P4** covers 4, 5, 6, 7, 12 → data 1, 0, 0, 0 = one 1 → **P4 = 1**
- **P8** covers 8, 9, 10, 11, 12 → data 1, 0, 1, 0 = two 1s → **P8 = 0**

**(c) Complete codeword (positions 1 → 12):**

| Position | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Bit | **0** | **0** | 1 | **1** | 1 | 0 | 0 | **0** | 1 | 0 | 1 | 0 |

**Transmitted codeword = 0011 1000 1010**

## How the receiver corrects an error (syndrome) SUPPLEMENTARY

Recompute each check over its group, including the received check bit. Write the failing checks as a binary number **C8 C4 C2 C1** (1 = fails). This number is the **position of the bad bit**; 0000 = no error.

*Example using HM2*: suppose bit 6 flips, so 0011 1100 1010 is received.

- C1 (1, 3, 5, 7, 9, 11): 0, 1, 1, 0, 1, 1 = four 1s → OK → 0
- C2 (2, 3, 6, 7, 10, 11): 0, 1, **1**, 0, 0, 1 = three 1s → FAIL → 1
- C4 (4, 5, 6, 7, 12): 1, 1, **1**, 0, 0 = three 1s → FAIL → 1
- C8 (8, 9, 10, 11, 12): 0, 1, 0, 1, 0 = two 1s → OK → 0
- **Syndrome C8C4C2C1 = 0110 = 6** → flip bit 6 → corrected ✔

## 8.8 ★ Cyclic Redundancy Check (CRC)

### Theory (the worksheet's worked example and links [ED]; supplementary explanation)

- Sender and receiver agree on a **generator G** of **r + 1 bits** (equivalently a **polynomial of degree r**), e.g. **$x^3 + 1$ ↔ 1001**, **$x^3 + x^2 + 1$ ↔ 1101**, **$x^3 + x + 1$ ↔ 1011**.
- **Polynomial ↔ bits**: write a 1 for every power present, from the highest power down to $x^0$. E.g. $x^3 + x + 1 = 1{\cdot}x^3 + 0{\cdot}x^2 + 1{\cdot}x^1 + 1{\cdot}x^0$ → **1011**.

**Sender** (the method used in the worksheet [ED]):

1. **Append r zeros** to the data (r = degree of G = number of bits in G − 1).
2. **Divide** by G using **modulo-2 long division** (XOR, no borrows).
3. The **r-bit remainder R is the CRC**.
4. **Transmit data + R** (R replaces the appended zeros). The transmitted frame is **exactly divisible by G**.

**Receiver**: divides the whole received frame by G. **Remainder 0 → accept; non-zero → error detected** ([ED]: *"Since the remainder is non-zero, we know that something went wrong!"*).

**Modulo-2 division rules**:

- If the leading bit of the current window is **1**, XOR it with G; if it is **0**, XOR with 0000 (i.e. just bring down the next bit).
- XOR: $0 \oplus 0 = 0$, $1 \oplus 1 = 0$, $0 \oplus 1 = 1 \oplus 0 = 1$.

### ★ Why modulo-2 arithmetic? (worksheet question [ED])

SUPPLEMENTARY: modulo-2 addition and subtraction are the **same operation (XOR)**, with **no carries or borrows**. This makes division:

- **simple and fast in hardware**: a CRC is just a **shift register with XOR gates**;
- **mathematically consistent**: polynomial arithmetic over GF(2), which gives CRC its strong detection properties (all single-bit errors, all double-bit errors with a suitable G, all odd-numbered errors if G has the factor (x + 1), and **all bursts of length ≤ r**).

### Worksheet example CR0 [ED] (verified)

*Message 11001001, G = $x^3 + 1$ = 1001.*

- Append 000 → 11001001**000**; dividing by 1001 gives **remainder 011** (quotient 1101 0011).
- **Transmit 11001001 011.**
- If the **leftmost bit flips** → received 01001001011 → remainder **010** (the worksheet writes it as "10") → non-zero → **error detected** ✔.

### ★ Worked example CR1 (Exam Q2(b), 4 marks): data 101110, G = 1001

r = 3 → append **000** → dividend **101110000**

```
1 0 1 1 1 0 0 0 0
1 0 0 1                 ← leading 1: XOR with 1001
─────────
0 0 1 0 1 0 0 0 0
    1 0 1 0             ← next window starting with 1
    1 0 0 1
    ─────────
    0 0 1 1 0 0 0
        1 1 0 0
        1 0 0 1
        ─────────
        0 1 0 1 0
          1 0 1 0
          1 0 0 1
          ─────────
          0 0 1 1   ← remainder (last 3 bits) = 011
```

Step by step (each XOR shown on a 4-bit window):

1. 1011 ⊕ 1001 = 0010 → working value 001010000
2. 1010 ⊕ 1001 = 0011 → 000011000
3. 1100 ⊕ 1001 = 0101 → 000001010
4. 1010 ⊕ 1001 = 0011 → 000000011

**Remainder R = 011. Transmitted bit string = 101110 011. Check**: 101110011 ÷ 1001 leaves remainder 000 ✔.

### ★ Worked example CR2 (T2-2026 Q3, 6 marks): data 110101, G = 1011 ($x^3 + x + 1$)

**(a)** Degree 3 → append **3 zeros** → **110101000** (1 mark).

**(b)** Division:

1. 1101 ⊕ 1011 = 0110 → 011001000
2. 1100 ⊕ 1011 = 0111 → 001111000
3. 1111 ⊕ 1011 = 0100 → 000100000
4. 1000 ⊕ 1011 = 0011 → 000001100
5. 1100 ⊕ 1011 = 0111 → 000000111

```
110101000
1011
────
 1100            (0110 → bring down: 1100)
 1011
 ────
  1111
  1011
  ────
   1000
   1011
   ────
     1100        (0011 → bring down two bits: 1100)
     1011
     ────
      111        ← remainder
```

**CRC remainder = 111.**

**(c)** Transmitted bit string = **110101 111**. Check: 110101111 ÷ 1011 → remainder 000 ✔.

## ★ Worked example CR3 (T1B Q2, 16 marks): message 10011010, G = $x^3 + x^2 + 1$

**(a)** Polynomial → bits: $x^3 + x^2 + 0{\cdot}x + 1$ → **1101** (r = 3). Append 000 → **10011010000**.

| Step | Window | ⊕ 1101 | Result | Working value |
|---|---|---|---|---|
| 1 | 1001 | 1101 | 0100 | 01001010000 |
| 2 | 1001 | 1101 | 0100 | 00100010000 |
| 3 | 1000 | 1101 | 0101 | 00010110000 |
| 4 | 1011 | 1101 | 0110 | 00001100000 |
| 5 | 1100 | 1101 | 0001 | 00000001000 |
| 6 | 1000 | 1101 | 0101 | 00000000101 |

```
10011010000
1101
————
 1001
 1101
 ————
  1000
  1101
  ————
   1011
   1101
   ————
    1100
    1101
    ————
       1000      (0001 → bring down 3 bits → 1000)
       1101
       ————
        101      ← remainder
```

**Remainder = 101** → **transmit 10011010 101**. Polynomial view: $M(x) = x^7 + x^4 + x^3 + x$; $x^3M(x) = x^{10} + x^7 + x^6 + x^4$, and the remainder mod $(x^3 + x^2 + 1)$ is $x^2 + 1 \leftrightarrow 101$ ✔.

**(b)** Leftmost bit inverted → received **00011010101**. Divide by 1101: the only step is 1101 ⊕ 1101 at positions 4–7 → 00000000101 → **remainder 101 ≠ 000** → the receiver knows an **error has occurred**, because a correct frame is **exactly divisible** by G (remainder 0). It then discards the frame and requests retransmission (ARQ).

### Worked example CR4: the worksheet CRC-8 [ED] (verified)

- CRC-8 polynomial: **$x^8 + x^2 + x + 1$** → **100000111**.
- Data 1011 0011 0101 0110 + 8 zeros → remainder **1101 0101** → transmit **1011 0011 0101 0110 1101 0101** ✔ (worksheet result confirmed).
- **7th bit flipped** → received 1011 0**0**01 0101 0110 1101 0101 → remainder **0010 1010 (101010)** ≠ 0 → **error detected** ✔ (matches the worksheet image).

### Worked examples CR5 (worksheet Kurose problems [ED], verified)

*G = 10011 (r = 4).*

| D | Remainder R | Transmitted D·R |
|---|---|---|
| **1010101010** | **0100** | 1010101010 0100 |

| D | Remainder R | Transmitted D·R |
|---|---|---|
| (a) 1000100101 | **1000** | 1000100101 1000 |
| (b) 0101101010 | **1111** | 0101101010 1111 |
| (c) 0110100011 | **1110** | 0110100011 1110 |

Division for the main case D = 1010101010 (append 0000 → 10101010100000):

1. 10101 ⊕ 10011 = 00110 → 00110010100000
2. 11001 ⊕ 10011 = 01010 → 00010100100000
3. 10100 ⊕ 10011 = 00111 → 00000111100000
4. 11110 ⊕ 10011 = 01101 → 00000011010000
5. 11010 ⊕ 10011 = 01001 → 00000001001000
6. 10010 ⊕ 10011 = 00001 → 00000000000100 → **R = 0100** ✔

### Worked example CR6 (worksheet [ED]): M(x) = 110001110110, G(x) = $x^5 + x^2 + x + 1$

- G = $x^5 + 0x^4 + 0x^3 + x^2 + x + 1$ → **100111** (r = 5) → append 00000.
- Steps (6-bit windows): 110001⊕100111 = 010110; 101101⊕100111 = 001010; 101010⊕100111 = 001101; 110111⊕100111 = 010000; 100000⊕100111 = 000111; 111000⊕100111 = 011111; 111110⊕100111 = 011001; 110010⊕100111 = 010101.
- **Remainder = 10101** → **T(x) = 110001110110 10101**.

## 8.9 Comparison of the four techniques (assembled)

| Method | Redundancy | Detects | Corrects | Strength | Weakness |
|---|---|---|---|---|---|
| Single parity | 1 bit | All **odd**-numbered errors (all single-bit) | No | Cheapest | Misses even-numbered errors (~50% of bursts) |
| 2-D parity | Row + column bits | All 1-, 2-, 3-bit errors; many bursts | **Single-bit** | Simple correction | Misses some 4-bit rectangles |
| Internet checksum | One word (16 bits in IP/TCP; 4 in exercises) | Most errors | No | Simple software (addition) | Misses reordered words and compensating errors |
| Hamming | r bits, 2^r ≥ m + r + 1 | Single (and double with an extra overall parity bit) | **Single-bit** | Forward correction, no retransmission | Weak against bursts |
| CRC | r bits (degree of G) | All single, all odd (with x + 1), **all bursts ≤ r** | No (detect + ARQ) | Very strong; simple hardware | Detection only |

**Where they appear in the module's technologies** [BT]: Bluetooth uses **1/3-rate FEC** (send each bit 3 times, majority vote) on the 18-bit header, a **2/3-rate FEC** that is a **(15, 10) shortened Hamming code** (corrects all single and detects all double errors), an **8-bit HEC** on the header, a **16-bit CRC** on payloads, and **ARQ**. Full detail in Part 3.

## 8.10 Examination focus — Topic 8

| Question | Paper | Marks |
|---|---|---|
| Hamming: 10101111, check bits + codeword | Exam Q2(a) | 4 |
| Hamming: 11001010 (inequality, placement, codeword) | T2-2026 Q2 | 8 |
| CRC: 101110, G = 1001 | Exam Q2(b) | 4 |
| CRC: 110101, G = 1011 | T2-2026 Q3 | 6 |
| CRC: 10011010, $x^3 + x^2 + 1$ + leftmost bit flipped | T1B Q2 | 16 |
| Checksum: 1001 1100 1010 0011 | Exam Q2(c) | 2 |
| Checksum: 1010 1100 0110 1001 + receiver result | T2-2026 Q4 | 4 |
| Single-bit vs burst; parity appropriateness | T1B Q1, T2-2026 Q5–6 | 10, 8 |

## 8.11 Topic 8 revision summary

- Detection: append check bits, recompute, compare. **ARQ** retransmits; **FEC** corrects.
- **Single-bit** = 1 bit changed; **burst** = ≥ 2 bits, length from first to last bad bit.
- **Parity** detects odd numbers of errors (all single-bit), misses even numbers (unreliable for bursts), **cannot correct**.
- **Checksum**: one's-complement sum with **end-around carry**, then **complement**. Receiver sum = **all 1s** if OK. Exam: **1011**; T2-2026: **1000** (receiver 1111).
- **Hamming**: **2^r ≥ m + r + 1** (8 data → 4 check bits, 12-bit codeword); P1, P2, P4, P8; syndrome = position of the error. 10101111 → **1010 0100 1111**; 11001010 → **0011 1000 1010**.
- **CRC**: append r zeros, XOR-divide by G, remainder = CRC; receiver remainder 0 = OK. 101110/1001 → **011**; 110101/1011 → **111**; 10011010/1101 → **101**.

# Part 3 — Detailed Study Notes: Topics 9–11

**Topics in this part** 9. Wireless LANs: IEEE 802.11 10. Bluetooth, IEEE 802.15 and ZigBee 11. Routing and Ad Hoc Network Routing (SUPPLEMENTARY: no lecture notes were provided)

### Source codes used in this part

| Code | File |
|---|---|
| **[WL n]** | *WLANS_v2.pptx*, slide n |
| **[FZ]** | *Chapter 15 WLANS by Forouzan.pdf* |
| **[BT-B]** | *BLUETOOTH AND IEEE 802.15_V2.pdf* (Beard & Stallings Ch. 12) |
| **[BT-S n]** | *Bluetooth and IEEE 802.15.pptx*, slide n (image-only slides, read visually) |
| **[BTX n]** | *Bluetooth_Exercises_Reordered_20_Questions.pdf*, question n |
| **[AT]** | *Fundamentals of Communications Access Technologies.pdf* |
| **[WCP]** | *Wireless Communication Problems.pdf* |

# TOPIC 9 — WIRELESS LANs (IEEE 802.11)

## 9.1 Introduction

A **LAN** interconnects computers within a small area (home, school, lab, office building) using network media. A **WLAN** links two or more devices using a **wireless medium** within such an area. It is sometimes called **wireless Ethernet**, and in the USA and elsewhere the public uses **WiFi** (*wireless fidelity*) as a synonym [WL 1]. Strictly, WiFi is a WLAN **certified by the WiFi Alliance**, a non-profit association of 300+ companies [FZ].

WLANs are a fast-growing market that brings the flexibility of wireless access into office, home and production environments. They are typically restricted to buildings, a campus or single rooms. **Goal: replace office cabling, and add flexibility for ad hoc communication** (e.g. group meetings) [WL 2].

**Exam weight**: very high. Hidden node, CSMA/CA, ACKs, two ISPs on channel 11, wired vs wireless LANs and co-channel interference together account for about **20 marks of the exam** and much of Test 1 and Test 2.

## 9.2 Key definitions

| Term | Definition | Source |
|---|---|---|
| **BSS (Basic Service Set)** | The building block of a WLAN: stationary or mobile wireless stations and an **optional** central base station, the **access point (AP)** | [WL 10], [FZ] |
| **Ad hoc BSS** | A BSS **without an AP**: a stand-alone network that **cannot send data to other BSSs**; stations send frames directly to each other | [WL 10], [FZ] |
| **Infrastructure BSS** | A BSS **with an AP**; each client associates with an AP that connects to other networks | [WL 11] |
| **ESS (Extended Service Set)** | Two or more BSSs with APs, connected through a **distribution system** (wired or wireless, usually Ethernet) | [WL 11], [FZ] |
| **Access point (AP)** | A radio that is a **wireless portal**: it **controls medium access** and acts as a **bridge** to other wireless or wired networks | [WL 5], [FZ] |
| **SSID** | **Service set identifier**: the logical **name** of an 802.11 network | [FZ] |
| **DCF** | **Distributed coordination function**: the fundamental 802.11 MAC mechanism, based on **CSMA/CA** | [WL 19] |
| **PCF** | **Point coordination function**: **optional, centralized, contention-free** polling by the AP (infrastructure only) | [WL 19, 40] |
| **NAV** | **Network allocation vector**: a timer showing how long other stations must wait before checking whether the channel is idle | [WL 37] |
| **IFS** | **Interframe space**: a waiting time before transmitting. Its length sets the **priority** (SIFS < PIFS < DIFS) | [WL 34] |

# 9.3 WLAN characteristics, design goals, wired vs wireless

## Advantages and disadvantages [WL 3]

| Advantages | Disadvantages |
|---|---|
| **Very flexible** within radio coverage | **Low bandwidth** compared to wired (~11–54 Mbit/s in the slide), because of radio limits, **higher error rates** from interference, and **higher delay/delay variation** from extensive error correction and detection |
| **Ad hoc networks without previous planning** | **Lower QoS** |
| Allows small, **independent devices** | **Proprietary solutions**; standards take time (slow standardisation) |
| **More robust against disasters** (earthquakes, fire) | Enhanced features only work in a **homogeneous** environment (same-vendor adapters) |
| | Must follow many **national restrictions**; global solutions are slow |

## Design goals [WL 4]

Global, seamless operation; **low power** for batteries (power-saving modes); **no special permissions or licences** (licence-free band); **robust** transmission; **easy to use**, simple management; **protection of investment in wired networks** (same data types and services); **security** (no one reads others' data), **privacy** (no user profiles collected), **safety** (low radiation).

## ★ Wired LAN vs wireless LAN [FZ] (Asked in: T2-2025 Q5, 10 marks)

| Aspect | Wired LAN | Wireless LAN |
|---|---|---|
| **Medium** | Wires; dedicated/switched links | **Air**; the signal is **broadcast**, all hosts **share** the medium (multiple access) |
| **Hosts** | Connected at a fixed point, with a link-layer address tied to the NIC; must be physically connected | **Not physically connected**; can **move freely** while using the network |
| **Isolated LAN** | Hosts connected via a **link-layer switch** | **Ad hoc network**: hosts communicate freely, **no AP** |
| **Connection to other networks** | Via a **router** | Via an **access point** that **glues two environments** (wireless ↔ wired); different role from a switch |
| **Moving between them** | Only the lower two layers differ. Change the NICs and replace the switch with an AP: **MAC addresses change, IP addresses stay the same** | |
| **Access control** | **CSMA/CD** | **CSMA/CA** (CD doesn't work in wireless) |
| **Characteristics** | Negligible attenuation and interference | **Attenuation, interference, multipath propagation, higher error rate** |

## ★ Unique concerns a wireless LAN designer must address (T2-2025 Q5, second half)

From the notes [FZ characteristics; WL 3–4; WL 16, 29; WCP]:

1. **Attenuation / path loss**: signal strength drops quickly with distance and through walls → place APs carefully, add APs for coverage [WCP path loss].
2. **Interference**: co-channel and adjacent-channel interference from other WLANs, Bluetooth and microwave ovens in the crowded **2.4 GHz ISM band** → channel planning, 5 GHz [WL 23, WCP].
3. **Multipath propagation and fading** → equalisation, antenna diversity, MIMO, OFDM, lower data rates [WCP].
4. **Higher error rate** → ACKs and retransmission, FEC, **fragmentation** [FZ].
5. **Medium access without collision detection**: hidden and exposed stations → CSMA/CA, RTS/CTS, NAV [FZ].
6. **Mobility**: hosts move between BSSs/ESSs (BSS-transition, ESS-transition mobility) → association, handoff [FZ].
7. **Limited power / battery** → power-saving modes [WL 4].
8. **Security and privacy**: broadcast medium, anyone in range can listen → authentication and encryption [WL 4, 27].
9. **Licence-free spectrum regulation**: power limits (≤ 1 W, typically 50 mW), national restrictions [WL 23, WL 3].
10. **QoS and capacity**: shared, lower bandwidth → rate adaptation, PCF for time-sensitive traffic [WL 6, 23, 40].

## 9.4 Infrastructure vs ad hoc networks [WL 5–9]

**Infrastructure-based**:

- Provides **access to other networks**. Communication is typically **only between wireless nodes and the AP**, not directly between nodes.
- The AP **controls medium access** and acts as a **bridge**.
- Several BSSs + a fixed network can form **one logical network** beyond radio coverage.
- Network functionality lies in the AP, so **clients stay simple**.
- If **only the AP controls access, no collisions** are possible, which is useful for **QoS** (the AP can **poll** nodes to guarantee data rates).
- **Loses flexibility**: can't be used for disaster relief if no infrastructure is left.

**Ad hoc**:

- **No infrastructure**, so the **greatest flexibility**; no AP controls access.
- **Each node is more complex**: it implements medium access and **forwards data**.
- Nodes communicate **only if within each other's radio range, or if other nodes forward the message**.
- **802.11 and HiperLAN2** are typically infrastructure-based but also support ad hoc. **Bluetooth is a typical ad hoc network.** (HiperLAN = European high-performance radio LAN standard, similar to WiFi.)

## 9.5 IEEE 802.11 architecture [WL 10–14, FZ]

- **BSS** (ad hoc or infrastructure) and **ESS** (BSSs connected by a **distribution system**). The ESS has **mobile stations** (normal stations inside a BSS) and **stationary stations** (APs that are part of the wired LAN).
- Stations within reach can communicate **without the AP**. Communication **outside the BSS goes via the AP**. This is like a cellular network: **BSS ≈ cell, AP ≈ base station** [WL 11 notes].

- A mobile station **can belong to more than one BSS** at the same time [FZ].

**Station types by mobility** [FZ]:

| Type | Meaning |
|---|---|
| **No-transition** | Stationary, or moving only inside one BSS |
| **BSS-transition** | Moves between BSSs **within one ESS** |
| **ESS-transition** | Moves from one ESS to another; **802.11 does not guarantee continuous communication** during the move |

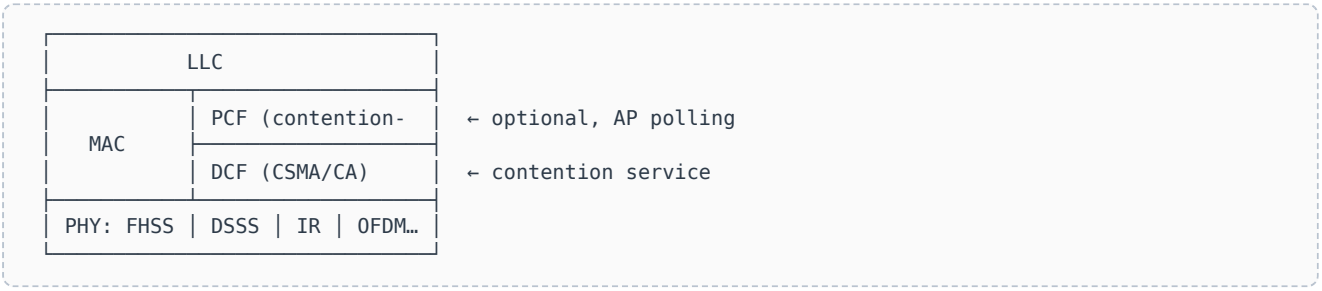
### Channels and association [WL 12–13]

- The spectrum is **divided into channels** at different frequencies. The **AP administrator chooses the AP's channel**. **Interference is possible: a neighbouring AP may choose the same channel!**
- An arriving host must **associate** with an AP:
  - it **scans channels** listening for **beacon frames** containing the AP's **SSID and MAC address**;
  - it **selects** an AP and may perform **authentication**.
- **Passive scanning**: (1) APs send beacon frames; (2) the host sends an **association request** to the selected AP; (3) the AP sends an **association response**.
- SUPPLEMENTARY: **active scanning**: the host broadcasts a **probe request**, APs reply with **probe responses**, then association request/response.

SUPPLEMENTARY (needed for the channel 1/11 question): the 2.4 GHz band (2.4–2.4835 GHz) has **11 channels** (US) spaced **5 MHz** apart, but each 802.11b/g signal is about **22 MHz wide** (see [WC-S 4]: a 22 MHz signal at 2.412 GHz occupies 2.401–2.423 GHz). So adjacent channels **overlap**, and only **channels 1, 6 and 11 are non-overlapping**.

## 9.6 Protocol architecture [WL 15–19, FZ]

- 802.11 covers the **physical and data-link layers**.
- The data-link layer is split into:
  - **LLC (logical link control)**: framing, flow and error control; a **common interface** between the network layer and the 802.11 MAC; identifies/multiplexes higher-layer protocols and **hides MAC details**.
  - **MAC (medium access control)**: decides **who transmits next**. 802.11 defines two MAC sublayers: **DCF** (bottom, contention, CSMA/CA) and **PCF** (top, optional, contention-free polling).
- **Three classes of multiple access protocols**: **channel partitioning** (incl. CDMA), **random access**, **taking turns**. 802.11's random access protocol is **CSMA/CA** [WL 15].



## 9.7 Physical layer [WL 21–26]

- Bands: **2.4–2.485 GHz** and **5.1–5.8 GHz**. The 2.4 GHz band is **unlicensed** and shared with phones and **microwave ovens**. At 5 GHz, range is **shorter** for a given power and **multipath** is worse, but there is **less interference**.
- All except infrared use **short-range radios in the ISM bands**: unlicensed, with a radiated power of **at most 1 W (50 mW typical)**. Garage door openers, cordless phones and microwaves compete for the same spectrum. **2.4 GHz is more crowded than 5 GHz.**
- **Rate adaptation**: each PHY defines **multiple rates**. A weak signal uses a **low rate**; a clear signal uses the **highest** rate.
- 802.11n, ac and ax use **MIMO**.

| Standard | Technique | Band | Modulation | Data rate |
|---|---|---|---|---|
| 802.11 FHSS | Frequency hopping (79 × 1 MHz sub-bands) | 2.4 GHz ISM | 2- or 4-level FSK | 1 or 2 Mbps |
| 802.11 DSSS | Direct sequence | 2.4 GHz ISM | BPSK or QPSK | 1 or 2 Mbps |
| 802.11 Infrared | Infrared light, 800–950 nm (not ISM) | — | **Pulse position modulation (PPM)** | 1 or 2 Mbps |
| **802.11a** | **OFDM** (with QAM) | **5 GHz** (5.725–5.850 GHz) | QAM | up to **54 Mbps** |
| **802.11b** | **HR-DSSS** | 2.4 GHz | — | **5.5 and 11 Mbps** |
| **802.11g** | **OFDM + FEC** | 2.4 GHz (compatible with b) | — | **22 or 54 Mbps** |
| **802.11n** | **MIMO** (+ OFDM) | 2.4/5 GHz | — | up to **600 Mbps** |

> **NOTES ERROR**: [WL 24–26] write the 2.4 GHz band as "2.400–**4.835** GHz". It should be **2.400–2.4835 GHz** (the same slides correctly say 2.4–2.485 GHz on slide 21).

## 9.8 MAC functions [WL 27–28]

The 802.11 MAC covers three areas: **reliable data delivery, access control, security**.

**Reliable data delivery**: the physical and MAC layers are **unreliable** (noise, interference, propagation effects), so many frames are lost. Reliability could be left to higher layers such as **TCP**, but 802.11 adds a **frame exchange protocol at the MAC layer**:

- **Two-frame exchange**: the receiver returns an **ACK**. If the source gets no ACK within a short time (the data or the ACK was damaged), it **retransmits**.
- **Four-frame exchange** (more reliable): **RTS → CTS → DATA → ACK**.
  - **RTS** alerts all stations **within range of the source** that an exchange is under way, so they refrain from transmitting.
  - **CTS** alerts all stations **within range of the destination**.

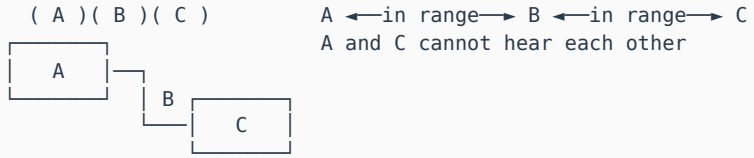
## 9.9 ★ Why not CSMA/CD in wireless? [WL 16, WL 29, FZ]

Wireless LANs **cannot implement CSMA/CD** for three reasons:

1. **Send and receive at the same time**: to detect a collision a station must transmit and listen simultaneously (full duplex). That means **costly stations and more bandwidth**. Battery-powered hosts don't have the power. Also, **received signals are much weaker than transmitted ones** [WL 16].
2. **Hidden station problem**: collisions may happen **without being detected**.
3. **Distance / fading**: stations can be far apart; **signal fading** may prevent a station at one end from hearing a collision at the other end.

## 9.10 ★ The hidden station (hidden terminal) problem [FZ, WL 30, 39, 44]

Using the exam's **Figure 2**: three nodes **A, B, C** in a row. Each circle is a node's transmission range. **B is within range of both A and C, but A and C are out of each other's range.**




- **Problem**: A senses the channel, hears nothing (C is out of range), and starts sending to B. C also senses an idle channel (it can't hear A) and starts sending to B. **Both frames collide at B**, but **neither A nor C can detect it**. A and C are **hidden from each other with respect to B**.
- **Consequences**: frames are lost, retransmissions follow, and **capacity is reduced** [FZ]. Hidden stations can also be caused by **obstacles** [FZ].
- **What happens if A and C send simultaneously** (Exam Q5(b), T1A Q8): both start transmitting because each senses the medium idle. Their signals **overlap at B**, so B receives a **garbled** frame and decodes neither. A and C keep transmitting their whole frames (no collision detection), wasting the channel. Neither gets an **ACK**, so both time out, **back off a random time and retransmit**.

### ★ Solution: RTS/CTS handshake [FZ, WL 39, 43]

- Before sending data, the source sends an **RTS** that includes the **duration** it needs. The destination replies with a **CTS** that also carries the duration.
- **Key point**: the **CTS** is heard by **everyone in range of the receiver**, including the hidden station. In Forouzan's figure (B sending to A, with C hidden from B): B's RTS reaches A but not C. Because both B and C are in range of A, **A's CTS reaches C**. C learns that a hidden station is using the channel and **refrains until that duration is over**.
- *"The CTS frame in the CSMA/CA handshake can prevent collision from a hidden station"* [WL 43].

## 9.11 ★ CSMA/CA and DCF: the complete procedure [WL 32–35, FZ]

**Interframe spaces** [WL 34]. A shorter IFS means **higher priority**:

| IFS | Length | Used for |
|---|---|---|
| **SIFS** (short IFS) | **Shortest** (highest priority) | **Immediate responses**: ACK after data, CTS after RTS, data after CTS |

| IFS | Length | Used for |
|---|---|---|
| **PIFS** (PCF IFS) | Middle | The **point coordinator (AP)** issuing **polls** in PCF; takes precedence over contention traffic |
| **DIFS** (DCF IFS) | **Longest** | **Ordinary (contention) traffic** |

**DCF algorithm** [WL 32–33, 35]:

1. A station with a frame **senses the medium** (checks the energy level at the carrier frequency [FZ]).
2. If **idle**, it waits **one IFS (DIFS)**. If still idle, it may transmit.
3. If **busy** (initially or during the IFS), it **defers** and monitors until the current transmission ends.
4. After that, it waits **another DIFS**. If idle, it waits a **random backoff** time and senses again.
5. **During backoff**, if the medium becomes busy, the **backoff timer freezes** and **resumes** when the medium becomes idle. When the counter reaches **zero**, the station transmits.
6. Optional **RTS → (SIFS) → CTS → (SIFS) → DATA → (SIFS) → ACK** exchange.
7. Other stations set their **NAV** and defer.
8. **No ACK (or no CTS) = assume a collision**: perform another backoff and retry.

### Frame exchange timeline [FZ Fig 15.7]

```
Source:      |DIFS| RTS |      | SIFS |   DATA   |      |
Destination:            |SIFS| CTS  |          |SIFS| ACK |
Others:                 |<========== NAV (no channel sensing) ========>|
```

### Network allocation vector (NAV) ★ [WL 37, FZ]

*How is collision avoidance accomplished? The key is NAV.* The RTS includes the **duration** the sender needs. Stations that hear it set a **NAV timer** showing how long to wait before even **checking the channel**. Before sensing the physical medium, each station **first checks whether its NAV has expired**. This is **virtual carrier sensing**.

### Collision during handshaking [WL 38, FZ]

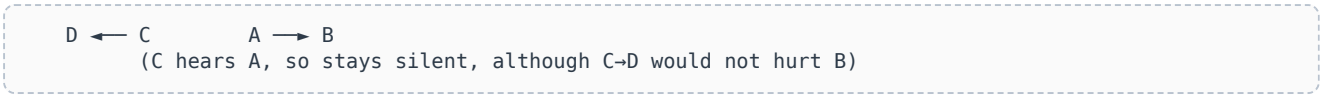
Two stations may send **RTS frames at the same time**, and they collide. There is **no collision detection**, so a sender **assumes a collision if no CTS arrives**, applies the **backoff strategy**, and tries again. (RTS frames are short, so a collision wastes little time.)

### ★ Applying CSMA/CA to Figure 2: A sends a frame to B (Exam Q5(c), T1A Q9)

1. A **senses** the medium (and its NAV). It is idle, so A waits **DIFS**, then (if still idle) completes a **random backoff**.
2. A sends an **RTS** to B with the duration of the whole exchange. **B** hears it (C does not).
3. B waits **SIFS** and replies with a **CTS** that also carries the duration. **Both A and C hear the CTS** (C is in B's range). **C sets its NAV** and stays silent, even though it can't hear A.
4. A waits **SIFS** and sends **DATA** to B.
5. B waits **SIFS** and returns an **ACK**. When the NAV expires, the medium is free and C may contend (DIFS + backoff).
6. If A gets **no CTS or no ACK**, it assumes a collision, **doubles its backoff window** (SUPPLEMENTARY: binary exponential backoff) and retries.

## 9.12 The exposed station problem [FZ, WL 31, 45–46]

- **A is transmitting to B. C wants to send to D**, which would **not** interfere with A→B. But C is **within A's range**, so it hears A's transmission and **refrains**. **C is too conservative and wastes capacity.**
- **RTS/CTS does not help**: C hears A's RTS and stays quiet, because it cannot know that its transmission wouldn't affect the zone between A and B.




## 9.13 Point coordination function (PCF) [WL 40–42, FZ]

- **Optional**; only in **infrastructure** networks (not ad hoc); implemented **on top of DCF**; used mostly for **time-sensitive** traffic.
- **Centralised polling**: the **AP (primary)** polls stations (**secondaries**) one after another, and they send any data they have **to the AP**. All exchanges go through the primary, which decides who uses the channel. **Poll and select** functions prevent collisions.
- **Drawback**: if the **primary station fails, the system goes down**.
- **Priority over DCF**: PCF uses **PIFS**, which is **shorter than DIFS**.
- **Repetition interval** [WL 42; SUPPLEMENTARY detail]: time is divided into repeating **superframes**. Each starts with a **beacon**, followed by a **contention-free period** (PCF polling) and then a **contention period** (DCF), so both kinds of traffic coexist.

## 9.14 Fragmentation [FZ]

The wireless environment is **very noisy**, so frames are often corrupted. 802.11 recommends **fragmentation**: dividing a large frame into smaller ones, because **resending a small frame is more efficient than resending a large one**.

## 9.15 ★ Why acknowledgements in 802.11 but not wired Ethernet? (Exam Q5(d), T1A Q10)

- In **CSMA/CD Ethernet**, collision detection serves two purposes: a **detected** collision means resend, and the **absence** of a collision acts as **a kind of acknowledgement** that the frame got through [FZ]. Wired links also have **very low error rates**.
- In **802.11**, stations **cannot detect collisions** (no simultaneous send/receive, hidden stations, fading) [WL 16, 29]. And wireless frames are often lost to **noise, interference and propagation effects** [WL 27]. *"Acknowledgment is needed in this protocol because the station does not have any means to check for the successful arrival of its data at the destination"* [FZ]. So the receiver **returns an ACK**, and **no ACK = retransmit** [WL 27].

## 9.16 ★ Co-channel interference in WiFi (T1A Q13, 4 marks)

Why many WiFi networks have a **high error rate** from co-channel interference:

- The AP administrator **chooses the channel**, and **neighbouring APs may choose the same channel** [WL 12].

- **2.4 GHz is crowded**: few non-overlapping channels (1, 6, 11 — SUPPLEMENTARY), shared with cordless phones, Bluetooth and **microwave ovens** [WL 21, 23].
- **Inward / co-channel interference** comes from products using **similar radio frequencies** nearby, which **block transmissions or cause bit errors** [WCP].
- **ISM users have no regulatory protection** and must tolerate interference [WC-S 10].
- Hidden stations in other BSSs on the same channel cause **collisions that CSMA/CA can't prevent**.
- **Result**: corrupted frames, failed ACKs, retransmissions, backoffs and lower throughput.

## 9.17 Topic 9 revision summary

- WLAN = wireless LAN = "wireless Ethernet" = WiFi (certified). **BSS** (ad hoc or infrastructure with AP), **ESS** = BSSs + distribution system; SSID; station mobility types.
- Association: scan for **beacons** (SSID, MAC) → association request/response. AP channel is chosen by the admin, and neighbours may clash.
- Stack: **LLC + MAC (DCF/CSMA/CA + optional PCF)**; PHY variants (FHSS, DSSS, IR/PPM, a/g OFDM 54 Mbps, b HR-DSSS 11 Mbps, n MIMO 600 Mbps). 2.4 GHz crowded; 5 GHz shorter range, less interference. **Rate adaptation.**
- **No CSMA/CD**: can't send+receive; hidden station; fading.
- **CSMA/CA**: sense → DIFS → random backoff (frozen while busy) → RTS/SIFS/CTS/SIFS/DATA/SIFS/ACK. **NAV** = virtual carrier sense. **SIFS < PIFS < DIFS.** No CTS/ACK → backoff and retry.
- **Hidden station**: A and C both reach B but not each other → collision at B. **RTS/CTS fixes it**: the CTS reaches the hidden node.
- **Exposed station**: C needlessly defers; RTS/CTS can't fix it.
- **PCF**: AP polling, PIFS, infrastructure only, single point of failure.
- **ACKs**: no collision detection plus high error rates.

# TOPIC 10 — BLUETOOTH, IEEE 802.15 AND ZIGBEE

## 10.1 Introduction: short-range communication and the IoT [BT-B 12.1, BT-S 2–4]

Short-range technologies (a few metres) focus on **low cost and low energy** for long battery life. Several standards come from the **IEEE 802.15 working group**. They provide two services:

1. **Personal area network (PAN)**: devices within a person's working space, especially **on their bodies**, share information and communicate with the user.
2. **Internet of Things (IoT)**: **device-to-device** communication for **data collection, analysis and automation**. Examples: heart-monitoring implants, biochip transponders on farm animals, cars with sensors, devices that assist fire-fighters [BT-S 3]. IoT flow: **sensor → connectivity → data processing → user interface**, and the user can also act back on the system (e.g. adjust cold-storage temperature from a phone) [BT-S 4].

## 10.2 Bluetooth overview [BT-B 12.2, BT-S 6–8]

- An **always-on, short-range radio hookup on a microchip**. Developed by **Ericsson in 1994** so laptops could make calls via a mobile phone. Originally published as **IEEE 802.15.1**; now the **Bluetooth SIG** (special interest group) publishes the standards. Named after King **Harald Blaatand** of Denmark.
- **Key attributes: robustness, low power consumption, low cost.**
- **2.4 GHz** (globally available, unlicensed, low power). Two devices **within 10 m** can share **up to 2.1 Mbps or 24 Mbps**.
- Supports an open-ended list of applications: data, audio, graphics, video.
- **Up to 8 devices in a piconet**; **10 piconets can coexist** in the same coverage range; each link is **encoded** against eavesdropping and interference.

**Three application areas** [BT-B, BT-S 8]:

1. **Data and voice access points**: real-time voice and data, wirelessly connecting portable and stationary devices.
2. **Cable replacement**: removes proprietary cables. Connections are instant and work **without line of sight**. Range **≈ 10 m**, extendable to **100 m** with an amplifier.
3. **Ad hoc networking**: a device can connect instantly to another Bluetooth radio as soon as it comes into range.

**Uses** (Table 12.1): hands-free headset, car stereo, PC input/output, synchronisation, cable replacement (RS-232), infrared remote replacement, low-rate data transfer, bridges (Ethernet interconnection), game consoles, medical devices, security (theft prevention when the link breaks).

### ★ Why Bluetooth is a WPAN rather than a WLAN (BTX 1)

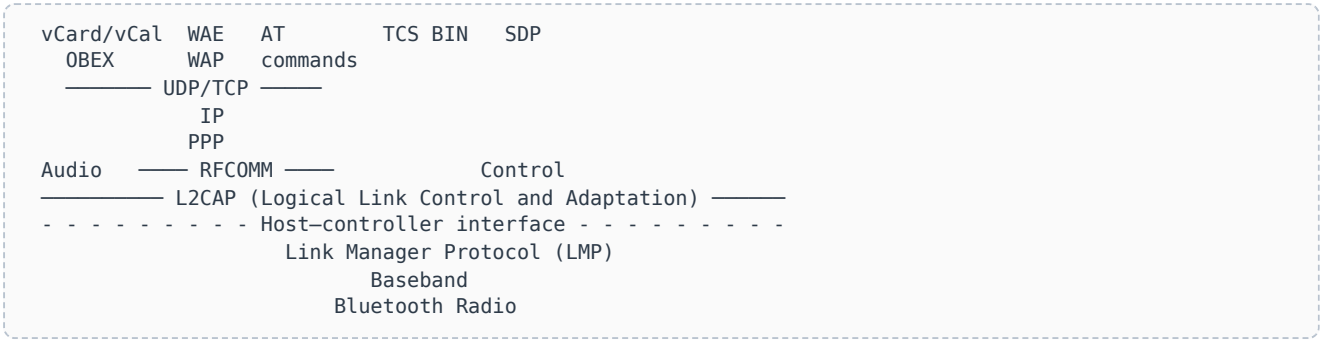
Its **range is about 10 m** (a person's personal space), it is designed for **low power and low cost** and **cable replacement** between a person's own devices, and its basic network (a piconet) holds only **8 active devices**. A WLAN covers a building or campus (tens to hundreds of metres), serves many users, and connects to wider networks.

## 10.3 Standards documents: core and profiles [BT-B]

The standards (2500+ pages) have two groups:

- **Core specifications**: details of the protocol layers, from the radio interface to link control, interoperability, testing and timers.
- **Profile specifications**: the use of Bluetooth to support **particular applications**. Each profile says which parts of the core are **mandatory, optional or not applicable** for a usage model. **Purpose: a standard of interoperability**, so products from different vendors that claim a usage model **work together**.

## 10.4 ★ Protocol architecture [BT-B, BT-S 10–20]




**Core protocols (five-layer stack)**:

| Layer | Role |
|---|---|
| **Radio** | Air interface: **frequency, frequency hopping, modulation, transmit power** |
| **Baseband** | **Connection establishment within a piconet, addressing, packet format, timing, power control** |
| **LMP** (link manager protocol) | **Link setup** and ongoing link management, incl. **security (authentication, encryption)** and negotiation of baseband packet sizes |
| **L2CAP** | **Adapts upper-layer protocols to the baseband**; provides **connectionless and connection-oriented** services |
| **SDP** (service discovery) | Device information, services and their characteristics can be **queried** so a connection can be set up |

**Cable replacement protocol: RFCOMM** presents a **virtual serial port**, making replacement of serial cables as transparent as possible. It provides binary data transport and **emulates EIA-232 (RS-232) control signals** over the baseband. **Telephony control protocol: TCS BIN** (telephony control specification, binary) is a **bit-oriented** protocol for **call control signalling** for speech and data calls, plus **mobility management** for groups of TCS devices. **Adopted protocols** (Bluetooth invents only what's necessary and reuses existing standards):

- **PPP**: carries IP datagrams over a point-to-point link.
- **TCP/UDP/IP**: the foundation protocols of TCP/IP.
- **OBEX**: object exchange; a session-level protocol from **IrDA**, similar to HTTP but simpler. Content formats include **vCard** (business card) and **vCalendar**.

## 10.5 ★ Profiles [BT-B, BT-S 21]

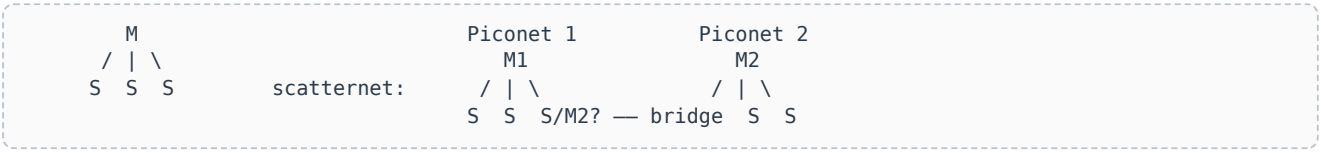
- **Over 40 profiles**. A profile is **a set of protocols that implement a particular Bluetooth-based application**; each defines the protocols and features that support a **usage model**.
- **All Bluetooth nodes support the Generic Access Profile**; profiles may depend on other profiles.
- **Example: the File Transfer Profile** transfers directories, files, documents, images and streaming media, and can browse folders on a remote device. It **depends on the Generic Object File Exchange, Serial Port and Generic Access profiles**, uses **OBEX** within Generic Object File Exchange, and interfaces with **L2CAP and RFCOMM**.

### ★★ How profiles benefit applications despite adding complexity (Asked in: Exam Q4(c), T2-2025 Q7, T2-2026 Q7, BTX 19)

1. **Interoperability**: the purpose of a profile is a *"standard of interoperability so that products from different vendors that claim to support a given usage model will work together"* [BT-B]. A headset from vendor X works with a phone from vendor Y, so users don't care about brands.
2. **Ready-made building blocks for developers**: the profile already specifies **which protocols and features to use** (mandatory/optional/not applicable). Developers **don't redesign** the protocol stack for each application; they implement the profile, e.g. the File Transfer Profile reuses OBEX, RFCOMM and L2CAP [BT-B].
3. **Only what's needed is implemented**: an application implements only the subset of the large core specification its usage model needs. This means **smaller, cheaper, low-power devices** (key Bluetooth attributes).
4. **Reuse through dependencies**: profiles build on others (all on the Generic Access Profile), so common functions (discovery, connection setup, security) are **consistent** across all applications.
5. **Plug-and-play user experience**: devices discover each other's services (**SDP**) and connect automatically, because both sides know exactly how the application behaves.
6. **Predictable behaviour and testing**: certification against profiles makes products reliable and easier to test.

## 10.6 ★ Piconets and scatternets [BT-B, BT-S 23]

- **Piconet**: the basic unit, **one master + one to seven active slaves**.
- The **master determines the channel (frequency-hopping sequence) and phase (timing offset, i.e. when to transmit)** for all devices, **using its own device address** as a parameter. Slaves **tune to the same channel and phase**.
- **A slave may only communicate with the master, and only when granted permission by the master.**
- **Scatternet**: a device in one piconet may also be part of **another** piconet, as **slave or master in each**. This overlap is a scatternet.




(A device can be **slave in both**, or **master in one and slave in the other**, but **not master in both**: see 10.12.)

- **Advantage**: many devices share the same physical area and use bandwidth efficiently.

- **Frequency hopping**: **1 MHz carrier spacing**, typically up to **80 frequencies (80 MHz)**. Without hopping, a channel = one 1 MHz band. With hopping, a **logical channel = a hopping sequence**. At any time the available bandwidth is **1 MHz, shared by at most 8 devices**. **Different logical channels share the same 80 MHz. Collisions occur when devices in different piconets happen to use the same hop frequency at the same time**; more piconets means more collisions and poorer performance.
- **Summary**: *the physical area and total bandwidth are shared by the scatternet; the logical channel and data transfer are shared by a piconet.*

### Can a piconet have more than eight stations? (BTX 5)

Only **8 can be active** (1 master + 7 active slaves): the active-member address **AM_ADDR is 3 bits** ($2^3$ = 8 values; 0 = broadcast, so 1–7 for slaves) [BT-S packet header]. SUPPLEMENTARY: additional devices can be **parked** (synchronised but inactive), up to 255, and activated by swapping with an active slave.

## 10.7 Radio specification [BT-B Table 12.2, BT-S 24–25]

| Parameter | Basic Rate (BR) | Enhanced Data Rate (EDR) |
|---|---|---|
| Topology | Up to 7 simultaneous links, logical star | Same |
| Modulation | **GFSK** | **π/4-DQPSK and 8DPSK** |
| Peak data rate | **1 Mbps** | **2 and 3 Mbps** |
| RF bandwidth | 220 kHz (−3 dB), 1 MHz (−20 dB) | Same |
| RF band | **2.4 GHz ISM** | Same |
| RF carriers | **23/79** | 23/79 |
| Carrier spacing | **1 MHz** | 1 MHz |
| Transmit power | 0.1 W | 0.1 W |
| Piconet access | **FH-TDD-TDMA** | Same |
| Hop rate | **1600 hops/s** | Same |
| Scatternet access | **FH-CDMA** | Same |

- **Power classes**: **Class 1**: up to **100 mW (+20 dBm)**, min 1 mW; **power control mandatory** (4–20 dBm); greatest range. **Class 2**: up to **2.5 mW (+4 dBm)**, min 0.25 mW (−6 dBm); power control optional. **Class 3**: **max 1 mW**, lowest power.
- **79 × 1 MHz physical channels** in most countries; **f = 2.402 + n MHz, n = 0…78** (US, Europe). Japan, Spain and France have 23 channels.
- **Power control** keeps devices from emitting more than necessary; implemented via **LMP** between master and slaves.
- **BR modulation: Gaussian FSK**. Binary 1 = positive frequency deviation, 0 = negative; **1 Msymbol/s × 1 bit = 1 Mbps**; minimum deviation 115 kHz.
- **EDR** (Bluetooth 2.0): the same 1 Msymbol/s with **π/4-DQPSK (2 bits) and 8DPSK (3 bits)** → 2 and 3 Mbps. The raw 3 Mbps gives **≈ 2.1 Mbps** after overhead.

**Answers to BTX 10–13**: the radio layer specifies the air interface (frequency, hopping, modulation, power). The actual bandwidth used is **1 MHz per hop channel** within the 79–80 MHz band. The spread-spectrum technique is **FHSS** (1600 hops/s). Modulation is **GFSK** (BR), with π/4-DQPSK/8DPSK for EDR. Bits become radio waves by shifting the carrier frequency up (1) or down (0).

## 10.8 ★ Baseband: frequency hopping and TDD [BT-B, BT-S 26–30]

**FH serves two purposes**:

1. **Resistance to interference and multipath** effects.
2. **Multiple access among co-located devices in different piconets.**

**FH workflow**: total bandwidth is divided into **79 physical channels of 1 MHz** → hopping jumps between them in a **pseudorandom sequence** → **the same sequence is shared by all devices in a piconet** (an "FH channel").

- **1600 hops/s**, so each channel is occupied for **0.625 ms = one slot**; slots are numbered.
- **Time division duplex (TDD)**: data go **one direction at a time**, alternating (master → slave, slave → master). With more than two devices, the access technique is **TDMA**. So piconet access = **FH-TDD-TDMA**.
- **Multislot packets**: packets can take **1, 3 or 5 slots**. The radio **stays on the same frequency** for the whole packet, then returns to the frequency its hopping sequence requires for the next slot, so **2 or 4 hop frequencies are skipped**.
- **TDD prevents crosstalk** between transmit and receive in the transceiver, which is essential for a **one-chip implementation**.
- The **FH sequence is set by the master** and is a function of the **master's Bluetooth address** (permutations and XOR).
- Different piconets have different masters, hence **different hop sequences**. Occasionally two piconets use the **same physical channel in the same slot → collision and lost data**. This is rare and is handled with **FEC and error detection/ARQ**. This is **FH-CDMA** between piconets.
- **Adaptive frequency hopping**: quality metrics let the hop sequence **avoid poor channels** (e.g. those used by WiFi). Fewer than 79 channels may be used, but **at least 20**.

**How can FH and TDD be combined? (BTX 14)**: each 625 µs slot is on a new hop frequency. The master transmits in even-numbered slots and the slave replies in the next (odd) slot **on the next hop frequency**. So **direction alternates slot by slot (TDD) while the frequency changes slot by slot (FH)**.

**Microwave ovens (BTX 16)**: yes, they can interfere. Ovens radiate around **2.45 GHz**, inside Bluetooth's **2.4 GHz ISM** band [WCP: microwave ovens operate at 2.4 GHz]. The effect is limited because Bluetooth **hops 1600 times/s across 79 channels**, so only hops landing near the oven's frequency are hit, and **adaptive frequency hopping** can avoid them. Lost packets are recovered by **FEC/ARQ**.

**MAC protocol in the baseband (BTX 17)**: **TDMA with master polling (FH-TDD-TDMA)**. Slaves send only in the slot after the master addresses them. **No contention within a piconet.**

## 10.9 ★ User logical links [BT-B, BT-S 31–35]

| Link | Description |
|---|---|
| **SCO** (synchronous connection-oriented) | **Fixed bandwidth**, point-to-point, master ↔ **one** slave. **Reserved slots at regular intervals**; the unit is **two consecutive slots** (one each way), so it is **symmetric**. A master supports **up to 3 SCO links**; a slave supports 3 to one master or 2 to different masters. **SCO packets are never retransmitted.** |
| **eSCO** (extended SCO) | Reserves slots like SCO, but **can be asymmetric** and **supports retransmission** |
| **ACL** (asynchronous connectionless) | **Point-to-multipoint** between the master and all slaves. Uses slots **not reserved for SCO/eSCO**; the master can exchange packets with any slave **per slot**. **Only one ACL link** exists. Most ACL packets are **retransmitted** if needed |

- **SCO/eSCO** carry **time-bounded data** that need a **guaranteed data rate but not guaranteed delivery**, e.g. **digitally encoded audio** tolerant of loss. The rate is guaranteed by **slot reservation**.
- **ACL** gives a **packet-switched** style connection: **no bandwidth reservation**; delivery may be guaranteed via error detection + retransmission. **A slave may send an ACL packet in the slave-to-master slot only if it was addressed in the preceding master-to-slave slot** (polling). 1-, 3- and 5-slot packets; unprotected or **2/3 FEC**. Maximum: a **5-slot unprotected** packet with asymmetric allocation gives **721 kbps forward and 57.6 kbps reverse** (Basic Rate). EDR **3-DH5** reaches **2178.1 kbps** asymmetric (Table 12.4).

## 10.10 Packet format [BT-B, BT-S 36–44]

```
|<--- 72 bits --->|<- 54 bits ->|<----- 0 to 2745 bits ----->|
|  Access code    |   Header    |          Payload           |

Access code: | preamble 4 | SYNC word 64 | trailer 4 |
Header (18 bits, sent with 1/3 FEC = 54):

| AM_ADDR 3 | Type 4 | Flow 1 | ARQN 1 | SEQN 1 | HEC 8 |
```

- **Access code**: timing synchronisation, offset compensation, paging, inquiry. Types: **CAC** (channel access code: identifies a piconet), **DAC** (device access code: paging and responses), **IAC** (inquiry access code: inquiry).
- **Header fields**:
  - **AM_ADDR (3 bits)**: active-mode address of a slave (at most 7 active slaves); **0 = broadcast** from the master.
  - **Type (4 bits)**: packet type (SCO/ACL types, error protection, number of slots). The **ID packet** (access code only, 68 bits) is used in inquiry and access procedures.
  - **Flow (1 bit)**: ACL flow control. 0 = stop sending, 1 = resume.
  - **ARQN (1 bit)**: acknowledgement for CRC-protected ACL traffic. **ACK = 1, NAK = 0**; no reply means a NAK is assumed.
  - **SEQN (1 bit)**: alternating 1/0 sequence bit to **filter out retransmissions** (duplicates caused by a failed ACK).
  - **HEC (8 bits)**: header error control, an **8-bit error-detection code** protecting the header.
- **Payload**: payload header (8 bits single-slot, 16 bits multislot), with **L_CH** (logical channel: LMP 11, start/unfragmented L2CAP 10, continuation 01, other 00), **Flow** (L2CAP-level flow control) and **Length** (bytes of data); the payload body; and a **16-bit CRC** on all data payloads (except AUX1).

## 10.11 ★ Error correction in Bluetooth [BT-S 45–50]

Three schemes:

1. **1/3-rate FEC**: used on the **18-bit header** and the voice field in **HV1** packets. It simply **sends three copies of each bit**, and **majority logic** decodes each triple.
2. **2/3-rate FEC**: used in all **DM** packets, the data field of **DV**, the **FHS** packet and **HV2**. It is a form of **Hamming code (15, 10)**: it **corrects all single errors and detects all double errors** in each codeword.
3. **ARQ**: used with **DM and DH** packets and the data field of DV. Its elements:
   - **Error detection** (CRC, supplemented by FEC)
   - **Positive acknowledgement** of error-free packets
   - **Retransmission after timeout**

- **Negative acknowledgement and retransmission** ACK/NAK are carried **in the next return packet** (piggybacked in ARQN).

## 10.12 ★ Master vs base station/AP, and one device as master of two piconets

### Bluetooth master vs 802.11 base station/AP (Asked in: Exam Q4(a), T2-2026 Q8, BTX 8)

From [BT-S 74] (802.15.3 slide comparing a PNC, an 802.11 AP and a Bluetooth master): "*The AP provides a link to other networks and acts as a relay point for all MAC frames. The Bluetooth master sets up piconets and acts as a relay point for all traffic.*" Combined with [BT-B] and [WL]:

| | Bluetooth master (central) | 802.11 AP / base station |
|---|---|---|
| **Medium access coordination** | **Fully controls access**: sets the hop sequence and timing; **polls** slaves (TDD/TDMA); **a slave transmits only when addressed by the master**, so **no contention or collisions within the piconet** | Stations normally **contend** using **CSMA/CA (DCF)**: random backoff, collisions possible. Polling (**PCF**) is **optional** |
| **Role in connecting to other networks** | Sets up the **piconet** (an ad hoc PAN) and relays traffic **inside** it; it is **not a gateway** to a wired network or the Internet (slaves talk only to the master; inter-piconet communication is via scatternet bridge devices) | **Bridges the wireless BSS to the distribution system/wired LAN/Internet**; relays all MAC frames; ESS roaming |
| **Number of devices** | **Max 7 active slaves** (3-bit AM_ADDR) | Many stations (no such small fixed limit) |
| **Nature** | Any Bluetooth device can become master (role is temporary/dynamic) | Dedicated, usually fixed infrastructure device |
| **Range / power** | ~10 m, low power | Tens to hundreds of metres |

### Can one device be the master of two piconets at the same time? (Asked in: Exam Q4(d), T2-2025 Q9, BTX 9)

**No.** A piconet **is defined by its master**: the hopping sequence (channel) and the timing (phase) are derived from the **master's own Bluetooth address and clock** [BT-B]. If one device were master of two "piconets", **both would use the same hopping sequence and timing**, so they would be **one and the same piconet**, not two. (Its slaves would simply all be in one piconet, limited to 7.) A device **can** be in two piconets as **slave in both**, or **master in one and slave in the other**, by **time-sharing** between the two hop sequences, which is how scatternets work [BT-B Fig 12.2].

## 10.13 ★ Hidden terminal problem in Bluetooth (Exam Q4(b), T2-2025 Q4)

**Within a piconet, no.** The hidden terminal problem arises with **contention-based** access like CSMA/CA, where stations decide on their own to transmit after sensing an idle channel. In Bluetooth the **master polls**: **a slave transmits only in the slot after the master has addressed it** [BT-B ACL rule], and SCO slots are **reserved**. Two slaves never transmit simultaneously by their own decision, so there is **nothing for a hidden terminal to collide with**. All slaves are within range of the master, which schedules every transmission (TDD/TDMA). **Between piconets** (scatternet), collisions **can** occur when two piconets happen to hop to the same frequency in the same slot. But this is **FH-CDMA collision**, not a hidden terminal problem caused by carrier sensing, and it is handled by FEC/ARQ [BT-B].

SUPPLEMENTARY (Kurose & Ross's answer to the same question): "No, the hidden terminal problem does not occur in Bluetooth because the master polls; slaves only transmit when polled."

## 10.14 ★ Scatternet collision probability (Asked in: Exam Q4(e), T2-2025 Q8, BTX 20)

*Two piconets, no CSMA, 80 frequencies, both always have packets to send. A successful transmission needs a successful poll (master → slave) **and** a successful reply (slave → master).*

**Reasoning** (uses [BT-B]: each piconet hops pseudorandomly and independently over the frequencies; a collision occurs when both use the same frequency in the same slot):

1. In any one slot, both piconets transmit (always busy). Piconet 2 is on a frequency that is effectively random relative to piconet 1. **P(same frequency in one slot) = 1/80.**
2. **P(one slot is collision-free) = 1 − 1/80 = 79/80.**
3. An exchange uses **two slots** (poll + reply) on **two different hop frequencies**, so the slots are **independent**: P(both succeed) = $(79/80)^2$ = **0.9752**.
4. **P(collision affecting the exchange) = $1 - (79/80)^2 = 159/6400 \approx 0.0248 \approx 2.5\%$.**

**Answer: about 2.5%** (≈ 1/80 per individual packet; ≈ 2/80 per poll-reply transaction).

SUPPLEMENTARY note on assumptions: if the question means "probability that a *single packet* collides", the answer is **1/80 = 1.25%**. State both, clearly: "per packet 1/80; per successful poll+response transaction $1 - (79/80)^2 \approx 2.5\%$". Mention that with **79** channels (real Bluetooth) the figures are 1/79 and ≈ 2.5%.

## 10.15 Bluetooth High Speed (3.0) and Bluetooth Smart (4.0) [BT-S 52–55]

**Bluetooth 3.0 + HS**: up to **24 Mbps**. It adds a **Wi-Fi connection** for higher rates: the **Alternative MAC/PHY (AMP)** (actually an 802.11 link). The Bluetooth radio still handles **device discovery, association, initial connection setup, profile configuration and connection maintenance**; AMP is used **only when higher rates are needed**. Each AMP uses a **Protocol Adaptation Layer (PAL)** to map Bluetooth protocols onto the 802.11 MAC/PHY. **(BTX 3)**: Bluetooth 3.0 reaches 24 Mbps by **using a co-located 802.11 (Wi-Fi) radio via AMP** for the data transfer.

**Bluetooth Smart = Bluetooth Low Energy (BLE), Bluetooth 4.0** (BTX 2):

- An **intelligent, power-friendly** version; runs for long periods on a **single battery** or scavenged energy; can talk to legacy Bluetooth devices and smartphones; very successful for **IoT** (e.g. health monitors).
- Same **2.4 GHz** ISM band, but **40 channels spaced 2 MHz apart** (**3 advertising + 37 data**), instead of 79 × 1 MHz.
- Devices may implement transmitter, receiver or both; **single-mode** (BLE only, **cheap**, compact) or **dual-mode** (also BR/EDR).
- **Design features**: **10 mW** output; **150 m** range in an open field; new **advertising** mechanism for efficient discovery; new **asynchronous connectionless MAC** for low delay, fast transactions; new **generic attribute protocol** (simple client/server model); new **generic attribute profile** to collect sensor data efficiently.

## 10.16 IEEE 802.15 [BT-B 12.5, BT-S 56–80]

- The IEEE working group for **WPANs**, formed to develop standards for **short-range wireless PANs**. In a PAN all devices are **owned by one person or family** (PCs, smartphones, tablets, health monitors, smart glasses, peripherals).

- **802.15.1**: MAC and PHY from the Bluetooth specification (approved 2002, updated 2005).
- **802.15.3**: devices that are **low cost and low power** compared with 802.11 but with **higher data rates** than 802.15.1 (issued 2003). **802.15.3c**: **gigabit rates at 60 GHz**. **802.15.3d**: **100 Gbps**.
- **802.15.4**: **very low cost, very low power**, data rates **lower** than 802.15.1 (2003).
- Architecture: 802.15.1, 802.15.3 and 802.15.4 share **LLC**, with different MAC and PHY layers.

| Standard | PHY / rates |
|---|---|
| 802.15.1 Bluetooth | 2.4 GHz; 1, 2 or 3 Mbps; 24 Mbps HS |
| 802.15.3c | 60 GHz; 1–6 Gbps |
| 802.15.3d | 100 Gbps (the slide's table labels it 60 GHz; SUPPLEMENTARY: the published standard actually uses ~252–325 GHz) |
| 802.15.4 / 4e | 868/915 MHz, 2.4 GHz; DSSS 20, 40, 100, 250 kbps; UWB 110 kbps–27 Mbps; CSS 250 kbps, 1 Mbps |

## 802.15.3: high data rate WPAN [BT-S 63–68]

- Examples: cameras/phones to printers or kiosks; laptop to projector; **5:1 surround sound** speakers; video from a set-top box; music to headphones; camcorder to TV; remote viewfinders.
- **Requirements**: short range (~10 m); **high throughput (> 20 Mbps)** for video and multichannel audio; low power; low cost; **QoS** (guaranteed data rate); **dynamic environment** (devices enter and leave the piconet; mobile devices < 7 km/h); simple connectivity; **privacy**.
- **MAC**: a network of **DEVs**; one DEV is the **piconet coordinator (PNC)**, which sends **beacons**. Time is divided into **superframes**: a beacon period, a **contention access period** (command/control, **CSMA/CA**) and a **channel time allocation period** (**TDMA with guaranteed time slots, GTSs**: QoS).
- ★ **PNC vs 802.11 AP vs Bluetooth master** [BT-S 64]: *the AP provides a link to other networks and acts as a relay point for all MAC frames; the Bluetooth master sets up piconets and acts as a relay point for all traffic;* ***the PNC controls access to the time resources of the piconet but is not involved in the exchange of data frames between DEVs***.
- **PHY**: 2.4 GHz; five modulation formats at **11 Mbaud** give **11–55 Mbps**; **trellis-coded modulation (TCM)**.
- **802.15.3c**: 60 GHz; a common ~5 GHz-wide band worldwide; **high power, doesn't penetrate walls**; **beamforming** helps; **four 2160 MHz channels**. Usage models: (1) uncompressed HD video (1920×1080 × 24 bit × 60 fps → **> 3.5 Gbps**); (2) multivideo streaming; (3) office desktop peripherals; (4) conference ad hoc; (5) kiosk file downloading. **Three PHY modes**: single carrier 1.5/3 Gbps; high-speed OFDM up to 5.775 Gbps; audio/visual OFDM 0.952/1.904/3.807 Gbps. The MAC supports coexistence of the modes, **frame aggregation**, and optional **beamforming in three stages** (sector-level coarse search, beam-level fine search, optional tracking).
- **802.15.3d**: **100 Gbps**, low cost and power; wireless offices, data-centre point-to-point links, wireless backhaul, intra-device; a few cm to 100 m.

## 802.15.4: low rate WPAN [BT-S 69–78]

- **TG4**: a low data rate solution with **multi-month to multi-year battery life** and **very low complexity**.
- PHYs: **868/915 MHz DSSS** (20, 40, 100, 250 kbps); **2.4 GHz DSSS** (250 kbps); **UWB** impulse radio (very short pulses, low spectral density below thermal noise; 851 kbps, optionally 110 kbps, 6.81 or 27.234 Mbps; **precision ranging to within 1 m**); **2.4 GHz chirp spread spectrum (CSS)** (1 Mbps, optionally 250 kbps).

- Applications: **sensors, interactive toys, smart badges, remote controls, home automation**.
- **MAC**: **CSMA-CA** channel access, automatic network establishment from a **coordinator node**, and a **fully handshaked** protocol for reliable delivery.
- Extensions: **4c/4d** (780 and 950 MHz PHYs); **4e** (industrial: time synchronisation, **channel hopping**, low duty cycles); **4f** (**active RFID**); **4g** (**smart utility networks/smart grid**, 40–1000 kbps); **4j** (**medical body area networks**, 2.4 GHz); **4k** (**LECIM**: critical infrastructure such as roads and bridges, multi-year battery); **4m** (**TV white space**); **4p** (**rail**: Positive Train Control).
- **Other 802.15**: **802.15.2** (**coexistence** of 802.15 with 802.11 in the same bands); **802.15.6** (**body area networks**: in-, on- and around-body; narrowband, UWB, body channel); **802.15.7** (**visible light communication**, 380–780 nm, 11.67 kbps–96 Mbps; challenges: flicker and dimming).

## 10.17 ZigBee [BT-B 12.6, BT-S 81–90]

- A suite of protocols **based on IEEE 802.15.4** for **small, low-power radios**, simpler than WPANs. Suited to **low data rate, long battery life, secure networking**. Data rates **20–250 kbps**; best for **periodic or intermittent** communication. Named after the **waggle dance of honey bees**. Standards since **2003** (the Alliance formed in 1998; ZigBee ratified Dec 2004; **ZigBee PRO** 2007).
- **Features**: low cost; low power; **mesh networking** (nodes route each other's messages: reliability and range); **star, tree and mesh** topologies; **one coordinator** creates, controls and maintains the network; **wakes in 30 ms** (vs Bluetooth up to 3 s), so nodes can sleep most of the time.
- **Four components added to 802.15.4**: **network layer (routing)**; **application support sublayer (APS)**; **ZigBee device objects (ZDOs)**, the most significant addition (device roles, join requests, device discovery, security); **manufacturer-defined application objects**.
- **Stack profiles**: Profile 1 (**ZigBee**: home, light commercial); Profile 2 (**ZigBee PRO**: multicasting, higher security).
- **Application profiles**: building automation, health care, home automation, input device, Light Link, network devices, retail, remote control, Smart Energy 1.1 and 2 (IP-based), telecom services. Other specs: **ZigBee IP** (IPv6 mesh); **ZigBee RF4CE** (consumer electronics remote control).
- ★ **Device types**: **Coordinator**: the most capable; **only one per network**; starts the network, is the root of a tree, maintains information such as **security keys**. **Router**: passes data on to other devices. **End device**: just enough function to talk to a router or coordinator; **cannot relay**; sleeps a lot; least memory, **cheapest**.
- **Routing**: ZigBee automatically builds low-speed **ad hoc networks**, using routing protocols to forward packets over **multiple hops**. **AODV (Ad hoc On-demand Distance Vector)** is used: it sends **route requests** and either responds or forwards them until someone responds; nodes **store and use** discovered routes and can form clusters. *(This is the only routing protocol mentioned anywhere in your notes; see Topic 11.)*
- **Networks**: **non-beacon** (unslotted CSMA/CA, routers always on, so they need permanent power) vs **beacon-based** (periodic beacons; superframe with a **contention access period** of 16 slots using CSMA/CA and an optional **contention-free period** with **GTSs**; nodes sleep between beacons, intervals 15.36 ms to 251.65824 s at 250 kbps). Beacon mode is typical for star/tree networks, non-beacon for mesh; most commercial systems use non-beacon.
- **Radio hardware**: **2.4 GHz** (16 channels, 5 MHz each, 250 kbps, **offset QPSK**); **915 MHz** (40 kbps, BPSK); **868 MHz** (20 kbps, BPSK). Range **10–75 m**, up to **1500 m with ZigBee PRO**.
- **Security**: strict key management; one **trust center**; the layer that creates a frame secures it; **128-bit keys** (network or link).

## 10.18 Topic 10 revision summary

- Bluetooth: Ericsson 1994; SIG; 2.4 GHz; 10 m (100 m); robust, low power, low cost; **core + profiles**. Stack: Radio, Baseband, LMP, L2CAP, SDP; RFCOMM; TCS BIN; PPP/TCP/IP/OBEX.
- **Profiles = interoperability + reuse + only-what's-needed** (asked 4 times).
- **Piconet**: 1 master + ≤ 7 active slaves; the master's address sets the hop sequence and timing; **slaves speak only when polled**. **Scatternet**: overlapping piconets; **FH-CDMA** between piconets; collision when the same hop in the same slot.
- Radio: 79 × 1 MHz, **1600 hops/s, 625 µs slots**, **GFSK 1 Mbps**, EDR 2/3 Mbps; classes 1/2/3.
- Baseband: **FH-TDD-TDMA**; 1/3/5-slot packets; adaptive FH (≥ 20 channels).
- Links: **SCO** (reserved, symmetric, no retransmission), **eSCO**, **ACL** (packet-switched, polled, retransmission).
- Packet: access code 72 / header 54 (18 × 3) / payload 0–2745; AM_ADDR, Type, Flow, ARQN, SEQN, HEC.
- Error control: **1/3 FEC (repetition), 2/3 FEC (Hamming 15,10), ARQ**.
- **Master vs AP**: polling (no contention) vs CSMA/CA; a piconet PAN vs a bridge to the wired network. **One master for two piconets is impossible**: the master defines the hop sequence. **No hidden terminal inside a piconet**: polling. **Scatternet collision ≈ $1 - (79/80)^2 \approx 2.5\%$.**
- BT 3.0 HS (AMP/802.11, 24 Mbps); BT 4.0 Smart/BLE (40 × 2 MHz channels, 10 mW, 150 m).
- 802.15.3 (high rate, PNC, superframes, GTS), .3c (60 GHz Gbps), .3d (100 Gbps), .4 (low rate, years of battery), ZigBee (coordinator/router/end device, AODV, mesh).

# TOPIC 11 — ROUTING AND AD HOC NETWORK ROUTING

**SUPPLEMENTARY TOPIC.** None of the uploaded files cover this topic. The course outline names *Ch. 3 Communication Networks, Ch. 4 Protocols and the TCP/IP Suite* and *Ch. 13 (Agrawal) Ad hoc Networks*, which were not provided. Everything below follows the standard textbook treatment used by these exact questions: the Figure 1 network and the count-to-infinity questions are from **Kurose & Ross, *Computer Networking*, Ch. 5**; the T2-2026 Figure is **Tanenbaum's Dijkstra example**; MANET categories follow **Agrawal & Zeng** and standard MANET literature. The only related item in your notes is **AODV**, mentioned on the ZigBee slide [BT-S 87].

This topic is worth **30% of the exam paper** (and 25–32% of the Test 2 papers). Learn the Dijkstra tables below perfectly.

## 11.1 Routing, forwarding and switching ★ (Asked in: T2-2026 Q1, 6 marks)

| Process | Purpose | Where / scope | Information used to decide |
|---|---|---|---|
| **Routing** | Determine the **end-to-end path** (route) packets take from source to destination | **Network-wide**, **control plane**; runs over seconds/minutes; routing algorithms (Dijkstra/link-state, Bellman-Ford/distance-vector; AODV in MANETs) | **Network topology and link costs** (exchanged with other routers via routing protocols); output = the routing/forwarding table |
| **Packet forwarding** | **Move an arriving packet from a router's input link to the correct output link** | **Router-local**, **data plane**; per packet, nanoseconds | The packet's **destination network-layer (IP) address**, looked up in the **forwarding table** (longest prefix match) |
| **Switching** | Move **frames** to the correct outgoing port **within a LAN** (link layer), or the transfer through a router's internal **switching fabric** | **Link layer** (layer 2) within one network; or inside the router | The frame's **destination MAC address**, looked up in the **switch table** (learned by **self-learning** from source addresses) |

**One-line distinction**: routing = **planning the trip** (map → route); forwarding = **taking one interchange exit** (look up the destination in the table); switching = the **same local job at layer 2** using MAC addresses.

## 11.2 Graph model of routing

A network is a **graph G = (N, E)**: N = routers (nodes), E = links (edges) with **costs c(x, y)** (delay, distance, cost, or inverse bandwidth). **c(x, y) = ∞** if x and y are not neighbours. A **least-cost path** minimises the sum of link costs.

Two classes of algorithms:

- **Global (link-state)**: every node knows the **complete topology and all link costs** → runs **Dijkstra**.
- **Decentralised (distance-vector)**: each node knows **only its neighbours and their distance estimates**; computes iteratively → **Bellman-Ford**.

## 11.3 ★ Link-state routing and Dijkstra's algorithm

**Link state**: each router **floods** link-state advertisements (its neighbours and link costs) to **all** routers, so all have an identical map. Each router then runs **Dijkstra** locally (OSPF, IS-IS).

**Dijkstra's algorithm** (for source u):

- **D(v)**: current least-cost estimate from u to v. **p(v)**: predecessor of v on that path. **N'**: set of nodes whose least-cost path is **definitively known**.

```
Initialisation:
  N' = {u}
  for all nodes v:  if v is a neighbour of u: D(v) = c(u,v)  else D(v) = ∞
Loop (until N' = all nodes):
  find w not in N' with the minimum D(w)
  add w to N'
  for every neighbour v of w not in N':
      D(v) = min( D(v),  D(w) + c(w,v) )      ← update if the path via w is cheaper
```

- **Greedy**: always fixes the closest unfinished node. **n iterations**; complexity **O(n²)** (O(n log n + E log n) with a heap). It needs **non-negative costs**.

**Exam method**: draw a table with one row per step: **Step | N' | D(v),p(v) for each node**. Mark a node's column finished once it enters N'. Finish with the **forwarding table** (least-cost path and first hop).

### Figure 1 (Exam Q3(c), T2-2025 Q2) link costs

**x–z 8, x–y 6, x–v 3, x–w 6, y–z 12, y–v 8, y–t 7, v–t 4, v–u 3, v–w 4, t–u 2, w–u 3.**

**Adjacency (read this off Figure 1 before you start, and write it in your answer):**

| Node | Neighbours (cost) |
|---|---|
| x | z 8, y 6, v 3, w 6 |
| y | z 12, x 6, v 8, t 7 |
| z | x 8, y 12 |
| v | x 3, y 8, w 4, u 3, t 4 |
| w | x 6, v 4, u 3 |
| u | v 3, w 3, t 2 |
| t | y 7, v 4, u 2 |

### ★ Worked example DJ1: shortest paths from x (T2-2025 Q2, 8 marks)

| Step | N' | D(t),p | D(u),p | D(v),p | D(w),p | D(y),p | D(z),p |
|---|---|---|---|---|---|---|---|
| 0 | x | ∞ | ∞ | **3,x** | 6,x | 6,x | 8,x |
| 1 | xv | 7,v | **6,v** | — | 6,x | 6,x | 8,x |
| 2 | xvu | 7,v | — | — | **6,x** | 6,x | 8,x |
| 3 | xvuw | 7,v | — | — | — | **6,x** | 8,x |
| 4 | xvuwy | **7,v** | — | — | — | — | 8,x |
| 5 | xvuwyt | — | — | — | — | — | **8,x** |
| 6 | xvuwytz | — | — | — | — | — | — |

*Step-by-step reasoning*:

- Step 0: x's neighbours v 3, w 6, y 6, z 8; t and u are ∞. Minimum = **v (3)**.
- Step 1 (via v, D = 3): t = 3 + 4 = **7**; u = 3 + 3 = **6**; w: 3 + 4 = 7 > 6, keep 6; y: 3 + 8 = 11 > 6, keep. Minimum among u 6, w 6, y 6 → pick **u** (tie broken alphabetically).
- Step 2 (via u, D = 6): t: 6 + 2 = 8 > 7, keep 7; w: 6 + 3 = 9 > 6, keep. Add **w (6)**.
- Step 3 (via w): no improvements (u, v done). Add **y (6)**.
- Step 4 (via y): t: 6 + 7 = 13 > 7; z: 6 + 12 = 18 > 8. Add **t (7)**.
- Step 5: add **z (8)**.

**Result (least-cost paths from x)**:

| Destination | Cost | Path | First hop (forwarding table) |
|---|---|---|---|
| v | 3 | x → v | v |
| u | 6 | x → v → u | v |
| w | 6 | x → w | w |
| y | 6 | x → y | y |
| t | 7 | x → v → t | v |
| z | 8 | x → z | z |

## ★ Worked example DJ2: shortest paths from t (Exam Q3(c)(i), 4 marks)

| Step | N' | D(u),p | D(v),p | D(w),p | D(x),p | D(y),p | D(z),p |
|---|---|---|---|---|---|---|---|
| 0 | t | **2,t** | 4,t | ∞ | ∞ | 7,t | ∞ |
| 1 | tu | — | **4,t** | 5,u | ∞ | 7,t | ∞ |
| 2 | tuv | — | — | **5,u** | 7,v | 7,t | ∞ |
| 3 | tuvw | — | — | — | **7,v** | 7,t | ∞ |
| 4 | tuvwx | — | — | — | — | **7,t** | 15,x |
| 5 | tuvwxy | — | — | — | — | — | **15,x** |
| 6 | tuvwxyz | — | — | — | — | — | — |

*Key updates*: via u (2): w = 2 + 3 = 5; v: 2 + 3 = 5 > 4, keep. Via v (4): x = 4 + 3 = **7**; w: 4 + 4 = 8 > 5; y: 4 + 8 = 12 > 7. Via w (5): x: 5 + 6 = 11 > 7. Via x (7): z = 7 + 8 = **15**; y: 7 + 6 = 13 > 7. Via y (7): z: 7 + 12 = 19 > 15.

| Destination | Cost | Path |
|---|---|---|
| u | 2 | t → u |
| v | 4 | t → v |
| w | 5 | t → u → w |
| x | 7 | t → v → x |
| y | 7 | t → y |
| z | 15 | t → v → x → z |

## ★ Worked example DJ3: shortest paths from w (Exam Q3(c)(ii), 4 marks)

| Step | N' | D(t),p | D(u),p | D(v),p | D(x),p | D(y),p | D(z),p |
|---|---|---|---|---|---|---|---|
| 0 | w | ∞ | **3,w** | 4,w | 6,w | ∞ | ∞ |
| 1 | wu | 5,u | — | **4,w** | 6,w | ∞ | ∞ |
| 2 | wuv | **5,u** | — | — | 6,w | 12,v | ∞ |
| 3 | wuvt | — | — | — | **6,w** | 12,v | ∞ |
| 4 | wuvtx | — | — | — | — | **12,v** | 14,x |
| 5 | wuvtxy | — | — | — | — | — | **14,x** |
| 6 | wuvtxyz | — | — | — | — | — | — |

*Key updates*: via u (3): t = 3 + 2 = 5; v: 3 + 3 = 6 > 4. Via v (4): y = 4 + 8 = 12; t: 4 + 4 = 8 > 5; x: 4 + 3 = 7 > 6. Via t (5): y: 5 + 7 = 12 (equal, keep v). Via x (6): z = 6 + 8 = 14; y: 6 + 6 = 12 (equal). Via y (12): z: 12 + 12 = 24 > 14.

| Destination | Cost | Path |
|---|---|---|
| u | 3 | w → u |
| v | 4 | w → v |
| t | 5 | w → u → t |
| x | 6 | w → x |
| y | 12 | w → v → y **(tie: w → x → y and w → u → t → y also cost 12)** |
| z | 14 | w → x → z |

> **Tie note**: y has **three equal-cost paths (12)**. Any one is correct; state the tie-breaking rule you used (e.g. "keep the first-found predecessor"). Mentioning the tie shows full understanding.

## ★ Worked example DJ4: T2-2026 Q9 (A to all nodes, 10 marks)

**Link costs read from the figure**: **A–B 2, A–G 6, B–C 7, B–E 2, E–G 1, E–F 2, F–C 3, F–H 2, G–H 4, C–D 3, H–D 2.** (This is Tanenbaum's classic Dijkstra example.)

| Node | Neighbours (cost) |
|---|---|
| A | B 2, G 6 |
| B | A 2, C 7, E 2 |
| C | B 7, F 3, D 3 |
| D | C 3, H 2 |
| E | B 2, F 2, G 1 |
| F | E 2, C 3, H 2 |
| G | A 6, E 1, H 4 |
| H | G 4, F 2, D 2 |

| Step | N' | B | C | D | E | F | G | H |
|---|---|---|---|---|---|---|---|---|
| 0 | A | **2,A** | ∞ | ∞ | ∞ | ∞ | 6,A | ∞ |
| 1 | AB | — | 9,B | ∞ | **4,B** | ∞ | 6,A | ∞ |
| 2 | ABE | — | 9,B | ∞ | — | 6,E | **5,E** | ∞ |
| 3 | ABEG | — | 9,B | ∞ | — | **6,E** | — | 9,G |
| 4 | ABEGF | — | 9,B | ∞ | — | — | — | **8,F** |
| 5 | ABEGFH | — | **9,B** | 10,H | — | — | — | — |
| 6 | ABEGFHC | — | — | **10,H** | — | — | — | — |
| 7 | ABEGFHCD | — | — | — | — | — | — | — |

*Step-by-step reasoning*:

- Step 0: neighbours of A: B 2, G 6. Pick **B (2)**.
- Step 1 (via B = 2): C = 2 + 7 = 9; E = 2 + 2 = **4**. Pick **E (4)**.
- Step 2 (via E = 4): G = 4 + 1 = **5** < 6 → update G to (5, E); F = 4 + 2 = 6. Pick **G (5)**.
- Step 3 (via G = 5): H = 5 + 4 = 9. Pick **F (6)**.
- Step 4 (via F = 6): C: 6 + 3 = 9 = 9 (**tie**, keep B); H = 6 + 2 = **8** < 9 → (8, F). Pick **H (8)**.
- Step 5 (via H = 8): D = 8 + 2 = 10. Pick **C (9)**.
- Step 6 (via C = 9): D: 9 + 3 = 12 > 10, keep. Pick **D (10)**.

**(b) Final costs and routes from A**:

| Destination | Cost | Complete shortest route |
|---|---|---|
| **B** | **2** | A → B |
| **C** | **9** | A → B → C *(equal-cost alternative: A → B → E → F → C, also 9)* |
| **D** | **10** | A → B → E → F → H → D |
| **E** | **4** | A → B → E |
| **F** | **6** | A → B → E → F |
| G | 5 | A → B → E → G (note: cheaper than the direct link A–G = 6) |
| H | 8 | A → B → E → F → H |

## 11.4 ★ Distance-vector routing and Bellman-Ford

**Bellman-Ford equation**: the least cost from x to y is

$$d_x(y) = \min \text{ over all neighbours v of } \{\ c(x,v) + d_v(y)\ \}$$

**Distance-vector algorithm** (RIP): each node keeps a **distance vector $D_x$ = [$D_x(y)$ for all y]** and its neighbours' vectors.

1. **Initialise**: $D_x(y)$ = c(x, y) for neighbours, ∞ otherwise; send $D_x$ to the neighbours.
2. **Loop**: wait for a **link-cost change** or a **neighbour's new vector**; recompute $D_x(y) = \min_v\{c(x, v) + D_v(y)\}$; **if $D_x$ changed, send it to the neighbours**.

It is **iterative** (until no more changes), **asynchronous** (no lock-step) and **distributed** (each node uses only neighbour information).

### ★ Link-state vs distance-vector (Exam Q3(a), 4 marks)

| | Link-state (LS) | Distance-vector (DV) |
|---|---|---|
| **Knowledge** | **Global**: every router has the **complete topology** | **Local**: only neighbours' cost estimates ("routing by rumour") |
| **Algorithm** | **Dijkstra** | **Bellman-Ford** |
| **What is sent, to whom** | Own **link costs**, **flooded to all** routers | Whole **distance vector**, **only to neighbours** |
| **Message complexity** | High: O(n·E) messages per change | Lower: exchanges only between neighbours |
| **Speed of convergence** | Fast: O($n^2$) computation, **no loops** once floods finish | **Slow**, variable; **routing loops** possible; **count-to-infinity** |
| **Robustness** | Better: each router computes its **own** table; a bad advertisement affects only its own link | Worse: a wrong cost **propagates through the whole network** |
| **Examples** | OSPF, IS-IS | RIP, (BGP is path-vector) |

**Similarities ("compare")**: both find **least-cost paths**, both produce a **forwarding table**, both are **dynamic** (adapt to cost changes) and both use **link costs**.

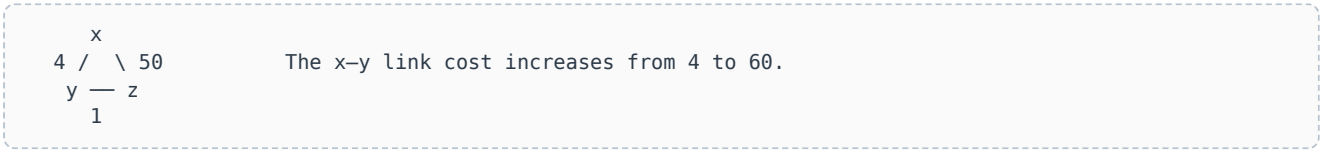
### ★ Essential difference: Dijkstra vs Bellman-Ford (Exam Q3(b), 4 marks)

- **Dijkstra** is a **centralised/global, greedy** algorithm: the node computing it needs the **entire graph**. It **permanently fixes one node per iteration** (the nearest unfinished node), so it finishes in **n steps**. It requires **non-negative** link costs.
- **Bellman-Ford** is **decentralised and iterative**: each node computes using **only its neighbours' current distance estimates** and **repeatedly relaxes** distances until nothing changes (at most n − 1 rounds). It **handles negative link costs** (and can detect negative cycles), is naturally **distributed**, but **converges more slowly** and can suffer count-to-infinity.
- **In one sentence**: *Dijkstra uses complete global knowledge and settles nodes greedily one at a time; Bellman-Ford uses only neighbour information and iteratively relaxes all estimates until they converge.*

## 11.5 ★ The count-to-infinity problem (Exam Q3(d), T2-2025 Q3)

**What it is**: in distance-vector routing, **bad news (a link-cost increase or failure) travels slowly**. Neighbours keep advertising **stale** routes that actually pass through the node that has just lost the route, which creates a **routing loop**. The nodes keep increasing their estimates step by step, "counting" up towards the true cost, or **towards infinity** if the destination is unreachable.

**Classic example** (Kurose & Ross): y–x cost 4, y–z 1, z–x 50. Initially D_y(x) = 4, D_z(x) = 5 (via y).



1. y sees c(y, x) = 60. y computes D_y(x) = min{60 + 0, c(y,z) + D_z(x)} = min{60, 1 + 5} = **6**. That is wrong: z's 5 is a route **through y**! Now y routes to x **via z**, and z routes to x **via y**: a **routing loop**.
2. y tells z; z updates D_z(x) = min{50, 1 + 6} = **7**.
3. z tells y; y updates to 8; z to 9; ...
4. This continues for **44 iterations**, until z's estimate via y exceeds 50 and z finally chooses its direct link (50); then y = 51. Packets **loop between y and z** the whole time. If the x–y link had **failed** (cost ∞) and z had no other route, they would **count to infinity**.

**Fixes**: **poisoned reverse** (if z routes to x through y, z tells y D_z(x) = ∞; this fixes loops between **two** nodes but **not** larger loops); **split horizon**; defining **infinity as a small number** (RIP: 16 hops); hold-down timers.

★ **Will it occur if we decrease the cost of a link? No.** Decreasing a cost is **good news**, and **good news travels fast**. Each node that hears of the lower cost computes a lower (correct) distance through it and sends it on. Distances only **decrease** and converge within a number of iterations bounded by the network diameter. No node relies on a stale path that is now too cheap, so no loop forms. (Kurose: *"No. Decreasing link cost won't cause a loop."*)

★ **What about connecting two nodes that have no link? No.** Adding a new link is the same as **decreasing its cost from ∞ to a finite value**: it is good news, which propagates quickly without counting to infinity.

## 11.6 ★ Ad hoc networks (MANETs) and why routing is complicated (Exam Q2(d))

A **MANET** (mobile ad hoc network) is a collection of **mobile nodes with no fixed infrastructure** (no AP or base station). Nodes communicate directly if in range, otherwise **other nodes forward** packets (multi-hop). **Every node is also a router.** This matches your WLAN notes: in ad hoc networks *"nodes can only communicate if they can reach each other physically... or if other nodes can forward the message"*, and *"the complexity of each node is higher: implement medium access mechanisms, forwarding data"* [WL 8–9].

**Why routing in ad hoc networks is complicated**:

1. **Dynamic topology**: nodes **move**, join and leave, so links **break and form constantly**; routes go stale quickly.
2. **No infrastructure or central control**: no dedicated routers; **every node must route**, with **distributed** decisions.
3. **Limited bandwidth**: the shared wireless channel is scarce, so routing control traffic (flooding updates) competes with data.
4. **Limited battery and processing** in mobile nodes, so frequent route computation and transmission drain energy.
5. **Unreliable wireless links**: fading, interference, **hidden/exposed terminals**, high error rates, sometimes **asymmetric (unidirectional)** links.
6. **Multi-hop**: many hops, so delays and failure points multiply.
7. **Scalability and security**: large networks generate huge overhead; no trusted infrastructure, so it is vulnerable to misbehaving nodes.

## 11.7 ★ The three categories of MANET routing protocols (Exam Q2(d) 10, T2-2025 Q1 12)

| | 1. Proactive (table-driven) | 2. Reactive (on-demand / source-initiated) | 3. Hybrid |
|---|---|---|---|
| **Principle** | Every node **continuously maintains routes to all destinations** in routing tables, updated by **periodic and triggered** broadcasts, whether or not routes are needed | Routes are discovered **only when a source needs one**: **route discovery** (flood a route request, the destination or an intermediate node replies) + **route maintenance** (repair broken routes) | **Combines both**: **proactive within a local zone** (nearby nodes), **reactive between zones** (distant nodes) |
| **Route availability** | **Immediately available**, so **low latency** | **Delay** while discovering the route | Fast locally; on-demand far away |
| **Overhead** | **High, constant** control traffic; wasted on unused routes; poor with high mobility | **Low** when traffic is light; but route-discovery floods are costly under heavy traffic | Balanced; tunable zone size |
| **Best for** | Small, low-mobility networks with frequent traffic | Large, highly mobile networks with occasional traffic | Large networks with mixed traffic |
| **Examples** | **DSDV** (Destination-Sequenced Distance Vector), **OLSR** (Optimized Link State Routing), WRP | **AODV** (Ad hoc On-demand Distance Vector, as used by **ZigBee** [BT-S 87]), **DSR** (Dynamic Source Routing), TORA | **ZRP** (Zone Routing Protocol) |

**AODV in brief** (your notes [BT-S 87] + standard detail): a source broadcasts a **RREQ (route request)**; intermediate nodes record the **reverse path** and rebroadcast; the destination (or a node with a fresh route) **unicasts an RREP (route reply)** back along the reverse path, which sets up forward entries; **sequence numbers** prevent loops and stale routes; **RERR** messages report broken links.

**DSDV in brief**: a table-driven distance-vector protocol (Bellman-Ford). Each entry carries a **destination sequence number** to avoid loops and count-to-infinity; full and incremental updates are broadcast periodically.

SUPPLEMENTARY: some textbooks (incl. Agrawal) name the categories **table-driven, source-initiated on-demand, and hybrid**. These are the same three. Others add **geographic/position-based** or hierarchical routing.

## 11.8 Topic 11 revision summary

- **Routing** (network-wide path, topology and costs, control plane) vs **forwarding** (router-local, destination IP + forwarding table) vs **switching** (layer 2, MAC + switch table).
- **Link-state**: flood link costs, everyone has the map, **Dijkstra** (greedy, global, $O(n^2)$, non-negative costs). **Distance-vector**: exchange vectors with neighbours, **Bellman-Ford** $d_x(y) = \min_v\{c(x,v) + d_v(y)\}$, iterative, asynchronous, distributed; slow; count-to-infinity.
- **Dijkstra answers**: from **x**: v3, u6, w6, y6, t7, z8. From **t**: u2, v4, w5, x7, y7, z15. From **w**: u3, v4, t5, x6, y12 (3-way tie), z14. **T2-2026 from A**: B2, E4, G5, F6, H8, C9 (tie), D10.
- **Count to infinity**: bad news travels slowly → loops (44 iterations in the 4/1/50 example). **Cost decrease or a new link: no** (good news travels fast). Fixes: poisoned reverse, split horizon, max hop count.

- **MANET routing is hard**: mobility, no infrastructure, bandwidth, energy, unreliable links. **Three categories: proactive (DSDV, OLSR), reactive (AODV, DSR), hybrid (ZRP).**

# Part 4 — Examination Pattern Analysis

This part analyses **all five past assessments** (450 marks in total). Every statement in sections 4.1–4.6 is **counted from the papers you uploaded**. Section 4.7 is clearly labelled as **prediction**. Nobody can know what will appear in your exam, but the patterns below are strong.

**Topic numbers**: T1 Transmission fundamentals · T2 Wireless channel · T3 Channel correction/capacity · T4 Signal encoding · T5 Multiple access · T6 Spread spectrum · T7 OFDM/OFDMA · T8 Error detection/correction · T9 WLAN 802.11 · T10 Bluetooth/802.15 · T11 Routing/MANET.

## 4.1 Question-by-question extraction

### Exam paper (green; title/year not visible; pages 2–4; 100 marks)

| Q | Question (abridged) | Marks | Topic | Type |
|---|---|---|---|---|
| 1(a) | Differences between path loss, multipath propagation, interference | 6 | T2 | Comparison |
| 1(b) | Spread spectrum principles of (i) FHSS (ii) DSSS | 4 | T6 | Explanation |
| 1(c) | Benefits of OFDMA approaches: adjacent, regularly spaced, randomly spaced subcarriers | 6 | T7 | Explanation |
| 1(d) | Describe CDMA; why difficult to jam | 4 | T5/T6 | Explanation |
| 2(a) | Hamming code for 10101111: check bits + codeword | 4 | T8 | **Calculation** |
| 2(b) | CRC: 101110, generator 1001 | 4 | T8 | **Calculation** |
| 2(c) | Internet checksum of 1001 1100 1010 0011 (4-bit) | 2 | T8 | **Calculation** |
| 2(d) | Why is ad hoc routing complicated? Three MANET routing categories + examples | 10 | T11 | Essay |
| 3(a) | Compare and contrast link-state and distance-vector | 4 | T11 | Comparison |
| 3(b) | Essential difference: Dijkstra vs Bellman-Ford | 4 | T11 | Comparison |
| 3(c)(i) | Dijkstra on Figure 1 from **t** | 4 | T11 | **Algorithm** |
| 3(c)(ii) | Dijkstra on Figure 1 from **w** | 4 | T11 | **Algorithm** |
| 3(d) | Count-to-infinity; does it occur on link-cost decrease? On connecting two unlinked nodes? | 4 | T11 | Explanation |
| 4(a) | Bluetooth master vs 802.11 base station | 4 | T10 | Comparison |
| 4(b) | Does the hidden terminal problem occur in Bluetooth? | 4 | T10 | Application |
| 4(c) | How can profiles be an advantage for applications? | 4 | T10 | Explanation |
| 4(d) | Can one device be controller of two piconets? | 4 | T10 | Application |
| 4(e) | Scatternet of 2 piconets: collision probability (80 freq.) | 4 | T10 | **Calculation** |
| 5(a) | Mobile node moving away: two BS actions | 2 | T3 | Application |
| 5(b) | Figure 2: hidden node; A and C send simultaneously | 4 | T9 | Scenario |
| 5(c) | Figure 2: CSMA/CA when A sends | 4 | T9 | Scenario |
| 5(d) | Why ACKs in 802.11 but not wired Ethernet? | 2 | T9 | Explanation |

| Q | Question (abridged) | Marks | Topic | Type |
|---|---|---|---|---|
| 5(e)(i) | Two ISPs' APs both on channel 11 | 4 | T9 | Scenario |
| 5(e)(ii) | Channels 1 and 11 | 4 | T9 | Scenario |

## Test 1, version A (T1A; 100 marks)

| Q | Question | Marks | Topic | Type |
|---|---|---|---|---|
| 1 | Three basic propagation mechanisms in a mobile system | 12 | T2 | Explanation |
| 2 | Main problems of signal propagation; why not straight lines; reflection useful and harmful | 12 | T2 | Essay |
| 3 | Path loss vs multipath vs interference | 9 | T2 | Comparison |
| 4 | FHSS and DSSS principles | 10 | T6 | Explanation |
| 5 | Benefits of the three OFDMA approaches | 9 | T7 | Explanation |
| 6 | CDMA; why difficult to jam | 4 | T5/T6 | Explanation |
| 7 | Mobile node moving away: two BS actions | 4 | T3 | Application |
| 8 | Figure 2: hidden node; A and C simultaneous | 4 | T9 | Scenario |
| 9 | Figure 2: CSMA/CA when A sends | 4 | T9 | Scenario |
| 10 | Why ACKs in 802.11 but not Ethernet | 4 | T9 | Explanation |
| 11(i) | Two ISPs on channel 11 | 4 | T9 | Scenario |
| 11(ii) | Channels 1 and 11 | 4 | T9 | Scenario |
| 12 | Receiving site: occasional loss of signal due to interference; steps to correct | 10 | T2/T9 | Scenario/essay |
| 13 | Why many WiFi networks have high error rate from co-channel interference | 4 | T9 | Explanation |
| 14 | 850 MHz, 125 mW, received 1 µW: distance? | 6 | T2 | **Calculation** |

## COM5231 Test 1, version B (T1B; 100 marks)

| Q | Question | Marks | Topic | Type |
|---|---|---|---|---|
| 1 | Single-bit vs burst errors; extent parity is appropriate | 10 | T8 | Explanation |
| 2(a) | CRC: 10011010, $x^3 + x^2 + 1$ (polynomial long division) | 10 | T8 | **Calculation** |
| 2(b) | Leftmost bit inverted: receiver's CRC result; how it knows | 6 | T8 | **Calculation** |
| 3 | Three basic propagation mechanisms | 12 | T2 | Explanation |
| 4 | Path loss vs multipath vs interference | 12 | T2 | Comparison |
| 5 | Three methods to increase capacity of a wireless system | 12 | T3 | Essay |
| 6 | FHSS and DSSS principles | 10 | T6 | Explanation |
| 7 | CDMA; why difficult to jam | 10 | T5/T6 | Explanation |
| 8 | Difference between OFDM and OFDMA | 10 | T7 | Comparison |
| 9 | SU-MIMO vs MU-MIMO | 8 | T3 | Comparison |

### Test 2 2025 (T2-2025; 100 marks)

| Q | Question | Marks | Topic | Type |
|---|---|---|---|---|
| 1 | Three categories of MANET routing protocols | 12 | T11 | Explanation |
| 2 | Dijkstra on Figure 1 from **x** | 8 | T11 | **Algorithm** |
| 3 | Count-to-infinity problem | 5 | T11 | Explanation |
| 4 | Hidden terminal problem; does it occur in Bluetooth? | 10 | T9/T10 | Explanation |
| 5 | Compare wired and wireless LANs; unique WLAN design concerns | 10 | T9 | Comparison/essay |
| 6 | Receiving site with interference: steps to correct | 10 | T2/T9 | Scenario |
| 7 | Bluetooth profiles: benefit to applications | 10 | T10 | Explanation |
| 8 | Scatternet collision probability | 10 | T10 | **Calculation** |
| 9 | One device as controller of two piconets? | 5 | T10 | Application |
| 10(i) | Two ISPs on channel 11 | 10 | T9 | Scenario |
| 10(ii) | Channels 1 and 11 | 10 | T9 | Scenario |

### COM5231 Test 2 2026 (T2-2026; 50 marks)

| Q | Question | Marks | Topic | Type |
|---|---|---|---|---|
| 1 | Routing vs packet forwarding vs switching: purpose and decision information | 6 | T11 | Comparison |
| 2(a)–(c) | Hamming for 11001010 (inequality, placement, codeword) | 8 | T8 | **Calculation** |
| 3(a)–(c) | CRC: 110101, G = 1011 | 6 | T8 | **Calculation** |
| 4(a)–(b) | Checksum 1010 1100 0110 1001 + receiver result | 4 | T8 | **Calculation** |
| 5 | Define single-bit and burst error; burst length | 3 | T8 | Definition |
| 6(a)–(c) | Even parity: single-bit? burst? can it correct? | 5 | T8 | Application |
| 7 | Two ways profiles benefit developers/users | 4 | T10 | Explanation |
| 8 | Bluetooth central vs 802.11 AP: medium access, connecting to other networks | 4 | T10 | Comparison |
| 9(a)–(b) | Dijkstra from A (new figure); final costs and routes | 10 | T11 | **Algorithm** |

## 4.2 Marks by topic (counted)

| Topic | Exam | T1A | T1B | T2-2025 | T2-2026 | **Total** | **% of 450** |
|---|---|---|---|---|---|---|---|
| T2 Wireless channel (incl. interference troubleshooting, path-loss calc) | 6 | 49 | 24 | 10 | — | **89** | **19.8%** |
| T9 WLAN 802.11 | 18 | 24 | — | 35 | — | **77** | **17.1%** |
| T11 Routing/MANET *(no notes)* | 30 | — | — | 25 | 16 | **71** | **15.8%** |
| T8 Error detection/correction | 10 | — | 26 | — | 26 | **62** | **13.8%** |

| Topic | Exam | T1A | T1B | T2-2025 | T2-2026 | Total | % of 450 |
|---|---|---|---|---|---|---|---|
| T10 Bluetooth/802.15 | 20 | — | — | 30 | 8 | **58** | **12.9%** |
| T3 Channel correction/capacity/MIMO | 2 | 4 | 20 | — | — | **26** | **5.8%** |
| T7 OFDM/OFDMA | 6 | 9 | 10 | — | — | **25** | **5.6%** |
| T6 Spread spectrum | 4 | 10 | 10 | — | — | **24** | **5.3%** |
| T5 Multiple access (CDMA) | 4 | 4 | 10 | — | — | **18** | **4.0%** |
| T1 Transmission fundamentals | — | — | — | — | — | **0** | 0% |
| T4 Signal encoding | — | — | — | — | — | **0** | 0% |
| **Total** | 100 | 100 | 100 | 100 | 50 | **450** | 100% |

**The exam paper itself (the best guide to your final exam)**: T11 routing **30**, T10 Bluetooth **20**, T9 WLAN **18**, T8 coding **10**, T2 **6**, T7 **6**, T6 **4**, T5 **4**, T3 **2**. The exam **samples every half of the module**: Q1 = Test 1 material (channel, SS, OFDMA, CDMA); Q2–Q3 = coding + routing; Q4 = Bluetooth; Q5 = WLAN.

## 4.3 Repeated and closely related questions (counted)

| # | Question (essentially identical wording) | Papers | Count |
|---|---|---|---|
| R1 | **Bluetooth profiles add complexity; how are they an advantage to applications?** | Exam 4(c), T2-2025 Q7, T2-2026 Q7, BT exercise 19 | **4** |
| R2 | **Path loss vs multipath vs interference** | Exam 1(a), T1A Q3, T1B Q4 | **3** |
| R3 | **Spread spectrum principles of FHSS and DSSS** | Exam 1(b), T1A Q4, T1B Q6 | **3** |
| R4 | **Describe CDMA; why difficult to jam** | Exam 1(d), T1A Q6, T1B Q7 | **3** |
| R5 | **Two ISPs' APs on channel 11; then channels 1 and 11** | Exam 5(e), T1A Q11, T2-2025 Q10 | **3** |
| R6 | **CRC calculation** (different numbers) | Exam 2(b), T1B Q2, T2-2026 Q3 (+ worksheet) | **3** |
| R7 | **One device as controller/master of two piconets** | Exam 4(d), T2-2025 Q9, BT exercise 9 | **3** |
| R8 | **Scatternet collision probability, 80 frequencies** | Exam 4(e), T2-2025 Q8, BT exercise 20 | **3** |
| R9 | **Bluetooth master vs 802.11 base station / AP** | Exam 4(a), T2-2026 Q8, BT exercise 8 | **3** |
| R10 | **Three basic propagation mechanisms** | T1A Q1, T1B Q3 (+ lecturer review Q) | **2** |
| R11 | **Benefits of OFDMA approaches (adjacent / regular / random)** | Exam 1(c), T1A Q5 (+ OFDM review 8.10) | **2** |
| R12 | **Hidden node with Figure 2; A and C transmit simultaneously** | Exam 5(b), T1A Q8 | **2** |
| R13 | **CSMA/CA with Figure 2, A sending** | Exam 5(c), T1A Q9 | **2** |
| R14 | **Why ACKs in 802.11 but not in Ethernet** | Exam 5(d), T1A Q10 | **2** |
| R15 | **Mobile node moving away: two actions** | Exam 5(a), T1A Q7 | **2** |

| # | Question (essentially identical wording) | Papers | Count |
|---|---|---|---|
| R16 | **Interference at a receiving site: steps to correct** | T1A Q12, T2-2025 Q6 | **2** |
| R17 | **Hamming code** (different data) | Exam 2(a), T2-2026 Q2 (+ worksheet) | **2** |
| R18 | **Internet checksum, 4-bit words** | Exam 2(c), T2-2026 Q4 (+ worksheet) | **2** |
| R19 | **Single-bit vs burst errors; parity** | T1B Q1, T2-2026 Q5–6 (+ worksheet) | **2** |
| R20 | **Three categories of MANET routing** | Exam 2(d), T2-2025 Q1 | **2** |
| R21 | **Dijkstra on the same Figure 1** (from x, t, w) | Exam 3(c), T2-2025 Q2 | **2** (+ T2-2026 Q9 new graph) |
| R22 | **Count-to-infinity** | Exam 3(d), T2-2025 Q3 | **2** |
| R23 | **Hidden terminal problem in Bluetooth** | Exam 4(b), T2-2025 Q4 | **2** |

**Key finding**: these **23 question types account for the large majority of the 450 marks.** Of the 24 sub-questions in the exam paper, **22** also appear (same or nearly the same wording) in a test, the worksheet or the Bluetooth exercise sheet. The only exam-only items are Q3(a) link-state vs distance-vector and Q3(b) Dijkstra vs Bellman-Ford. The lecturer **reuses questions heavily**, often word for word, and changes only the numbers in calculations.

**Questions copied word for word from sources**:

- **Test 1A Q14** = *Wireless Communication Problems*, **Problem 2.7** (same numbers).
- **Exam 2(a)** (Hamming 10101111) and **Exam 2(c)** (checksum 1001 1100 1010 0011) = the **error-detection worksheet** exercises.
- **Exam Q4(a), (c), (d), (e)** = **Bluetooth exercises 8, 19, 9, 20**.
- **T1B Q1** = the last worksheet question; **T1B Q2** = the worksheet's CRC example with a different message.
- **Exam Q3 / T2-2025 Q2** Figure 1 = Kurose & Ross Ch. 5 network; **T2-2026 Q9** = Tanenbaum's Dijkstra figure.
- **T1B Q9, T1A Q1** = the lecturer's own review list on *The Wireless Channel* slide 72.

## 4.4 Common question formats

| Format | Typical wording | Typical marks | Examples |
|---|---|---|---|
| **Compare / differences** | "What are the differences...", "Compare and contrast...", "Distinguish..." | 4–12 | R2, R9, OFDM vs OFDMA, SU vs MU-MIMO, LS vs DV, wired vs wireless LAN |
| **Explain principle** | "Explain the principles of...", "Describe..." | 4–12 | FHSS/DSSS, CDMA, propagation mechanisms |
| **Scenario / figure** | "Use Figure 2 to...", "Suppose two ISPs..." | 2–10 | Hidden node, CSMA/CA, ISP café, interference site |
| **Yes/No + justify** | "Does ... occur? Explain", "Is there any reason why..." | 4–10 | Hidden terminal in BT; two piconets; count-to-infinity |

| Format | Typical wording | Typical marks | Examples |
|---|---|---|---|
| **Calculation** | "Show...", "What is the value..." | 2–16 | Hamming, CRC, checksum, path loss/distance, collision probability |
| **Algorithm trace** | "Apply Dijkstra... step-by-step" | 4–10 | Figure 1, T2-2026 figure |
| **Essay** | "Discuss the steps...", "State and explain the three..." | 10–12 | Interference troubleshooting, capacity methods, MANET categories |

## 4.5 Calculation and algorithm questions (counted)

| Calculation | Appears | Marks available |
|---|---|---|
| **CRC** | Exam, T1B, T2-2026 | 4 + 16 + 6 = **26** |
| **Dijkstra** | Exam (×2), T2-2025, T2-2026 | 8 + 8 + 10 = **26** |
| **Hamming** | Exam, T2-2026 | 4 + 8 = **12** |
| **Scatternet probability** | Exam, T2-2025 | 4 + 10 = **14** |
| **Internet checksum** | Exam, T2-2026 | 2 + 4 = **6** |
| **Free-space distance/power** | T1A | **6** |

**≈ 90 of 450 marks (20%) are pure calculation/algorithm.** These are the **most reliable marks**: the method never changes, only the numbers.

## 4.6 Depth of knowledge required

- Most 4-mark questions want **2–4 precise, explained points** (≈ one point per 1–2 marks).
- 10–12 mark questions want **structure**: a definition, then 3–5 explained points or a comparison table, then a conclusion.
- Scenario questions reward **applying** the theory (CSMA/CA steps, NAV, CTS) to the named nodes (A, B, C), not reciting it.
- Calculations reward **showing every step** (formula, substitution, each XOR or table row, the final answer with units). Several questions explicitly say *"clearly presenting your step-by-step process"*.
- **No questions** in any paper on **Topic 1 (Nyquist/Shannon)** or **Topic 4 (ASK/FSK/PSK/QAM)**. They are still in the module and the lecturer's exercises, so don't skip them, but rank them lower.

## 4.7 PREDICTION (not fact): priorities for the final exam

These are **predictions** based on the patterns above. They are not knowledge of the exam.

**Very high probability (appeared in the exam AND in tests; prepare word-perfect answers):**

1. Bluetooth profiles advantage (R1); master vs AP (R9); two piconets (R7); scatternet probability (R8); hidden terminal in BT (R23).
2. Two-ISP café scenario (R5); hidden node and CSMA/CA with Figure 2 (R12, R13); ACKs (R14).
3. Dijkstra (any graph; Figure 1 from any node); LS vs DV; Dijkstra vs Bellman-Ford; count-to-infinity; MANET categories.
4. Hamming, CRC and checksum with **new numbers**.
5. Path loss vs multipath vs interference; FHSS/DSSS; CDMA jamming; OFDMA approaches.

**Medium probability (appeared in tests only):** three propagation mechanisms; propagation problems; capacity methods; OFDM vs OFDMA; SU vs MU-MIMO; wired vs wireless LAN; interference troubleshooting; co-channel interference; routing/forwarding/switching; parity and burst errors; path-loss distance calculation.

**Lower probability but examinable (in notes/review questions, never yet examined):** Nyquist/Shannon calculations; modulation (ASK/FSK/PSK/QAM); FDMA/TDMA details and formulas; cellular generations; SC-FDMA; 802.15.3/.4 and ZigBee; Bluetooth packet format and links; PCF/IFS details; channel correction mechanisms (FEC, equalization, AMC).

**Recommended study time split** (prediction-based): Routing 20% · WLAN 20% · Bluetooth 20% · Error coding 15% · Wireless channel/SS/CDMA/OFDMA 20% · Everything else 5%.

# Part 5 — Model Answers: Exam Paper and Test 1 (Version A)

### How these answers are written

- Original numbering and marks are preserved. Answer length is matched to the marks (about one well-explained point per 1–2 marks).
- **Sources** are given in brackets after each answer. Codes as in Parts 1–3: [WC-S]/[WC-B] wireless channel slides/book; [WCP] wireless problems; [MA] multiple access; [AT] access technologies article; [MX] multiplexing notes; [SS] spread spectrum; [OF] OFDM; [ED] error-detection worksheet; [WL] WLAN slides; [FZ] Forouzan WLAN; [BT-B]/[BT-S] Bluetooth book/slides; [BTX] Bluetooth exercises.
- **SUPPLEMENTARY** marks material not in your notes (mainly all of the routing questions).
- No memorandum was provided. These are **model answers**, not the lecturer's marking guide.

# EXAM PAPER (100 marks)

## QUESTION 1 (20 marks)

### 1(a) What are the differences between the following types of wireless channel impairments: path loss, multipath propagation, and interference from other sources? (6)

**Path loss** is the **reduction in signal power** as the signal travels from transmitter to receiver. In free space it happens because energy **spreads over an ever-larger sphere**, so $Pt/Pr = (4\pi d/\lambda)^2$: loss grows with the **square of distance and of frequency**. Real environments add obstruction and absorption, described by the **path-loss exponent** n (2 in free space, 2.7–3.5 urban, 4–6 obstructed indoors). Path loss is **predictable and deterministic**. It sets the transmit power, receiver sensitivity and SNR needed, and it is countered by **higher power, better antennas or more access points** [WC-B 5.2; WC-S 30–34; WCP §9].

**Multipath propagation** happens when **reflection, diffraction and scattering** from buildings, walls, the ground and small objects make **several copies of the same signal arrive with different delays and phases**. The copies add **constructively or destructively**, causing **fading** (drops of 20–30 dB over half a wavelength) and **intersymbol interference**. It **varies randomly with time and position** as the user moves, and gives "high signal strength but poor quality". It is countered by **equalization, diversity/MIMO, OFDM and spread spectrum** [WC-B 5.3; WCP §7].

**Interference from other sources** is caused by **other transmitters**, not the wanted signal's path. Examples are **co-channel** interference (other WLANs on the same channel, microwave ovens at 2.4 GHz), **adjacent-channel** interference, harmonics and atmospheric noise. It **depends on other users' activity**, lowers the SINR, and causes **bit errors and blocked transmissions**. It is countered by **channel/frequency planning, spread spectrum, directional antennas and removing the source** [WCP §10; WC-S 10].

| | Path loss | Multipath | Interference |
|---|---|---|---|
| Cause | Distance/frequency spreading | Reflection, diffraction, scattering of **own** signal | **Other** transmitters |
| Behaviour | Deterministic, smooth | Random, time-varying (fading, ISI) | Random/bursty, depends on others |
| Fix | Power, antennas, more APs | Equalization, diversity, OFDM | Channel planning, spread spectrum |

### 1(b) Explain the spread spectrum principles of (i) FHSS and (ii) DSSS. (4)

Spread spectrum expands the bandwidth B needed by a signal to **Bss ≫ B** using a **pseudorandom spreading code that is independent of the signal**. This gives privacy and resistance to jamming and interference [SS].

**(i) FHSS**: a **pseudorandom code generator** produces a k-bit pattern every **hopping period**. A **frequency table** maps it to one of **$M = 2^k$ carrier frequencies**, and a **frequency synthesizer** generates that carrier, which the data modulate. The signal **hops from carrier to carrier**, using one narrow frequency at a time but all M over time. A jammer or eavesdropper who doesn't know the sequence can only hit or catch **one hop** [SS].

**(ii) DSSS**: **each data bit is replaced by n chips** from a spreading code (e.g. the **11-chip Barker sequence 10110111000**), so the **chip rate = n × bit rate** and the bandwidth grows n times. The receiver correlates

with the same code to recover the data. Without the code the signal looks like noise, and narrowband interference is spread out and suppressed [SS].

### 1(c) What are the benefits of each of the following OFDMA approaches: (i) adjacent subcarriers, (ii) regularly spaced subcarriers, (iii) randomly spaced subcarriers? (6)

In OFDMA, subcarriers are grouped into **subchannels** that are allocated to users. The three ways of forming subchannels each have benefits [OF §8.2]:

**(i) Adjacent subcarriers** (a contiguous block): all subcarriers in the block have **about the same SINR**. This creates an **opportunity**: the system can **choose, from many possible blocks, the best block for each user**, optimising **channel efficiency, user requirements, priority and fairness**. It exploits frequency-selective fading for multi-user gain. (The cost is that it needs accurate, up-to-date **channel knowledge across the whole bandwidth**, via pilots and prediction; used by LTE and WiMAX.)

**(ii) Regularly spaced subcarriers** ("distributed"/"comb"): spreading a user's subcarriers across the band gives **frequency diversity**, so **enough good subcarriers are obtained whichever are chosen**, even in frequency-selective fading. The **channel-estimation burden is much lower** (used by LTE).

**(iii) Randomly spaced subcarriers**: the **same frequency-diversity benefit** as regular spacing, **plus reduced adjacent-cell interference**, because neighbouring cells' random patterns rarely coincide (used by WiMAX).

### 1(d) Describe CDMA and explain why it is difficult to jam. (4)

**CDMA** lets **all users transmit over the entire bandwidth at the same time**. Each user is separated by a **unique pseudorandom (orthogonal) code** that **spreads** its baseband data at a much higher **chip rate**. The spread signal is transmitted **below the noise level**. The receiver uses a **correlator with the same code to despread** the wanted signal, which then passes a narrowband filter. Other users' signals are **not despread** and remain low-level noise. **Precise power control** keeps every transmitter at the minimum power [MA p7–8].

**Difficult to jam because**:

1. The energy is **spread over a very wide band at very low power density**, so a jammer must cover the whole band to have an effect.
2. **Despreading spreads the jammer**: the correlator collapses the wanted signal back to narrowband but **spreads the jamming signal** (it doesn't carry the user's code), and the **narrowband filter removes most of it**.
3. The **pseudorandom code is unknown** to the jammer, so it cannot concentrate its energy where the signal is.

That is why spread-spectrum CDMA was used in **WWII to foil enemy jamming** [MX], and why its stated uses include **suppressing interference and making interception difficult** [MA p7].

## QUESTION 2 (20 marks)

### 2(a) An 8-bit byte 10101111 is encoded using an even-parity Hamming code. How many check bits are needed to detect and correct single-bit errors? What is the codeword? (4)

**Check bits**: we need r with **$2^r \geq m + r + 1$**, with m = 8.

- r = 3: $2^3 = 8 \geq 12$? No.
- r = 4: $2^4 = 16 \geq 13$? Yes → **4 check bits** (codeword length 12).

**Convention**: positions 1–12 numbered from the left; check bits at **1, 2, 4, 8**; data bits in order in 3, 5, 6, 7, 9, 10, 11, 12.

| Pos | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | P1 | P2 | 1 | P4 | 0 | 1 | 0 | P8 | 1 | 1 | 1 | 1 |

- P1 (3, 5, 7, 9, 11) = 1, 0, 0, 1, 1 → three 1s → **P1 = 1**
- P2 (3, 6, 7, 10, 11) = 1, 1, 0, 1, 1 → four 1s → **P2 = 0**
- P4 (5, 6, 7, 12) = 0, 1, 0, 1 → two 1s → **P4 = 0**
- P8 (9, 10, 11, 12) = 1, 1, 1, 1 → four 1s → **P8 = 0**

**Codeword = 1 0 1 0 0 1 0 0 1 1 1 1 = 101001001111.** [ED worksheet exercise; method SUPPLEMENTARY per the UNB Hamming page linked in ED]

### 2(b) A bit stream 101110 is transmitted using the standard CRC method with generator 1001. Show the actual bit string transmitted. (4)

G = 1001 has degree **3**, so append **000** → dividend **101110000**. Modulo-2 division (XOR):

```
101110000
1001
————
0010 → bring down → 1010
         1001
         ————
         0011 → bring down → 1100
                   1001
                   ————
                   0101 → bring down → 1010
                             1001
                             ————
                              011   ← remainder
```

Steps: 1011⊕1001 = 0010; 1010⊕1001 = 0011; 1100⊕1001 = 0101; 1010⊕1001 = 0011. **CRC = 011** → **transmitted string = 101110011**. (Check: 101110011 ÷ 1001 gives remainder 000.) [ED method]

### 2(c) Message 1001 1100 1010 0011 transmitted using the Internet checksum (4-bit words). Value of the checksum? (2)

One's-complement addition with end-around carry:

- 1001 + 1100 = 1 0101 → wrap → 0101 + 1 = **0110**
- 0110 + 1010 = 1 0000 → wrap → 0000 + 1 = **0001**
- 0001 + 0011 = **0100**

**Sum = 0100 → checksum = one's complement = 1011.** (The receiver adds all words + 1011 = 1111, meaning no error.) [ED exercise]

### 2(d) Why is routing in ad-hoc networks complicated? State and explain the three categories of MANET routing protocols, with an example of each. (10)

*SUPPLEMENTARY (no lecture notes provided), plus [WL 8–9] and [BT-S 87].*

**Why it is complicated (4 marks):**

1. **Dynamic topology**: nodes are **mobile**, so links break and form continuously and routes go stale quickly.

2. **No infrastructure or central controller**: there are no dedicated routers or APs. **Every node must also act as a router** and forward others' packets [WL 8–9], and decisions are distributed.
3. **Limited, shared wireless bandwidth**: routing control traffic competes with data, so flooding updates is expensive.
4. **Limited battery and processing power** in mobile nodes.
5. **Unreliable links**: fading, interference, **hidden/exposed terminals**, high error rates and asymmetric links; plus multi-hop delay and security concerns.

**Three categories (6 marks):**

1. **Proactive (table-driven)**: every node **maintains routes to all destinations at all times** using **periodic/triggered updates**. Routes are **available immediately** (low latency), but **control overhead is high and constant**, and wasted on routes never used and under high mobility. *Example:* ***DSDV*** *(Destination-Sequenced Distance Vector); also OLSR.*
2. **Reactive (on-demand / source-initiated)**: routes are found **only when a source needs one**, through **route discovery** (flood a route request; the destination replies) and **route maintenance**. **Low overhead** when traffic is light, but a **discovery delay** before the first packet. *Example:* ***AODV*** *(Ad hoc On-demand Distance Vector), which ZigBee uses: it "sends out route requests and either responds or forwards the messages to other nodes until someone responds" [BT-S 87]; also DSR.*
3. **Hybrid**: **proactive within a local zone** around each node and **reactive between zones**, which balances latency and overhead and scales to large networks. *Example:* ***ZRP*** *(Zone Routing Protocol).*

## QUESTION 3 (20 marks) — *All SUPPLEMENTARY (no routing notes provided)*

### 3(a) Compare and contrast link-state and distance-vector routing algorithms. (4)

**Similarities**: both compute **least-cost paths** over a graph of link costs, build a **forwarding table**, and adapt dynamically to cost changes.

**Differences**:

| | Link-state | Distance-vector |
|---|---|---|
| Knowledge | **Global**: complete topology at every router | **Local**: only neighbours' distance estimates |
| Algorithm | **Dijkstra** | **Bellman-Ford** |
| Advertisement | Own link costs **flooded to all** routers | Whole distance vector sent **only to neighbours** |
| Convergence | **Fast**, no loops | **Slow**; routing loops; **count-to-infinity** |
| Robustness | Errors stay local (each computes its own table) | A wrong cost **propagates network-wide** |
| Example | OSPF | RIP |

### 3(b) What is the essential difference between Dijkstra's algorithm and the Bellman-Ford algorithm? (4)

**Dijkstra** is a **global, greedy** algorithm: the node running it must know the **entire network graph**. It **permanently settles one node per iteration**, always the closest unsettled node, so it finishes in **n iterations** ($O(n^2)$). It requires **non-negative** costs. **Bellman-Ford** is **decentralised and iterative**: each node computes $\mathbf{d_x(y) = \min_v\{c(x,v) + d_v(y)\}}$ using **only its neighbours' current estimates**, and **repeatedly relaxes** all distances until nothing changes. It is naturally **distributed and asynchronous** (distance-vector routing) and **handles negative costs**, but **converges slowly** and can **count to infinity**.

**Essence**: *Dijkstra = complete global knowledge, nodes settled one at a time; Bellman-Ford = neighbour-only information, iterative relaxation until convergence.*

## 3(c) Apply Dijkstra's algorithm to Figure 1, showing the step-by-step process.

**Link costs from Figure 1**: x–z 8, x–y 6, x–v 3, x–w 6, y–z 12, y–v 8, y–t 7, v–t 4, v–u 3, v–w 4, t–u 2, w–u 3. Notation: D(n),p(n) = current least cost and predecessor; "—" = already in N'.

### (i) Shortest paths from t (4)

| Step | N' | D(u),p | D(v),p | D(w),p | D(x),p | D(y),p | D(z),p |
|---|---|---|---|---|---|---|---|
| 0 | t | **2,t** | 4,t | ∞ | ∞ | 7,t | ∞ |
| 1 | tu | — | **4,t** | 5,u | ∞ | 7,t | ∞ |
| 2 | tuv | — | — | **5,u** | 7,v | 7,t | ∞ |
| 3 | tuvw | — | — | — | **7,v** | 7,t | ∞ |
| 4 | tuvwx | — | — | — | — | **7,t** | 15,x |
| 5 | tuvwxy | — | — | — | — | — | **15,x** |
| 6 | tuvwxyz | — | — | — | — | — | — |

Updates: via u: w = 2 + 3 = 5. Via v: x = 4 + 3 = 7 (y via v = 12 > 7; w via v = 8 > 5). Via x: z = 7 + 8 = 15 (y via x = 13 > 7). Via y: z = 7 + 12 = 19 > 15.

| To | u | v | w | x | y | z |
|---|---|---|---|---|---|---|
| Cost | 2 | 4 | 5 | 7 | 7 | 15 |
| Path | t-u | t-v | t-u-w | t-v-x | t-y | t-v-x-z |

### (ii) Shortest paths from w (4)

| Step | N' | D(t),p | D(u),p | D(v),p | D(x),p | D(y),p | D(z),p |
|---|---|---|---|---|---|---|---|
| 0 | w | ∞ | **3,w** | 4,w | 6,w | ∞ | ∞ |
| 1 | wu | 5,u | — | **4,w** | 6,w | ∞ | ∞ |
| 2 | wuv | **5,u** | — | — | 6,w | 12,v | ∞ |
| 3 | wuvt | — | — | — | **6,w** | 12,v | ∞ |
| 4 | wuvtx | — | — | — | — | **12,v** | 14,x |
| 5 | wuvtxy | — | — | — | — | — | **14,x** |
| 6 | wuvtxyz | — | — | — | — | — | — |

Updates: via u: t = 3 + 2 = 5 (v via u = 6 > 4). Via v: y = 4 + 8 = 12 (x via v = 7 > 6; t via v = 8 > 5). Via t: y = 5 + 7 = 12 (tie). Via x: z = 6 + 8 = 14; y = 6 + 6 = 12 (tie). Via y: z = 24 > 14.

| To | u | v | t | x | y | z |
|---|---|---|---|---|---|---|
| Cost | 3 | 4 | 5 | 6 | 12 | 14 |
| Path | w-u | w-v | w-u-t | w-x | w-v-y* | w-x-z |

*y has **three equal-cost paths of 12**: w-v-y, w-x-y and w-u-t-y.

### 3(d) What is the "count to infinity" problem? Will it occur if we decrease a link cost? Why? What if we connect two nodes that have no link? (4)

**Count to infinity**: in distance-vector routing, **bad news travels slowly**. When a link cost increases or a link fails, a node may switch to a neighbour's **stale** advertised route, which in fact passes **back through itself**. This creates a **routing loop**: the two nodes repeatedly re-advertise, each raising its estimate by a small amount per exchange, "**counting up**" to the true cost, or **to infinity** if the destination is unreachable. *Example*: y–x = 4, y–z = 1, z–x = 50. If y–x rises to 60, y picks the path via z (1 + 5 = 6), z picks via y (7), and so on for 44 exchanges, with packets looping between y and z. (Fixes: poisoned reverse, split horizon, a maximum hop count.)

**Decrease a link cost: No.** It is **good news, and good news travels fast**. Each node recomputes a **lower** cost using the cheaper link and advertises it, so distances only decrease and converge in a few iterations. No node is left relying on an outdated, too-cheap path, so no loop forms.

**Connect two nodes that had no link: No.** Adding a link is the same as **decreasing its cost from ∞ to a finite value**: good news, which propagates quickly.

## QUESTION 4 (20 marks)

### 4(a) What are the differences between a master device in a Bluetooth network and a base station in an 802.11 network? (4)

1. **Medium access**: the **Bluetooth master controls all access**. It sets the **hopping sequence and timing** from its own address, and slaves may transmit **only when polled** by the master (FH-TDD-TDMA), so there is **no contention within a piconet**. In 802.11, stations **contend using CSMA/CA (DCF)** with random backoff, and the AP only polls in the **optional PCF** mode [BT-B; WL 32–40].
2. **Connection to other networks**: the **802.11 AP provides a link to other networks** (it bridges the BSS to the wired distribution system/Internet) and relays all MAC frames. The **Bluetooth master sets up the piconet and relays traffic inside it**, an ad hoc PAN that doesn't itself bridge to infrastructure [BT-S 64; FZ].
3. **Size and range**: a master serves at most **7 active slaves** within **~10 m**; an AP serves many stations over tens to hundreds of metres [BT-B].
4. **Role**: any Bluetooth device can be master (a dynamic role); an AP is a dedicated infrastructure device.

### 4(b) Does the hidden terminal problem occur in Bluetooth? Explain. (4)

**No, not within a piconet.** The hidden terminal problem arises in **contention-based** access (CSMA), where two stations that can't hear each other both sense an idle channel and transmit to a common receiver. In Bluetooth, **the master polls**: *"a slave is permitted to return an ACL packet in the slave-to-master slot if and only if it has been addressed in the preceding master-to-slave slot"* [BT-B], and SCO slots are reserved. **Slaves never decide on their own to transmit**, and all are in range of the master that schedules them, so two slaves cannot collide at the master. (Packets from **different piconets** can collide when they hop to the same frequency in the same slot, but that is an FH-CDMA collision handled by FEC/ARQ, not a hidden terminal problem.)

### 4(c) Profiles add significant complexity. How can they be an advantage from the perspective of the applications? (4)

1. **Interoperability**: a profile is *"a standard of interoperability so that products from different vendors that claim to support a given usage model will work together"* [BT-B]. An application (e.g. a headset or file transfer) works with any compliant device.
2. **Ready-made, reusable design**: each profile defines exactly **which protocols and features are mandatory, optional or not applicable** for a usage model, so application developers don't have to design their own use of the 2500-page core. E.g. the **File Transfer Profile** reuses **OBEX, RFCOMM and L2CAP** [BT-B].
3. **Only the needed subset is implemented**: smaller, cheaper, lower-power devices.
4. **Consistent common functions**: all devices support the **Generic Access Profile** (discovery, connection, security), so applications behave predictably and plug-and-play [BT-B].

### 4(d) A Bluetooth device can be in two piconets at the same time. Is there any reason why one device cannot be the controller in both of them at the same time? (4)

**Yes. A device cannot be master of two piconets.** A piconet is **defined by its master**: the **frequency-hopping sequence (channel) and the phase (timing)** used by all its members are computed **from the master's own Bluetooth address and clock** [BT-B]. If one device were master of both, **both would follow the identical hopping sequence and timing, i.e. they would be the same piconet**, not two separate ones (and together they would be limited to 7 active slaves). A device **can** belong to two piconets as **slave in both** or **master in one and slave in the other**, time-sharing between the two hop sequences. That is how scatternets are formed [BT-B Fig 12.2].

### 4(e) Scatternet of 2 piconets, no CSMA. What is the probability of a collision between packets of the two piconets? Successful transmission needs a successful poll (master → slave) and a successful reply (slave → master). 80 frequencies; all piconets always have packets. (4)

**Assumptions** (from [BT-B]): each piconet hops **independently and pseudorandomly** over the 80 frequencies; both transmit in every slot; a collision occurs when both use the **same frequency in the same slot**.

1. P(collision in one slot) = **1/80**.
2. P(a slot is collision-free) = **79/80**.
3. A transaction = **poll + reply** = 2 slots on 2 independent hops: P(success) = $(79/80)^2$ = 6241/6400 = **0.9752**.
4. **P(collision) = $1 - (79/80)^2$ = 159/6400 ≈ 0.0248 ≈ 2.5%.**

(Per individual packet the probability is 1/80 = 1.25%. Because a successful exchange needs **two** collision-free packets, the transaction collision probability is about **2.5%**.)

## QUESTION 5 (20 marks)

### 5(a) As a mobile node moves farther from a base station, what two actions could the BS take so that the loss probability of a transmitted frame does not increase? (2)

1. **Increase its transmission power**, to compensate for the larger path loss ($\propto d^2$) and keep the SNR up [WC-B 5.2].

2. **Reduce the transmission rate**: switch to a **more robust modulation and coding scheme** (adaptive modulation and coding / 802.11 rate adaptation), which tolerates the lower SNR [WC-S 51; WL 23].

### 5(b) Use Figure 2 to explain the "hidden node" problem. What happens when A and C start sending simultaneously? (4)

In Figure 2, **B is within range of both A and C, but A and C are outside each other's range**. When A transmits to B, **C cannot hear A**, so C's carrier sense finds the channel idle. A and C are **hidden from each other with respect to B** [FZ]. **If A and C send at the same time** (e.g. both to B): both sense an idle medium and transmit. Their signals **overlap at B, so B receives a collision** and can decode neither frame. Since wireless stations **cannot detect collisions**, A and C keep sending their **whole frames**, wasting the channel. Neither receives an **ACK**, so both time out, **back off a random time and retransmit**. Hidden nodes therefore **reduce network capacity** [FZ; WL 29].

### 5(c) Use Figure 2 to explain how CSMA/CA is realised in IEEE 802.11. Assume A wants to send a frame to the destination (B). (4)

1. **Carrier sense**: A checks its **NAV** and senses the medium. When idle, it waits **DIFS**, then counts down a **random backoff** (the counter freezes if the medium becomes busy) [WL 32–35].
2. **RTS**: A sends an RTS containing the **duration** of the whole exchange. B hears it (C doesn't).
3. **CTS**: after **SIFS**, B replies with a CTS that also carries the duration. **C is in B's range, so it hears the CTS, sets its NAV and defers**, even though it can't hear A. This is how the hidden node is handled [FZ; WL 39, 43].
4. **DATA and ACK**: after SIFS, A sends the data. After SIFS, B returns an **ACK**. If A gets no CTS or no ACK, it assumes a collision, backs off and retries [WL 35, 38].

### 5(d) Why are acknowledgements used in 802.11 but not in wired Ethernet? (2)

In wired Ethernet (CSMA/CD) the sender **detects collisions**, and *"if a collision is not detected, it is a kind of acknowledgment that the frame was received"*; wired error rates are also very low [FZ]. In 802.11, a station **cannot detect collisions** (it can't send and receive at once; hidden stations; fading), and frames are frequently lost to **noise, interference and fading**. So *"the station does not have any means to check for the successful arrival of its data"*, and an **explicit ACK** (with retransmission on timeout) is required [FZ; WL 27].

### 5(e) Two ISPs provide WiFi in a café, each with its own AP and IP address block.

#### (i) By accident both APs use channel 11. Will 802.11 completely break down? What happens when two stations, each associated with a different ISP, transmit at the same time? (4)

**No, it does not break down.** The two BSSs simply **share one channel as a single collision domain**. Every station still runs **CSMA/CA**: it senses the channel and its NAV, so stations from **both** ISPs **defer to each other** and take turns, as if they were one bigger BSS. (The different IP blocks don't matter at the MAC layer; each AP only accepts frames addressed to its own BSS.) If two stations from different ISPs **transmit at the same moment** (e.g. their backoff counters expire together, or they are hidden from each other), their frames **collide**. Neither AP receives a correct frame, **no ACKs are returned**, both stations **back off a random time** (with a larger contention window) and **retransmit**. The result is **more collisions and lower throughput for both ISPs** (each gets roughly half the channel capacity), but the network keeps working.

### (ii) Now one AP uses channel 1 and the other channel 11. How do the answers change? (4)

Channels **1 and 11 do not overlap** (in the 2.4 GHz band, signals are ~22 MHz wide [WC-S 4] and channels are 5 MHz apart, so 1 and 11 are 50 MHz apart; SUPPLEMENTARY). The two BSSs now form **two independent collision domains**:

- Stations of different ISPs **don't hear each other's transmissions** in carrier sense, so they **don't defer to each other**.
- **Two stations can transmit at the same time without colliding**: each frame reaches its own AP successfully and is ACKed.
- **Each ISP gets the full capacity of its own channel**: aggregate throughput roughly **doubles**, with fewer collisions and backoffs.

# TEST 1, VERSION A (T1A)

## Q1. Explain the three basic propagation mechanisms in a mobile communication system. (12)

In a mobile system, the receiver often has **no clear line of sight** to the base station. The signal still arrives because of three mechanisms, which also cause **multipath** [WC-B 5.3, Fig 5.5]:

**1. Reflection (4 marks)**: occurs when the EM wave meets a surface that is **large compared with its wavelength**, e.g. the ground, walls, buildings and furniture [WCP]. Part of the energy bounces off. In a mobile system the **ground-reflected wave** and the line-of-sight wave can **cancel each other**, causing high signal loss. Because the mobile antenna is lower than most buildings, many reflected waves arrive, and they combine **constructively or destructively** at the receiver. Reflection is the most significant effect even when LOS exists.

**2. Diffraction (4 marks)**: occurs at the **edge of an impenetrable body that is large compared with the wavelength** (a sharp edge such as a building corner or rooftop). The edge acts as a **new source**: **secondary waves** spread out in different directions, including **into the shadow behind the obstacle**. Signals can therefore be received **even when there is no unobstructed LOS** from the transmitter [WC-B 5.3; WCP].

**3. Scattering (4 marks)**: occurs when the obstacle is **on the order of the wavelength or smaller** and there are **many such objects**: lamp posts, traffic signs, foliage, rough surfaces. The incoming signal is **scattered into many weaker outgoing signals** in all directions. At cellular frequencies such objects are everywhere, so **scattering is hard to predict** [WC-B 5.3].

**Overall effect**: with a clear LOS, diffraction and scattering are minor and reflection matters most. With **no LOS** (e.g. at street level in a city), **diffraction and scattering are the primary means of reception**. Together the three produce **multipath**: several delayed copies that cause **fading and ISI** [WC-B 5.3].

## Q2. What are the main problems of signal propagation? Why do radio waves not always follow a straight line? Why is reflection both useful and harmful? (12)

**Main problems (6 marks)** [WC-S 18, 39; WCP §2.3; WC-B 5.2–5.3]:

1. **Attenuation / path loss / free-space loss**: power falls with distance ($\propto d^2$) and is greater at higher frequencies.
2. **Noise**: thermal (cannot be eliminated), intermodulation, crosstalk, impulse noise (the main source of digital errors).
3. **Atmospheric absorption**: water vapour and oxygen attenuate the signal.
4. **Multipath propagation**: reflection, diffraction and scattering create delayed copies.
5. **Fading**: fast/slow and flat/frequency-selective variation in received power; shadowing.
6. **Intersymbol interference**: delayed copies overlap the next symbols.
7. **Refraction**: bending of the wave.
8. **Interference** from other transmitters (co-channel, adjacent-channel), plus the shared medium and a higher BER.

**Why waves don't always travel in straight lines (3 marks)**:

- **Refraction**: wave speed depends on the **density of the medium**, so at the boundary between media (or through the varying atmosphere) the wave **bends**. Sky waves (3–30 MHz) are **refracted by the ionosphere** back to Earth [WC-S 15, 19].
- **Ground waves** (below ~3 MHz) **follow the curvature of the Earth** [WC-S 14].
- **Reflection, diffraction and scattering** redirect the wave around and off obstacles [WC-B 5.3].

**Reflection: useful and harmful (3 marks)**:

- **Useful**: reflected waves let the signal **reach receivers with no line of sight** (indoors, behind buildings, in streets). The extra copies can be **combined by diversity/MIMO** receivers to strengthen the signal [WC-B 5.3–5.4].
- **Harmful**: reflected copies arrive with **different delays and phases**, so they can **cancel** the direct signal (the ground reflection vs LOS), causing **fading** (20–30 dB dips), **distortion** and **intersymbol interference**, which raises the bit error rate [WC-B 5.3].

## Q3. Differences between path loss, multipath propagation and interference from other sources. (9)

*Same content as Exam Q1(a), expanded to 3 marks per impairment.* Use the three paragraphs and the table in Exam 1(a), and add one extra point to each:

- **Path loss**: example: indoors it increases ~**20 dB per 100 ft**; the main countermeasure is **additional access points** [WCP §9]. Expressed as $Lp = (4\pi d/\lambda)^2 = Pt/Pr$.
- **Multipath**: it is especially severe **indoors**; vendors use **equalization and antenna diversity** and may need **lower data rates** [WCP §7]. It produces **Doppler** effects when the receiver moves.
- **Interference**: two directions: **inward** (co-channel, from other systems on similar frequencies, e.g. microwave ovens at 2.4 GHz) and **outward** (adjacent-channel: your WLAN disrupting others, e.g. aircraft navigation). It is uncommon in spread-spectrum WLAN bands because power is low (< 1 W), so devices must be close and in the same band [WCP §10].

## Q4. Explain the spread spectrum principles of (i) FHSS and (ii) DSSS. (10)

**General principle (2 marks)**: spread spectrum is designed for wireless, where stations share the air and need protection from **eavesdropping and jamming**. It adds **redundancy** by expanding bandwidth **B to Bss ≫ B**. Two principles: (1) the allocated bandwidth is **much larger than needed**; (2) the spreading is done **after** the signal is created, by a process **independent of the signal**, using a **spreading code** that looks random but is a pattern [SS].

**(i) FHSS (4 marks)**:

- Uses **M carrier frequencies**. At each moment the source modulates **one carrier**, and the next moment another.
- A **pseudorandom code generator** produces a **k-bit pattern each hopping period Th** ($M = 2^k$); a **frequency table** maps it to a frequency; a **frequency synthesizer** generates it; the **modulator** applies the data.
- *Example*: M = 8, k = 3; the pattern 101, 111, 001, ... selects 700 kHz, 900 kHz, 300 kHz, ... and repeats after 8 hops.
- **Privacy**: an intruder can only capture a **small piece** without the sequence. **Anti-jamming**: a jammer can hit **one hop, not the whole transmission**. **Sharing**: M stations can share Bss, like FDM with the allocation changing every hop [SS].

**(ii) DSSS (4 marks)**:

- **Each data bit is replaced by n bits called chips**, using a spreading code. The **chip rate is n times the bit rate**, so the bandwidth grows n times.
- *Example*: the **Barker sequence** (n = 11, **10110111000**) used in wireless LANs. With polar NRZ, the spread signal = data × chips (a 1 sends the code, a 0 sends its inverse). The required bandwidth is **11×** the original.

- **Privacy** if the intruder doesn't know the code; **immunity to interference** if each station uses a different code. Bandwidth can be **shared** only if special (orthogonal) codes are used, i.e. **CDMA** [SS].

### Q5. Benefits of the OFDMA approaches: adjacent, regularly spaced, randomly spaced subcarriers. (9)

*As in Exam Q1(c), with 3 marks each.* Add:

- **Adjacent**: all SINRs are about equal, so whole blocks can be compared. The scheduler can pick **each user's best block** (optimising efficiency, requirements, priority, fairness). It requires **pilot symbols, channel knowledge across the full band, and prediction**, because information may be out of date. Used by **LTE and WiMAX** [OF].
- **Regularly spaced**: "distributed"/"comb"; **frequency diversity**, so there are enough good subcarriers **regardless of which are chosen**; **low channel-estimation burden**. Used by **LTE**.
- **Randomly spaced**: the **diversity benefits** of regular spacing **plus reduced adjacent-cell interference**. Used by **WiMAX**.

### Q6. Describe CDMA and explain why it is difficult to jam. (4)

*Identical to Exam Q1(d).*

### Q7. As a mobile node gets farther from a base station, what two actions could the BS take so that the frame loss probability does not increase? (4)

1. **Increase transmit power** (2 marks): received power falls as **$1/d^2$** ($Pr = Pt/(4\pi d/\lambda)^2$). Raising Pt keeps Pr and the SNR above the receiver's required threshold [WCP Eq 2.1–2.2].
2. **Lower the data rate / use more robust modulation and coding** (2 marks): with **adaptive modulation and coding** (802.11 **rate adaptation**), the BS switches to a lower-order modulation and stronger FEC, which need a **lower SNR for the same bit error rate** [WC-S 51; WL 23]. (Also acceptable: **beamforming** a directional beam toward the node [WC-B 5.4].)

### Q8. Use Figure 2 to explain the hidden node problem. What happens when A and C send simultaneously? (4)

*Identical to Exam Q5(b).*

### Q9. Use Figure 2 to explain how CSMA/CA is realised in 802.11 (A sends). (4)

*Identical to Exam Q5(c).*

### Q10. Why are acknowledgements used in 802.11 but not in wired Ethernet? (4)

*Exam Q5(d) answer, expanded (4 marks):*

1. **No collision detection in wireless**: a station can't transmit and receive at the same time (the received signal is far weaker; it would need costly full-duplex hardware) [WL 16, 29].
2. **Hidden stations**: even if it could listen, a collision at the receiver may be **inaudible to the sender** [FZ].
3. **High error rate**: noise, interference, fading and multipath corrupt many frames, so the MAC and PHY are **unreliable** [WL 27].
4. **In Ethernet**, a **detected** collision triggers a resend and **no collision acts as an implicit acknowledgement**, on a low-error cable [FZ]. So 802.11 uses **explicit ACKs after SIFS**, and

retransmits when no ACK arrives [WL 27, 34].

### Q11. Two ISPs in a café. (i) Both on channel 11. (4) (ii) Channels 1 and 11. (4)

*Identical to Exam Q5(e)(i) and (ii).*

### Q12. A wireless receiving site is experiencing occasional loss of signal due to interference. Discuss the steps you would take to correct this problem. (10)

A systematic approach (about 1 mark per explained step) [WCP §9–10; WC-S 9–10; WL 12, 21–23; WC-B 5.4]:

1. **Characterise the problem**: record **when** it happens (time of day, which events), **where** (which areas), and **which channels/frequencies**. "Occasional" loss suggests an **intermittent source**, e.g. a **microwave oven at 2.4 GHz** or a neighbouring network's bursts.
2. **Survey the spectrum**: use a **spectrum analyser / WLAN site-survey tool** to identify interfering signals and **co-channel or adjacent-channel** networks. Decide whether it is **inward** interference (others → us) or something we cause (outward).
3. **Identify and locate the source**: microwave ovens, cordless phones, Bluetooth devices, neighbouring APs, other ISM equipment, faulty electrical equipment (impulse noise).
4. **Remove, switch off, shield or relocate** the interfering device, or **increase the separation** between it and the receiver. Interference needs the devices to be **close and in the same band** [WCP §10].
5. **Change channel**: move the AP/link to a **clear, non-overlapping channel** (e.g. 1, 6 or 11 in 2.4 GHz), or move to the **less crowded 5 GHz band** [WL 23].
6. **Improve the antenna system**: reposition/raise the antenna to restore LOS; use **directional antennas** that point at the wanted transmitter and **reject** interference from other directions; use **antenna/space diversity** [WC-B 5.4].
7. **Improve the link margin**: increase the **wanted signal's power or antenna gain** within regulatory limits, so the **SINR** stays above threshold during interference bursts.
8. **Use robust transmission techniques**: **spread spectrum / (adaptive) frequency hopping**, **FEC**, **adaptive modulation and coding / lower data rates**, and ARQ to ride out bursts [WC-B 5.4].
9. **Network design**: add APs or reduce cell size; coordinate channel and power plans with neighbouring networks.
10. **Verify and monitor**: re-test after each change, and keep monitoring to confirm the problem has gone.

### Q13. Explain why many WiFi networks can have a high error rate caused by co-channel interference. (4)

1. **Few clean channels**: 2.4 GHz has only **three non-overlapping channels** (1, 6, 11; SUPPLEMENTARY), and the **AP administrator chooses the channel**, so *"interference [is] possible: channel can be same as that chosen by neighboring AP"* [WL 12]. Dense deployments put many BSSs on the same channel.
2. **Crowded, unprotected ISM band**: 2.4 GHz is shared with cordless phones, Bluetooth and **microwave ovens**. ISM users **must tolerate interference** and have **no regulatory protection** [WL 21, 23; WC-S 10].
3. **Same-frequency signals add directly to the wanted signal**: co-channel (inward) interference **blocks transmissions or causes bit errors** [WCP §10], lowering the SINR.
4. **Hidden stations in neighbouring BSSs** on the same channel cause collisions that carrier sensing can't prevent. The result is lost frames, missing ACKs, retransmissions and lower throughput.

## Q14. Transmitter at 850 MHz, power 125 mW; received power 1 µW. Distance between transmitter and receiver? (6)

**Given**: Pt = 125 mW = 125 × $10^{-3}$ W; Pr = 1 µW = 1 × $10^{-6}$ W; f = 850 × $10^{6}$ Hz; c = 3 × $10^{8}$ m/s.

**Formula**: Lp = Pt/Pr = $(4\pi d/\lambda)^2$ [WCP Eq 2.1, 2.3].

1. **Path loss**: Lp = Pt/Pr = (125 × $10^{-3}$)/(1 × $10^{-6}$) = **125 × $10^{3}$ = 125 000**
2. **Wavelength**: λ = c/f = (3 × $10^{8}$)/(850 × $10^{6}$) = **0.3529 m**
3. **Substitute**: 125 000 = $(4\pi d/0.3529)^2$
4. **Square root**: √125 000 = **353.55** = 4πd/0.3529
5. **Solve**: d = (353.55 × 0.3529)/(4π) = 124.77/12.566 = **9.93 m**

**Answer: d ≈ 9.93 m** (this is Problem 2.7 in your notes. Note that the notes' printed step "125 × $10^{-3}$" should read 125 × $10^{3}$).

# Part 6 — Model Answers: Test 1B, Test 2 2025, Test 2 2026, Bluetooth Exercises and Error-Detection Worksheet

Conventions as in Part 5. Where a question is identical to one already answered in Part 5, the answer says so and gives any extra depth the higher mark allocation needs.

# COM5231 TEST 1, VERSION B (T1B; 100 marks)

## Q1. Explain the difference between single-bit and burst errors and the extent to which parity checking is appropriate as an error-detecting technique in both cases. (10)

*Definitions SUPPLEMENTARY (standard; the worksheet [ED] lists this question but gives no theory). Causes from [WC-B].*

**Single-bit error (2 marks)**: **only one bit** of a data unit (byte, frame) is changed, from 0 to 1 or 1 to 0. E.g. 0000 0010 sent, 0000 1010 received. These are relatively **rare** in practice, because noise usually lasts longer than one bit time.

**Burst error (3 marks)**: **two or more bits** in the data unit are changed. They need not be consecutive. The **burst length** runs from the **first** corrupted bit to the **last**, inclusive. Bursts are the **common case** in wireless: **impulse noise** (*"the primary source of error in digital data... a 0.01 s spike... would wash out about 10 000 bits at 1 Mbps"* [WC-B 5.2]) and **fading** dips of 20–30 dB [WC-B 5.3] corrupt many consecutive bits.

**Parity checking** (even parity: one bit appended so the number of 1s is even; the receiver recounts):

- **Single-bit errors (2 marks)**: parity **always detects** them. One flipped bit changes the count of 1s from even to odd. It is **cheap** (one extra bit) and simple, so it is **appropriate** where errors are isolated. It **cannot correct** them, though: it can't tell **which** bit flipped, so retransmission is needed.
- **Burst errors (3 marks)**: parity detects a burst **only if an odd number of bits** is flipped. Any **even** number of errors (2, 4, ...) leaves the parity unchanged and goes **undetected**. E.g. 10110010 → 1**10**10010: two errors, still four 1s, so it passes. Roughly **half of all bursts escape**, so single parity is **not appropriate** for bursty wireless channels. **Two-dimensional parity** improves this (it detects many bursts and corrects single errors), but **CRC** is preferred, since it detects **all bursts of length ≤ r** (the number of check bits).

## Q2. Transmit the message 10011010 protected by the CRC polynomial $x^3 + x^2 + 1$.

### (a) Use polynomial long division to determine the message that should be transmitted. (10)

1. **Generator**: $x^3 + x^2 + 0{\cdot}x + 1$ → **1101**; degree r = **3**.
2. **Append 3 zeros**: $M(x){\cdot}x^3$ → **10011010 000**.
3. **Divide by 1101 (modulo-2)**:

| Step | Leading window | XOR 1101 | Result | Remaining dividend |
|---|---|---|---|---|
| 1 | 1001 | 1101 | 0100 | 0100 1010 000 |
| 2 | 1001 | 1101 | 0100 | 0010 0010 000 |
| 3 | 1000 | 1101 | 0101 | 0001 0110 000 |
| 4 | 1011 | 1101 | 0110 | 0000 1100 000 |
| 5 | 1100 | 1101 | 0001 | 0000 0001 000 |
| 6 | 1000 | 1101 | 0101 | 0000 0000 101 |

```
1 0 0 1 1 0 1 0 0 0 0
1 1 0 1
-------
  1 0 0 1
  1 1 0 1
  -------
    1 0 0 0
    1 1 0 1
    -------
      1 0 1 1
      1 1 0 1
      -------
        1 1 0 0
        1 1 0 1
        -------
              1 0 0 0
              1 1 0 1
              -------
                1 0 1   = remainder
```

4. **Remainder (CRC) = 101.** Polynomial check: $x^{10} + x^7 + x^6 + x^4 \bmod (x^3 + x^2 + 1) = \mathbf{x^2 + 1} \leftrightarrow 101$ ✔.
5. **Transmitted message = 10011010 101**. Check: 10011010101 ÷ 1101 → remainder **000** ✔.

**(b) The leftmost bit is inverted by noise. What is the result of the receiver's CRC calculation? How does the receiver know an error occurred? (6)**

- Received: **0** 0011010101 = **00011010101**.
- The receiver divides by 1101. The leading zeros need no XOR; the first 1 is at position 4: 1101 ⊕ 1101 = 0000 → **00000000101**.
- **Remainder = 101 (≠ 000).**
- A correctly received frame is **exactly divisible by G** (remainder 0), because the sender made it so. A **non-zero remainder** therefore proves the frame was corrupted. The receiver **detects the error**, discards the frame and requests retransmission (ARQ). (This matches the worksheet's own example: *"Since the remainder is non-zero, we know that something went wrong!"* [ED].)

## Q3. Explain the three basic propagation mechanisms in a mobile communication system. (12)

*Identical to T1A Q1 (Part 5).*

## Q4. Differences between path loss, multipath propagation and interference. (12)

*Use T1A Q3 (Part 5): the Exam 1(a) paragraphs + table + the extra points. That gives 4 marks per impairment.*

## Q5. Discuss three methods to increase the capacity of a wireless communication system. (12)

Capacity = how many users or bits/s a system can carry in its spectrum. Three methods (4 marks each):

**1. Frequency reuse with smaller cells (network densification / SDMA)**

- Limit each transmitter's **range** and **reuse the same frequencies** in cells far enough apart that interference is acceptable. Traditional **macro cells** (km wide) are complemented by **small cells**: indoor **femtocells**, outdoor **picocells/relays**. Every reuse multiplies capacity, hence "network densification" [WC-B 5.4; WC-S 52].
- **SDMA/sectorization** strengthens this: directional antennas reduce co-channel interference, so the reuse pattern can be **tightened**. Moving from a **7-cell to a 4-cell pattern nearly doubles capacity**, and **intracell reuse** of the same channel by spatial separation becomes possible [MA p9–10].

**2. Multiple-antenna techniques (MIMO)**

- **SU-MIMO multilayer transmission** sends **parallel data streams** between one Tx–Rx pair, for a **higher data rate in the same bandwidth**.
- **MU-MIMO** forms **directional beams to several users simultaneously** (spatial multiple access), giving **more users per cell**.
- Diversity and beamforming improve link quality and coverage. 4×4 and 8×8 MIMO are in use; massive MIMO is coming [WC-B 5.4, Fig 5.9].

**3. Bandwidth expansion**

- **Carrier aggregation**: combine channels (802.11n/ac: 20 → 40/80/160 MHz; LTE aggregates carriers).
- **Millimetre-wave bands (30–300 GHz)**: much wider channels are available (at the cost of obstruction and absorption problems). These are needed because Shannon limits the bps/Hz achievable for a given SNR [WC-B 5.4; WC-S 52].

*(Other valid choices: **efficient multiple access**: CDMA gives 4–5× the capacity of other digital systems [MA p8] and OFDMA allows tight subcarrier packing and opportunistic scheduling [OF]; **adaptive modulation and coding** [WC-S 51].)*

## Q6. Explain the spread spectrum principles of FHSS and DSSS. (10)

*Identical to T1A Q4 (Part 5).*

## Q7. Describe CDMA and explain why it is difficult to jam. (10)

*Expanded version of Exam Q1(d) for 10 marks:*

**Description (6 marks)**:

1. **Spread-spectrum multiple access**: all users **share the entire bandwidth all the time**; there are no separate frequencies or time slots [MA p7; MX].
2. **Unique pseudorandom (orthogonal) code per user**, shared by the mobile and the base station [MA p8].
3. **Spreading**: data are XORed/multiplied with the code at a much higher **chip rate**. E.g. IS-95: **1.2288 Mchip/s** spreading 13 kbps voice over **1.25 MHz**, with up to **64 codes/users** per channel [AT].
4. **Despreading**: the receiver's **correlator** uses the same code. Only the wanted signal is despread and passes the **narrowband filter**; other users' signals remain noise-like [MA p8].
5. **Power control**: all users transmit at the minimum required power, to limit mutual interference (the near-far problem) and save battery [MA p8].
6. **Advantages**: greatest spectrum efficiency (8–10× analogue, 4–5× other digital), **soft handoff**, simplified frequency planning, privacy (Walsh codes). Used in IS-95 and 3G [MA p8].

**Why it is difficult to jam (4 marks)**:

1. The signal occupies a **very wide band at a power density below the noise floor**. A jammer would need enormous power across the whole band.
2. **Processing gain**: despreading **spreads the jammer** over the wide band while concentrating the wanted signal. The narrowband filter then rejects most of the jamming energy (SUPPLEMENTARY: gain ≈ chip rate/bit rate, e.g. ≈ 21 dB for IS-95).
3. The jammer **doesn't know the pseudorandom code** (or hop sequence), so it cannot correlate with or mimic the signal. In FH, *"a malicious sender may jam one hopping period, but not the whole period"* [SS].

4. **Designed for anti-jamming**: spread spectrum's purposes include suppressing interference and making interception difficult [MA p7]. CDMA was a **military technology used in WWII to foil German jamming** [MX].

## Q8. Describe the difference between OFDM and OFDMA. (10)

**OFDM (4 marks)**: a **multicarrier modulation/transmission technique**. A high-rate stream (R bps) is split by a serial-to-parallel converter into **N lower-rate substreams (R/N)**, carried on **N orthogonal subcarriers** (spacing fb = a multiple of 1/T, e.g. 15 kHz in LTE). Orthogonality lets the spectra **overlap without interference** (≈ 6× more signals than FDM in the notes' example). Longer symbols (N·Ts) **defeat ISI** and **frequency-selective fading** hits only some subcarriers. **In OFDM all subcarriers carry the data of a single source** [OF §8.1].

**OFDMA (4 marks)**: a **multiple access scheme built on OFDM**: a combination of **FDMA and TDMA**, where **different users are assigned subsets of the subcarriers (subchannels / LTE resource blocks of 12 subcarriers) at different times**. Subchannels can be formed from **adjacent, regularly spaced or randomly spaced** subcarriers, and are allocated by **opportunistic scheduling** (efficiency, proportional fairness $r_i/\bar{r}_i$, requirements, priority). In the uplink, a user **concentrates power** on its own subchannels, saving battery [OF §8.2; AT].

**Key differences (2 marks)**:

| | OFDM | OFDMA |
|---|---|---|
| Type | Modulation / transmission technique | Multiple access technique |
| Users per symbol | One user uses all subcarriers | Many users share the subcarriers simultaneously |
| Multiple access | Needs another method (e.g. **CSMA** in 802.11, TDMA) | Built in (frequency + time) |
| Examples | 802.11a/g/n | **LTE downlink**, WiMAX |

*"Not all technologies that use OFDM use OFDMA. Some forms of 802.11 use OFDM for the signal but CSMA for multiple access"* [OF §8.2].

## Q9. What distinguishes single-user MIMO from multiple-user MIMO? (8)

**MIMO** uses multiple antennas at transmitter and receiver, for diversity, beamforming, multi-user MIMO and multilayer transmission [WC-B 5.4].

**SU-MIMO (4 marks)**: all spatial streams go between the base station/AP and **one user** at a time. **Multilayer transmission** sends **multiple parallel data streams** over the same frequency and time between that single Tx–Rx pair. **Goal: a higher data rate for that user in a given bandwidth.** The user device needs **multiple antennas** (number of streams ≤ min(Tx, Rx antennas)).

**MU-MIMO (4 marks)**: *"with enough MIMO antennas, directional antenna beams can be established to **multiple users simultaneously**"* [WC-B 5.4]. The BS **beamforms** separate streams to **different users** on the same time/frequency: **spatial division multiple access**. **Goal: higher system capacity, i.e. more users per cell.** Users can have single antennas, but the BS needs **many antennas** and **knowledge of each user's channel** to steer the beams and cancel inter-user interference.

**Summary**: SU-MIMO increases **one user's throughput**; MU-MIMO increases **the number of users served simultaneously**, i.e. cell capacity [WC-B Fig 5.9].

# TEST 2 2025 (T2-2025; 100 marks)

## Q1. State and explain the three categories of routing protocols in MANETs. (12)

*Use Exam Q2(d) (Part 5), with 4 marks per category.* For 12 marks, add to each category:

- **Proactive**: works like **distance-vector/link-state routing adapted to mobility**. DSDV adds **destination sequence numbers** to avoid loops and count-to-infinity, with full and incremental table dumps. **OLSR** uses multipoint relays to reduce flooding. **Best for** small networks with low mobility and frequent traffic.
- **Reactive**: **AODV**: RREQ broadcast → reverse path set up → RREP unicast back → forward route installed; **RERR** on link breaks; sequence numbers keep routes fresh [BT-S 87 for the outline]. **DSR**: the **whole source route** is carried in each packet; route caching. **Best for** large, highly mobile networks with occasional traffic.
- **Hybrid**: **ZRP**: each node has a **zone** (radius in hops). Intra-zone routing is proactive (IARP); inter-zone routing is reactive (IERP), querying only **border nodes** ("bordercasting"). Combines low latency nearby with low overhead far away.

## Q2. Dijkstra on Figure 1 from x to all nodes. (8)

| Step | N' | D(t),p | D(u),p | D(v),p | D(w),p | D(y),p | D(z),p |
|---|---|---|---|---|---|---|---|
| 0 | x | ∞ | ∞ | **3,x** | 6,x | 6,x | 8,x |
| 1 | xv | 7,v | **6,v** | — | 6,x | 6,x | 8,x |
| 2 | xvu | 7,v | — | — | **6,x** | 6,x | 8,x |
| 3 | xvuw | 7,v | — | — | — | **6,x** | 8,x |
| 4 | xvuwy | **7,v** | — | — | — | — | 8,x |
| 5 | xvuwyt | — | — | — | — | — | **8,x** |
| 6 | xvuwytz | — | — | — | — | — | — |

Updates: via v (3): t = 3 + 4 = 7; u = 3 + 3 = 6; w via v = 7 > 6 and y via v = 11 > 6 (no change). Ties at 6 (u, w, y) broken alphabetically. Via u: t via u = 8 > 7; w via u = 9 > 6. Via y: t = 13, z = 18 (no change).

| To | v | u | w | y | t | z |
|---|---|---|---|---|---|---|
| Cost | 3 | 6 | 6 | 6 | 7 | 8 |
| Path | x-v | x-v-u | x-w | x-y | x-v-t | x-z |
| First hop | v | v | w | y | v | z |

## Q3. What is the count-to-infinity problem in distance-vector routing? (5)

*Use the first paragraph of Exam Q3(d) (Part 5)*, plus the cause and the fix (5 marks):

- **Cause**: DV nodes know only their **neighbours' distance estimates**, not the paths behind them. After a link failure or cost increase, a node can adopt a neighbour's route that **goes back through itself**.
- **Effect**: a **routing loop**. Estimates rise by small steps with each exchange (**"bad news travels slowly"**), up to the true cost or to **infinity** if unreachable. Packets loop meanwhile. In the y/z/x example (4, 1, 50 → 60), it takes **44 iterations**.

- **Fixes**: **poisoned reverse** (advertise ∞ to the neighbour you route through), **split horizon**, a **maximum metric** (RIP: 16 = infinity).

### Q4. Explain the hidden terminal problem. Does it occur in Bluetooth? (10)

**Hidden terminal problem (6 marks)**: in a wireless LAN, station **B** is in range of both **A** and **C**, but **A and C cannot hear each other** (distance or an obstacle) [FZ Fig 15.3].

- A sends to B. C senses the medium, **hears nothing** and concludes it is idle, so it also transmits (to B). The two frames **collide at B**.
- Neither A nor C can **detect** the collision: wireless stations can't send and listen at once, and they can't hear each other anyway. So the whole frames are wasted, followed by ACK timeouts and retransmissions. **Capacity drops** [FZ].
- This is one of the three reasons **CSMA/CD can't be used** in wireless [WL 29].
- **Solution in 802.11**: the **RTS/CTS handshake**. The **CTS from B** (with the duration) reaches C, which sets its **NAV** and defers [FZ; WL 39, 43].

**In Bluetooth (4 marks)**: **No, not within a piconet.** Access is **centrally scheduled by the master** (FH-TDD-TDMA polling). A slave transmits **only in the slot immediately after the master addresses it**, and SCO slots are **reserved** [BT-B]. Since no device transmits on its own after sensing the channel, two hidden slaves can never start transmitting at the same time to the master. (Collisions between **different piconets** happen only when they hop to the same frequency in the same slot. That is a separate FH-CDMA effect handled by FEC/ARQ [BT-B].)

### Q5. Compare wired and wireless LANs. What unique concerns must the designer of a wireless LAN address? (10)

**Comparison (5 marks)**: use the wired-vs-wireless table in Part 3 §9.3 [FZ]. Medium (wires vs broadcast air); hosts (fixed attachment vs free movement); isolated LAN (switch vs ad hoc); connection to other networks (router vs AP gluing wireless to wired); moving between them (only NICs and switch↔AP change; MAC addresses change, IP addresses stay the same); access method (CSMA/CD vs CSMA/CA); characteristics (wireless adds attenuation, interference, multipath and a higher error rate). Also from [WL 3]: wireless gives flexibility, ad hoc operation and disaster robustness, but lower bandwidth, lower QoS and higher delay.

**Unique concerns (5 marks; 1 mark each, any five)**:

1. **Attenuation/path loss and coverage**: AP placement; more APs [WCP].
2. **Interference** in the crowded, unprotected **ISM band**: channel planning, 5 GHz [WL 23].
3. **Multipath and fading**: diversity, MIMO, OFDM, rate adaptation [WCP; WL 26].
4. **Higher error rates**: ACKs, retransmission, FEC, **fragmentation** [FZ; WL 27].
5. **Medium access without collision detection**: **hidden and exposed stations**, so CSMA/CA, RTS/CTS and NAV [FZ].
6. **Mobility/roaming** between BSSs and ESSs; association [FZ; WL 12].
7. **Power management** for battery devices [WL 4].
8. **Security and privacy** on a broadcast medium [WL 4, 27].
9. **Regulation**: licence-free but power-limited (≤ 1 W); national restrictions [WL 3, 23].
10. **QoS** for time-sensitive traffic: PCF, rate adaptation [WL 40].

## Q6. A wireless receiving site has occasional loss of signal due to interference. Discuss the steps you would take. (10)

*Identical to T1A Q12 (Part 5).*

## Q7. While Bluetooth's profiles add significant complexity, how can they benefit applications from the application's perspective? (10)

*Exam Q4(c) answer, expanded to 10 marks.* (2 marks per developed point.)

1. **Guaranteed interoperability between vendors**: the profile specification exists *"to define a standard of interoperability so that products from different vendors that claim to support a given usage model will work together"* [BT-B]. A user's headset, car kit or keyboard works with any phone or PC supporting the same profile.
2. **A complete recipe for each usage model**: a profile is *"a set of protocols that implement a particular Bluetooth-based application"*. It states which parts of the core specification are **mandatory, optional and not applicable** [BT-B]. Application developers don't have to interpret the 2500+ page core themselves; the design decisions are made for them.
3. **Reuse of existing protocols and other profiles**: profiles build on each other. E.g. the **File Transfer Profile depends on the Generic Object Exchange, Serial Port and Generic Access profiles**, using **OBEX** over **RFCOMM** and **L2CAP** [BT-B]. Developers reuse proven components rather than writing new ones.
4. **Smaller, cheaper, lower-power implementations**: a device implements **only the profiles it needs** (a mouse doesn't need file transfer), keeping Bluetooth's key attributes of **low cost and low power** [BT-B].
5. **Consistent behaviour and easy setup**: every device supports the **Generic Access Profile**, so discovery, connection and security work the same everywhere. With **SDP**, devices can query each other's services and connect automatically. This gives plug-and-play operation and simpler testing and certification [BT-B].

## Q8. Scatternet of 2 piconets, no CSMA, 80 frequencies: probability of collision? (10)

*Exam Q4(e) answer, written out in full for 10 marks:*

**Model and assumptions (3 marks)**:

- Each piconet's master sets a **pseudorandom hop sequence** from its own address, so the two piconets hop **independently** [BT-B].
- **80** equally likely frequencies; one hop per slot.
- Both piconets **always transmit** (always have packets), with **no carrier sensing**.
- A **collision** occurs when both use the **same frequency in the same slot**.
- A **successful transmission** = poll (master → slave) **and** reply (slave → master), so **2 slots on 2 different hops**.

**Calculation (5 marks)**:

1. P(same frequency in a given slot) = 80 × (1/80)(1/80) = **1/80 = 0.0125**
2. P(no collision in a slot) = 1 − 1/80 = **79/80 = 0.9875**
3. The two slots are independent, so P(poll AND reply both collision-free) = $(79/80)^2$ = **0.97516**
4. **P(collision during the transaction) = $1 - (79/80)^2 = 159/6400 \approx 0.0248$**

**Answer (2 marks)**: **about 2.5%** of poll/response transactions suffer a collision (1.25% for any single packet). Performance degrades as more piconets are added, as the notes say: *"as the number of*

*piconets in an area increases, the number of collisions increases"* [BT-B]. SUPPLEMENTARY: with k other piconets, P(slot OK) = $(79/80)^k$.

### Q9. A Bluetooth device can be in two piconets at the same time. Is there any reason one device cannot be the controller in both at the same time? (5)

*Identical to Exam Q4(d) (Part 5)*, plus one more mark: a device in two piconets must **time-share**. It follows one piconet's hop sequence and timing, then the other's. As master it would have to **impose its own hop sequence on both**, which would merge them into one.

### Q10. Two ISPs in a café. (i) Both on channel 11. (10) (ii) Channels 1 and 11. (10)

*Use Exam Q5(e) (Part 5), expanded:* **(i) (10 marks)**

- **Does it break down?** No. 802.11's **CSMA/CA is distributed**: every station senses the channel before transmitting, whatever its ISP or AP. The two BSSs simply become **one shared collision domain** on channel 11 [WL 32–35].
- **Normal operation**: when a station of ISP 1 is transmitting, stations of ISP 2 that hear it (or its RTS/CTS) **defer**. Their **NAV** is set from the duration fields, so the channel is **time-shared** between the two BSSs. Each AP only processes frames carrying **its own BSSID**; the other ISP's frames are ignored after reception. The separate IP address blocks are irrelevant at the MAC layer.
- **Simultaneous transmission by two stations** (one per ISP): if both finish DIFS + backoff in the same slot, or they are **hidden** from each other, their frames **collide** at the receiving AP(s). With no collision detection, both frames are lost; **no ACK** returns; each station assumes a collision, **doubles its contention window**, picks a new random backoff and **retransmits** [WL 33, 35, 38].
- **Consequences**: more collisions, retransmissions and backoff delay; the channel's capacity is **shared**, so each ISP gets roughly **half the throughput**; **co-channel interference** raises the error rate [WL 12; WCP].

**(ii) (10 marks)**

- Channel 1 (2.412 GHz) and channel 11 (2.462 GHz) are **50 MHz apart**. Each ~22 MHz-wide signal [WC-S 4] does not overlap the other: they are **non-overlapping channels** (SUPPLEMENTARY).
- The two BSSs are now **independent collision domains**. A station's carrier sense on channel 1 does **not** detect transmissions on channel 11, and vice versa.
- **Two stations of different ISPs transmitting at the same time do not collide.** Both frames are received correctly by their own APs and ACKed.
- Each ISP gets the **full capacity** of its own channel, so aggregate throughput roughly **doubles**, with fewer collisions, backoffs and retransmissions and **no co-channel interference** between the networks.
- Stations of the same ISP still contend with each other as normal under CSMA/CA.

# COM5231 TEST 2 2026 (T2-2026; 50 marks)

## Q1. Distinguish between routing, packet forwarding and switching. For each, state its purpose and the information used to make its decision. (6)

*SUPPLEMENTARY (2 marks each).*

- **Routing**: **purpose**: determine the **end-to-end path** that packets take from source to destination across the network (a network-wide, control-plane function performed by routing algorithms such as Dijkstra/link-state and Bellman-Ford/distance-vector). **Information used**: the **network topology and link costs**, learned by exchanging routing messages with other routers. The output populates the **routing/forwarding table**.
- **Packet forwarding**: **purpose**: inside one router, **move each arriving packet from its input port to the correct output port/link** (a local, per-packet, data-plane action). **Information used**: the packet's **destination IP (network-layer) address**, looked up in the **forwarding table** (longest-prefix match).
- **Switching**: **purpose**: move **frames** to the correct outgoing port **within a LAN** at the link layer (also the term for transferring packets across a router's internal switching fabric). **Information used**: the frame's **destination MAC address**, looked up in the **switch table**, which the switch builds by **self-learning** from the source MAC addresses of arriving frames.

## Q2. Data byte 11001010; even-parity Hamming code; positions numbered from 1 at the left; check bits at 1, 2, 4, 8; data bits left to right in the remaining positions; no overall parity bit.

### (a) Minimum number of check bits; show the inequality and verify. (2)

Inequality: **2^r ≥ m + r + 1**, with m = 8 data bits.

- r = 3: $2^3 = 8 \geq 8 + 3 + 1 = 12$? **No**.
- r = 4: $2^4 = 16 \geq 8 + 4 + 1 = 13$? **Yes**. **Minimum r = 4** check bits, giving a 12-bit codeword.

### (b) Data-bit placement and value of each check bit, with the positions covered. (4)

| Position | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Bit | P1 | P2 | **1** | P4 | **1** | **0** | **0** | P8 | **1** | **0** | **1** | **0** |

| Check bit | Positions covered | Data bits | Number of 1s | Even-parity value |
|---|---|---|---|---|
| **P1** | 1, 3, 5, 7, 9, 11 | 1, 1, 0, 1, 1 | 4 | **0** |
| **P2** | 2, 3, 6, 7, 10, 11 | 1, 0, 0, 0, 1 | 2 | **0** |
| **P4** | 4, 5, 6, 7, 12 | 1, 0, 0, 0 | 1 | **1** |
| **P8** | 8, 9, 10, 11, 12 | 1, 0, 1, 0 | 2 | **0** |

### (c) Complete transmitted codeword (positions 1 → 12). (2)

**0 0 1 1 1 0 0 0 1 0 1 0 → 001110001010**

## Q3. CRC: data 110101, generator 1011 ($G(x) = x^3 + x + 1$), modulo-2 division.

### (a) Number of zeros appended and the resulting bit string. (1)

Degree 3 → **append 3 zeros** → **110101000**.

### (b) Show the division and the CRC remainder. (3)

| Step | Window | ⊕ 1011 | Result | Working value |
|---|---|---|---|---|
| 1 | 1101 | 1011 | 0110 | 011001000 |
| 2 | 1100 | 1011 | 0111 | 001111000 |
| 3 | 1111 | 1011 | 0100 | 000100000 |
| 4 | 1000 | 1011 | 0011 | 000001100 |
| 5 | 1100 | 1011 | 0111 | 000000111 |

```
110101000
1011
----
 1100
 1011
 ----
  1111
  1011
  ----
   1000
   1011
   ----
     1100
     1011
     ----
      111   ← remainder
```

**CRC remainder = 111.**

### (c) Complete transmitted bit string. (2)

**110101111** (data 110101 followed by CRC 111). Check: 110101111 ÷ 1011 → remainder 000 ✔.

## Q4. Simplified Internet checksum, 4-bit words. Message 1010 1100 0110 1001; one's-complement addition with end-around carry.

### (a) Show the addition and calculate the checksum. (3)

| Step | Addition | Raw result | End-around carry | Running sum |
|---|---|---|---|---|
| 1 | 1010 + 1100 | 1 0110 | 0110 + 1 | **0111** |
| 2 | 0111 + 0110 | 1101 | none | **1101** |
| 3 | 1101 + 1001 | 1 0110 | 0110 + 1 | **0111** |

**Sum = 0111. Checksum = one's complement = 1000.** (Decimal check: 10 + 12 + 6 + 9 = 37; mod 15 → 7 = 0111 ✔.)

### (b) 4-bit result expected when the receiver adds all message words and the checksum, assuming no error. (1)

0111 + 1000 = **1111** (all ones; complementing it gives 0000, meaning no error detected).

## Q5. Define a single-bit error and a burst error. Explain how the length of a burst error is measured. (3)

- **Single-bit error**: only **one bit** of the data unit changes (0 → 1 or 1 → 0). (1)

- **Burst error**: **two or more bits** in the data unit change (not necessarily consecutive). (1)
- **Burst length**: measured from the **first corrupted bit to the last corrupted bit, inclusive**. Bits between them may or may not be corrupted. E.g. errors at bit positions 3 and 6 → burst length 6 − 3 + 1 = **4**. (1)

## Q6. Even-parity scheme with one parity bit per block. Is it guaranteed to detect each of the following? Justify.

### (a) A single-bit error. (2)

**Yes, always.** Even parity makes the number of 1s even. Flipping **one** bit changes the count by exactly one, making it **odd**, so the receiver's check fails and the error is detected.

### (b) A burst error. (2)

**No, not guaranteed.** Parity detects only an **odd** number of bit errors. A burst that flips an **even** number of bits (2, 4, ...) leaves the count of 1s even, so the block **passes** undetected. E.g. 1011 0010 → 1**10**1 0010: two errors, still four 1s. A burst is detected only if it happens to flip an odd number of bits.

### (c) Can a single parity bit identify and correct the erroneous bit? (1)

**No.** It only shows that **an odd number of bits** changed, not **which** bit. Correction needs more redundancy (e.g. Hamming code or 2-D parity) or retransmission.

## Q7. Bluetooth profiles add complexity. Explain two ways in which profiles benefit application developers or users. (4)

1. **Interoperability (users)** (2): profiles define a *"standard of interoperability so that products from different vendors that claim to support a given usage model will work together"* [BT-B]. A user can pair any compliant headset, car kit or keyboard with any phone without compatibility worries.
2. **Ready-made, reusable implementation (developers)** (2): a profile specifies **exactly which protocols and features are mandatory, optional or not applicable** for a usage model, and builds on other profiles. E.g. the File Transfer Profile reuses OBEX, RFCOMM, L2CAP and the Generic Access Profile [BT-B]. Developers implement a known recipe (and only the subset they need), which saves effort and keeps devices small, cheap and low-power.

## Q8. Compare the Bluetooth Classic central (master) with an access point in an 802.11 infrastructure WLAN. Explain one difference in medium-access coordination and one in their roles in connecting devices to other networks. (4)

- **Medium-access coordination (2)**: the **Bluetooth central schedules all access**. It defines the hop sequence and timing, and **polls** slaves, who may transmit **only in the slot after being addressed** (FH-TDD-TDMA), so there is **no contention or collision inside the piconet** [BT-B]. In **802.11**, stations **contend using CSMA/CA (DCF)**: sense, DIFS, random backoff, RTS/CTS, NAV. **Collisions are possible**, and the AP does not normally schedule access (polling via PCF is optional) [WL 32–40].
- **Connecting to other networks (2)**: the **802.11 AP is a bridge/portal**: it *"provides a link to other networks and acts as a relay point for all MAC frames"*, connecting the BSS to the wired distribution system/Internet [BT-S 64; FZ]. The **Bluetooth central only sets up the piconet and relays traffic within it** [BT-S 64]. Slaves can reach other piconets only via **scatternet bridge devices**, and a piconet is a short-range PAN, not an infrastructure gateway.

## Q9. Dijkstra's shortest-path algorithm on the given network.

**Link costs read from the figure**: A–B 2, A–G 6, B–C 7, B–E 2, E–G 1, E–F 2, F–C 3, F–H 2, G–H 4, C–D 3, H–D 2.

### (a) Shortest path from A to all nodes (step by step). (6)

| Step | N' | B | C | D | E | F | G | H |
|---|---|---|---|---|---|---|---|---|
| 0 | A | **2,A** | ∞ | ∞ | ∞ | ∞ | 6,A | ∞ |
| 1 | AB | — | 9,B | ∞ | **4,B** | ∞ | 6,A | ∞ |
| 2 | ABE | — | 9,B | ∞ | — | 6,E | **5,E** | ∞ |
| 3 | ABEG | — | 9,B | ∞ | — | **6,E** | — | 9,G |
| 4 | ABEGF | — | 9,B | ∞ | — | — | — | **8,F** |
| 5 | ABEGFH | — | **9,B** | 10,H | — | — | — | — |
| 6 | ABEGFHC | — | — | **10,H** | — | — | — | — |
| 7 | ABEGFHCD | — | — | — | — | — | — | — |

Working:

- Step 0: A's neighbours are B 2 and G 6. Smallest = **B**.
- Step 1: via B (2): C = 2 + 7 = 9; E = 2 + 2 = 4. Smallest = **E (4)**.
- Step 2: via E (4): G = 4 + 1 = **5** (< 6, update); F = 4 + 2 = 6. Smallest = **G (5)**.
- Step 3: via G (5): H = 5 + 4 = 9. Smallest = **F (6)**.
- Step 4: via F (6): C = 6 + 3 = 9 (equal to 9, keep B); H = 6 + 2 = **8** (< 9, update). Smallest = **H (8)**.
- Step 5: via H (8): D = 8 + 2 = 10. Smallest = **C (9)**.
- Step 6: via C (9): D = 9 + 3 = 12 > 10 (no change). Smallest = **D (10)**. Done.

### (b) Final shortest-path cost and complete route from A to B, C, D, E and F. (4)

| Destination | Cost | Route |
|---|---|---|
| B | **2** | A → B |
| C | **9** | A → B → C *(equal-cost alternative A → B → E → F → C = 9)* |
| D | **10** | A → B → E → F → H → D |
| E | **4** | A → B → E |
| F | **6** | A → B → E → F |

*(Also: G = 5 via A → B → E → G; H = 8 via A → B → E → F → H.)*

# BLUETOOTH EXERCISES (20 questions) [BTX]

**1. Why is Bluetooth called a WPAN, not a WLAN?** Its range is ~**10 m** (a person's personal space), it targets **low power, low cost cable replacement** between a user's own devices, and a piconet holds only **8 active devices**. A WLAN covers a building or campus, serves many users and connects to wider networks [BT-B; BT-S 56].

**2. Alternate name for Bluetooth Smart, and which version? Bluetooth Low Energy (BLE)**, introduced in **Bluetooth 4.0** [BT-S 53].

**3. How does Bluetooth 3.0 reach 24 Mbps?** By the **Alternative MAC/PHY (AMP)**: Bluetooth 3.0 + HS uses a co-located **802.11 (Wi-Fi) link** for high-rate data. The Bluetooth radio still does discovery, association, connection setup, profile configuration and maintenance; a **Protocol Adaptation Layer (PAL)** maps Bluetooth protocols onto the 802.11 MAC/PHY [BT-S 52; BT-B].

**4. Compare a piconet and a scatternet. Piconet**: the basic unit, **1 master + up to 7 active slaves**, sharing **one hopping sequence** set by the master; slaves talk only to the master. **Scatternet**: **two or more overlapping piconets** linked by devices that belong to more than one (as slave in both, or master in one and slave in the other). **The piconet shares the logical channel and data transfer; the scatternet shares the physical area and total bandwidth** (FH-CDMA) [BT-B].

**5. Can a piconet have more than eight stations? Not more than eight active ones**: 1 master + **7 active slaves**, because the **AM_ADDR** field is **3 bits** (8 values, with 0 reserved for broadcast) [BT-S header]. SUPPLEMENTARY: more devices can be **parked** (synchronised but inactive) and swapped in.

**6. Relationship between master and slave?** The **master determines the hopping sequence (channel) and phase (timing)** from its own address. **Slaves tune to it**, may **communicate only with the master**, and **only when granted permission** (polled) [BT-B].

**7. Types of links between master and slave. SCO** (synchronous connection-oriented: reserved symmetric slots, fixed bandwidth, **no retransmission**, used for voice); **eSCO** (extended SCO: reserved slots, **asymmetric** allowed, **retransmission** supported); **ACL** (asynchronous connectionless: **point-to-multipoint**, packet-switched, uses unreserved slots, **retransmission**, polled) [BT-B].

**8. Master vs base station in 802.11.** See Exam Q4(a) (Part 5) and T2-2026 Q8.

**9. Can one device be controller of two piconets?** No. See Exam Q4(d) (Part 5).

**10. Role of the radio layer?** It specifies the **air interface**: the 2.4 GHz ISM band, **79 × 1 MHz channels**, **frequency hopping**, **GFSK** (π/4-DQPSK/8DPSK for EDR) modulation, and **transmit power classes 1–3** [BT-B].

**11. Actual bandwidth used?** At any instant a piconet uses **1 MHz** (one hop channel). Over time it hops across the **79 MHz (≈ 80 MHz)** band, so the total spread bandwidth is ~79 MHz [BT-B].

**12. Spread spectrum technique? Frequency hopping spread spectrum (FHSS)**, **1600 hops/s** (0.625 ms slots), with optional **adaptive frequency hopping** [BT-B].

**13. Modulation technique in the radio layer? Gaussian frequency-shift keying (GFSK)**: binary 1 = **positive** frequency deviation, 0 = **negative** deviation from the carrier; 1 Msymbol/s × 1 bit = **1 Mbps**. EDR uses **π/4-DQPSK (2 bits/symbol, 2 Mbps)** and **8DPSK (3 bits/symbol, 3 Mbps)** [BT-B].

**14. How can FH and TDD be combined?** Each **625 µs slot** uses the **next frequency** in the hop sequence (FH), **and** the direction alternates between slots: master → slave, then slave → master (TDD). Multislot packets (3 or 5 slots) stay on one frequency, then jump to the frequency the sequence requires. This gives FH-TDD-TDMA [BT-B Fig 12.4–12.5].

**15. How does FH-CDMA differ from DS-CDMA? FH-CDMA**: users are separated by **different hopping sequences**. Each uses a narrow channel at a time, and collisions happen only when two users land on the same frequency in the same slot (Bluetooth scatternets). **DS-CDMA**: all users transmit **simultaneously over the whole wide band**, spread by **different chip codes**, and separated by **correlation** at the receiver (IS-95, WCDMA) [BT-B; MA; AT].

**16. Do Bluetooth devices and microwave ovens interfere? Yes, they can.** Microwave ovens radiate around **2.45 GHz**, inside Bluetooth's 2.4 GHz ISM band [WCP §10]. The effect is **limited**: Bluetooth hops over 79 channels 1600 times/s, so only hops near the oven's frequency suffer, **adaptive frequency hopping** can avoid bad channels, and lost packets are recovered by **FEC/ARQ** [BT-B].

**17. MAC protocol in the baseband? TDMA with master-controlled polling over FH and TDD (FH-TDD-TDMA)**. There is no contention within a piconet; slaves transmit only when addressed [BT-B].

**18. Role of L2CAP?** It **adapts upper-layer protocols to the baseband**, providing **connectionless and connection-oriented** services. SUPPLEMENTARY detail: protocol multiplexing, segmentation and reassembly of large packets, and QoS [BT-B].

**19. How are profiles an advantage to applications?** See Exam Q4(c) and T2-2025 Q7.

**20. Scatternet collision probability.** See Exam Q4(e) and T2-2025 Q8: **$\approx 1 - (79/80)^2 \approx 2.5\%$**.

# ERROR-DETECTION WORKSHEET EXERCISES [ED]

**W1. Message 11001001, CRC $x^3 + 1$ (1001).** Remainder **011** → transmit **11001001011**. Leftmost bit flipped → received 01001001011 → remainder **010** → non-zero → error detected. *(Worked in the worksheet; verified correct.)*

**W2. CRC-8 ($x^8 + x^2 + x + 1$ = 100000111) on 1011 0011 0101 0110.** Remainder **11010101** → transmit **1011 0011 0101 0110 1101 0101**. **7th bit flipped** → remainder **00101010** (101010) → error detected. *(Worksheet values verified.)*

**W3. G = 10011, D = 1010101010: R? R = 0100.** (a) 1000100101 → **1000**; (b) 0101101010 → **1111**; (c) 0110100011 → **1110**. *(Full division in Part 2, §8.8 CR5.)*

**W4. Internet checksum of 1001 1100 1010 0011 (4-bit). 1011**. (Exam Q2(c).)

**W5. Hamming for 10101111: check bits and codeword. 4 check bits; codeword 101001001111.** (Exam Q2(a).)

**W6. M(x) = 110001110110, G(x) = $x^5 + x^2 + x + 1$: frame T(x)? Why modulo-2?** G = **100111**, append 5 zeros, remainder **10101** → **T(x) = 110001110110 10101**. *(Division in Part 2, §8.8 CR6.)* **Why modulo-2**: addition and subtraction become the **same XOR operation with no carries or borrows**. That makes CRC computation trivial in hardware (**shift register + XOR gates**) and corresponds to **polynomial arithmetic over GF(2)**, which gives CRC its guaranteed detection properties.

**W7. Single-bit vs burst errors and parity.** See T1B Q1.

# Part 7 — Additional Practice Questions with Model Answers

> **These are GENERATED practice questions, not taken from past papers.** They are built from your lecture notes and the patterns in Part 4, with high-frequency topics weighted most. Every numerical answer has been computed and checked. Suggested marks are shown in brackets.
>
> **How to use**: cover the answer, write your own answer in exam conditions, then mark it against the model.

## SECTION A — Short definition questions (2 marks each)

**A1. Define multiple access.** *(2)* The technique that lets **many mobile users share a finite amount of radio spectrum simultaneously and efficiently**, so the system handles **as many calls as possible in a given bandwidth** (call-handling capacity) [MA p2].

**A2. Define the path loss exponent.** *(2)* A parameter **n** describing **how fast a radio signal's power decreases with distance** in a given environment ($Pr \propto 1/d^n$): 2 in free space, 2.7–3.5 in an urban cellular area, 4–6 obstructed in a building [WC-S 33–34].

**A3. Define coherence time and state when a channel is fast fading.** *(2)* **Coherence time Tc** is the time over which the channel stays relatively constant. The channel is **fast fading** when Tc is **not much greater** than the bit time Tb (Tc < Tb, Tc ≈ Tb or only slightly greater) [WC-B 5.3].

**A4. What is intersymbol interference?** *(2)* **Delayed copies of a pulse (from multipath) arriving at the same time as the primary pulse of a subsequent bit**, acting as noise to it and making detection difficult [WC-S 40].

**A5. Define the NAV.** *(2)* The **network allocation vector**: a timer set by stations that hear an RTS/CTS (which carry the duration of the exchange), showing **how long they must wait before checking the channel for idleness**. It is virtual carrier sensing [WL 37].

**A6. What is a piconet?** *(2)* The basic Bluetooth network unit: **one master and one to seven active slaves**, sharing the master's **frequency-hopping sequence and timing**. Slaves communicate only with the master, when permitted [BT-B].

**A7. Define chip rate.** *(2)* The **rate of a spreading code** in spread spectrum/CDMA. It is **much higher than the data bit rate** (e.g. 11 chips per bit with the Barker code), and is called the chip rate rather than the bit rate [MA p8; SS].

**A8. What is the purpose of a guard band in FDMA?** *(2)* An **unused frequency slot between adjacent channels** that **minimises crosstalk/adjacent-channel interference**, at the cost of wasted bandwidth [MA p4].

**A9. Define a burst error and its length.** *(2)* A burst error is **two or more bits changed** in a data unit. Its **length** is measured **from the first to the last corrupted bit, inclusive** (SUPPLEMENTARY).

**A10. What is orthogonality in OFDM?** *(2)* Two signals are orthogonal if **the average over a bit time of their product is zero**. For OFDM subcarriers spaced at multiples of 1/T, **each subcarrier's spectral peak falls where every other subcarrier's power is zero**, so they overlap without interfering [OF].

**A11. What is a resource block in LTE?** *(2)* The smallest group of subcarriers (and time slots) that OFDMA assigns to a user: **12 subcarriers** of 15 kHz [AT; OF].

**A12. Name the three power classes of Bluetooth.** *(2)* **Class 1**: 100 mW (+20 dBm), power control mandatory, greatest range. **Class 2**: 2.5 mW (+4 dBm). **Class 3**: 1 mW [BT-B].

## SECTION B — Conceptual explanation questions

**B1. Explain three reasons why CSMA/CD cannot be used in wireless LANs. (6)**

1. **Simultaneous send and receive**: collision detection requires a station to transmit and listen at the same time. Received signals are much weaker than transmitted ones, so this needs costly full-duplex hardware and more bandwidth, and battery-powered stations lack the power [WL 16, 29; FZ].
2. **Hidden station problem**: a collision at the receiver may be **inaudible to the sender**, because the other transmitter is out of its range [FZ].
3. **Distance and fading**: stations can be far apart, so **signal fading** may stop a station at one end from hearing a collision at the other [WL 29]. Therefore 802.11 uses **CSMA/CA** with ACKs [FZ].

**B2. Explain why forward error correction is preferred to backward error correction in wireless systems. (4)**

- In backward error correction (ARQ) the receiver only detects errors and asks for a **retransmission**. In **satellite** links the **delay** makes this undesirable; in **mobile** links **error rates are so high that the retransmitted block is likely to be corrupted too** [WC-S 45].
- FEC adds redundant bits so the receiver **corrects errors itself**, using only the received data. This costs bandwidth (a total/data ratio of 2–3) but avoids retransmission delay [WC-B 5.4].

**B3. Explain the role of the master's Bluetooth address in a piconet. (4)** The master's address (and clock) determines the **frequency-hopping sequence (channel) and the phase (timing)** for all piconet members. Slaves tune to it [BT-B]. Because different masters have different addresses, **different piconets use different hop sequences**, which gives **FH-CDMA** separation in a scatternet. This is also why one device **cannot be master of two piconets**.

**B4. Describe the three types of diversity and the two ways of using the received signals. (6) Space diversity**: multiple antennas spaced far enough apart (or collocated directional antennas) experience independent fading. **Frequency diversity**: spread the signal over a wider band or multiple carriers (**OFDM, spread spectrum**). **Time diversity**: spread data over time so a noise burst hits fewer bits (**interleaving, Rake receiver**). Use: **selection diversity** (pick the best signal) or **combining diversity** (co-phase and add the signals) [WC-S 48; WC-B 5.4].

**B5. Explain the exposed station problem and why RTS/CTS cannot solve it. (4)** A transmits to B, and C wants to send to D. C's transmission would **not** interfere at B, but C is **within A's range**, hears A, and **refrains**: it is too conservative and **wastes capacity**. RTS/CTS doesn't help: C hears A's RTS and still defers, because **it cannot know that its transmission would not affect the zone between A and B** [FZ].

**B6. Why does OFDM reduce the need for equalizers? (4)** OFDM divides the data into **N parallel substreams**, so each subcarrier's **symbol time is N·Ts** instead of Ts. N is chosen so that N·Ts is **much greater than the channel's RMS delay spread**, so multipath delays occupy only a small fraction of each symbol and **ISI is drastically reduced**. Equalizers, which are complex and expensive and grow with ISI severity, **may not be needed** [OF §8.1].

**B7. What is AODV and where is it used in your notes? (4) Ad hoc On-demand Distance Vector**, a **reactive** MANET routing protocol. A source **sends route requests**, which nodes either **answer or forward** until someone (the destination or a node with a route) responds. Nodes **store and use** the routes they discover [BT-S 87]. Your notes mention it as the routing protocol **ZigBee** uses to build multi-hop ad hoc networks.

# SECTION C — Comparison questions

### C1. Compare FDMA and TDMA under: principle, synchronisation, efficiency, battery life, equalization. (10)

| | FDMA | TDMA |
|---|---|---|
| Principle | Spectrum divided into **frequency bands with guard bands**; one user per channel **100% of the time** | Same frequency shared by dividing **time into slots**; users transmit in **bursts** |
| Synchronisation | **Not severe** (continuous transmission) | **Required**: lost sync means slots collide |
| Spectrum efficiency | **Inefficient**: idle channels can't be shared; guard bands waste bandwidth | **More efficient**: more users in the same spectrum |
| Battery / handoff | Continuous transmission | **Longer battery life**; **simpler handoff** (listen in idle slots); but **high peak power** on the uplink |
| Equalization | **Not required** | **Required** (high transmission rates) |
| Duplexer | Needed (forward and reverse frequencies) | **Not required** (different slots for Tx/Rx) |
| Examples | 1G AMPS, FM radio, walkie-talkies | GSM, IS-136, DECT |

[MA p3–7]

### C2. Compare ZigBee coordinator, router and end device. (6)

| Device | Role | Notes |
|---|---|---|
| **Coordinator** | Most capable; **only one per network**; **starts the network**, is the root of a tree, **maintains network information such as security keys** | Usually mains powered |
| **Router** | **Passes data on** to other devices (multi-hop) | Must stay awake in non-beacon networks |
| **End device** | Just enough function to talk to a router/coordinator; **cannot relay** | **Sleeps most of the time**, least memory, **cheapest** |

[BT-S 85–86]

### C3. Compare SCO and ACL links. (6)

| | SCO | ACL |
|---|---|---|
| Topology | Point-to-point (master ↔ one slave) | **Point-to-multipoint** (master ↔ all slaves) |
| Slots | **Reserved at regular intervals** (2 consecutive, symmetric) | Any **unreserved** slot, per-slot polling |
| Bandwidth | **Guaranteed** fixed rate | **No reservation** (packet-switched) |
| Retransmission | **Never** | **Yes** (ARQ) |
| Typical traffic | **Time-bounded voice** | **Data** |
| Max number | Master: 3 SCO links | **One** ACL link |

[BT-B]

**C4. Compare ASK, BFSK and BPSK in terms of noise immunity and bandwidth. (6) ASK**: data in amplitude; **most susceptible to noise** and sudden gain changes; bandwidth ∝ baud rate. **BFSK**: data in frequency; **less error-prone than ASK**; **needs more bandwidth** (two carriers separated by 2Δf). **BPSK**: data in phase (0°/180°); **less susceptible to noise than ASK** (noise changes amplitude more easily than phase); **same bandwidth as ASK, less than BFSK**; needs more sophisticated hardware [SE; WC-S 59–64].

## SECTION D — Calculation and problem-solving questions

**D1. Hamming code. Encode the data byte 10110011 using an even-parity Hamming code (positions numbered from 1 at the left, check bits at 1, 2, 4, 8). (a) How many check bits? (b) Calculate each check bit. (c) Write the codeword. (8) (a)** 2^r ≥ m + r + 1 with m = 8: r = 3 gives 8 < 12 ✘; r = 4 gives 16 ≥ 13 ✔ → **4 check bits** (12-bit codeword). **(b)**

| Pos | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | P1 | P2 | 1 | P4 | 0 | 1 | 1 | P8 | 0 | 0 | 1 | 1 |

- P1 (3, 5, 7, 9, 11) = 1, 0, 1, 0, 1 → three 1s → **1**
- P2 (3, 6, 7, 10, 11) = 1, 1, 1, 0, 1 → four 1s → **0**
- P4 (5, 6, 7, 12) = 0, 1, 1, 1 → three 1s → **1**
- P8 (9, 10, 11, 12) = 0, 0, 1, 1 → two 1s → **0 (c)** Codeword = **1 0 1 1 0 1 1 0 0 0 1 1 = 101101100011**.

**D2. Hamming error correction. The 12-bit codeword 101101100011 (from D1) is received as 101101101011. Find and correct the error. (4)** Received: positions 1–12 = 1 0 1 1 0 1 1 0 **1** 0 1 1 (bit 9 changed).

- C1 (1, 3, 5, 7, 9, 11) = 1, 1, 0, 1, 1, 1 → five 1s → fails → **1**
- C2 (2, 3, 6, 7, 10, 11) = 0, 1, 1, 1, 0, 1 → four 1s → OK → **0**
- C4 (4, 5, 6, 7, 12) = 1, 0, 1, 1, 1 → four 1s → OK → **0**
- C8 (8, 9, 10, 11, 12) = 0, 1, 0, 1, 1 → three 1s → fails → **1** Syndrome C8C4C2C1 = **1001 = 9** → **bit 9 is wrong** → flip it → **101101100011** ✔.

**D3. CRC. Data 1101011011, generator $x^4 + x + 1$. Find the transmitted frame. (8)**

- G = $x^4 + 0x^3 + 0x^2 + x + 1$ = **10011** (r = 4); append 0000 → **11010110110000**.
- Division (5-bit windows):
  1. 11010 ⊕ 10011 = 01001
  2. 10011 ⊕ 10011 = 00000
  3. 10110 ⊕ 10011 = 00101
  4. 10100 ⊕ 10011 = 00111
- **Remainder = 1110** → **transmitted frame = 1101011011 1110**.
- Check: dividing 11010110111110 by 10011 gives remainder 0000 ✔.

**D4. CRC. Data 100100, generator 1101. Find the CRC and the transmitted string. (6)** Append 000 → 100100000.

1. 1001 ⊕ 1101 = 0100
2. 1000 ⊕ 1101 = 0101
3. 1010 ⊕ 1101 = 0111
4. 1110 ⊕ 1101 = 0011
5. 1100 ⊕ 1101 = 0001 **Remainder = 001** → transmit **100100001**.

**D5. Internet checksum (4-bit words). Message 0110 1011 1001 0101. Find the checksum and show the receiver's check. (4)**

| Step | Addition | Raw | After end-around carry |
|---|---|---|---|
| 1 | 0110 + 1011 | 1 0001 | 0001 + 1 = **0010** |
| 2 | 0010 + 1001 | 1011 | **1011** |
| 3 | 1011 + 0101 | 1 0000 | 0000 + 1 = **0001** |

**Sum = 0001 → checksum = 1110.** Receiver: 0001 + 1110 = **1111** → no error. (Decimal check: 6 + 11 + 9 + 5 = 31 = 2 × 15 + 1 → 0001 ✔.)

**D6. Free-space received power. A 2.4 GHz access point transmits 100 mW. Find the received power at 50 m in watts and dBm. (6)**

- $\lambda = c/f = 3 \times 10^8 / 2.4 \times 10^9 =$ **0.125 m**
- $4\pi d/\lambda = 4\pi \times 50 / 0.125 =$ **5026.5** → squared = **$2.527 \times 10^7$**
- $Pr = Pt / (4\pi d/\lambda)^2 = 0.1 / 2.527 \times 10^7 =$ **$3.96 \times 10^{-9}$ W (3.96 nW)**
- In dBm: $10 \log(3.96 \times 10^{-9} / 10^{-3}) =$ **−54.0 dBm** (Equivalently, L_dB = 20 log f + 20 log d − 147.56 = 187.60 + 33.98 − 147.56 = 74.0 dB, and 20 dBm − 74 dB = −54 dBm ✔.)

**D7. Shannon capacity. A 20 MHz channel has SNR = 30 dB. Find the Shannon capacity. How many signal levels would Nyquist require for this rate? (6)**

- SNR = 10^(30/10) = **1000**
- $C = B \log_2(1 + SNR) = 20 \times 10^6 \times \log_2(1001) = 20 \times 10^6 \times 9.967 =$ **199.3 Mbps** (≈ 200 Mbps)
- Nyquist: $199.3 \times 10^6 = 2 \times 20 \times 10^6 \times \log_2 M \rightarrow \log_2 M = 4.98 \rightarrow$ **M ≈ 32 levels**

**D8. FDMA channels. A system has Bt = 25 MHz, guard bands Bg = 100 kHz at each edge, Bc = 200 kHz. How many channels? If each channel carries 8 TDMA slots, how many users? (4)**

- N = (Bt − 2Bg)/Bc = (25 000 − 200)/200 = 24 800/200 = **124 channels**
- Users = 124 × 8 = **992** [MA p4, p6]

**D9. Scatternet probability. Three piconets share 80 frequencies (always transmitting, no CSMA, independent hops). What is the probability that a poll + reply transaction in piconet 1 suffers a collision? (6)**

- In one slot, piconet 1 avoids collision only if **both** other piconets choose a different frequency: $(79/80)^2$ per slot.
- The transaction needs 2 slots: P(success) = $((79/80)^2)^2 = (79/80)^4 =$ **0.9509**.
- **P(collision) = $1 - (79/80)^4 \approx 0.049 \approx 4.9\%$**. Adding piconets increases collisions, as the notes say [BT-B].

**D10. Dijkstra. Using Figure 1 (the exam network), compute the shortest paths from u to all nodes. (8)**
Link costs: x–z 8, x–y 6, x–v 3, x–w 6, y–z 12, y–v 8, y–t 7, v–t 4, v–u 3, v–w 4, t–u 2, w–u 3.

| Step | N' | D(t),p | D(v),p | D(w),p | D(x),p | D(y),p | D(z),p |
|---|---|---|---|---|---|---|---|
| 0 | u | **2,u** | 3,u | 3,u | ∞ | ∞ | ∞ |
| 1 | ut | — | **3,u** | 3,u | ∞ | 9,t | ∞ |
| 2 | utv | — | — | **3,u** | 6,v | 9,t | ∞ |
| 3 | utvw | — | — | — | **6,v** | 9,t | ∞ |
| 4 | utvwx | — | — | — | — | **9,t** | 14,x |

| Step | N' | D(t),p | D(v),p | D(w),p | D(x),p | D(y),p | D(z),p |
|---|---|---|---|---|---|---|---|
| 5 | utvwxy | — | — | — | — | — | **14,x** |
| 6 | utvwxyz | — | — | — | — | — | — |

Updates: via t: y = 2 + 7 = 9 (v via t = 6 > 3). Via v: x = 3 + 3 = 6 (y via v = 11 > 9; w via v = 7 > 3). Via w: x via w = 9 > 6. Via x: z = 6 + 8 = 14 (y via x = 12 > 9). **Results**: t 2 (u-t); v 3 (u-v); w 3 (u-w); x 6 (u-v-x); y 9 (u-t-y); z 14 (u-v-x-z).

**D11. Bellman-Ford. Node x has neighbours y (cost 2) and z (cost 7). y advertises D_y(z) = 1 and D_y(w) = 5; z advertises D_z(w) = 3. Compute D_x(z) and D_x(w). (4)**

- D_x(z) = min{ c(x,y) + D_y(z), c(x,z) + D_z(z) } = min{2 + 1, 7 + 0} = **3 (via y)**
- D_x(w) = min{ c(x,y) + D_y(w), c(x,z) + D_z(w) } = min{2 + 5, 7 + 3} = **7 (via y)**

**D12. Path loss. A 900 MHz cellular link has a receiver 2 km from the base station. Find the free-space path loss as a ratio and in dB. (4)**

- $\lambda = 3 \times 10^8 / 9 \times 10^8 =$ **0.333 m**
- $L_p = (4\pi d/\lambda)^2 = (4\pi \times 2000 / 0.333)^2 = (75\,398)^2 =$ **$5.68 \times 10^9$**
- $L_{dB} = 10 \log(5.68 \times 10^9) =$ **97.5 dB** (check: $20 \log(9 \times 10^8) + 20 \log(2000) - 147.56 = 179.08 + 66.02 - 147.56 = 97.5$ ✔)

## SECTION E — Application and scenario questions

**E1. A university library's WiFi slows down every lunchtime, and users report many retransmissions. The APs use 2.4 GHz channel 6. A staff kitchen with microwave ovens is next door. Diagnose and recommend. (10)**

- **Diagnosis**: microwave ovens radiate around **2.45 GHz** in the same **ISM band** as 2.4 GHz WiFi [WCP §10]. At lunchtime the ovens cause **inward/co-channel interference**, which **raises the bit error rate**. Corrupted frames are not ACKed, so stations retransmit and back off (CSMA/CA), and throughput falls [WL 27, 33]. ISM users have **no regulatory protection** [WC-S 10].
- **Recommendations**: (1) move the APs to the **5 GHz band** (less crowded, no ovens) [WL 23]; (2) change the 2.4 GHz channel to one away from the oven's emissions (e.g. channel 1); (3) **relocate APs** away from the kitchen or add **shielding**; use **directional antennas** pointing away; (4) **add APs / reduce cell size** so clients have a stronger signal (higher SINR); (5) allow **rate adaptation** and **fragmentation** (smaller frames are less likely to be hit) [FZ]; (6) **verify** with a spectrum analyser before and after.

**E2. Five laptops in a meeting room with no AP want to share files. Which 802.11 architecture applies? What limitation do they face? (4)** An **ad hoc BSS (independent BSS)**: stations form a network **without an AP**, locating each other and agreeing to be part of a BSS [FZ]. **Limitations**: it is a **stand-alone network that cannot send data to other BSSs** or the Internet [WL 10; FZ]. Stations communicate only if **within each other's range** (or if others forward) [WL 9]. PCF is not available (infrastructure only) [WL 40].

**E3. A pedestrian (Tc ≈ 70 ms) and a car on the highway (Tc ≈ 5 ms) both use a 1 kbps telemetry link. Classify the fading for each. (4)** Bit time Tb = 1/1000 = **1 ms**.

- **Pedestrian**: Tc = 70 ms ≫ 1 ms → **slow fading** (the channel hardly changes during a bit).
- **Car**: Tc = 5 ms, which is only 5× Tb, **not much greater** → by the notes' definition it is treated as **fast fading** [WC-B 5.3].

**E4. A mobile user moves to the cell edge in an LTE network. Using your OFDMA notes, explain how the scheduler can still serve this user fairly. (6)** Pure efficiency scheduling would favour users with the **highest SINR**, starving cell-edge users [OF]. **Proportional fairness** compares each user's **achievable rate now ($r_i$) to its own average rate ($\bar{r}_i$)**: the metric is $r_i/\bar{r}_i$. When the edge user's channel momentarily improves **relative to its own average**, its metric rises and it is scheduled, even if its absolute rate is lower than central users'. Subchannel choice (adjacent blocks matched to good frequencies), **AMC** and higher power on its subchannels also help. In the uplink, **subchannelization concentrates the device's power** on its allocated subchannels [OF §8.2].

## SECTION F — Long-form examination questions

**F1. "Wireless channels are hostile." Discuss the main impairments of the wireless channel and the mechanisms modern systems use to overcome them. (20)** *Model answer outline (≈ 1 mark per developed point):*

1. **Introduction (2)**: the wireless channel is shared, time-varying and unpredictable. The received signal differs from the transmitted one because of impairments, causing random degradation (analog) or bit errors (digital) [WC-B 5.2].
2. **Impairments (9)**:
   - **Free-space loss/path loss**: $(4\pi d/\lambda)^2$; path loss exponent n [WC-S 30–34].
   - **Noise**: thermal (irreducible), intermodulation, crosstalk, impulse (the main digital error source) [WC-S 36–38].
   - **Atmospheric absorption and refraction** [WC-S 39].
   - **Multipath** via reflection, diffraction and scattering → **destructive addition (20–30 dB)** and **ISI** [WC-B 5.3].
   - **Fading**: large-scale (shadowing) and small-scale; **fast/slow** (coherence time) and **flat/frequency-selective** (coherence bandwidth); Rayleigh/Rician [WC-B 5.3].
   - **Interference**: co-channel and adjacent channel; the unprotected ISM band [WCP §10].
3. **Mechanisms (8)**:
   - **FEC** (redundancy; ARQ is impractical) [WC-S 45].
   - **Adaptive equalization** (combats ISI) [WC-S 47].
   - **Diversity** (space, frequency, time; selection or combining) [WC-S 48].
   - **MIMO**: diversity, beamforming, MU-MIMO, multilayer [WC-B 5.4].
   - **AMC** (adapt hundreds of times per second) [WC-S 51].
   - **OFDM** (long symbols, selective fading on few subcarriers) [OF].
   - **Spread spectrum** (100× bandwidth, anti-interference) [SS].
   - **Bandwidth expansion** (carrier aggregation, reuse, mmWave) [WC-S 52].
4. **Conclusion (1)**: real systems **combine** all of these (e.g. LTE and 802.11ac use OFDM + MIMO + AMC + FEC) [WC-B 5.4].

**F2. Discuss medium access control in IEEE 802.11. Include the reason CSMA/CD is not used, DCF with RTS/CTS and NAV, the role of IFS, PCF, and the hidden and exposed station problems. (20)** *Outline*:

- MAC functions: reliable delivery, access control, security [WL 27] (2)
- Why not CSMA/CD: three reasons [WL 29] (3)
- DCF/CSMA/CA steps: sense, DIFS, backoff (freeze), RTS/CTS/DATA/ACK with SIFS, no ACK → backoff and retry [WL 32–35] (5)
- NAV and collisions during handshaking [WL 37–38] (2)

- IFS: SIFS < PIFS < DIFS and the priority each gives [WL 34] (2)
- PCF: AP polling, PIFS priority, infrastructure only, superframe, single point of failure [WL 40–42] (3)
- Hidden station (CTS solves it) and exposed station (not solved) [FZ] (3)

**F3. Bluetooth is described as robust, low-power and low-cost. Explain how its architecture and protocols achieve these goals. (15)** *Outline*:

- **Robustness**: FHSS at 1600 hops/s over 79 channels (resists interference and multipath); adaptive FH; 1/3 and 2/3 FEC; CRC + ARQ; HEC; SEQN duplicate filtering [BT-B; BT-S 45–50] (5)
- **Low power**: short range (10 m), power classes and **power control via LMP**, TDD on one chip, BLE (10 mW, long battery life) [BT-B; BT-S 53–55] (5)
- **Low cost**: one-chip transceiver (TDD prevents crosstalk), unlicensed ISM band, reuse of adopted protocols (PPP, TCP/IP, OBEX), profiles so devices implement only what they need, single-mode BLE chips [BT-B] (5)

**F4. Discuss routing in mobile ad hoc networks: why it is hard, the three categories of protocols with examples, and how distance-vector problems such as count-to-infinity are avoided in DSDV and AODV. (20)** *(SUPPLEMENTARY) Outline*: challenges (6: mobility, no infrastructure, bandwidth, energy, unreliable links, security); proactive (4: DSDV, OLSR); reactive (4: AODV, DSR; RREQ/RREP/RERR); hybrid (3: ZRP); loop avoidance (3): **destination sequence numbers** in DSDV and AODV mark route freshness, so a node never accepts an older (stale) route. This prevents the stale-route loops behind count-to-infinity.

# Part 8 — Final Examination Revision Guide

Use this as your **last-week and night-before** document. Every item points back to the full notes (Parts 1–3) and answers (Parts 5–6).

## A. High-priority topics (from the counted patterns in Part 4)

| Priority | Topic | Why | Where |
|---|---|---|---|
| ★★★ | **Bluetooth Q-bank**: profiles, master vs AP, two piconets, hidden terminal, scatternet probability | Exam Q4 (20 marks) + Tests; profiles asked **4 times** | Part 3 §10; Part 5 Q4 |
| ★★★ | **802.11 MAC scenarios**: hidden node and CSMA/CA (Figure 2), ACKs, two ISPs channel 11 / 1 + 11 | Exam Q5 (18 marks); ISP question asked **3 times** | Part 3 §9; Part 5 Q5 |
| ★★★ | **Routing**: Dijkstra (Figure 1 from any node; new graphs), LS vs DV, Dijkstra vs BF, count-to-infinity, MANET categories | Exam Q2(d)+Q3 = **30 marks** | Part 3 §11; Part 5 Q2–3 |
| ★★★ | **Error coding calculations**: Hamming, CRC, checksum; parity vs bursts | In **every** paper except T1A | Part 2 §8 |
| ★★ | **Wireless channel**: path loss vs multipath vs interference; three propagation mechanisms; propagation problems; interference troubleshooting; distance calculation | Test 1 core; Exam Q1(a) | Part 1 §2 |
| ★★ | **Spread spectrum and CDMA**: FHSS/DSSS principles; CDMA anti-jamming | Asked **3 times each** | Part 2 §5–6 |
| ★★ | **OFDM/OFDMA**: three subcarrier approaches; OFDM vs OFDMA | Exam Q1(c), T1A, T1B | Part 2 §7 |
| ★★ | **Capacity and MIMO**: three ways to increase capacity; SU vs MU-MIMO; mobile node moving away | T1B, Exam Q5(a) | Part 1 §3 |
| ★ | Nyquist/Shannon, modulation, FDMA/TDMA formulas, generations, 802.15.3/.4, ZigBee, packet formats | In notes, never yet examined | Parts 1–3 |

## B. Essential definitions (reproduce these word-perfect)

| Term | Definition |
|---|---|
| Channel capacity | The maximum rate at which data can be transmitted over a given communication path, or channel, under given conditions |
| Bandwidth (effective) | The narrow band of frequencies in which most of the signal's energy is contained |
| SNR | Ratio of signal power to noise power at a point in the transmission; SNR_dB = 10 $\log_{10}(S/N)$ |
| Free space loss | Attenuation because the signal spreads over an ever-larger sphere with distance: Pt/Pr = $(4\pi d/\lambda)^2$ |
| Path loss exponent | Parameter describing how fast received power decreases with distance in an environment (n = 2 in free space) |
| Reflection | Occurs when a wave meets a surface **large** relative to its wavelength |

| Term | Definition |
|---|---|
| Diffraction | Occurs at the **edge** of an impenetrable body large compared to the wavelength; secondary waves reach shadowed areas |
| Scattering | Occurs when obstacles are **on the order of the wavelength or smaller**; the signal splits into many weaker signals |
| Multipath | Multiple copies of a signal with varying delays received due to reflection/diffraction/scattering |
| ISI | Delayed copies of a pulse arrive with the primary pulse of a subsequent bit |
| Fading | Variation in received signal strength over time/position due to multipath, shadowing and movement |
| Flat vs frequency-selective fading | Flat: all frequency components fluctuate in the same proportion (Bc ≫ Bs). Selective: only part of the band is attenuated |
| Fast vs slow fading | Fast: coherence time not much greater than bit time. Slow: Tc ≫ Tb |
| FEC | Receiver corrects errors using only information in the received transmission (redundant bits) |
| Diversity | Providing multiple independent copies of the signal, because channels fade independently |
| MIMO | Multiple antennas at transmitter and receiver, used for diversity, beamforming, MU-MIMO, multilayer transmission |
| AMC | Dynamically adjusting modulation and coding rate to current channel conditions |
| Multiple access | Many users sharing finite spectrum simultaneously and efficiently, to maximise capacity |
| FDMA / TDMA / CDMA / SDMA | Divide by frequency (+ guard bands) / time slots / codes over the whole band / space (beams, sectors, reuse) |
| Spread spectrum | Expanding bandwidth B to Bss ≫ B with a pseudorandom code independent of the signal, for privacy and anti-jamming |
| FHSS | Carrier hops among M frequencies selected by a pseudorandom k-bit pattern each hop period |
| DSSS | Each data bit is replaced by n chips; chip rate = n × bit rate |
| OFDM | Multicarrier modulation: data split into N lower-rate streams on orthogonal subcarriers, all from one source |
| OFDMA | FDMA + TDMA on OFDM: users get subsets of subcarriers (subchannels) at different times |
| SC-FDMA | OFDMA-like scheme with an extra DFT; lower PAPR; LTE uplink |
| Single-bit / burst error | One bit changed / two or more bits changed; burst length from first to last bad bit |
| BSS / ESS | Stations + optional AP / two or more BSSs with APs joined by a distribution system |
| CSMA/CA | Sense, wait DIFS, random backoff, optional RTS/CTS, data, ACK |
| NAV | Timer from RTS/CTS durations telling stations how long to wait before checking the channel |
| Hidden station | Two stations out of each other's range both transmit to a common receiver, causing a collision there |
| Exposed station | A station refrains from transmitting although its transmission would not cause a collision |
| Piconet / scatternet | 1 master + ≤ 7 active slaves / overlapping piconets sharing area and bandwidth |

| Term | Definition |
|---|---|
| Profile | A set of protocols implementing a particular Bluetooth application; a standard of interoperability |
| Routing / forwarding / switching | Network-wide path selection / router-local input→output move by destination IP / layer-2 frame move by MAC address |
| Count-to-infinity | DV routing loop where estimates rise slowly after bad news ("bad news travels slowly") |

# C. Formulas and procedures

## C1. Formula sheet

| Formula | Use when |
|---|---|
| **$T = 1/f$; $\lambda = c/f$** ($c = 3 \times 10^8$ m/s) | Period, wavelength |
| **$C = 2B \log_2 M$** (Nyquist) | Max bit rate, **noiseless** channel; or find M |
| **$C = B \log_2(1 + SNR)$** (Shannon) | Max bit rate, **noisy** channel. **SNR as a ratio** |
| **$SNR_{dB} = 10 \log_{10}(S/N)$; S/N = 10^(dB/10)** | dB conversion |
| **$Pt/Pr = (4\pi d/\lambda)^2 = (4\pi fd/c)^2$** | Free-space loss, power, distance, frequency |
| **$Pr = Pt/(4\pi d/\lambda)^2$; $d = (\lambda/4\pi)\sqrt{(Pt/Pr)}$; $f = c\sqrt{Lp}/(4\pi d)$** | Rearrangements |
| **$L_{dB} = 20 \log f + 20 \log d - 147.56$** | Free-space loss in dB (f in Hz, d in m) |
| **SNR = Pr/Pn; Lp = Pt/Pr** | WCP equations 2.2–2.3 |
| **$Tc = 0.423c/(vf)$** | Coherence time |
| **$N = (Bt - 2Bg)/Bc$** | FDMA channels |
| **Users = channels × slots per channel** | TDMA users |
| **$S = N/r$, $r = \log_2 L$** | Baud vs bit rate |
| **$f_i = fc + (2i - 1 - M)fd$** | MFSK frequencies |
| **2^r ≥ m + r + 1** | Hamming check bits |
| **P(collision) = 1 − (1 − 1/F)^(2k)** | Scatternet: F frequencies, k other piconets, 2 slots per poll+reply (k = 1, F = 80 → 2.5%) |
| **$r_i / \bar{r}_i$** | Proportional fairness metric |
| **$d_x(y) = \min_v\{c(x,v) + d_v(y)\}$** | Bellman-Ford |

## C2. Procedures (step lists)

### Hamming (even parity, 8 data bits)

1. Inequality: r = 4 (16 ≥ 13).
2. Positions 1–12; P at 1, 2, 4, 8; data in 3, 5, 6, 7, 9, 10, 11, 12 left to right.
3. P1: 3, 5, 7, 9, 11 · P2: 3, 6, 7, 10, 11 · P4: 5, 6, 7, 12 · P8: 9, 10, 11, 12. Each = XOR of its bits (make the group's 1-count even).
4. Write the 12-bit codeword. To correct: recompute checks, read C8C4C2C1 as the bad-bit position.

### CRC

1. G bits from the polynomial (degree r = length − 1).
2. Append r zeros.
3. XOR-divide: whenever the leading bit is 1, XOR with G; otherwise bring down.
4. The last r bits = remainder = CRC; transmit data + CRC.
5. Receiver: divide the whole frame; remainder 0 = OK, non-zero = error.

**Internet checksum (k-bit words)**

1. Add words one at a time; if a carry exits the top bit, **wrap it around** (add 1).
2. Checksum = **complement** of the final sum.
3. Receiver: add everything including the checksum → **all 1s** = OK.

**Dijkstra**

1. Table: Step | N' | D(n),p(n) for every node.
2. Initialise with the source's neighbours; others ∞.
3. Repeat: add the min-D node to N'; relax its neighbours (D = min(D, D(w) + c)).
4. Show every row; mark settled nodes "—".
5. Finish with a table of cost + full path (+ first hop). **Mention ties.**

**Free-space problems**

1. Convert units. 2. λ = c/f. 3. Choose the rearrangement. 4. Compute carefully (square root before dividing). 5. Units and a sanity check.

## D. Frequently repeated questions: exam-ready short answers

**D1. Profiles as an advantage (asked 4×).** Interoperability across vendors (the stated purpose of profiles); a ready-made recipe of mandatory/optional features for each usage model; reuse of protocols and other profiles (e.g. File Transfer uses OBEX/RFCOMM/L2CAP + Generic Access); devices implement only the profiles they need (cheap, low power); consistent discovery, connection and security via the Generic Access Profile and SDP.

**D2. Path loss vs multipath vs interference (3×).** Path loss: deterministic power decrease with distance and frequency, $(4\pi d/\lambda)^2$; fix with power, antennas, more APs. Multipath: own signal arrives via reflection/diffraction/scattering with different delays; random fading and ISI; fix with equalization, diversity, OFDM. Interference: other transmitters (co-/adjacent-channel, microwave ovens); fix with channel planning, spread spectrum, directional antennas.

**D3. FHSS and DSSS (3×).** SS expands B → Bss ≫ B with a pseudorandom code independent of the signal. FHSS: PN generator → frequency table → synthesizer; the carrier hops among $M = 2^k$ frequencies every hop period; a jammer hits only one hop. DSSS: each bit → n chips (Barker 11: 10110111000); bandwidth × n; privacy and interference immunity.

**D4. CDMA and jamming (3×).** All users share the whole band all the time, separated by unique pseudorandom codes at the chip rate; the correlator despreads the wanted signal; others stay as noise; power control. Hard to jam: wide band and low power density; despreading spreads and filters out the jammer; the code is unknown; designed for anti-jamming (WWII).

**D5. Two ISPs on channel 11 / channels 1 and 11 (3×).** Ch 11: no breakdown; one shared collision domain; CSMA/CA makes them defer to each other; simultaneous transmissions collide, no ACK, backoff and retransmit; each gets about half the throughput. Ch 1 + 11: non-overlapping, independent; simultaneous transmissions succeed; each gets the full channel.

**D6. CRC (3×).** See procedure C2. Past answers: 101110/1001 → **011**; 110101/1011 → **111**; 10011010/1101 → **101** (leftmost flip → remainder 101 ≠ 0).

**D7. Master of two piconets (3×).** No: the master's address and clock define the hop sequence and timing, so both would be the same piconet. A device can be slave/slave or master/slave.

**D8. Scatternet probability (3×).** 1/80 per slot; poll + reply need 2 clean slots: $1 - (79/80)^2 \approx$ **2.5%**.

**D9. Master vs AP (3×).** Master polls (no contention), sets the hop sequence, ≤ 7 slaves, relays inside the PAN. AP: stations contend with CSMA/CA (PCF optional), and the AP bridges to the wired network / other networks.

**D10. Three propagation mechanisms (2×).** Reflection (large surfaces; ground reflection may cancel LOS); diffraction (sharp edges; reaches shadow areas); scattering (small objects; many weak signals, unpredictable). Without LOS, diffraction and scattering dominate.

**D11. OFDMA approaches (2×).** Adjacent: equal SINR, choose the best block per user (needs full CSI). Regular: frequency diversity, low estimation burden. Random: diversity + reduced adjacent-cell interference.

**D12. Hidden node, Figure 2 (2×).** A and C can't hear each other but both reach B; both sense idle, transmit, collide at B; undetected; no ACKs; backoff and retransmit; capacity lost.

**D13. CSMA/CA, Figure 2 (2×).** A: NAV/sense → DIFS → backoff → RTS → B: CTS (C hears it, sets NAV) → A: DATA → B: ACK (each after SIFS). No CTS/ACK → backoff and retry.

**D14. ACKs (2×).** Ethernet detects collisions; no collision = implicit success. 802.11 can't detect collisions (half duplex, hidden stations, fading) and has high error rates, so it needs explicit ACKs.

**D15. Mobile node moving away (2×).** Increase transmit power; lower the rate / more robust modulation and coding (AMC).

**D16. Interference troubleshooting (2×).** Characterise → spectrum survey → locate the source → remove/relocate/shield → change channel/band → better antenna placement, directional antennas → raise SINR → robust techniques (SS, FEC, AMC) → more APs → verify.

**D17. Hamming (2×).** 10101111 → **101001001111**; 11001010 → **001110001010**.

**D18. Checksum (2×).** 1001 1100 1010 0011 → **1011**; 1010 1100 0110 1001 → **1000** (receiver sum 1111).

**D19. Parity (2×).** Detects all single-bit (odd) errors; misses even-numbered errors (bursts); cannot correct.

**D20. MANET categories (2×).** Proactive/table-driven (DSDV, OLSR); reactive/on-demand (AODV, DSR); hybrid (ZRP). Hard because of mobility, no infrastructure, bandwidth, energy, unreliable links.

**D21. Dijkstra Figure 1 (2×).** From x: v3 u6 w6 y6 t7 z8. From t: u2 v4 w5 ×7 y7 z15. From w: u3 v4 t5 ×6 y12 z14.

**D22. Count-to-infinity (2×).** DV loop after bad news, slowly counting up. A cost decrease or a new link: no, because good news travels fast.

**D23. Hidden terminal in Bluetooth (2×).** No within a piconet: master polling, slaves transmit only when addressed.

## E. Common mistakes (and how to avoid them)

1. **Putting dB into Shannon.** Convert first: 24 dB → 251, not 24.
2. **Mixing Pt in watts with path loss.** Lp = Pt/Pr is a **ratio** (e.g. $125 \times 10^3$), not $125 \times 10^{-3}$ (your notes contain this exact slip).

3. **Forgetting to square-root** when solving $(4\pi d/\lambda)^2 = Lp$ for d, or to **square** when finding Lp.
4. **Units**: mW → $10^{-3}$, µW → $10^{-6}$, MHz → $10^{6}$, km → $10^{3}$. Write them every time.
5. **CRC: appending the wrong number of zeros.** Append **degree** = (number of bits in G) − 1, not the number of bits in G.
6. **CRC: doing ordinary subtraction.** It is **XOR**: no borrows.
7. **CRC: writing only the remainder** when asked for the transmitted string. Give **data + remainder**.
8. **Checksum: forgetting the end-around carry**, or forgetting to **complement** the sum.
9. **Hamming: using 3 check bits for 8 data bits.** Always test the inequality: $2^3 = 8 < 12$.
10. **Hamming: mixing up the coverage sets.** P4 covers 4–7 and 12; P8 covers 8–12.
11. **Parity: saying it "corrects" errors or detects all bursts.** It detects only odd numbers of errors and corrects none.
12. **Dijkstra: skipping rows or not showing predecessors.** Marks are for the step-by-step table. Remember G = 5 via E (not 6 direct) in T2-2026, and the **ties** (y from w; C from A).
13. **Dijkstra: forgetting to update** an already-labelled node when a cheaper path appears.
14. **Count-to-infinity: saying a cost decrease causes it.** It doesn't: good news travels fast.
15. **Hidden node: describing the exposed node instead** (or claiming RTS/CTS solves exposed stations).
16. **CSMA/CA: putting DIFS between RTS and CTS.** SIFS separates RTS–CTS–DATA–ACK; DIFS comes before contention.
17. **Two ISPs: saying 802.11 "breaks down".** It doesn't; the channel is shared and collisions increase.
18. **Bluetooth master: claiming it can master two piconets**, or that there are 8 slaves (it's 7 active slaves + 1 master).
19. **Scatternet probability: giving only 1/80.** The question stresses poll **and** reply, so use $1 - (79/80)^2$, and you may mention 1/80 per packet.
20. **OFDM vs OFDMA: treating them as the same.** OFDM = modulation (one user); OFDMA = multiple access (many users).
21. **Saying 4G uses CDMA** (as one table in the notes does). LTE uses **OFDMA/SC-FDMA**.
22. **Three propagation mechanisms: listing "free space, transmission...".** In a **mobile system** the three are **reflection, diffraction, scattering**.

## F. Examination answering strategies

1. **Read the mark value and budget**: ~1 developed point per 1–2 marks; ~1 minute per mark (Test 2 2026: 50 marks in 60 minutes).
2. **Definition first**: open long answers with a one-sentence definition. It earns a mark and frames the answer.
3. **Use the lecturer's keywords**: "guard band", "chip rate", "pseudorandom", "NAV", "SIFS/DIFS", "piconet", "FH-TDD-TDMA", "orthogonal", "coherence time". Markers look for them.
4. **Tables for "compare/differences"**: 3–5 rows, then one sentence of conclusion.
5. **Name the nodes in scenarios**: "A sends RTS to B; C hears B's CTS and sets its NAV...". Don't just recite generic theory.
6. **Yes/No questions**: state Yes or No in the **first word**, then justify with the mechanism.
7. **Calculations**: write the **formula**, **substitute with units**, show **every** XOR/addition/table row, **box the final answer**, and add a one-line **check** (e.g. remainder 000; receiver sum 1111). State conventions (Hamming positions; tie-breaking).
8. **Dijkstra**: draw the full table; then a final table of cost + path. Mention equal-cost alternatives.

9. **Long essays (10–12 marks)**: intro → 4–6 headed points, each explained with "because..." → short conclusion.
10. **Answer the easy repeated questions first** (profiles, ISP, CRC, Dijkstra) to bank marks, then return to the rest.
11. **Never leave a calculation blank**: the method marks are available even if the arithmetic slips.

## G. Final revision checklist

Tick each item when you can do it **from memory**, without notes.

### Transmission and channel

- ☐ T = 1/f, λ = c/f; analog vs digital signals; amplifier vs repeater
- ☐ Nyquist and Shannon; the 3–4 MHz, 24 dB → 8 Mbps → M = 16 example
- ☐ Ground / sky / LOS propagation modes with frequency ranges
- ☐ Five propagation mechanisms; **three mobile mechanisms explained**
- ☐ Free-space loss formula; **solve Test 1A Q14 (9.93 m)** and the WCP problems
- ☐ Path loss exponent table; four noise types
- ☐ Multipath effects; fading types (fast/slow, flat/selective, large/small scale)
- ☐ **Path loss vs multipath vs interference table**
- ☐ Ten wireless problems; **interference troubleshooting steps**; co-channel interference

### Correction, capacity and encoding

- ☐ FEC vs ARQ; why ARQ is impractical; equalization; diversity types
- ☐ Four uses of MIMO; **SU-MIMO vs MU-MIMO**
- ☐ AMC; three bandwidth-expansion approaches; **three capacity methods**
- ☐ ASK/FSK/PSK/QPSK/QAM; baud vs bit rate; BER curves

### Multiple access, SS, OFDM

- ☐ FDMA/TDMA/CDMA/SDMA: principle, pros, cons; **comparison table**; formulas
- ☐ **CDMA + why hard to jam**
- ☐ **FHSS and DSSS principles**; Barker code; FH-CDMA vs DS-CDMA
- ☐ OFDM orthogonality and benefits; **OFDM vs OFDMA**; **three OFDMA approaches**; proportional fairness; SC-FDMA
- ☐ 1G–5G key facts

### Error coding

- ☐ Single-bit vs burst; burst length; **parity capabilities and limits**
- ☐ **Hamming** (inequality, positions, coverage, codeword, syndrome)
- ☐ **CRC** (append, XOR division, receiver check, polynomial ↔ bits)
- ☐ **Internet checksum** (end-around carry, complement, receiver 1111)

### WLAN

- ☐ BSS/ESS/AP/SSID; infrastructure vs ad hoc; association
- ☐ Why no CSMA/CD (3 reasons); **CSMA/CA steps**; IFS; NAV; RTS/CTS
- ☐ **Hidden and exposed station** (with Figure 2); **ACKs**; PCF

- ☐ **Two ISPs (channel 11; channels 1 and 11)**; wired vs wireless LAN; design concerns
- ☐ 802.11 PHY variants (a/b/g/n)

**Bluetooth / 802.15**

- ☐ Stack layers and roles; RFCOMM, TCS BIN, adopted protocols
- ☐ **Profiles advantage**; piconet/scatternet; radio parameters; FH-TDD-TDMA
- ☐ SCO/eSCO/ACL; packet fields; 1/3 FEC, 2/3 FEC, ARQ
- ☐ **Master vs AP; two piconets; hidden terminal in BT; scatternet probability**
- ☐ BT 3.0 HS, BLE; 802.15.3 (PNC), 802.15.4, ZigBee devices and AODV

**Routing (supplementary)**

- ☐ Routing vs forwarding vs switching
- ☐ **Dijkstra tables** (Figure 1 from x, t, w, u; T2-2026 from A)
- ☐ LS vs DV; Dijkstra vs Bellman-Ford; Bellman-Ford equation
- ☐ **Count-to-infinity**, plus the cost-decrease and new-link cases
- ☐ Why MANET routing is hard; **proactive/reactive/hybrid with examples**

**Good luck. If you can tick every item, you can answer every question that has appeared in your past papers.**